JMP® Essentials

An Illustrated Guide for New Users

Second Edition

Curt Hinrichs and Chuck Boiler

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JMP® Essentials: An Illustrated Step-by-Step Guide for New Users, Second Edition

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Preface

JMP Essentials was written for the new user of JMP software who needs to get the right results right away. If you have data and problems to solve, JMP will help you make sense of them to gain understanding and arrive at a good decision. It is the goal of this book to help you with this task.

We often find new users of JMP simply trying to complete a specific task with their data. Perhaps you just need to generate a certain graph for a Microsoft PowerPoint presentation or to quickly see how patterns in your data will lead you to an important discovery. If these scenarios sound close to home, you've come to the right place. This book is task-oriented and will help you access your data, identify that graph or statistic you need, and quickly and easily create and share it with others.

JMP software is built around the workflow of the problem solver. One of its outstanding features is that it consistently provides the correct graphs and statistics for the data you're working with (something we refer to as JMP's smart interface). JMP will lead you down the right path to the right result, provided your data is properly classified and you have an idea of what questions you're trying to answer.

JMP is easy to use. In most cases, generating a graph or result will take you seconds, maybe minutes, to complete but never hours or days. Much of this efficiency is due not only to the small number of steps required but to the ability to navigate intuitively toward the right solution quickly, rather than through repetitive and time-consuming trial-and-error. This book provides you with the essential knowledge to get to your solution even faster.

Though JMP contains state-of-the-art visualization and advanced statistical and data mining tools, we also believe it is the right tool for the user who might be less confident in his or her analytic abilities. While there is a time and place for every feature in JMP, we have tried to include only those topics that a typical new user—a manager or analyst, for example—would need. If it's been a while since you've studied statistics or dealt with its terminology, do not worry. We will present the key ideas and fill in the gaps as needed.

Audience

While this second edition of *JMP Essentials* is designed for new users, we have also focused on the most fundamental and commonly used features of JMP so that it may also serve as a valuable reference to the occasional user too. This focus is what we like to think of as the 20% of JMP that is used 80% of the time. Throughout the book, we have provided the necessary instruction to generate results quickly. Each key step in this process is illustrated with screenshots to help you see the result and develop your confidence in using JMP. We don't assume any formal background in statistics. Instead we emphasize the intuition of concepts over statistical theory. If you require deeper statistical understanding, we recommend the documentation included in JMP and some excellent textbooks in the Bibliography.

Most new users of JMP have one or more of the following distinct needs:

1. They need to get their data into an appropriate and structured format so it may be effectively analyzed or visualized. (See Chapter 2).
2. They have a good idea of what graph they need and recognize it by its illustration or name and simply want to create it (see Chapter 3).
3. They really don't know what the data will say, and need help exploring or summarizing it (see Chapter 4).
4. They need to interpret or answer some specific questions about the data they have (see Chapters 5 and 6).

We also believe that the complete book is well suited as a reference guide to the following groups of users:

- **Spreadsheet users** who are looking for a convenient way to produce nice visualizations of their data or to supplement a spreadsheet's statistical capabilities. JMP reads and writes data from a variety of programs including Microsoft Excel. This book provides a quick and easy way to make your spreadsheet data come alive and allows you to fully and interactively explore that data.
- **Students enrolled in introductory statistics courses** who may need JMP instruction. JMP is the ideal tool for students because its navigation reinforces the basic assumptions taught in an introductory course. This book provides an overview of the JMP tools needed in most first-year courses.
- **SAS users** who want to take advantage of JMP's data visualization tools. We've provided Appendix A to illustrate the features of how JMP integrates with SAS.

Approach and Features

We have found that the best way to learn JMP is by using it and getting value out of it quickly. Our goal is to present the materials in this book in the most user-oriented approach possible. So, we have made every effort to organize the presentation around the new user's common needs and questions and the most direct and concise means to answer them. We also recognize that the most basic use of data is in generating graphs of data rather than performing more complex statistical analyses. The following features are included for this purpose:

- We present the material with a show-and-tell approach. In most cases, we show you what the results look like alongside the conditions and steps required to produce them. We think this approach is especially useful for JMP users who have a good idea of what they want from JMP and just need the steps to create it.
- When appropriate, we provide an example-driven context for each JMP platform that explains its use, value, and general application to problems. We have tried to distill these contexts down to typical or easy-to-understand cases.
- We organize the contents into easily manageable chunks of information. While the entire book is designed to cover a fairly complete overview of the basics, each chapter represents one family of tasks (such as importing data, creating graphs, and sharing graphs).
- We hope you will keep this book near your computer. Within each chapter or tab, we have designed each page—whenever possible—to be self-contained (beginning and ending a task in a page or page spread), allowing you to quickly find and execute the required steps you're looking for without having to flip pages.
- No matter what your professional background, this book assumes only that you have a basic working knowledge of Microsoft Windows. Virtually all of the information in this book applies to using JMP on the Macintosh operating system, but only the Windows version of JMP is used in the examples.

Software Used to Develop This Content

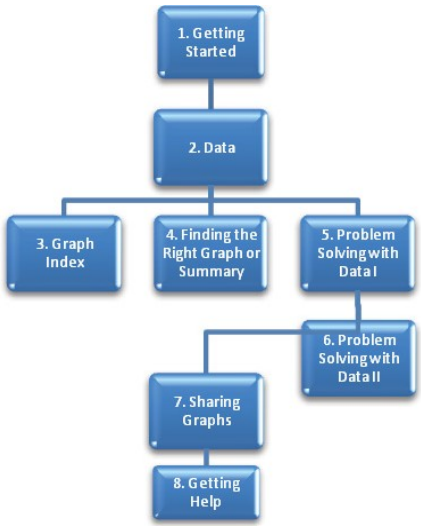
This edition of JMP Essentials was developed with JMP 11 and pre-release versions of JMP 12. Users of JMP Pro, JMP Student Edition or earlier releases of JMP will find nearly all of the instructions in this book suitable for their needs.

Organization

This book is designed like a cookbook. Find what you need, and follow the steps. We have organized the contents of this book to reflect both the process of analyzing data (getting data, analyzing it, and sharing the results) and the progression from the very basic features in JMP to more specialized ones. We hope this organization offers the most value to the reader. Much of our judgment in this regard comes from our experience working one-on-one with new JMP users.

- Chapter 1 covers the preliminary material you'll need for the rest of the book. The chapter identifies the conventions we use and introduces you to JMP menus, windows, and preferences.
- Chapter 2 covers the first step in any analysis: getting your data into JMP. With the exception of some material in Chapter 2, other chapters are self-contained, and you can read them in any order.
- Chapters 3 through 6 cover graphing and analysis:
  - Chapter 3 is for the user who knows what graph he or she wants.
  - Chapter 4 is for the user who does not know what the data says and needs to explore it to find an appropriate graph or summary. This chapter introduces maps.
  - Chapters 5 and 6 are for the user who needs to solve a problem and answer questions using analytics and graphs.
- Chapter 7 covers topics related to sharing your graphs or results in a presentation, document, or through a browser.
- Chapter 8 covers additional resources that are available within JMP, online, and from outside resources, such as training, books, and user groups.

Figure 1 Chapter Organization



New to the Second Edition

It is gratifying to hear that the first edition has been useful to many new users of JMP. Our goal in writing this second edition is two-fold; first, to improve upon the core features of the first edition and to that end, we have made many refinements, corrections and additions throughout the book based on feedback from users, colleagues and reviewers. Secondly, in the 4 years since the first edition was published, JMP has added several important new features that we believe are *Essential* to the new user. While the first edition still works fine for most of its content, JMP has developed many new and useful features for the new or occasional users that we thought should be included. We have added 8 new sections to the book in addition to many subsections:

- **Elements of Using JMP** (Section 1.3). This new section provides an overview of the 4 common components to most JMP analyses.
- **The Excel Add-In** (Section 1.6). The JMP add-in for Excel is an optional means to launch common JMP platforms from within Microsoft Excel environment.
- **Filtering Data** (Section 2.5). This section covers the important subject of drilling down and filtering data with the concept of Row States.
- **Shape Files & Background Maps** (Section 2.7). Creating maps is easy when you understand a little bit about how the underlying data works. This section covers the essentials of the special data files and columns you need to create them.
- **Creating Maps** (Section 3.2). Our section about **Interactive Maps** illustrates the steps required to create thematic maps and utilizing background maps
- **Using Graph Builder to Create Maps** (Section 4.2). This section illustrates the steps required to visualize data on street-level maps.
- **Informative Missing** (Section 6.2). We've thoroughly revised section 6.2 to introduce the important concept of Informative Missing which is now included in the Partition platform (and many other modeling platforms in JMP Pro). If your rows contain missing values, this feature can help you use these rows to yield better models.
- **Sharing Dynamic Graphs with HTML 5** (Section 7.4). HTML 5 is an emerging standard for the purpose of sharing interactive graphs with people who don't have JMP.
- **Combining Windows into Dashboards** (Section 7.7). One of the great features of JMP is its dynamically linked graphs. You can now combine several different windows/platforms into one window or dashboard. These are referred to as Instant Applications in JMP.

Authors' Website

We have created an authors' website for the book that provides updates, corrections, enhancements, and additional examples for new releases and offers a means to contact us with any suggestions or comments you have. We also provide links to helpful resources outlined in Chapter 8. Go to http://support.sas.com/publishing/authors/hinrichs.html and http://support.sas.com/publishing/authors/boiler.html for more information.

We'd love to hear from you.

About These Authors

Curt Hinrichs is Senior Manager of Marketing at SAS Institute Inc. Curt joined SAS in 2006 to develop and launch the JMP Academic Program that provides faculty, researchers, and students with easy access to JMP software and learning resources. Curt and his team are charged with developing JMP users among the next generation of data analytics professionals. They work directly with faculty, authors, and administrators and partner with leading academic societies, publishers, and service providers to support the effective use of JMP software in the classroom. Prior to joining SAS, Curt worked as an acquisitions editor for Thomson Learning, and later worked as Publisher and Vice President/Editor in Chief for its Mathematics and Statistics Publishing Group. He holds a bachelor's degree in economics from San Diego State University.

Chuck Boiler is the U.S. Systems Engineer Manager for JMP, a business unit of SAS. Since joining SAS in 1997, he has held management roles with JMP and helped develop solutions for conducting many types of analysis, including design of experiments for the semiconductor industry, quality control for pharmaceutical manufacturing, and marketing applications for survey analysis using JMP. Boiler currently works with customers and U.S. field engineering staff to help them solve business problems and discover hidden opportunities in data. Earlier in his career he worked for Abacus Concepts, makers of StatView software, as technical services manager and software quality assurance manager. A member of the American Society of Quality, Boiler received a bachelor's degree in education from the University of Oregon and has done graduate work in ancient philosophy at the Graduate Theological Union in Berkeley, California. He is a graduate of Gallup University's Great Manager Program.

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Acknowledgments

We'd first like to thank John Sall for creating a great product in JMP. Who knew that statistics could be so fun? If you are new to JMP, we hope it inspires you as it has us. Thanks also to Jon Weisz, Dave Richardson, and Diana Levey for supporting the idea for this second edition.

We have been very fortunate to work with the outstanding professionals at SAS Press. In particular, we'd like to thank Stephenie Joyner who in both the first and second editions, has been our principal source of advice and support throughout the development process. To Brenna Leath who helped us get to the essentials of what needed to be said. Thanks also to Julie Platt, Denise Jones, Robert Harris, Stacy Suggs, Dan Valente, and Shelley Sessoms for making it all work and look good, and to Cindy Puryear, Brenda Hairston and Lou Metzger for their encouragement and bringing this work to market.

The manuscript for this book was substantially improved by the insightful suggestions of our reviewers. The book contains many new features and refinements due to their input. Our reviewers include:

Chris Albright, Indiana University
Mark Bailey, SAS Institute
Peter Bruce, Statistics.com
Laura Higgins, SAS Institute
Bob Lamphier, SAS Institute
Paul Marovich, SAS Institute
Gail Massari, SAS Institute
Tonya Mauldin, SAS Institute
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Di Michelson, SAS Institute
Chris Olsen, Grinnell College
Jeff Perkinson, SAS Institute
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Heath Rushing, SAS Institute
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Sue Walsh, SAS Institute
Scott Wise, SAS Institute
Annie Zangi, SAS Institute
Richard Zink, SAS Institute

We also wish to thank Sam Savage of Stanford University for testing the first edition in his class, and David Shultz and Mary Loveless for using the book with customer training.

This book began with a desire to help the new JMP user and evolved into a labor of love. But without the love and incredible support of our families, friends and colleagues this project would have never materialized. Thank you!

Curt Hinrichs
Chuck Boiler

San Francisco, CA October 2014

Chapter 1 Getting Started

1.1 Using JMP Essentials

1.2 Launching JMP

1.3 JMP Menus

1.4 Elements of using JMP

1.5 JMP Launch Dialog Windows

1.6 The Excel Add-In (Optional)

1.7 JMP Preferences

1.8 Summary

JMP was developed to help people with questions about their data get the answers they need through the use of graphs and numerical results. For most people, memories of statistics can be a very unpleasant, if not forgotten, part of their education. If you see yourself as a new, occasional, or even reluctant user of data analysis, we want you to know that we have written this book for you.

It is important to note that throughout the historical development of statistics as a scientific discipline, people had real problems they needed to solve and developed statistical techniques to help solve them. Statistics can be thought of as sophisticated common sense, and JMP takes a practical, common sense approach to solving data-driven problems.

JMP was designed around the workflow of analyzing data rather than as a collection of tools only a statistician can understand. When you think about your data analysis problem, try to formulate the questions that might help you address it. For example, do you need to describe the variation in selling prices of homes in a city or understand the relationship of customer satisfaction with service waiting times? With this mindset, you will find the menus and navigation in JMP to be very compatible with the types of questions you are trying to answer.

Displaying graphs (or pictures) of data is one of JMP's strengths. For most people, an effective graph can convey more information more quickly than a table of numbers or statistics. In any JMP analysis, graphs are presented first and then the appropriate numerical results follow. This is by design. JMP also provides a **Graph** menu that contains additional visualization tools that are independent of numerical results. The goal of this chapter is to introduce you to JMP and its basic

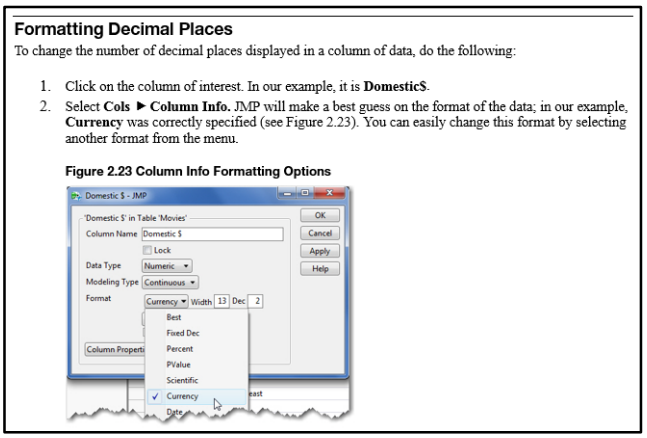
navigation. We cover the menus and windows and introduce you to the conventions used throughout the book.

1.1 Using JMP Essentials

All but one chapter in this book (Chapter 3, "Index of Graphs") is laid out in a consistent manner to help you generate results quickly. The format of the book has been designed to be used alongside your computer where JMP is installed. After an introduction to the concept, we have designed each section to be self-contained. That is, with few exceptions, the steps required to produce a result begin and end without having to flip through several pages.

We provide numbered steps that generate the result illustrated in the figure that follows (see Figure 1.1).

Figure 1.1 Book Layout



Note: This edition of JMP Essentials was written with JMP 11 and pre-release versions of JMP 12. However, the methods covered in this book are mostly basic and have not substantially changed since the earliest releases of the software. Thus, you will find most instructions contained in this book compatible with earlier and future JMP releases.

Conventions

We are confident that, having made it this far, you know the basic terminology associated with operating a computer, including click, right-click, double-click, drag, select, copy, and paste. We use these terms and they appear in numbered steps (see Figure 1.2). When there is a single or self-evident step, these instructions are included in the body of the text. Each step or action appears in bold type.

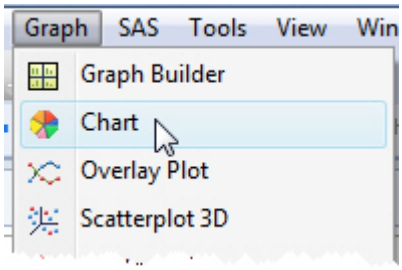
Figure 1.2 Selection Path Example

1. Select **File ▶ Open**.
   The **Big Class.xls** file, which is illustrated here, can be found by selecting **C: ▶ Program Files ▶ SAS ▶ JMP ▶ 11 ▶ Samples ▶ Import Data ▶ Big Class.xls**.
2. From the **Files of Type** drop-down menu, select **Excel Files**.
3. Select the file that you want, then select **Open** which will launch the Excel Import Wizard dialog with a view of your data. If it looks correctly structured, select **Import**.

In writing this book, we have adopted the same conventions contained in JMP documentation to ease your transition to using the documentation.

Menu items such as **Graph** are associated with a JMP command such as **Chart**. We use the greater than (▶) symbol to indicate the next step in an operation. Thus, **Graph ▶ Chart** indicates that you should select the **Chart** command (or platform) from the **Graph** menu (see Figure 1.3).

Figure 1.3 Menu Conventions



Book Features

Most chapters feature one or more examples to illustrate the procedures within that chapter (see Figure 1.4). All of the examples have corresponding data tables that are included in JMP's built-in Sample Data directory (**Help ▶ Sample Data**).

Figure 1.4 Data Table Description

Example 2.1 Big Class

We will be using the Big Class.jmp data file to illustrate the steps in this chapter. This data set consists of 40 middle-school students and their name, height, weight, gender, and age. You can access this data set in the Sample Data folder that is installed with JMP:

File ▶ Open ▶ C: ▶ Program Files ▶ SAS ▶ JMP ▶ 11 ▶ Samples ▶ Data ▶ Big Class.jmp

Important definitions are in bold for easy reference (see Figure 1.5).

Figure 1.5 Definitions

Data
refers to any values placed in the cell of a JMP data grid. Examples include numeric and/or text descriptions: 3.6, $2500, Female, Somewhat Likely, or 11/14/13.

Data type
refers to the nature of the data. The data type can be either numeric (numbers) or character (often words and letters but sometimes also numbers).

Modeling type
refers to how the data within a column should be used in an analysis or a graph. JMP uses three distinct modeling types: continuous, nominal, and ordinal.

We include notes, tips, and cautions where appropriate to point out relevant or important information (see Figure 1.6).

Figure 1.6 Note and Tips

Note: Once you've selected a new value, you can replace that value in the same column, create a new column with these values, or even create a formula column. Be careful! If you select **In Place**, these values cannot be changed because the Recode command replaces values in that column.

The appendices offer reference material including Appendix C (see Figure 1.7), a JMP 11 Quick Guide that provides essential menu steps to perform a specific analysis (if you know what you're after), Appendix B, a glossary of terms used in this book, and Appendix A, an introduction to using JMP and SAS together.

Figure 1.7 JMP Quick Guide

Task	Menu Selection
Adding Labels	*Click on column heading;* **Cols > Label/ Unlabel**
ANOVA -One Way	**Analyze > Fit Y by X; ▼ > Means/ Anova**
-Two or More Factors	**Analyze > Fit Model**
Bar Chart .	**Graph > Chart**

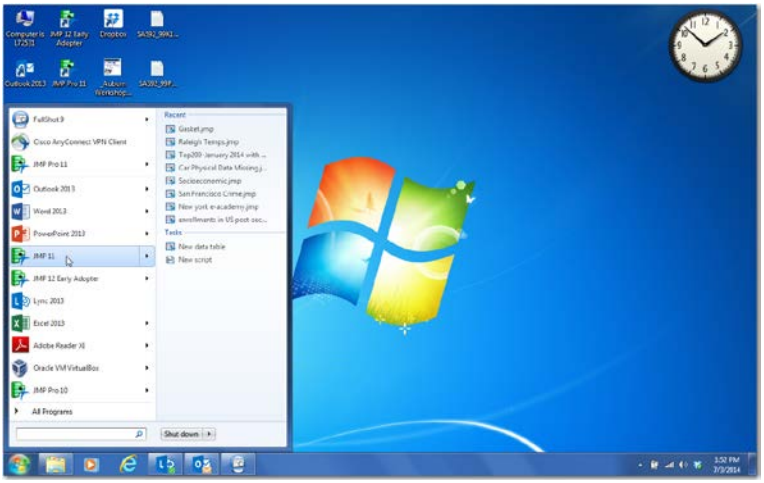
1.2 Launching JMP

Let's begin by launching JMP. To launch JMP from the Microsoft Windows Start menu:

1. Select the **Start** menu.
2. Select All Programs.
3. Select **JMP 11 ▶ JMP 11** (see Figure 1.8).

Note: Windows 8 users will begin with the Start Screen.

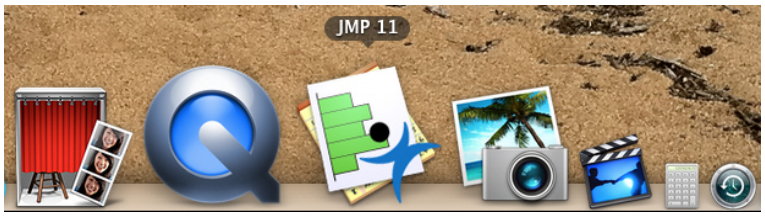
Figure 1.8 Opening JMP in Windows



Note: JMP is offered in two versions: JMP and JMP Pro. JMP Pro contains more advanced predictive modeling tools that are beyond the scope of this book. Thus, you will find the steps we cover in this book identical to both versions. The only minor exception is here: Select **Start ▶ All Programs ▶ JMP Pro 11 ▶ JMP Pro 11.**

Macintosh users can click on the JMP icon (see Figure 1.9) to launch JMP from the application dock. If the icon does not appear on the dock, select **Finder ▶ Applications ▶ JMP 11.**

Figure 1.9 Accessing JMP on the Mac

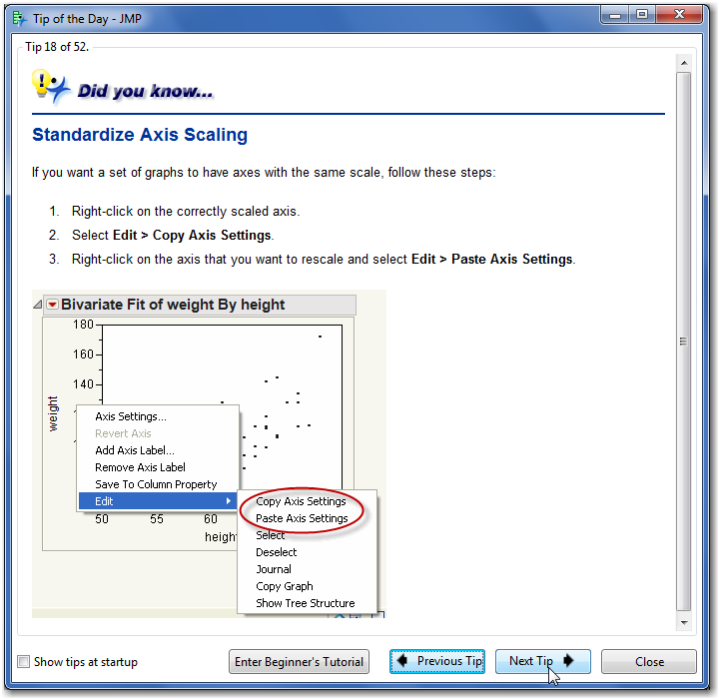


After JMP has launched, you might notice that two windows have also opened: Tip of the Day and JMP Home Window.

Tip of the Day

The Tip of the Day window is the first thing you see because it addresses the most common questions that new users ask, such as, “How do I do X?” Well, the X in these common questions is represented and answered in 52 different Tip of the Day windows. You can scroll through them by clicking **Next Tip** at the bottom of the window (Figure 1.10). Some of the Tip boxes contain important and basic navigational hints, while others only apply to more advanced features in JMP.

Figure 1.10 Tip of the Day



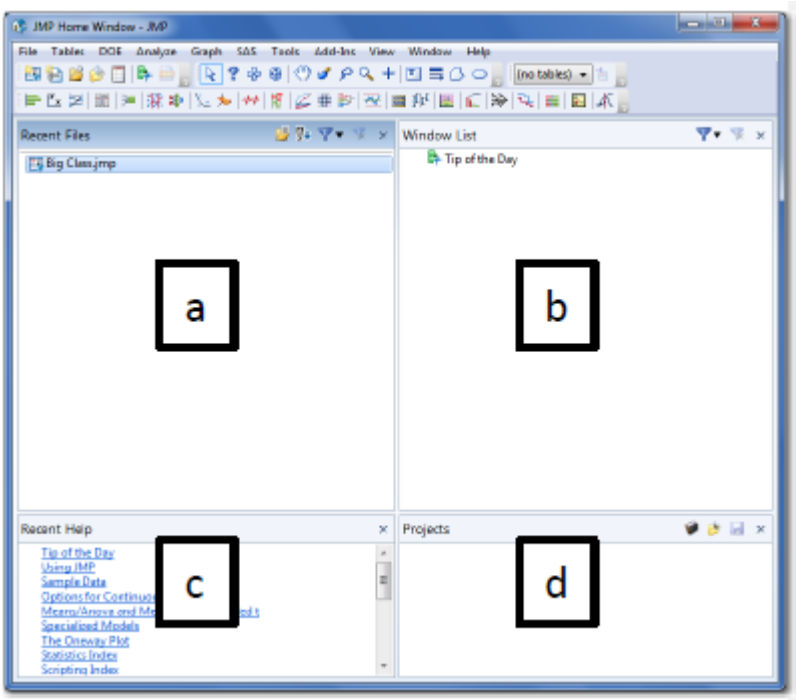
Note the **Enter Beginner’s Tutorial** button. This tutorial walks you through a basic analysis of data, from opening data tables to creating graphs and results. JMP contains several other tutorials that are directed toward more specific types of problems and are found in the **Help** menu.

Note: If you do not want to see the Tip of the Day window every time you launch JMP, you can simply uncheck the **Show tips at startup** box in the lower left corner of the window.

The JMP Home Window

When you launch JMP, the Home Window appears (Figure 1.11). The Home Window first appeared in Windows in JMP 9, but now also appears in a similar format in the Macintosh version, beginning in JMP 11. The Home Window organizes and helps you navigate data tables, documentation, and open files and any results that you've generated. If you tend to have several data tables and analyses running at the same time, the Home Window provides a convenient way to quickly navigate to what you want.

Figure 1.11 The JMP Home Window



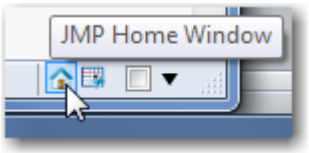
The Home Window is divided into four panels, which are:

a. The upper left panel contains recent files that you've accessed, listed from the most recently opened. If you are opening JMP for the first time, this panel should be blank.
b. The upper right panel titled "Window" contains a list of open data tables files and their associated results. In JMP, you can have any number of data tables and results open, but only one active data file may be analyzed at any one time. You may double click on any item in this panel to activate it and bring it to the forefront.
c. The lower left panel contains Recent Help. JMP includes extensive documentation built right into the software. This panel lists the documentation you've accessed with the most recent at the top.

d. The lower right panel lists projects that are a special type of JMP file that allow you to package a number files, slide decks, and so forth within a single file.

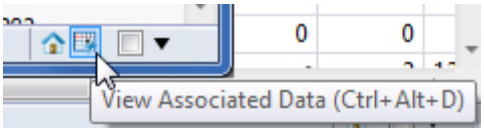
While the Home Window allows you to navigate directly to a file or result, each data table and results window also provides shortcuts back to the Home Window. At the lower right of each window, select the icon that looks like a house to return to the Home Window (Figure 1.12).

Figure 1.12 Shortcut Back to the Home Window



A results window has a second icon, which is also the same icon used to denote ".jmp" formatted files which we call JMP Data Tables. Click on a Data Table icon and you will be taken to the corresponding data table for that results window (Figure 1.13). Note that if you are looking at a data table, you will not see this second icon because you are already in the data table window.

Figure 1.13 Shortcut Back to the Data Table

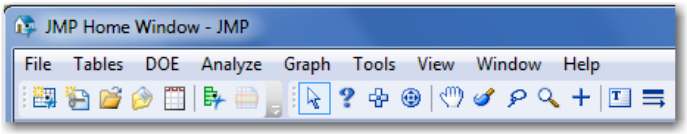


The check box with the down arrow button next to it allows you to combine multiple results windows or graphs into a single window or "dashboard". We will discuss creating dashboards in Chapter 7.

1.3 JMP Menus

At the top of the Home Window, you see a series of menus (**File, Edit, Tables,** and so on). These are the menus we use to illustrate the concepts in this book. They are also the same menus we refer to as JMP's native menus because they have been present in JMP since its first release.

These menus serve to open or import data, to edit or structure it, and to create graphs and analyses of your data. They are also a valuable source for assistance through the **Help** menu, which is discussed later. The menus are logically sequenced from left to right.



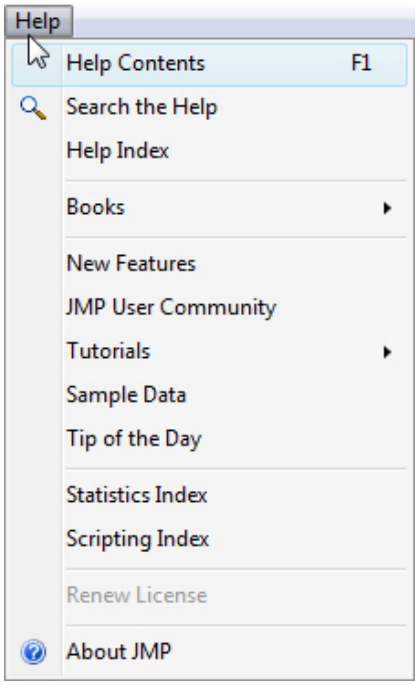
- **File** is where you go to open or import data and to save, print, or exit JMP. It is also where you can customize the appearance or settings within JMP through **Preferences** (explained in Section 1.5).
- **Edit** will appear when needed and provides the usual cut, clear, copy, paste, and select functions, as well as undo, redo, and special JMP functions.
- **Tables** provides the tools to manage, summarize, and structure your data (see Section 2.6).
- **DOE** contains the Design of Experiments tools, which we will not cover in this book. For more information, see **Help ▶ Books ▶ JMP Design of Experiments Guide**.
- **Analyze** contains the analysis tools that generate both graphs and statistics and serves as the home for all of JMP's statistical tools from simple to advanced (Chapters 5 & 6).
- **Graph** contains graph tools that are independent of statistics (at least initially). Graphs in this menu include basic charts to advanced multivariable and animated visualization tools and maps (Chapters 3 & 4).
- **Tools** allows you to transform your cursor into a help tool, a brushing tool, a selection or scrolling tool, and much more (Section 7.2).
- **View** provides options to control which windows, menus and toolbars are visible including the JMP Starter (Section 8.3).
- **Window** helps you manage windows within JMP.
- **Help** provides resources for learning and using JMP. Let's start with an introduction to the **Help** menu.

Note: Additional menu items including "Add-ins" and "SAS" may appear if and when you have tools of these types installed.

The Help Menu

The **Help** menu (see Figure 1.14) provides access to learning resources you can use as you expand your knowledge of JMP features, learn about statistics, and learn how to interpret results. These resources include searchable indexes, guided tutorials, tips of the day, and printable books including *Using JMP*. Data tables employed in this book and in all JMP documentation are included in the Sample Data directory. Chapter 8 covers the features of the **Help** menu in greater detail.

Figure 1.14 The Help Menu



JMP also features context-specific help, meaning that when you use the JMP Help Tool in any graph or statistical result, you are directed to the right spot in the documentation to assist you in understanding the result. For more information on the JMP Help Tool, see section 8.1. In statistical results, JMP provides Hover Help that reveals context-specific interpretation of statistical results. See Chapter 5 for more information.

Interpretation can be straightforward for descriptive graphs or basic summary statistics, but as you dig deeper into an analysis or employ more advanced methods, it is vitally important that you understand the meaning of the results, particularly when they are shared or presented. The documentation under Help ▶ Books includes over 4,300 pages of reference material in fourteen books that address the needs of professional statisticians and analysts. If you encounter results that you do not understand, however, we strongly recommend that you seek assistance from experienced data analysts.

The Analyze and Graph Menus

Because most graphs or statistical results begin with the **Analyze** and **Graph** menus, let's explore the structure within these two menus a little bit more.

Click on the **Analyze** menu at the top of the window. Glance at the choices on the menu. Top-down, the platforms are organized from the basic to more advanced tools. Next, click on the **Graph** menu at the top of the window. Glance at the graph choices. The menus in JMP–specifically the **Analyze** and **Graph** menus (see Figures 1.15a and 1.15b)–are designed to provide both a description and visual cues for analyzing, graphing, and exploring data.

Figure 1.15a The Analyze Menu

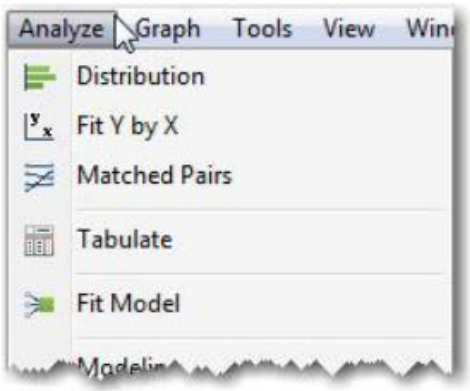
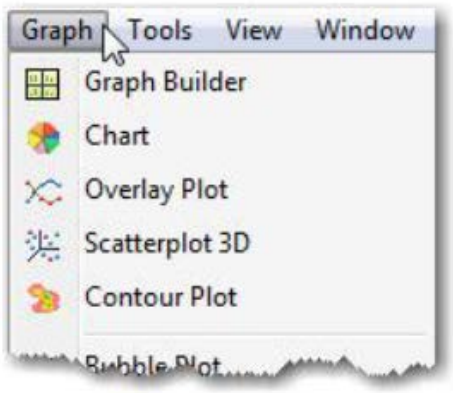
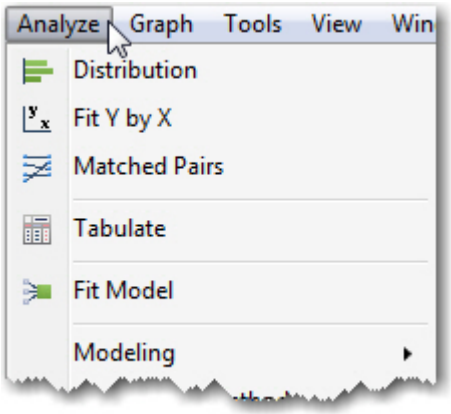


Figure 1.15b The Graph Menu



Note that each entry under these menus has both a name and an icon (on the Mac, the icons will not appear). The icons next to the **Graph** menu options give you a preview of each graph. From the **Analyze** menu, the icons depict the description or relationships you will see in graphs and statistical results (Figure 1.16).

Figure 1.16 Visual Cues Provided for Basic Analysis

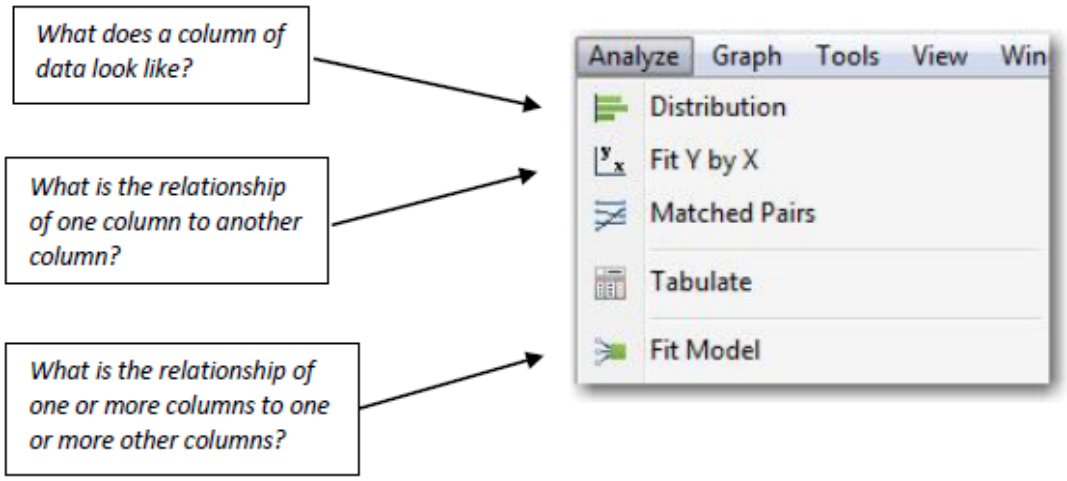


Note: The **Analyze** menu items produce both graphs and statistical results, while the **Graph** menu items produce only graphs.

Framework of the Analyze Menu

There is a problem-solving framework to the **Analyze** menu that we will discuss in detail in Chapter 5. As mentioned in the introduction, your exploratory objective will translate to these menu items. This structure streamlines the analysis process; in order to select the correct menu item, you only need to count how many columns you are interested in and know whether you are trying to describe, compare, or understand their relationship (see Figure 1.17).

Figure 1.17 Framework of the Analyze Menu



This framework cues you to the correct analysis choice on the menu without exposing you to many statistical terms until you need them. Make no mistake; you still get the statistics when you want them, but you do not have to know all the statistical terms or assumptions in order to access them.

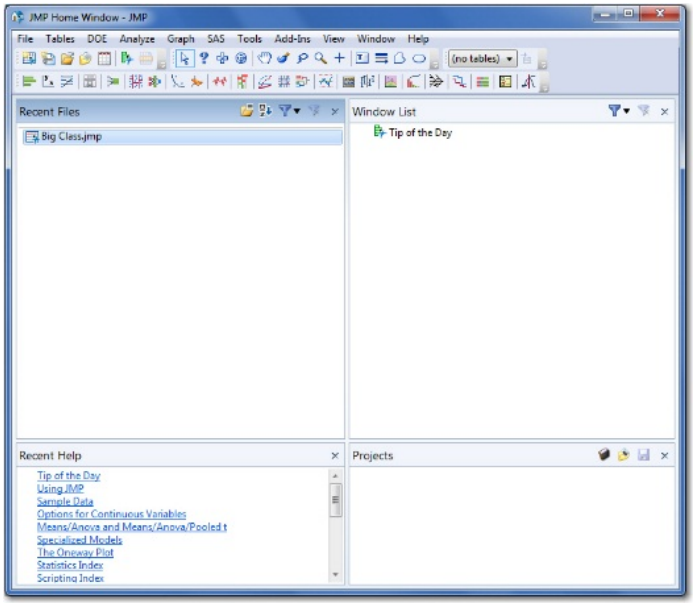
Note: JMP's **Analyze** menu contains terms such as Distribution and Fit Y by X that might be unfamiliar, but the ideas behind them are very straightforward. We describe them in simple terms as needed throughout the book. Many items under the **Analyze** and **Graph** menus are referred to as platforms or commands through this book. For example, Distribution and Fit Y by X are referred to as platforms.

1.4 Elements of using JMP

Before we launch JMP for the first time, let's look at the four common elements of a JMP analysis. All JMP analyses contain these and they follow a consistent process.

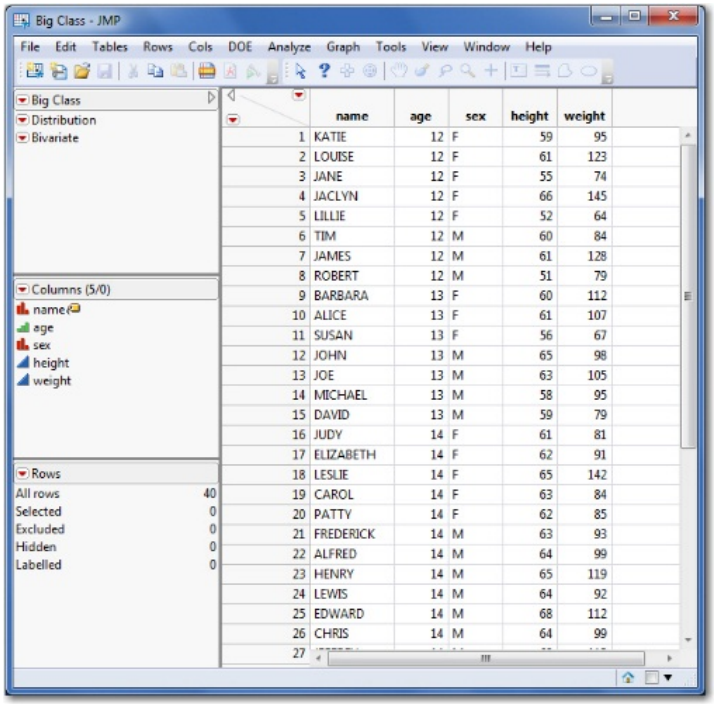
1. The first is the **JMP Home Window**, where you begin a JMP session (Figure 1.18). This is your mission control center. As described earlier in this chapter, from here you can open or create a data table or easily navigate between data tables, results, and help.

Figure 1.18 The JMP Home Window



2. The second element is a **Data Table** where your data reside, which you may have imported or opened through the Home Window (Figure 1.19). The data table is also where you will usually initiate an analysis or graph described next. We will cover the Data Table in Chapter 2.

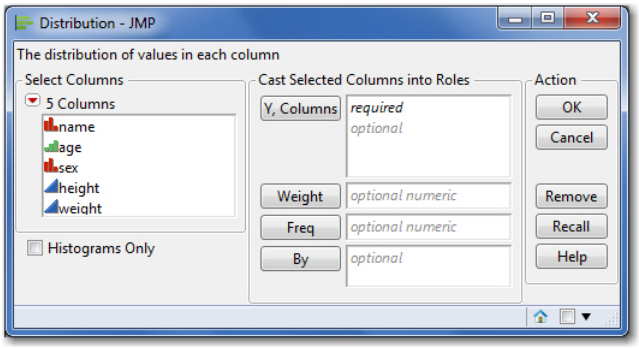
Figure 1.19 A JMP Data Table



	name	age	sex	height	weight
1	KATIE	12	F	59	95
2	LOUISE	12	F	61	123
3	JANE	12	F	55	74
4	JACLYN	12	F	66	145
5	LILLIE	12	F	52	64
6	TIM	12	M	60	84
7	JAMES	12	M	61	128
8	ROBERT	12	M	51	79
9	BARBARA	13	F	60	112
10	ALICE	13	F	61	107
11	SUSAN	13	F	56	67
12	JOHN	13	M	65	98
13	JOE	13	M	63	105
14	MICHAEL	13	M	58	95
15	DAVID	13	M	59	79
16	JUDY	14	F	61	81
17	ELIZABETH	14	F	62	91
18	LESLIE	14	F	65	142
19	CAROL	14	F	63	84
20	PATTY	14	F	62	85
21	FREDERICK	14	M	63	93
22	ALFRED	14	M	64	99
23	HENRY	14	M	65	119
24	LEWIS	14	M	64	92
25	EDWARD	14	M	68	112
26	CHRIS	14	M	64	99
27					

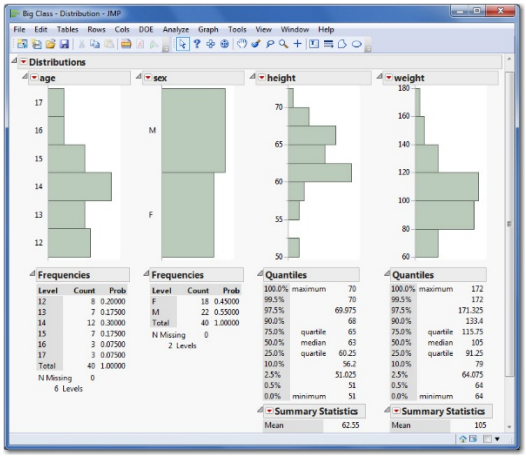
3. Once you have a data table open in JMP, you'll want to select a task through the JMP menus. These tasks (or commands as we call them in JMP) generate a **Launch Window** to execute your desired command (Figure 1.20). You will notice that the columns or variables from your data table are pre-populated in the Launch window. Chapters 3 through 6 will explore these tasks and their results.

Figure 1.20 A Launch Window



4. The result of any executed command is called the **Report Window,** which contains the graphs and statistics you've asked JMP to glean from your data (Figure 1.21). We will be seeing Report Windows throughout this book as we illustrate JMP's features, but Chapter 7 will focus on how to share these graphs and reports with others.

Figure 1.21 A Report Window



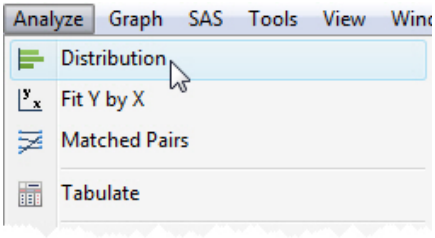
1.5 JMP Launch Dialog Windows

Throughout this book, each set of instructions used to create a graph or an analysis is prompted by a launch window that follows a consistent format and execution. To launch a window, however, you must first open a data table.

For purposes of illustration, we will open the Equity.jmp data table:

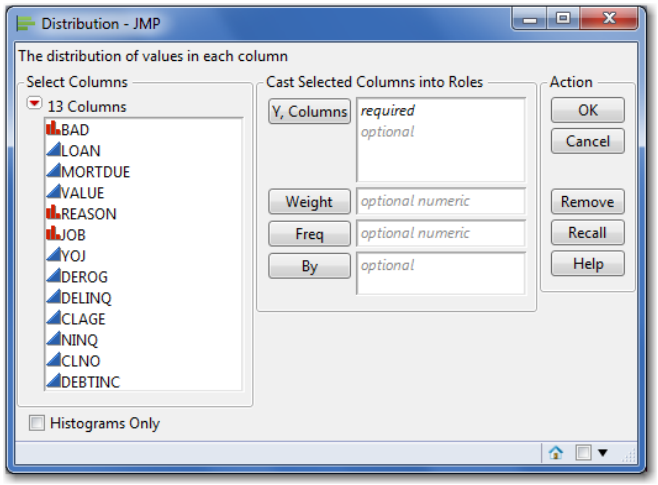
1. Select **Help ▶ Sample Data ▶ Open the Sample Data Directory ▶ Equity.jmp**.
2. Select **Analyze ▶ Distribution** (see Figure 1.22).

Figure 1.22 Selecting the Distribution Platform



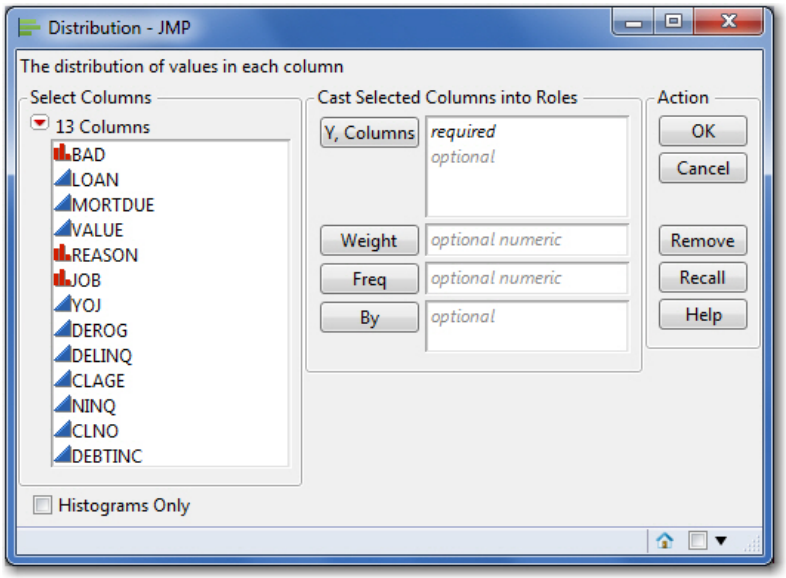
3. This generates the Distribution window with the columns (variables) from the Equity.jmp data table populated under the Select Columns window (see Figure 1.23).

Figure 1.23 The Distribution Launch Window



Most JMP launch windows consist of three main elements, organized from left to right (see Figure 1.24):

Figure 1.24 Launch Window Basics



1. **Available columns** (or variables) of data to analyze from your data table. These appear on the left under **Select Columns**.
2. **Roles** that you want to place (or cast) on the column(s). In this area, you see buttons and empty areas under **Cast Selected Columns into Roles**. Within these empty areas, you are given a hint in italics about which columns are required and which are optional to run the analysis.
3. **Action buttons** to execute commands.

To use this Distribution window or almost any other in JMP, click on a column and select the role (or click and drag the column into that role's empty space). Once you are satisfied with your selections, select **OK**.

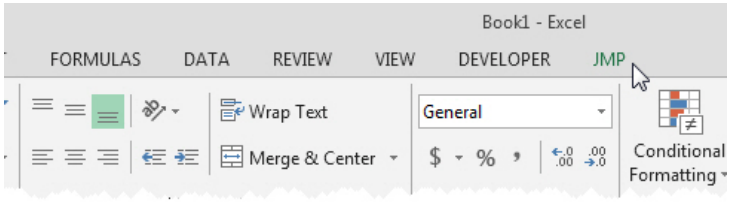
Almost every analysis and graph window in JMP appears in this way. Now that you've learned this format, you are ready to handle just about any command window in JMP.

Note: The **Y, Columns** role refers to what column you want to place on the vertical, or y, axis. In other windows, such as Fit Y by X, you also have an **X** role to select that corresponds to the horizontal, or x, axis. The Weight, Freq, and By roles are more specialized, but can streamline your analyses often without the requirement of reshaping your data (For more information, see **Help ▶ Books ▶ Using JMP ▶ Launch Windows**).

1.6 The Excel Add-In (Optional)

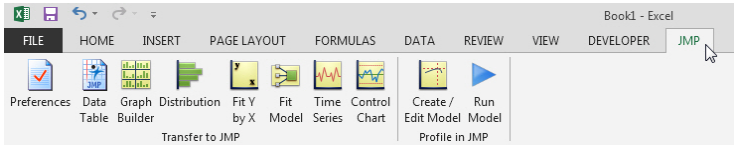
We find that many new users of JMP are often Microsoft Excel users too. JMP can easily import Excel data, which we will describe in greater detail in Chapter 2, but one feature that Excel power users may appreciate is the JMP add-in for Excel. The Excel add-in is a convenient Windows-only way to launch JMP platforms from within the Excel environment. If Excel is installed on your Windows computer and you then install JMP, the add-in should appear as a new tab along the top of your Excel window (see Figure 1.25). If it does not, go to 'Add-ins" within Excel and select the check box next to the JMP add-in item.

Figure 1.25 The JMP Add-In Tab in Excel



Selecting the JMP tab will reveal a JMP ribbon providing a good selection (but not all) of the commonly used JMP platforms (see Figure 1.26).

Figure 1.26 The JMP Ribbon In Excel

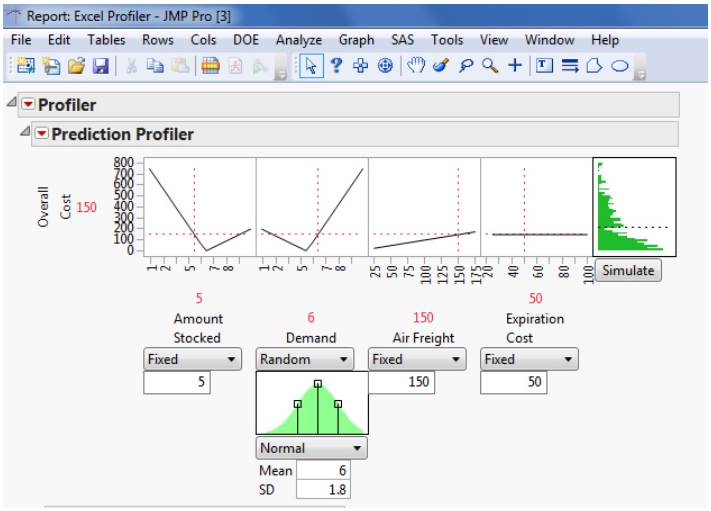


Because the JMP environment offers dynamic and visual exploration of your data, each JMP platform option will launch JMP, convert your Excel worksheet into a JMP data table, and set up the corresponding Launch window within the JMP environment. Let's briefly summarize their functions.

1. **Preferences** help to bring your data to JMP in the right format. Here, you can specify the number of header rows in your Excel worksheet and whether to bring over hidden rows or columns.
2. **Data Table** automatically converts your Excel worksheet into a JMP Data Table. Note that it will utilize the preferences you've set. If your data does not transfer correctly, change your preferences accordingly or utilize the Excel Import Wizard discussed in Chapter 2.
3. **Graph Builder** is an easy-to-use data visualization platform. Selecting this option will convert your worksheet into a JMP data table, launch the Graph Builder platform, and populate the dialog with your variables or columns so that you are ready to visualize your data.

4. **Distribution**, **Fit Y by X**, **Fit Model**, **Time Series,** and **Control Chart** will again, convert your worksheet into a JMP data table and launch the corresponding platform with your variables ready to be assigned into roles.
5. **Create/Edit Model** and **Run Model** allow you to visualize your spreadsheet models using JMP's profiler. If you are interested in performing "what-if" analysis on your spreadsheet models, the profiler allows you to do so visually. This is a great tool for presenting models because you can interact with the model and immediately visualize the effect of change. It also contains Monte Carlo simulation to explore how uncertainty will affect your model and fine-tune it to achieve desired results (see Figure 1.27).

Figure 1.27 The Excel Profiler



1.7 JMP Preferences

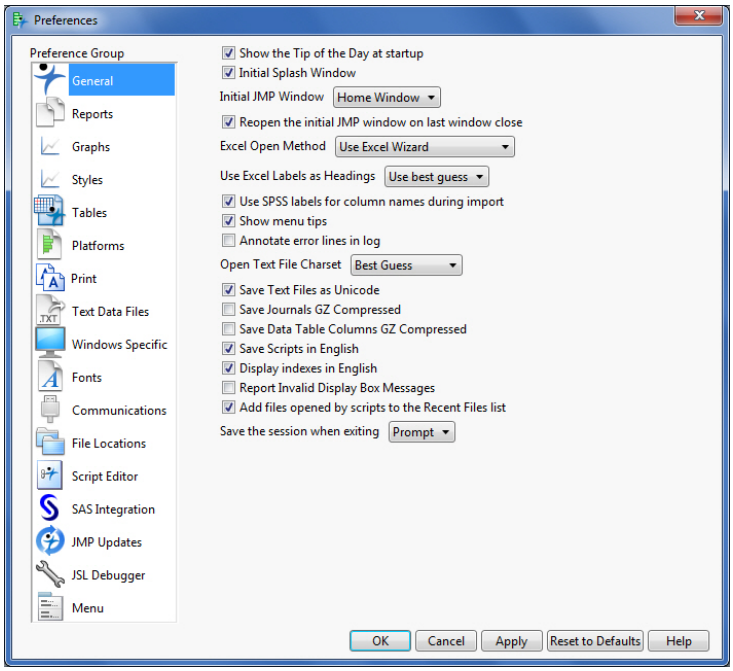
JMP's **Preferences** determine the way JMP appears or behaves on your machine. JMP has been carefully crafted to support the workflow of the data analyst. Its defaults have been selected to reflect common use, which we use in this book. However, JMP also provides options to tailor the software to corporate standards or individual tastes. In this section, we will explore how one can customize the look, feel, and options that appear in JMP. Preferences (**File ▶ Preferences**) are the primary means of setting or changing the defaults in JMP that you will see each time you operate the software–think global settings here. Virtually any function in JMP can be set as a default, including specific tests within any platform, the look of graphs, color schemes, font sizes and styles, and how JMP works with other products such as SAS.

To view the preferences, choose **File ▶ Preferences** (see Figure 1.28).

Figure 1.28 Accessing Preferences from the File Menu

This opens the Preferences window (see Figure 1.29), containing 17 main categories on the left and options within those categories on the right. You can change preferences by checking or unchecking the boxes within the categories on the right or by selecting items from drop-down menus. Changing preferences may affect such things as the graph or result format, the font, the location of a file, and much more, each and every time you use those features in JMP. If you are unsure about making a change to the preferences, we recommend that you wait until you have a need to do so.

Figure 1.29 The Preferences Dialog Window

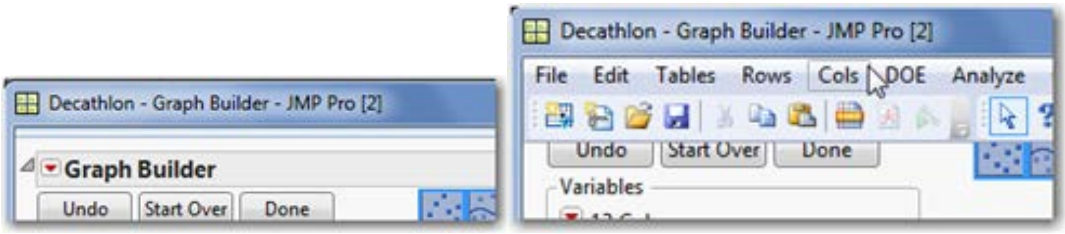


Note: If you need to make a change within a single graph or result, note that JMP also provides many of these formatting options within the graphs themselves.

Let's see how this works. New users often prefer to "turn-off" the menu auto-hide option (which by design, provides a little more window real estate for graphics and statistics power users), making it a little easier to find the menu options described in this book.

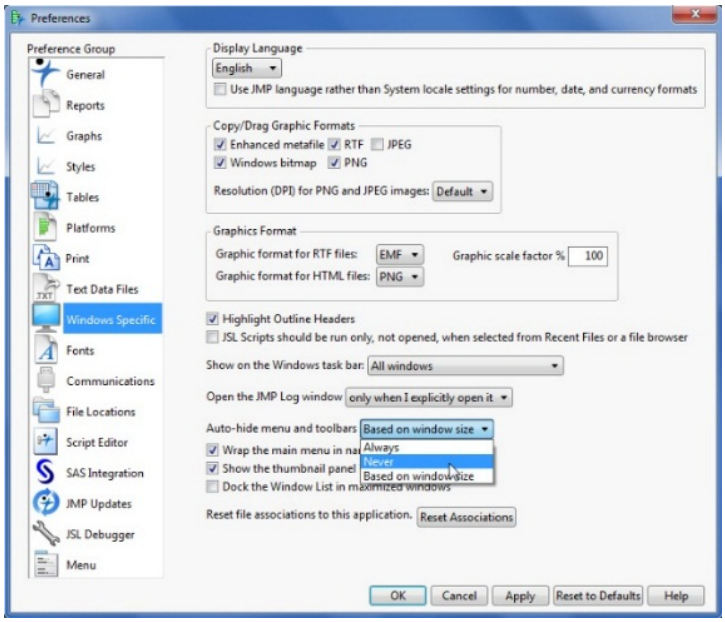
Below we have an illustration of the menu hidden and un-hidden (Figure 1.30). Notice the "File", "Edit", etc menus appear when they are not hidden.

Figure 1.30 Illustration of Menu Hidden and Unhidden



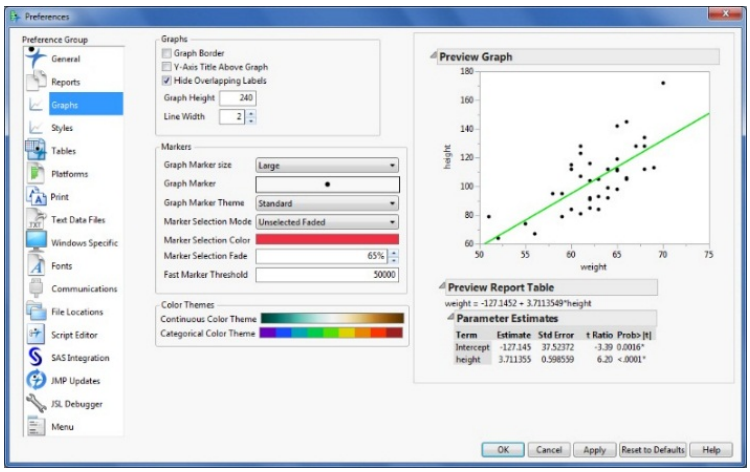
To change this "auto-hide" default to always show the menus, select **File ▶ Preferences ▶ Windows Specific ▶ Autohide menus and toolbar ▶ Never** (Figure 1.31).

Figure 1.31 Removing Menu Auto-Hide



If you wish to change the default marker size, style, or color themes employed in graphs, select **File ▶ Preferences ▶ Graphs**. Included is a handy preview to see how your selections will appear Figure 1.32).

Figure 1.32 Graph Preferences



1.8 Summary

JMP was developed to help the business professional, scientist, or engineer get answers to the questions and problems they encounter. The navigation and menus within JMP provide a natural extension of your problem-solving and a direct means to explore your data and generate the results you need. This book uncovers the structure of JMP's menus and provides easy steps for producing results. The standardized format of the windows in JMP prompts you through most analysis and graphing. Results can be customized using global detailed preferences.

Chapter 2 Data

2.1 Getting Data into JMP

2.2 The JMP Data Table

2.3 Data and Modeling Types

2.4 Cleaning and Formatting Data

2.5 Selecting, Highlighting, and Filtering Data: Row States

2.6 Adding Visual Dimension to Your Data

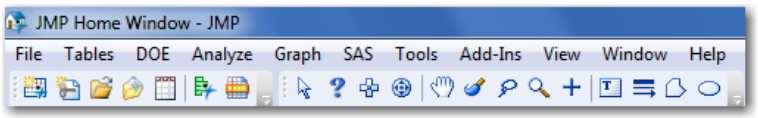
2.7 Shape Files and Background Maps

2.8 The Tables Menu

2.9 Summary

The first step in creating a graph or analysis is to get your data into JMP. With JMP, you can easily import data from many different sources such as Microsoft Excel or Access, or you can enter your data directly into a JMP data table. Because most readers already have data in one form or another, this section focuses on getting that data into JMP from another application. Sometimes data isn't in the best condition when you import it. Later in this chapter, we discuss what you can do to format data or deal with missing data. JMP also now supports shape files that can be used to create maps; we will describe the special requirements of using these file types.

As mentioned in the previous chapter, we use Windows as our default system to illustrate JMP. JMP instructions for Windows and Macintosh are basically the same, though some operating system differences are noted when they occur.



Example 2.1 Big Class

We will be using the Big Class.jmp data file to illustrate the steps in this chapter. This data set consists of 40 middle-school students and their name, height, weight, gender, and age. You can access this data set in the Sample Data folder that is installed with JMP: **File ▶ Open ▶ C: ▶ Program Files ▶ SAS ▶ JMP ▶ 11 ▶ Samples ▶ Data ▶ Big Class.jmp**

2.1 Getting Data into JMP

Getting your data into JMP is a familiar process. Like many other desktop applications, you can simply select **File ▶ Open** to import your data into JMP. JMP can handle many different data formats. Table 2.1 shows the default formats JMP recognizes. Other previously installed applications could contain proprietary formats that might also appear as import options. You can import files with these formats as well.

In this section, we show you how to open JMP data tables and how to import Microsoft Excel spreadsheets and text files in JMP. Each of these file formats follows the same basic procedure, but each has special options that allow you to import exactly what you want. JMP interfaces with databases using Open DataBase Connectivity standard (ODBC). Through the Database Open Table dialog box, you can query your data using SQL. We illustrate only the essential connectivity here; more information about querying your data is available in the JMP documentation (**Help ▶ Search ▶ SQL**). At the end of this section, we show you how to create a new data table in JMP.

Table 2.1

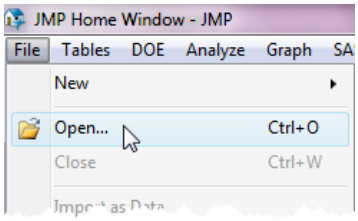
File Type	File Extension
JMP Files	.jmp, .jsl, .jrn, .jrp, .jmpprj, .jmpmenu, .jmpaddin, .jmpapp
JMP Data Tables	.jmp
Excel Files	.xls, .xlsx, .xlsm
Text Files	.txt, .csv, .dat, .tsv
SAS Data sets	.sas7bdat, .xpt, .stx
SAS Program files	.sas
R Code	.r, .R
MATLAB Code	.m, .M
HTML Files	.htm, .html
FACS Files	.fcs
SPSS Data Files	.sav
xBase Data Files	.dbf
Shapefiles	.shp
Minitab Worksheets	.mtp
Teradata Database	.trd

Opening a JMP File

Let's start by opening a JMP data table. At the top left of the JMP window is the **File** menu:

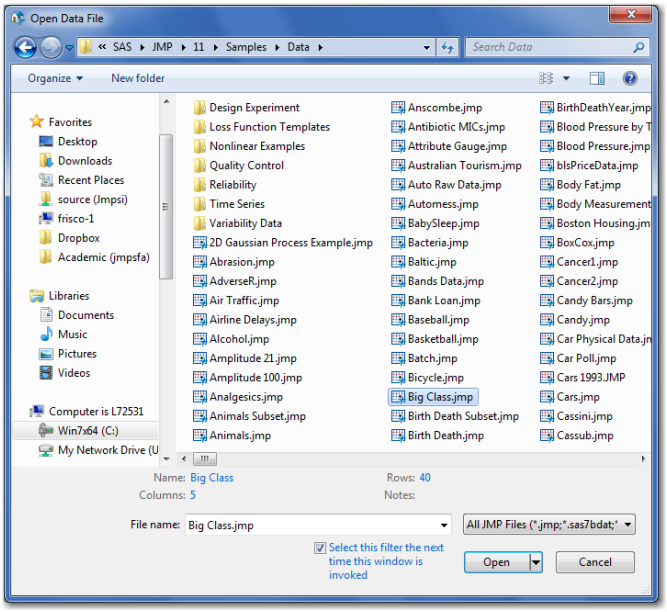
1. Select **File** ▶ **Open** (see Figure 2.1). A familiar dialog box opens (on a Mac, select **File** ▶ **Open** and locate your file in the appropriate folder).

Figure 2.1 Opening a File



We will use the Big Class data table described earlier. Click on the **Big Class.jmp** file and select **Open** (see Figure 2.2).

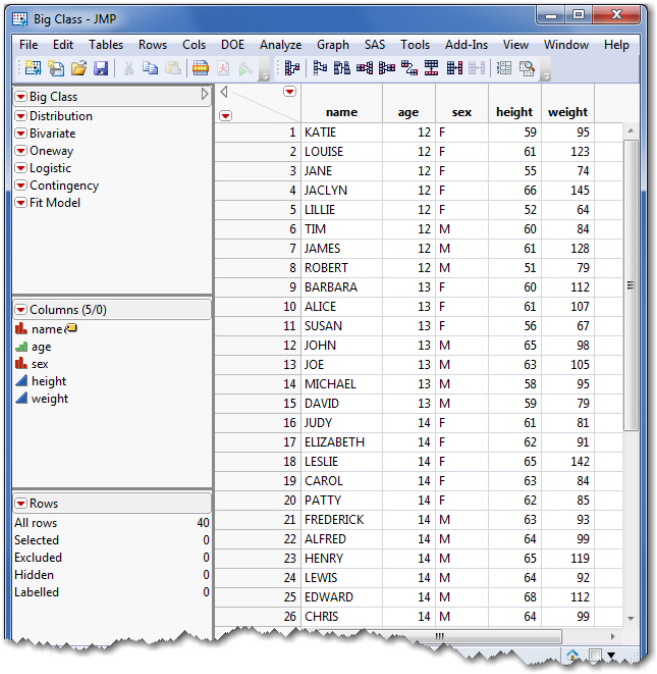
Figure 2.2 Open File Dialog



Note: To locate this file for the first time, select **File** ▶ **Open** ▶ **C:** ▶ **Program Files** ▶ **SAS** ▶ **JMP** ▶ **11** ▶ **Samples** ▶ **Data** ▶ **Big Class.jmp**. Alternatively, you can also select **Help** ▶ **Sample Data** ▶ **Open the Sample Data Directory** ▶ **Big Class.jmp**.

These steps open the JMP data table Big Class (see Figure 2.3). With these simple steps, you are now ready to analyze or visualize this data.

Figure 2.3 The JMP Data Table



	name	age	sex	height	weight
1	KATIE	12	F	59	95
2	LOUISE	12	F	61	123
3	JANE	12	F	55	74
4	JACLYN	12	F	66	145
5	LILLIE	12	F	52	64
6	TIM	12	M	60	84
7	JAMES	12	M	61	128
8	ROBERT	12	M	51	79
9	BARBARA	13	F	60	112
10	ALICE	13	F	61	107
11	SUSAN	13	F	56	67
12	JOHN	13	M	65	98
13	JOE	13	M	63	105
14	MICHAEL	13	M	58	95
15	DAVID	13	M	59	79
16	JUDY	14	F	61	81
17	ELIZABETH	14	F	62	91
18	LESLIE	14	F	65	142
19	CAROL	14	F	63	84
20	PATTY	14	F	62	85
21	FREDERICK	14	M	63	93
22	ALFRED	14	M	64	99
23	HENRY	14	M	65	119
24	LEWIS	14	M	64	92
25	EDWARD	14	M	68	112
26	CHRIS	14	M	64	99

This spreadsheet-like table is referred to as the *JMP data table,* which is JMP's common data format regardless of where the data comes from. Section 2.2 discusses the components of the data table.

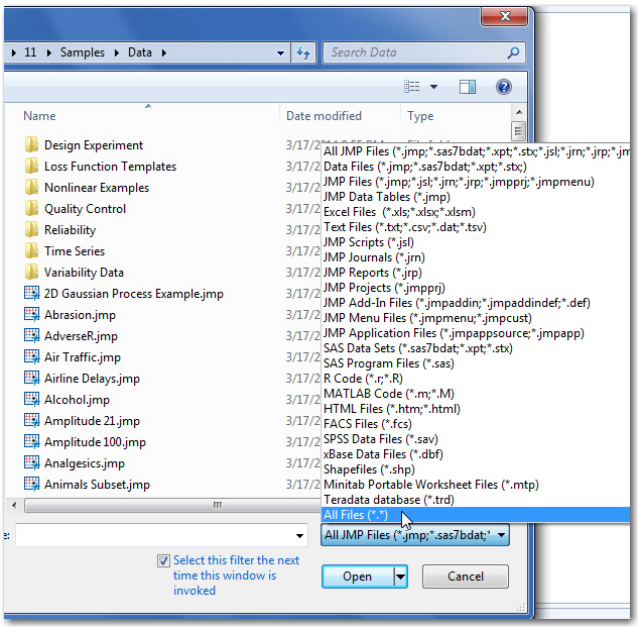
Note: In the Big Class example, the variables (name, age, sex, height, and weight) are located as column heads and the individual sets of observations appear in rows. This structured format of the data is required. The importing examples that follow assume that your data already exists in this format. Section 2.6 introduces some tools to use if your data does not adhere to this structure.

Importing Data into JMP

Importing data into JMP from another file format is similar to opening a JMP file. Within the **File ▶ Open** pop-up window, the **Files of Type** drop-down menu indicates **All JMP Files** as the default.

If you are importing another file type, simply click on the down arrow and select the right type. You can also select **All Files** from the drop-down menu (see Figure 2.4). Select the file that you want, and then click **Open**. On the Mac, select **File ▶ Open** and available files will be highlighted.

Figure 2.4 Selecting All File Types

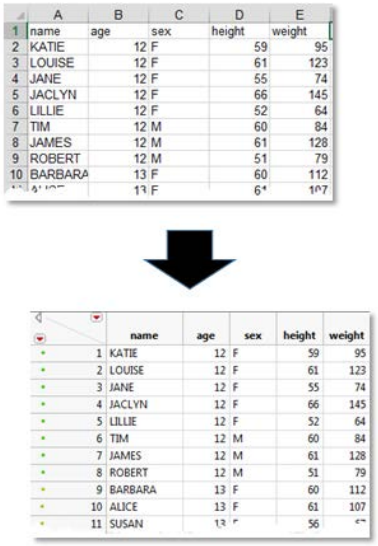


Note: If you know the format of your data, first select the correct format from the **Files of Type** drop-down menu. You will see the available files of that type within the folder. Once you've located the right file, select the file and click **Open**.

Importing an Excel File

Importing an Excel file is easy, as long as your variables are in columns and your cases or sets of observations are in rows. Ideally, any variable names should appear in the row directly above the first row of data, as shown in Figure 2.5. The import process automatically opens and converts the data into a JMP data table and uses your variable names as column headings:

Figure 2.5 Importing an Excel File



	A	B	C	D	E
1	name	age	sex	height	weight
2	KATIE	12	F	59	95
3	LOUISE	12	F	61	123
4	JANE	12	F	55	74
5	JACLYN	12	F	66	145
6	LILLIE	12	F	52	64
7	TIM	12	M	60	84
8	JAMES	12	M	61	128
9	ROBERT	12	M	51	79
10	BARBARA	13	F	60	112

	name	age	sex	height	weight
1	KATIE	12	F	59	95
2	LOUISE	12	F	61	123
3	JANE	12	F	55	74
4	JACLYN	12	F	66	145
5	LILLIE	12	F	52	64
6	TIM	12	M	60	84
7	JAMES	12	M	61	128
8	ROBERT	12	M	51	79
9	BARBARA	13	F	60	112
10	ALICE	13	F	61	107
11	SUSAN	13	F	56	

1. Select **File ▶ Open**.

   The **Big Class.xl**s file, which is illustrated here, can be found by selecting **C: ▶ Program Files ▶ SAS ▶ JMP ▶ 11 ▶ Samples ▶ Import Data ▶ Big Class.xls**.

2. From the **Files of Type** drop-down menu, select **Excel Files**.
3. Select the file that you want, then select **Open** which will launch the Excel Import Wizard dialog with a view of your data. If it looks correctly structured, select **Import**.

   Note: If your data isn't structured correctly or have multiple worksheets within an Excel workbook, JMP provides additional controls in the Excel Import Wizard dialog box to select individual worksheets and to specify that your headings are placed in the right location within the JMP data table. Important note to Mac users: The Excel Import Wizard is a new feature available in JMP 12. Mac users on JMP 11 and earlier versions may open older Excel files formats with .xls or .csv file extensions, but not the newer xlsx formats (without downloading a driver to do so). You may also simply save your newer Excel files in the older .xls format and then import them.

   Shortcut: If you have JMP open and an Excel worksheet or workbook on your desktop, you can simply drag the file over the JMP shortcut icon on your desktop to launch the Excel import wizard.

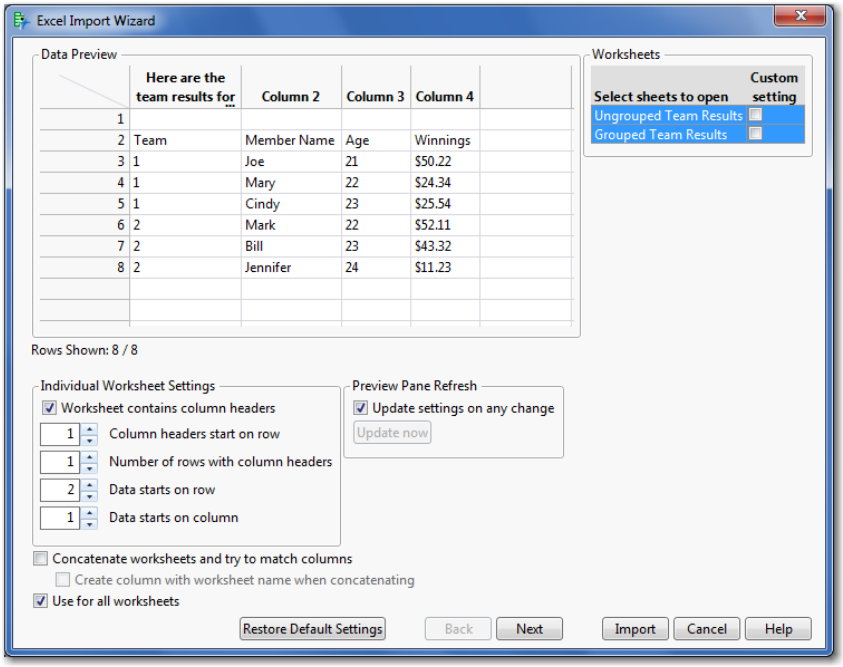
The Excel Import Wizard

While the previous example was simple and straightforward, a common characteristic of Excel worksheets is that data does not always adhere to the essential column/row structure that is required by JMP. For example, you may have multiple nested headers where one row might represent year and the next row contain months within that year. The new Excel Import Wizard has made importing this worksheet and maintaining the month within year structure much easier.

This wizard also provides options to specify which rows should be headers or columns, to specify hidden or merged columns, and to replicate these settings or merge data from multiple worksheets within a workbook.

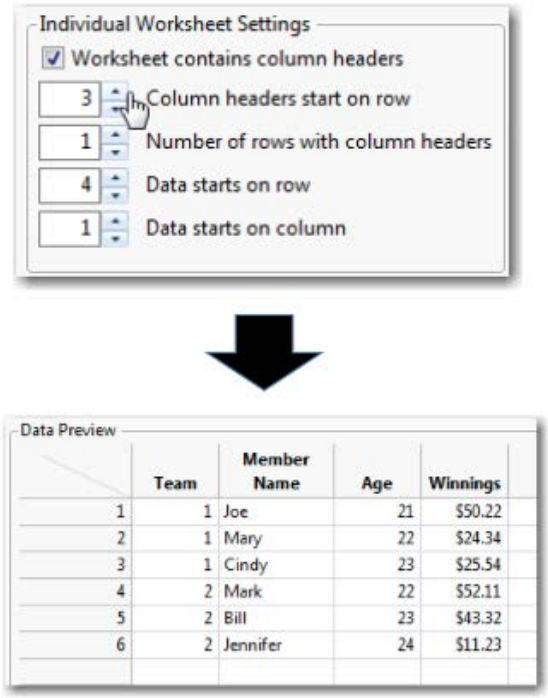
To illustrate this feature, let's try another example with one of these characteristics. We will use the Team Results.xls worksheet from the Import Data folder. This worksheet has headers/column names that appear to be in the second row (see Figure 2.6). The Excel Import Wizard will help JMP decide how to import this data.

Figure 2.6 The Excel Import Wizard



1. Select **File ▶ Open**, then select an Excel workbook. In this example we are using **Team Results.xls** from the **C: ▶ Program Files ▶ SAS ▶ JMP ▶ 11 ▶ Samples ▶ Import Data** folder.
2. Select **Team Results.xls ▶ Open** to launch the Excel Import Wizard with an initial display of your data in the window (Figure 2.6).
3. As you can see, the column headers begin in row 3 (when you include the note that appears as a header in the preview) and the first set of observations in row 4. To get this into the right format, **adjust the Individual Worksheet Settings** as we've done in Figure 2.7.

Figure 2.7 Adjusting the Worksheet Settings



4. Once you've adjusted these settings, you should see your data take its proper shape in the window. Once you are satisfied with the adjustments you've made, select **Import**.

Importing a Text File

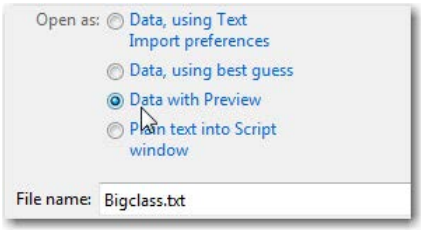
If you are importing a text file, another handy wizard is included in the **Data with Preview** file option. Like the Excel Import Wizard, this wizard allows you to view your data and specify how you want it to appear before importing it into a JMP data table. It also provides options to convert your text file if it is delimited by commas, tabs, or spaces:

1. Select **File ▶ Open**.

   The **Big Class.txt** file, illustrated here, can be found by selecting **C: ▶ Program Files ▶ SAS ▶ JMP ▶ 11 ▶ Samples ▶ Import Data ▶ Big Class.txt**.

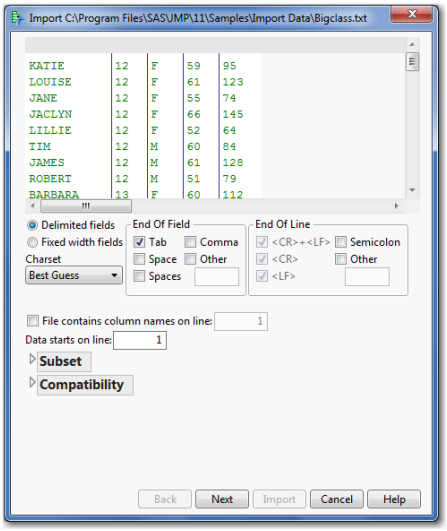
2. Select **Text Files (*.txt, *.csv, *.dat, *.tsv)** in the **Files of Type** drop-down menu. Select the file.
3. Select **Data with preview** (see Figure 2.8).

Figure 2.8 Data with Preview



4. Select **Open.**
5. Choose the settings you need (see Figure 2.9). Click **Next** and then **Import**.

Figure 2.9 Text Data Preview

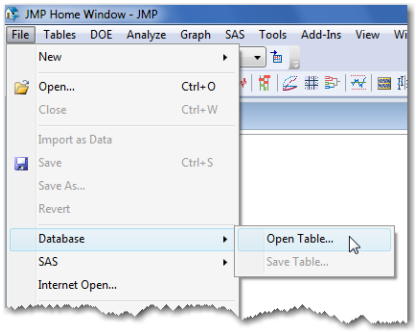


Importing a Database File

Options to import data extracted from a database are available through ODBC within JMP. To access this data, first connect to the database (the data source should already be defined) and then specify the table of interest. You can also query your data using Query Builder (in JMP 12) or the **Advanced** button. If you need more help defining your data source, select **Help ▶ Books ▶ Using JMP ▶ Ch. 3 Import Data from a Database.**

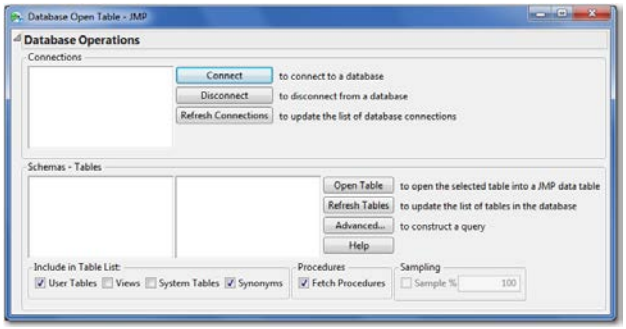
1. Select **File ▶ Database ▶ Open Table** (see Figure 2.10).

Figure 2.10 Database Open Menu



2. The Database Open Table window appears (see Figure 2.11). It prompts you to connect to your database and either to open a data table or to specify a query. Clicking the **Connect** button launches the Select Data Source window to locate and connect to your database.

Figure 2.11 Database Open Dialog



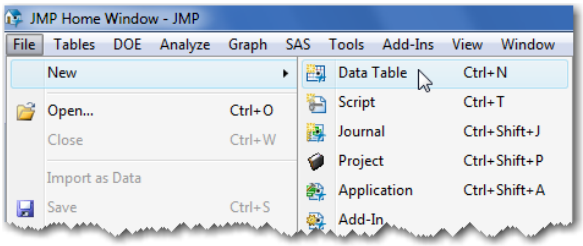
3. Locate the table of interest, highlight it, and click **Open Table** to import the data.

Note: You have the option of directly importing database files by selecting **File ▶ Open** as previously discussed (assuming these programs are installed). Using this more direct option allows you to import only a single table.

Creating a JMP Data Table from Scratch

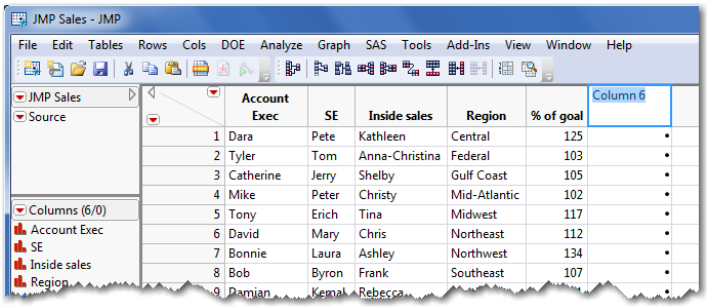
Select **File** ▶ **New** ▶ **Data Table** to create a new data table (see Figure 2.12):

Figure 2.12 Creating a New Data Table



1. Double-click on the first column's heading and type the column name (the variable name).
2. Press **Enter** and type the data into the first cell directly below the heading. Press **Enter** again, type the data, and repeat as needed. Rows within JMP are consecutively numbered as observations or cases (see Figure 2.13).

Figure 2.13 Add a New Column



3. To create another column, **double-click** on the next column's heading and enter the data as you did before.

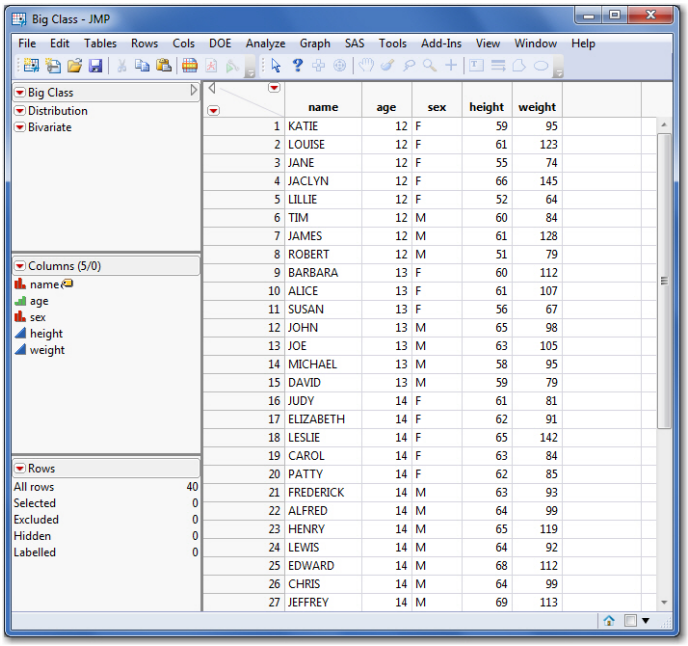
If it is more practical for you to enter a series of data for each row as you build your data table, set up all of your column headings first and then use the **Tab** key to move from the left columns to the right. When each column has been filled, the **Tab** key moves down to the beginning of the next row.

Note: JMP will recognize the type of data you are entering and assign a data type to the column, either numeric or character. It also assigns an icon next to the columns (or variables) in the box on the left. These icons are discussed in Section 2.3.

2.2 The JMP Data Table

The JMP Data Table looks very much like any spreadsheet (see Figure 2.14). In JMP, column headings indicate variables (what you've measured or counted) and rows indicate individual cases or sets of observations. JMP requires your data to be structured this way. If it is not, JMP can help you reformat your data (see Section 2.6).

Figure 2.14 The JMP Data Table



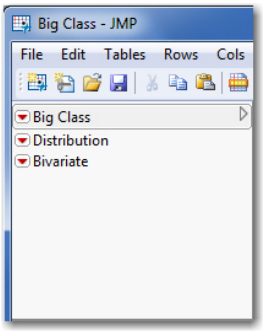
	name	age	sex	height	weight
1	KATIE	12	F	59	95
2	LOUISE	12	F	61	123
3	JANE	12	F	55	74
4	JACLYN	12	F	66	145
5	LILLIE	12	F	52	64
6	TIM	12	M	60	84
7	JAMES	12	M	61	128
8	ROBERT	12	M	51	79
9	BARBARA	13	F	60	112
10	ALICE	13	F	61	107
11	SUSAN	13	F	56	67
12	JOHN	13	M	65	98
13	JOE	13	M	63	105
14	MICHAEL	13	M	58	95
15	DAVID	13	M	59	79
16	JUDY	14	F	61	81
17	ELIZABETH	14	F	62	91
18	LESLIE	14	F	65	142
19	CAROL	14	F	63	84
20	PATTY	14	F	62	85
21	FREDERICK	14	M	63	93
22	ALFRED	14	M	64	99
23	HENRY	14	M	65	119
24	LEWIS	14	M	64	92
25	EDWARD	14	M	68	112
26	CHRIS	14	M	64	99
27	JEFFREY	14	M	69	113

Data Table refers to the spreadsheet-like grid where your data resides. The three panels on the left of the data table contain information about your data (metadata). The data grid can contain any number of columns (your variables) or rows (observations or cases). In this sense, we refer to data within the JMP data table as structured data.

In addition to the data grid, notice the three panels to the left of the data table. These panels provide vital information about your data as well as options to streamline and save your analyses.

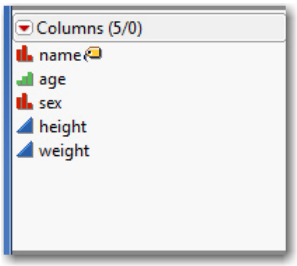
The first and upper-most panel contains the name of the data table (see Figure 2.15). This panel stores references and/or scripts. Scripts allow you to save, automate, and customize analyses. If you perform a regular analysis or a scheduled task, you will want to learn more about JMP scripts (see the JMP Scripting Guide at **Help ▶ Books ▶ JMP Scripting Guide**).

Figure 2.15 The Table Panel



The Columns panel (see Figure 2.16) is where your column names (or variables) appear. Each column has an icon in front of it.

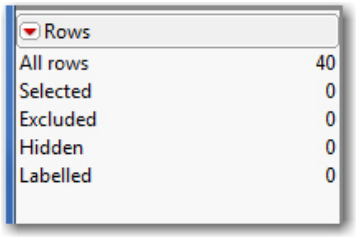
Figure 2.16 The Columns Panel



These icons correspond to the modeling type of the data in each column. As discussed in the next section, this is vitally important. JMP produces only the graphs or statistics that are appropriate for a column's modeling type. In most cases, you can change the modeling type by simply clicking on the icon and selecting another appropriate format.

The bottom panel is the Rows panel (see Figure 2.17). The Rows panel indicates how many rows (sets of observations) are in your data table. This panel also indicates the number of selected, hidden, or excluded rows, if any.

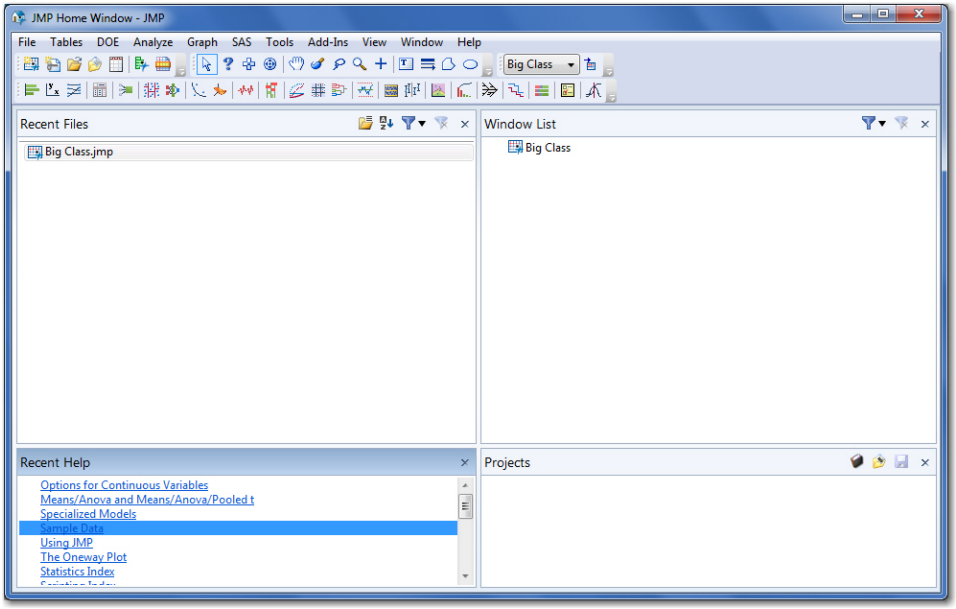
Figure 2.17 The Rows Panel



Note: When rows are hidden, the observations are not included in graphs. When rows are excluded, they are not included in analyses. This row state is effective when you want to see or analyze a subset of your data. You can also both hide and exclude specific rows, which effectively removes the row(s) from your analysis or graph, but not from your data table. Section 2.5 provides more information on row states including hiding and excluding rows.

Multiple data tables can be open at any time, but only one active data table can be analyzed at a time. If you have multiple data tables open within JMP and you want to switch to another open data table, go to the Home Window (see Figure 2.18) and select the desired data table under Windows List.

Figure 2.18 The Home Window



Another special type of data table are shape files which are used to create thematic maps. These data tables consist of two tables including a "Name" and corresponding "Boundary" table. These are stored in the Maps folder: **C: Program Files ▶ SAS ▶ JMP ▶ 11 ▶ Maps**. Section 2.7 covers some of the basics about shape files.

Note: There is no practical limit on the size of the data table you can analyze. However, because JMP runs in your computer's local memory, the amount of RAM you have determines the upper size limit of your data table. Your computer should be equipped with at least twice as much memory as the size of the data table. Thus if you have just 2 GB of RAM, you can analyze about a 10-variable data set with 1 million rows! JMP also provides 64 bit versions (if your computer is so equipped) allowing much larger data sets to be analyzed with greater memory. More details on JMP system requirements can be found at www.jmp.com.

2.3 Data and Modeling Types

One of JMP's great features is the ability to produce graphs and statistics that make sense for the data you are analyzing. This feature assumes that your data is correctly classified in the data table. So, what do we mean by data type and modeling type? Let's define a few terms.

Data
refers to any values placed in the cell of a JMP data grid. Examples include numeric and/or text descriptions: 3.6, $2500, Female, Somewhat Likely, or 11/14/13.

Data type
refers to the nature of the data. The data type can be either numeric (numbers) or character (often words and letters but sometimes also numbers).

Modeling type
refers to how the data within a column should be used in an analysis or a graph. JMP uses three distinct modeling types: continuous, nominal, and ordinal.

- *Continuous data* (also referred to as *quantitative, ratio* or *interval scale data*) takes a numeric form and is often thought of as some type of measurements. For example, home selling prices, income earned, costs per square foot, and dates are all examples of continuous data. As a rule of thumb, continuous data can be used in calculations. For example, calculating the average cost per square foot would be meaningful.
- *Nominal data* is categorical data (also referred to as *qualitative, discrete, count,* or *attribute data)* and can take on either a character or numeric form. Nominal data fits into categories or groups such as car type, gender, department, and sales territory and also includes indicator variables like yes/no or 0/1. In nominal data, it is helpful to count the frequency of the occurrence of values, but otherwise, nominal data is not used in calculations. For example, calculating the average car type would not be meaningful.
- *Ordinal data* is categorical data that has an inherent order or hierarchy. For instance, Likert scales (such as levels of satisfaction) in a survey and grade levels in school (freshman, sophomore, junior, senior) are examples of ordinal data. That is, they represent categories that have some sequence or order that should be retained in any analysis. Ordinal data is less common than continuous and nominal data, but there are a few analyses designed specifically for it. In most JMP analyses, nominal and ordinal data are treated the same way.

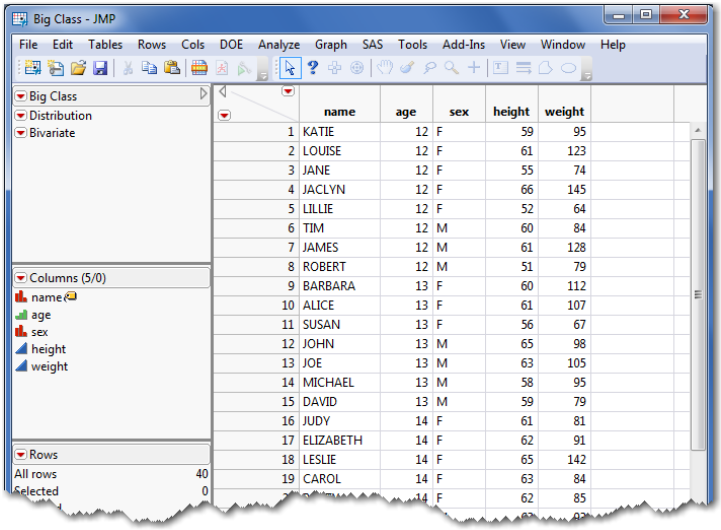
		Data Type	
		Numeric	Character
Modeling Type	Continuous	Yes	No
	Nominal	Yes	Yes
	Ordinal	Yes	Yes

Table Summary of Data and Modeling Types

Note: Numeric data is right justified in the data table whereas character data is left justified. This can be a useful to check whether data contains errors.

In our example, **Big Class** contains five variables (or columns) representing each of these modeling types (see Figure 2.19). Let's briefly explain why they are classified by their data and modeling types:

Figure 2.19 Understanding Modeling Types in the Data Table



Name is nominal because it is a character data type and the student's name is arbitrary.

Age is ordinal because the values are rounded down and we want to retain the six ordered age groups (12 to 17) in our analysis.

Note: Age could also be considered continuous because the values are numeric, but this would treat age differently and yield different results.

Sex is nominal because its data type is character (M or F) and it has no order.

Height and **weight** are continuous because they are both numeric and represent a measurement.

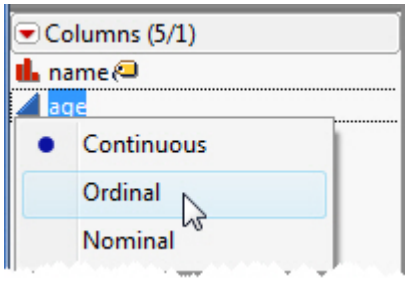
Note: Row State is a third data type that allows you to store and manage information about a row of data (See Section 2.5). For more information, select **Help ▶ Books ▶ Using JMP ▶ Ch. 4, Enter and Edit Data ▶ Assign Characteristics to Rows and Columns.**

Changing the Modeling Type

When you import data, the JMP default selects and assigns one of two modeling types based on whether the data is numeric or character. Numeric data becomes continuous and character data becomes nominal. Sometimes you might want to change the default modeling type of your data to generate results that are more meaningful.

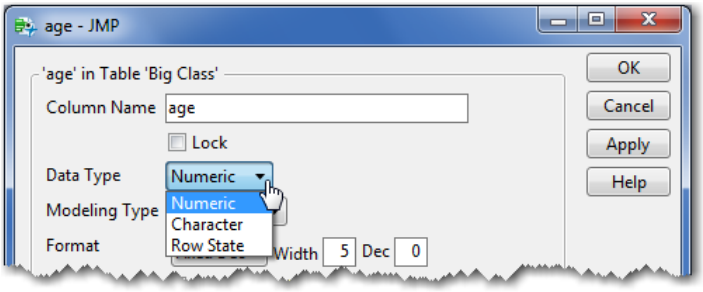
For example, if we imported the **Big Class** data from Excel, age as numeric data would be imported as a continuous column. We might want to change that to ordinal. Changing the modeling type is simple in JMP. Click the column's corresponding icon in the Columns panel in the data table and select the correct type (see Figure 2.20).

Figure 2.20 Changing the Modeling Type



If the **Continuous** option is grayed out, your data type is classified as character. To change the data type, double-click on the column heading and change the data type to numeric (see Figure 2.21). In this window, you can also change the modeling type along with a host of other formatting options, which are described in the next section.

Figure 2.21 Changing the Data Type



For more information, select **Help ▶ Books ▶ Using JMP ▶ Chapter 5, Set Column Properties ▶ About Data and Modeling Types**.

2.4 Cleaning and Formatting Data

Sometimes data isn't in the best shape or in the right form when it is imported. Fortunately, JMP has extensive column formatting abilities. This section focuses on the most common features, including:

- Cleaning up your data format, such as decimal places, dates, times, and currency. We will use the Column Info window to accomplish these tasks.
- Introducing the Formula Editor, which allows you to create new columns from old ones, add IF statements, and transform data using basic or more advanced functions.
  We will introduce a basic example in this section. For more information, select **Help ▶ Books ▶ Using JMP.**
- Learning to use the RECODE command, which is a handy way to merge similar categorical responses into a single category. For example, if you have Woman, Female, and Girl as responses, you can merge these into a single response: Female.

Example 2.2 Movies

We will use the **Movies.jmp** data table to illustrate the concepts in this section. This data table consists of the 277 top-grossing movies released between 1937 and 2003. The columns are:

Movie name of movie

Type genre/category of movie (for example, comedy, family)

Rating US movie rating system (for example, general audience [G], adult [R])

Year year of movie release (for example, 1937)

Domestic $ US domestic revenue in $ earned by the movie in that year

Worldwide $ Worldwide revenue in $ earned by the movie in that year

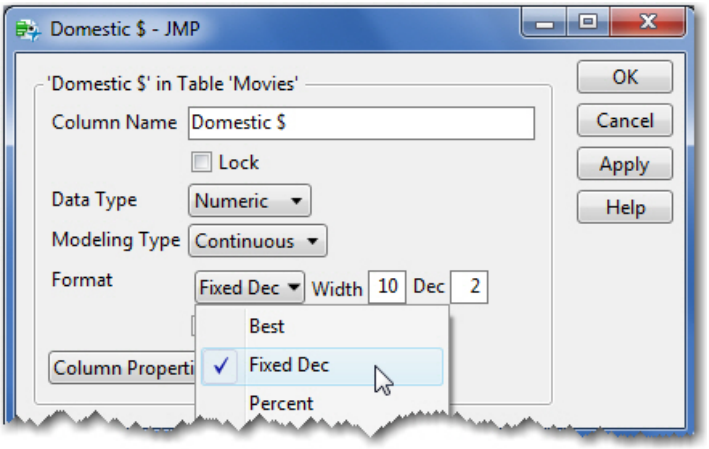
Director director of movie

You can access this data table in the **Sample Data** folder that is installed with JMP by selecting **File ▶ Open ▶ C: ▶ Program Files ▶ SAS ▶ JMP ▶ 11 ▶ Samples ▶ Data ▶ Movies.**

Getting your data into a standard format is done through the Column Info window, which is accessed from the Cols menu. Options to format your data are driven by the data and modeling types specified for that column of data. You can change these types, if necessary, to meet the requirements of your analysis. Recall that changing these types affects the graphs or statistics you can generate from that column (see the previous section). Let's begin by opening the **Movies.jmp** data table:

1. Open the **Movies.jmp** data table.
2. Select the **Domestic $ column**, and then select **Cols ▶ Column Info.**
3. Because Domestic$ is a numeric value, you see the **Format** drop-down menu (see Figure 2.22), which leads to several options. It is also our starting point for the next items we'll discuss. Note that if you select a Character column, the **Format** menu does not appear in the Column Info window.

Figure 2.22 Column Info Format Menu—Continuous Variables



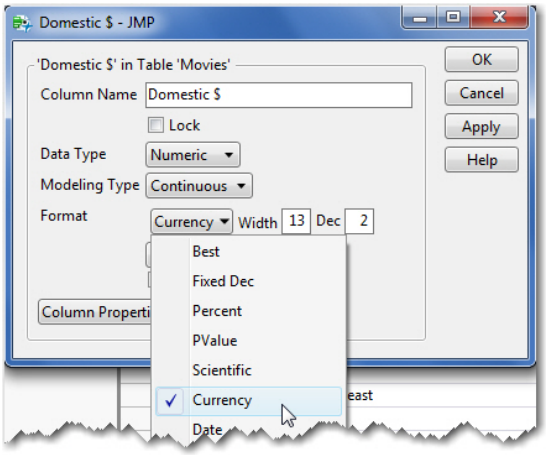
Note: You can also either double-click on the column name as mentioned in the previous section, or right-click on the column head and select **Column Info** from the menu.

Formatting Decimal Places

To change the number of decimal places displayed in a column of data, do the following:

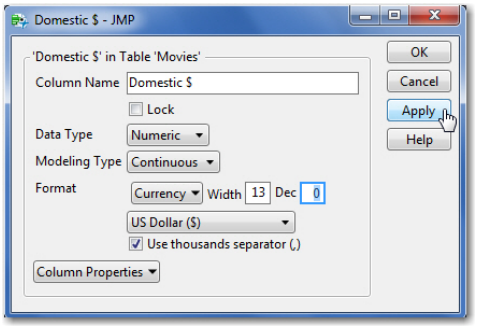
1. Click on the column of interest. In our example, it is **Domestic$**.
2. Select **Cols ▶ Column Info.** JMP will make a best guess on the format of the data; in our example, **Currency** was correctly specified (see Figure 2.23). You can easily change this format by selecting another format from the menu.

Figure 2.23 Column Info Formatting Options



3. To the right of the **Format** menu are two boxes, **Width** and **Dec**. **Width** refers to the number of characters that can be in the column, and **Dec** refers to the number of decimals right of the point. In our example, type "0" in the **Dec** box, then select **Apply** (see Figure 2.24).

Figure 2.24 Formatting Decimal Places



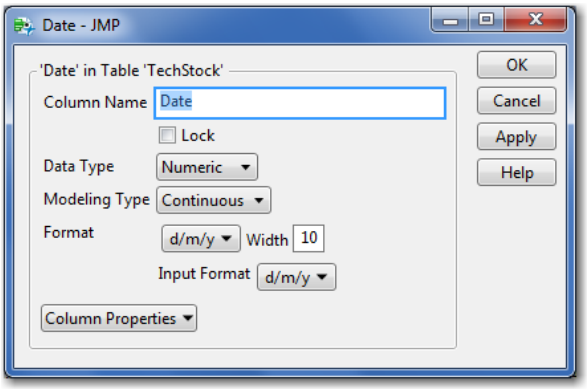
Note: This procedure applies to both **Percent** and **Fixed Dec**.

Formatting Dates, Time, and Duration

Dates are numeric values in JMP, which allows them to be transformed into other date formats and calculated for duration or elapsed time. If you are importing data that contains dates, ensure that the data type is numeric.

The Column Info (**Cols ▶ Column Info**) window provides the column's information (see Figure 2.25) and several date format options, as seen on the following page. When a date is selected from the **Format** menu, a secondary drop-down menu for the display format appears, along with a similar drop-down menu for the input format of your imported data. The format of your imported data needs to match one of JMP's input format options, which can then be transformed into any format among the display format options.

Figure 2.25 Column Info Window for a Date C



Let's walk through a new example, TechStock, to illustrate this concept.

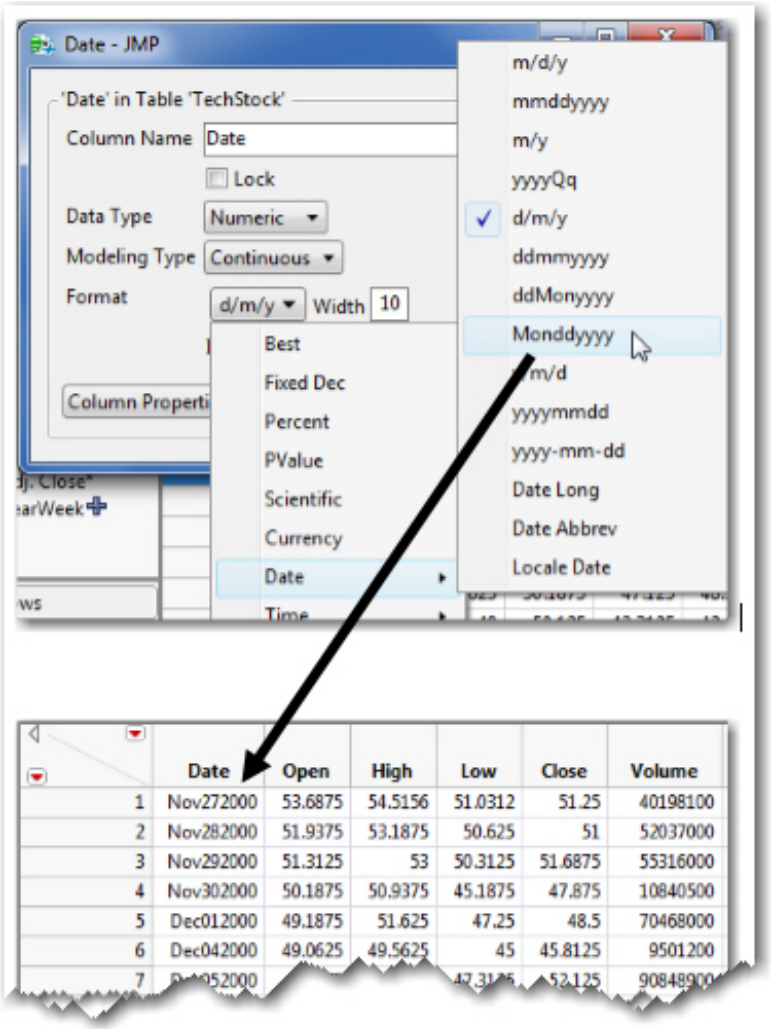
Example 2.3 TechStock

We will use the TechStock data table to illustrate dates in this section. This data set contains the stock price of the NASDAQ 100 (QQQ) at the high, low, and close for each trading day during the period 11/27/2000 to 2/26/2001. You can access this data set in the **Sample Data** folder that is installed with JMP: **File ▶ Open ▶ C: ▶ Program Files ▶ SAS ▶ JMP ▶ 11 ▶ Samples ▶Data ▶ Techstock.jmp.**

1. Open the TechStock.jmp data table.
2. Click on the **Date** column name**.**
3. Select **Cols ▶ Column Info**. Open the **Format** drop-down menu and select **Date**, which displays how the dates will appear in the data table.

4. It is currently displayed as d/m/y (see Figure 2.26), as indicated by the check mark. Change the format to **Monddyyyy.** Click **Apply** or **OK.** The date is now displayed as abbreviated month, day, and year in the Date column (see lower part of Figure 2.26).

Figure 2.26 Changing the Display Format of Dates

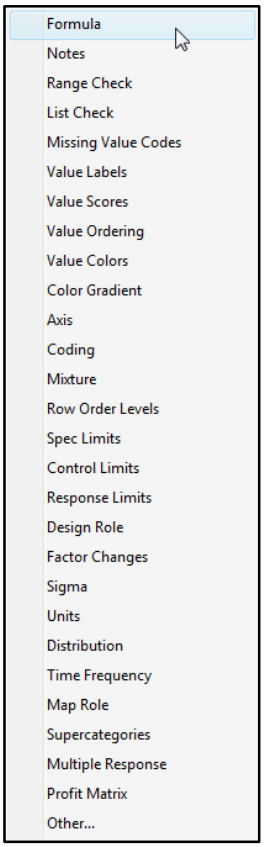


You can also format time and duration from this window.

Column Properties Menu

Column Properties, another useful tool in the Column Info window (see Figure 2.27), allows you to add formulas, check ranges of values for auditing, and assign customized ordering to the data, among other tasks. These functions are described in detail in the *Using JMP* book (**Help ▶ Books ▶ Using JMP ▶ Ch. 5 ▶ Set Column Properties**).

Figure 2.27 Column Properties



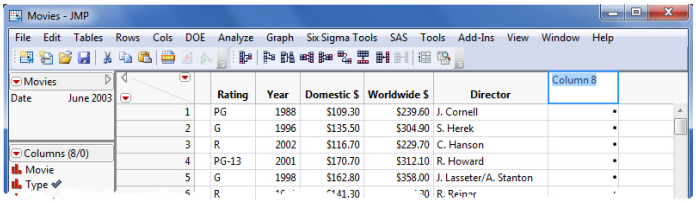
Formula Editor

JMP's formula editor is handy and flexible. Use it when you need to create a new column that contains values that are calculated or derived from existing columns in your data table. You can also transform your data, add conditional statements, and much more. Due to the advanced nature of these features, we'll cover only the most basic features here. For more information, see *Using JMP* at **Help ▶ Books ▶ Using JMP ▶ Ch. 5 ▶ Assign Column Properties**.

One of the common operations performed with the Formula Editor is creating a new column of data that contains a calculation from existing columns. To illustrate this feature, let's return to our **Movies.jmp** data table. For example, say we want to obtain the international revenues from these movies by subtracting the domestic revenues (Domestic $) from the worldwide revenues (Worldwide $).

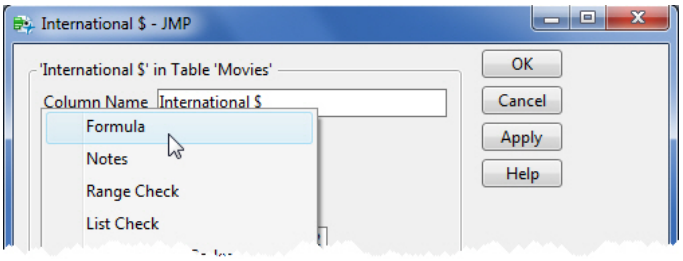
1. First, we need to create a new column. Double-click in the column head to the right of our last populated column (Director) (See Figure 2.28). Type **"International $"** in the heading, and press Enter. Click in the column head to highlight.

Figure 2.28 Creating a New Column



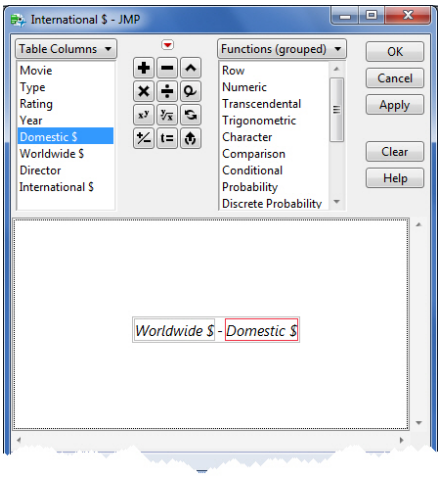
2. Select **Cols ▶ Column Info**. The Formula window appears in the **Columns Properties** menu (see Figure 2.29). Select **Formula** and **Edit Formula**. You see a list of columns or variables on the left side of the window.

Figure 2.29 Opening the Formula Editor



3. Click on the **Worldwide $** column, then the "**-**" symbol in the center palette, and then the **Domestic $** column. You see your formula take shape in the preview window* (see Figure 2.30).

Figure 2.30 Creating a Formula



4. When you click **OK**, the calculated values appear in the column of your data table.

Value Ordering

Value ordering allows you to specify an order to the values of a categorical column. JMP's built-in defaults order common ordinal columns such as months of the year or days of the week, but there are other instances when you'd like to arrange responses (values) in some logical order for graphs and analyses. For example, some surveys have a range of responses from "Not Satisfied" to "Very Satisfied," with a few intermediate responses in between.

In the **Movies.jmp** data table, we want to reorder the rating of the movies to display in this order: G, PG, PG-13, and R.

1. Click on the **Rating** column. Select **Cols ▶ Column Info**, and then select **Column Properties**.
2. Select **Value Ordering** (see Figure 2.31). A new window appears with the available responses from the Rating column. Select a response and move it up or down or reverse the order, whichever is appropriate. Select **Reverse** for this example (see Figure 2.32).

Figure 2.31 Value Ordering--Column Properties Menu

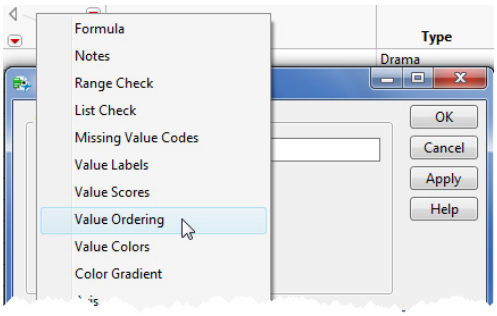


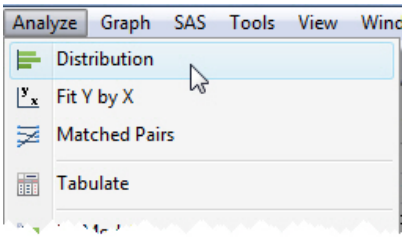
Figure 2.32 Specifying the Order of Categorical Variables

3. When you are satisfied with the order, click **Apply** and **OK**.

Now let's see the results of this exercise by employing the distribution platform, which is discussed in Chapters 3 and 5. Here's a preview:

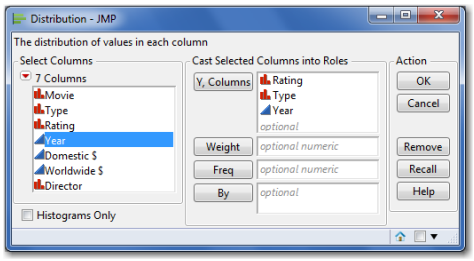
4. Go to **Analyze ▶ Distribution** (see Figure 2.33).

Figure 2.33 Launching the Distribution Platform



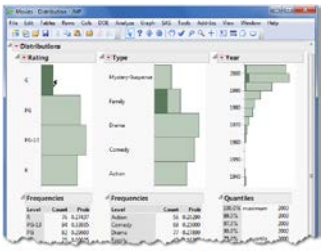
5. In the Distribution window, select **Rating**, **Type**, and **Year**, and click **Y, Columns**. Click **OK** (see Figure 2.34).

Figure 2.34 The Distribution Launch Dialog



6. Three bar graphs appear side by side. Although "R" was the first response listed in the Value Ordering window, it appears last at the bottom of the Distribution graph shown at right. If we did not make this change, the order of these responses would be reversed.
7. Click on the green **G** bar under Rating (see Figure 2.35). The G responses are now highlighted in the G bar as well as those same responses reflected by Type and Year. This dynamic visual feature is available in all JMP graphs.

Figure 2.35 The Distribution Results Window



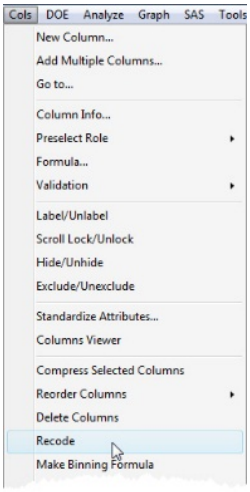
Recode

The Recode command is useful when you have a column of data containing values that you'd like to rename or consolidate. For example, if you have data labeled iPod, Nano, iTouch, and Shuffle, you might want to consolidate these into one response: iPod. Recoding assigns a specified new value to all of the existing responses of the original name or value.

In the **Movies.jmp** data table, we want to replace PG-13 movies with a PG rating because many movies made before 1985 only contained ratings of G, PG, and R.

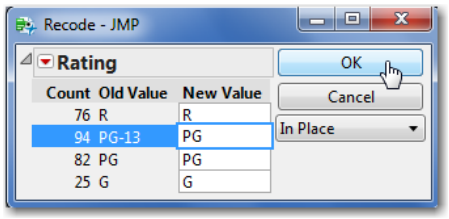
1. Select the column you want to recode, **Ratings**. Select **Cols ▶ Recode** (see Figure 2.36).

Figure 2.36 The Recode Option



2. This command generates an input window of current and unique responses, with an area to the right to specify a new value. In the box to the right of PG-13, type **PG**, then click **OK** (see Figure 2.37).

Figure 2.37 The Recode Dialog Window



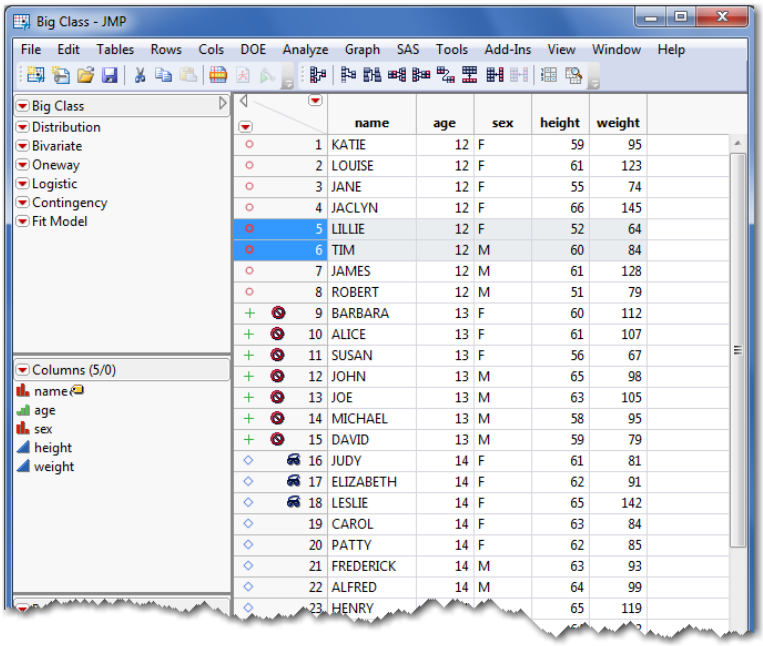
Note: Once you've selected a new value, you can replace that value in the same column, create a new column with these values, or even create a formula column. Be careful! If you select **In Place**, these values cannot be changed back because the Recode command replaces values in that column.

2.5 Selecting, Highlighting, and Filtering Data: Row States

So far we have focused on column properties; let's now look at rows or the observations in your data. In the process of exploring or analyzing data, it is often valuable to *drill-down* or to see and compare subsets of your data or rows. JMP makes this task seamless and simple through a concept called Row State that assigns one or more of the following six conditions to one or more rows (or sets of observations) within a data table; Selected, Hidden, Excluded, Labeled, Marked or Colored.

- **Selected** will appear as a highlighted row which will correspond to a highlighted point in any corresponding graph, you can also easily subset highlighted rows.
- **Hidden** means that the row will not appear in any graph. A blindfold icon appears next to hidden rows because you will not see that row in any graph.
- **Excluded** means that the row will not be used in any calculated result. A red circle with a line through it next to its row number indicates an excluded row.
- **Labels** provide the columns value when the observation is selected within an appropriate graph. Labels look like a price tag and appear next to the column in the column panel.
- **Markers** are distinguishing symbols typically used to represent a group within a categorical variable.
- **Colors** are used to represent different groups in a categorical variable or as a gradient in a continuous variable within graphs. Both Colors and Markers are covered in the next section.

Figure 2.38 Row States Indicated in a Data Table



Hiding and Excluding Data: Using Data Filter

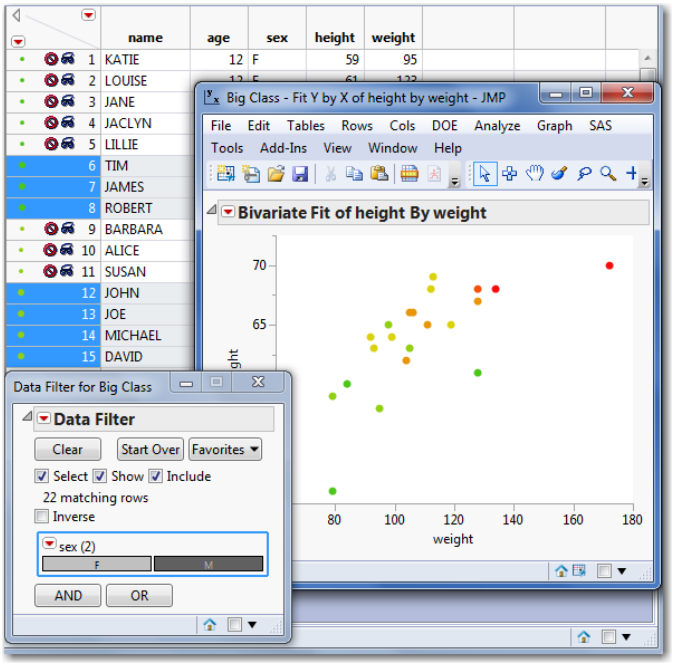
Hiding a row prevents that data from appearing in any graph (but is not excluded from any analysis). Conversely, excluding a row will remove the row from any calculated result but will still show the point in a graph. If you prefer to both hide and exclude a row, you can directly select a row or rows in your data table with your mouse, then go to **Rows ▶ Hide and Exclude**. This is an effective approach when you are dealing with few specific points like outliers.

When exploring data, however, it can be more efficient to hide and/or exclude entire groups or ranges within your data table, but rather than thinking about what you DON'T want to see or analyze, it is more natural to think about what you DO want to see. That is, to Show (rather than Hide) and Include (rather than Exclude).

JMP's Data Filter tool (**Rows ▶ Data Filter**) provides this capability and can be applied to any graph or analysis platform. The Data Filter allows you to dynamically show, include, or select groups within a column and toggle between them, or to specify a custom range within a continuous variable and create a slider to filter the graph or analysis. The Data Filter automatically hides and/or excludes values that are not selected.

In Figure 2.39, we have launched and asked the Data Filter to "Show" and "Include" the male students in our Big Class data graph. Notice that it has automatically hidden and excluded the female students in data table in the background. Data Filter will be illustrated in Chapter 6.

Figure 2.39 Using the Data Filter to Select a Group

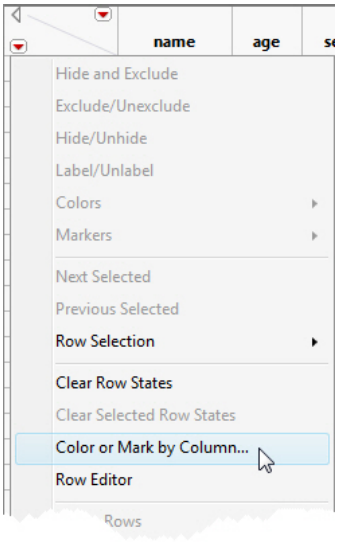


2.6 Adding Visual Dimension to Your Data

JMP is designed to be visual. Its many useful tools help you visualize or communicate your data effectively. For example, you can use colors or unique markers to signify a range or value of another column in any appropriate graph. Any color or marker assigned to your data can be saved and used in any number of graphs. You can change these colors or markers at any time. We return to our **Big Class.jmp** data table to illustrate this feature:

1. The **Rows** menu provides access to these features. Another way to access these features is through the "Rows" red triangle in the upper left side of the data grid. (Red triangles appear throughout JMP and provide context-specific options.) In this section, we will utilize the red triangle to access these features, but note that it or the Rows menu will provide the same access (see Figure 2.40).

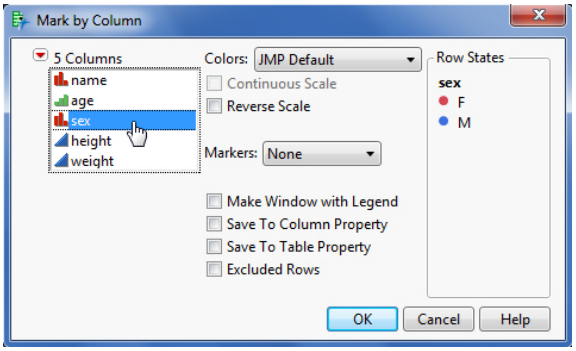
Figure 2.40 Color or Mark by Column



2. Select **Color or Mark by Column**.

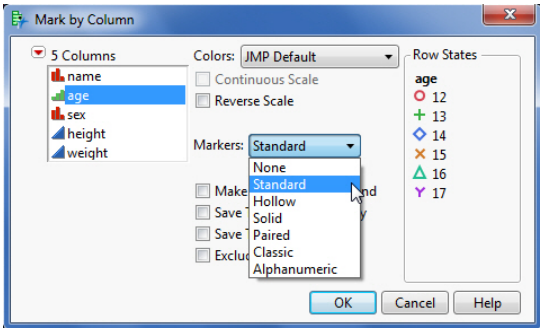
3. Select the column you'd like to distinguish with color (sex, in this example). You can see how JMP will express these values in color on the right side of the window (see Figure 2.41). Once you are satisfied, click **OK** and you will see colored markers preceding the row numbers in the data table.

Figure 2.41 Color by a Column



4. Alternatively in the same window, you can also distinguish points by using unique markers (for example, symbols). Like colors, unique markers can be assigned to categorical or continuous columns. Click on the **Markers** drop-down menu (see Figure 2.42). JMP provides many different marker types, and a submenu allows you to view and select the desired type.

Figure 2.42 Mark by a Column



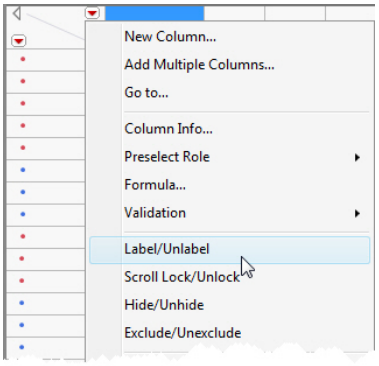
Note: You can color by one column and use distinct markers for a second column by first completing this process to apply markers on one column, then repeating this process to apply color on another column.

Adding Labels to Data

Sometimes in the process of exploring your data, it is useful to identify a point by a name, territory, or product type rather than its row number in the data table. Adding labels allows you to see these identifiers in a graph by simply clicking on a point of interest. For example, in the **Big Class.jmp** data table, we want to see the name of the student (rather than a row number) in a graph:

1. First, select a column by clicking on the column name in the data table. This selection activates the **Label option** when selecting the Columns red triangle. Select **Label/Unlabel** (see Figure 2.43).

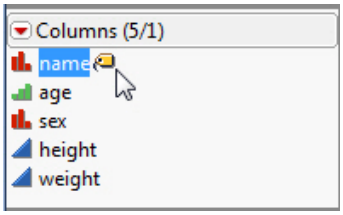
Figure 2.43 Adding Labels to a Column



2. You then see a label or what might look like a price tag next to that column in the Columns panel of the data table (see Figure 2.44). When creating a graph, labels with the name of the student (rather than the row number) are displayed when they are selected with the mouse.

   Note: You may add labels for more than one column. For example, we might want to have a label containing both the name and age of the student. Simply highlight all the columns you'd like to label first or repeat this process and add a label to "age".

Figure 2.44 A Column to Display Labels in Graphs



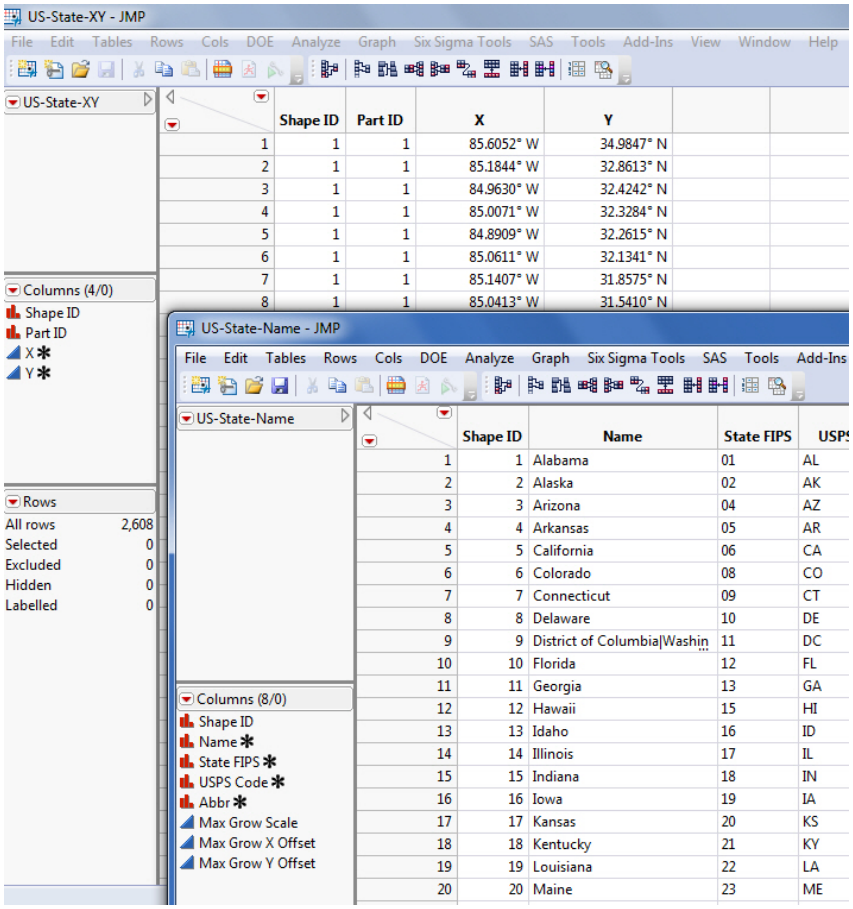
2.7 Shape Files and Background Maps

Creating thematic maps is easy to do and explained step-by-step in Chapters 3 and 4. In this section, we will describe a special type of data table required to create thematic maps called shape files.

JMP includes about ten common shape files such as States and Counties of the US and Countries of the World and others. You can import or create new shape files such as sales territories, and these need not be geographic maps. Shape files could represent any space: for example, a football stadium, an assembly plant, or an office building.

Shape files consist of two data tables, a boundary file, and a name file that share a common Shape IDs column. The boundary file (or XY file) provides the outline of the shape as a polygon that corresponds to each Shape ID. The name file provides the name (or abbreviations of the name) of the shape ID. Figure 2.45 provides an illustration of these special data tables.

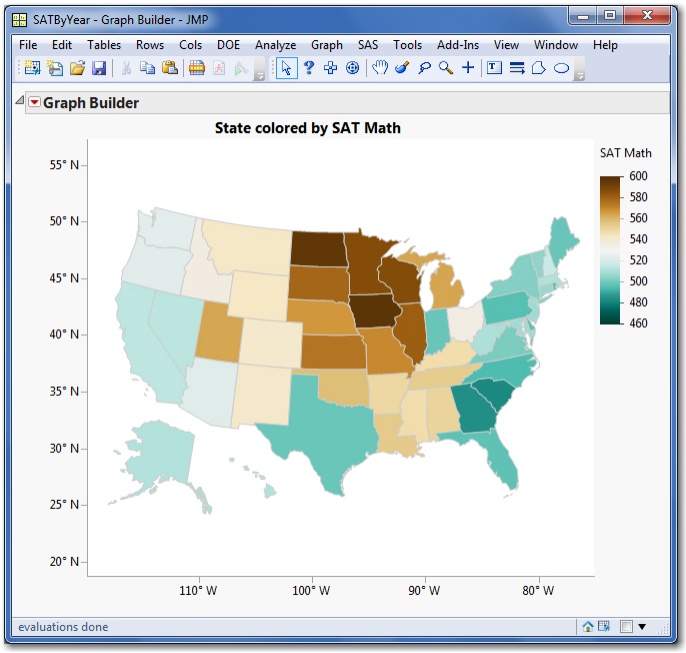
Figure 2.45 Shape Files Contain Both a Name and XY Data Table



Built-in shape files are included in the Maps folder: **C: ▶ Program Files ▶ SAS ▶ JMP ▶ 11 ▶ Maps.** Should you wish to create or import new shape files, they must be placed in this same Maps folder.

When a column containing names (corresponding with a Shape ID) is dragged into the Shape/Map box in Graphbuilder, it will render the boundaries/shapes (see Figure 2.46).

Figure 2.46 Using Shape to Create Thematic Maps



Note: In addition to shape files, JMP contains a variety of background or reference maps that provide the base map that you can plot data upon, provided that the points you wish to plot on the background contain latitude and longitude information. OpenStreet Maps was recently added to allow you to impose points or graphs at the street address level (see section 4.2).

2.8 The Tables Menu

The Tables menu is a collection of JMP tools you'll need to manage your data, whether you're sorting it, transposing it, or joining multiple data tables. Put another way, if your data is not structured in a manner that fits the JMP analysis framework, you need to use these commands to improve the structure. To keep things simple, we'll cover just a few of these features, including sorting, joining, and dealing with missing data. In this section, we learn:

- How to structure your imported data into a form you would like to see or that JMP will recognize.
- What to do when you have missing data.

Using the **Big Class.jmp** data table, let's first take a quick look at the **Summary** option under the **Tables** menu. This command allows you to obtain a variety of summary statistics for any column.

- Select **Tables** ▶ **Summary** (see Figure 2.47). Choose **height** from Select Columns and **Mean** from the **Statistics** menu (see Figure 2.48). Click **OK.** This action will generate a new data table, *Summary of Big Class,* with a mean height of 62.6.

Figure 2.47 Summary Platform from the Tables Menu

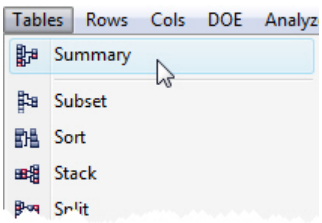
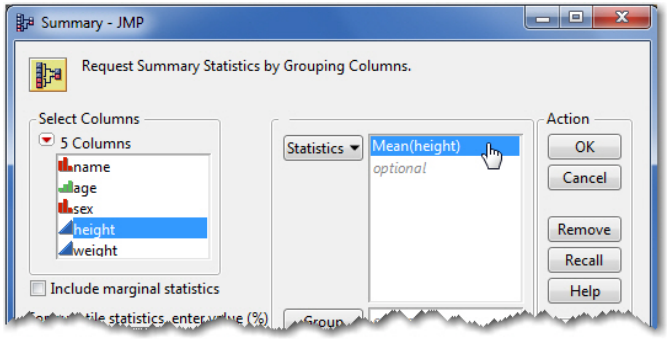


Figure 2.48 Summary Dialog Window

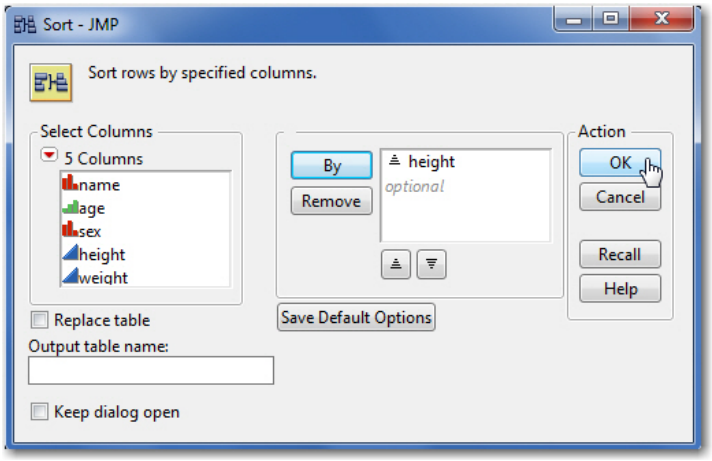


Sorting

You can sort numeric columns from highest to lowest or lowest to highest. With character columns, you can sort character data by alphabetical or reverse alphabetical order. Using JMP's sorting option keeps the rows (your sets of observations) intact. Sorting also creates a new JMP data table with the sorted values (if you check **Replace table**, the sorted values replace the existing data table).

1. Select **Tables ▶ Sort**. In the resulting window, identify which column you want to sort. Select **height** and click **By.**
2. Click on the column(s) you want to sort in the right window (height, in our example) to highlight the column.
3. Select the way you want to sort them, highest to lowest or lowest to highest, using the corresponding triangle icon (see Figure 2.49). Click **OK.** Each entire row is sorted according to the conditions you apply.

Figure 2.49 The Sort Launch Dialog



More information on sorting is available in **Help ▶ Books ▶ Using JMP ▶ Ch. 6 Reshape Data**.

Joining

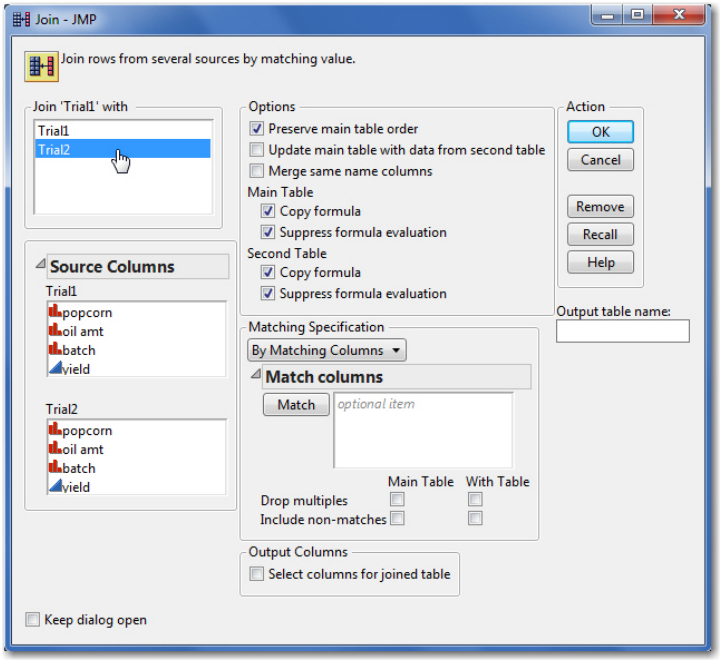
The **Join** option from the **Tables** menu allows you to combine or merge two or more different data tables into one. If some of the columns in your original data have the same name and type, this is a simple process. If not, there are some handy JMP tools to help you select how two different data tables can be joined. Let's look at a simple example:

1. First open the data tables you'd like to join. Select **Tables** ▶ **Join**.

   Trial1.jmp and **Trial2.jmp** can be found at **Help** ▶ **Sample Data** ▶ **Open the Sample Data Directory** ▶ **Trial1.jmp** and repeat for **Trial2.jmp**.

2. The window indicates your active data table and prompts you to select another data table that you want to merge or join. Select the data table(s) you want to join (see Figure 2.50). The column headings of each appear in the Source Columns windows.

Figure 2.50 The Join Launch Dialog Window



3. Decide how you want to join the data under the **Matching Specification** drop-down menu.
   a. **By Row Number** joins your data side-by-side by its row number.
   b. If your data has different column headings or you want to select a subset of columns, use **By Matching Columns**. Click on a column from each of the **Source Columns** windows you'd like to match and click **Match**. You now see each of those selected columns in the Match columns window with an "=" symbol between them.

4. If you want to name the new data table, enter the name in the **Output** table name box (otherwise, it will be named Untitled), and click **OK** (see Figure 2.51). A new data table appears (see Figure 2.52).

Figure 2.51 Join by Row Number

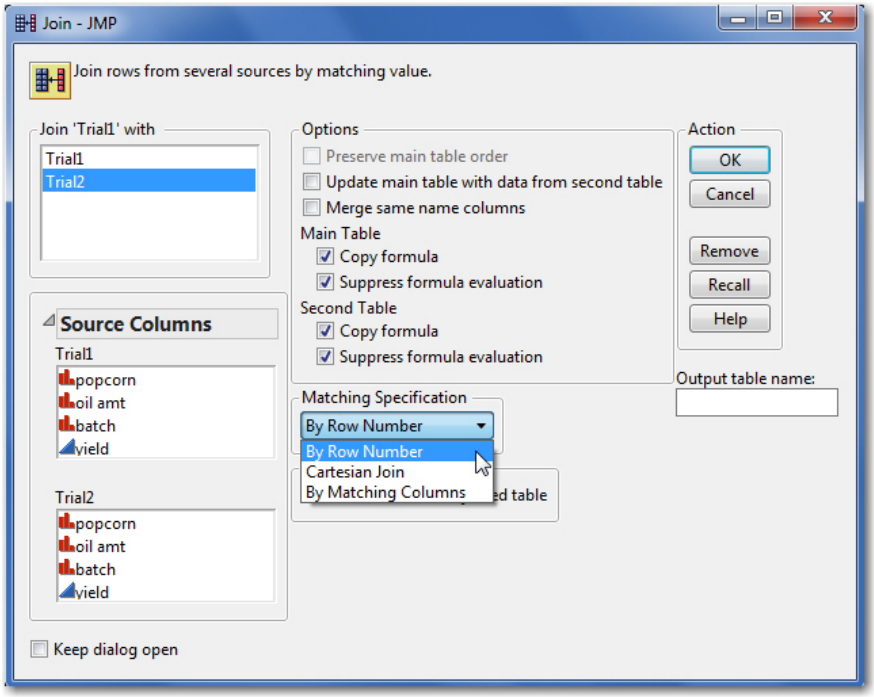
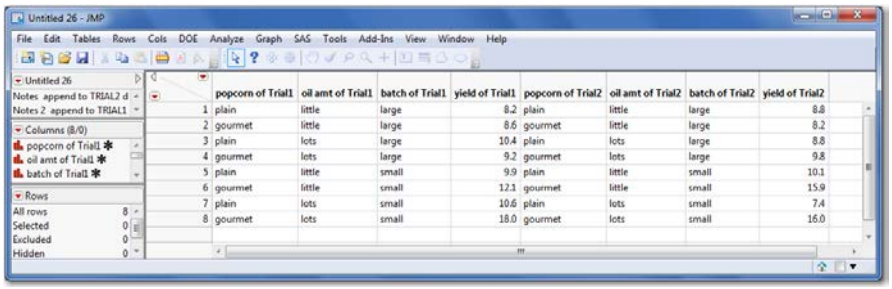


Figure 2.52 The Joined Data Table

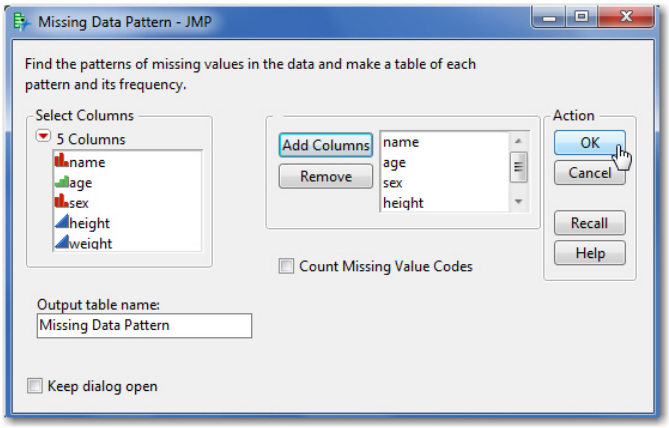


Missing Data

The Missing Data Pattern window can help you identify the quantity of missing data or whether any patterns exist due to non- response, data importing, or data entry errors. The **Missing Data Pattern** feature under the **Tables** menu (illustrated on the left) searches your specified columns and summarizes the frequencies of missing data. To explore this feature, some values from the **Big Class.jmp** data table have been removed.

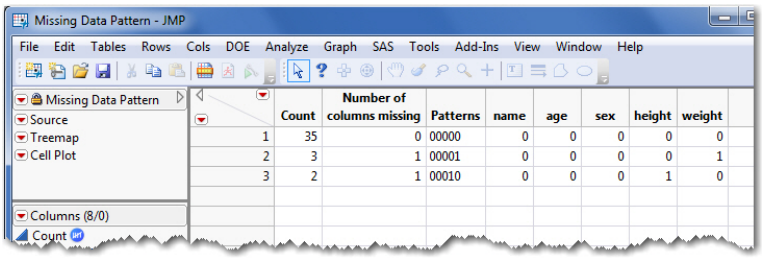
1. Select **Tables** ▶ **Missing Data Pattern**. This generates the window on the right (see Figure 2.53).

Figure 2.53 Missing Data Pattern Dialog



2. Select the columns in the left panel that you want to search. Select **Add Columns**, and then click **OK**.
3. This command generates a new Missing Data Pattern table that contains a count of rows that have missing values and a count of rows that have the same missing values among the same column(s) (see Figure 2.54).

Figure 2.54 Missing Data Pattern Results



	Count	Number of columns missing	Patterns	name	age	sex	height	weight
1	35	0	00000	0	0	0	0	0
2	3	1	00001	0	0	0	0	1
3	2	1	00010	0	0	0	1	0

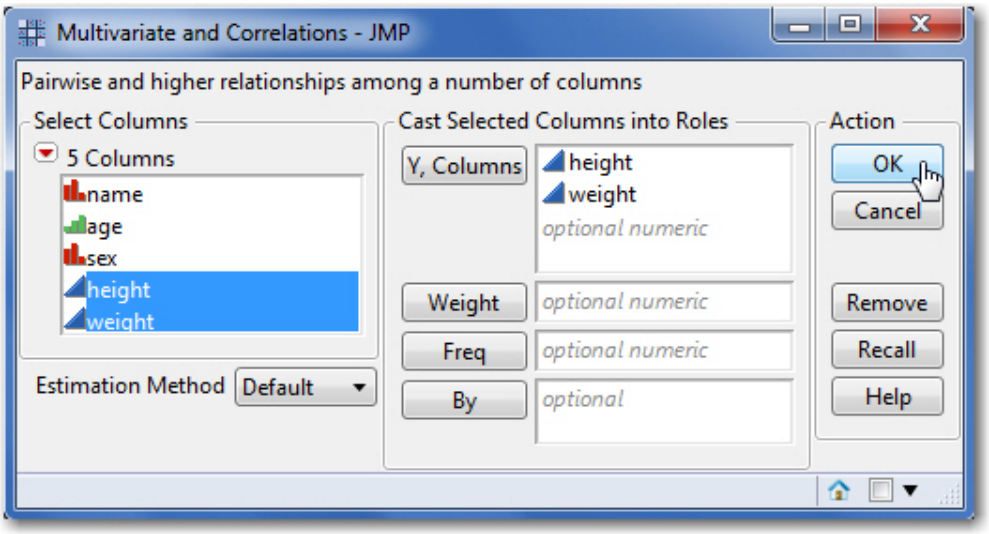
Note: You can proceed without addressing missing values, but JMP will by default ignore (or exclude) any rows containing missing values in most analysis platforms. JMP 11 includes “Informative Missing”

options in a couple platforms that utilize rows that would otherwise be ignored (JMP Pro has this feature in several platforms). This is important if you have lots of missing data because more usable observations will generally lead to better statistical models.

One solution to missing continuous data is to impute them. *Imputing* analyzes similar values in other columns and rows to estimate the missing value. JMP has an imputation feature under the red triangle within the Multivariate window. To illustrate, some values from the height and weight columns in **Big Class.jmp** have been removed:

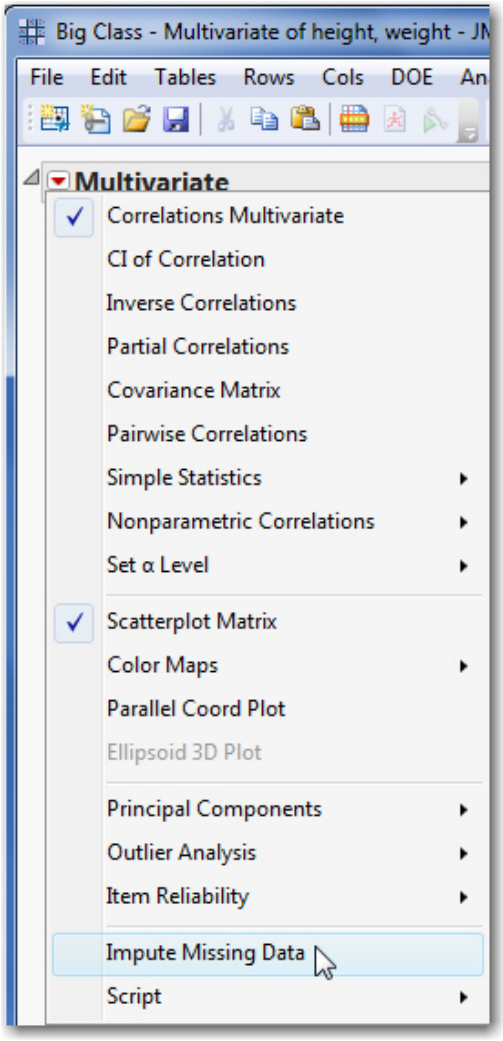
1. First, run the multivariate platform. Select **Analyze ▶ Multivariate Methods ▶ Multivariate**. Select **height** and **weight** (the continuous columns), in the **Y, Columns** window, and then click **OK** (see Figure 2.55).

Figure 2.55 The Multivariate Platform Offers Imputation



2. Click the red triangle next to **Multivariate** and select **Impute Missing Data** (see Figure 2.56). **Note:** Like many menu options in JMP, Impute Missing Data only appears in the menu when appropriate--in this case, when data is missing. A new data table is generated with the estimated missing values in place.

Figure 2.56 Launching the Imputation Option



3. Because you can only impute continuous values, cut and paste these columns into your original data table, which might contain other data types. (Alternatively, use **Update** from the **Tables** menu.)

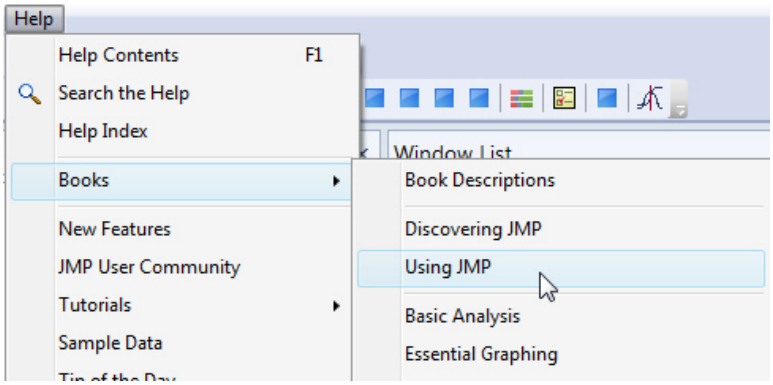
   Note: Methods of handling missing values are best selected with the help of an expert.

2.9 Summary

In this chapter, we covered a wide range of topics on getting your data into JMP and learning how to manage it. Because the data table not only stores your data but also stores key information that drives the appropriate analysis and graphs, it serves as the critical starting point for all exploration and visualization within JMP.

Analyzing and visualizing data often requires special features, and there are many advanced features in JMP that we didn't address. As we've indicated, your copy of JMP includes extensive documentation, which you can access through the **Books** section under the **Help** menu. We recommend the *Using JMP* book for a complete discussion of data, data tables, and the **Tables** menu. Select **Help ▶ Books ▶ Using JMP** (see Figure 2.57).

Figure 2.57 The Using JMP Book included in Help



Chapter 3 Index of Graphs

3.1 Basic Charts

3.2 Thematic Maps

3.3 Control Charts, Pareto, Variability and Overlay Plots

3.4 Graphs of One Column

3.5 Graphs Comparing Two Columns

3.6 Graphs Displaying Multiple Columns

3.7 Summary

This chapter is a quick reference to commonly used graphs in JMP. The format of this chapter differs from other chapters for good reason. In this chapter, you can peruse the graphs much like you'd look through a cookbook—find what you want and follow the recipe. A picture of the graph, brief description, required data conditions, usage description, and the steps required to generate the graph immediately follow. This chapter is not intended to be a complete index of graphs available in JMP, but we have tried to pick out those that we see used most often. This chapter is for the user who knows what graph they want and can pick it out by how it looks or what it's called.

You will find that many graph windows have additional options that allow you to further enhance your graphical result. However, we have focused on the steps to generate the base case of each graph, which are illustrated in the figures that accompany each graph.

Some of the graphs illustrated in this chapter are accessed from the **Graphs** menu, while others are accessed from the **Analyze** menu.

Statistical output is provided with graphs generated from the **Analyze** menu. For instructions on sharing or printing graphs and on surfacing graphs into other applications such as Microsoft Word or PowerPoint or interactive HTML, see Chapter 7, sections 3 and 4.

There is more than one way to generate many of the graphs in this chapter. A selection of graphing methods will be presented including newer methods available in the more intuitive Graph Builder and Control Chart Builder platforms, as well as instructions for the legacy Chart and legacy Control Chart platforms. You can decide which method works best for you. The preferred method is presented first.

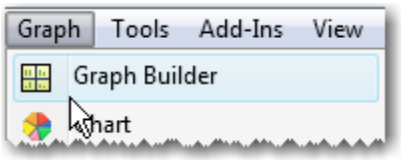
You can customize the appearance of any graph (including colors, markers, axes, legends, and fonts) by interacting with simple palettes and controls or by simply right-clicking on the area or item you want to change. See Chapter 2, sections 5 and 6, for more details on customizing the appearance of your graphs.

3.1 Basic Charts

The Graph Builder platform produces dozens of charts for general purposes. The platform responds differently depending on the modeling types of the data and the roles of the columns you select. In the context of those modeling types, it visually alerts you to charting options that come with your selections. This section is designed to help you understand some of the more commonly used charts that can be produced with this multi-purpose platform and how to produce them.

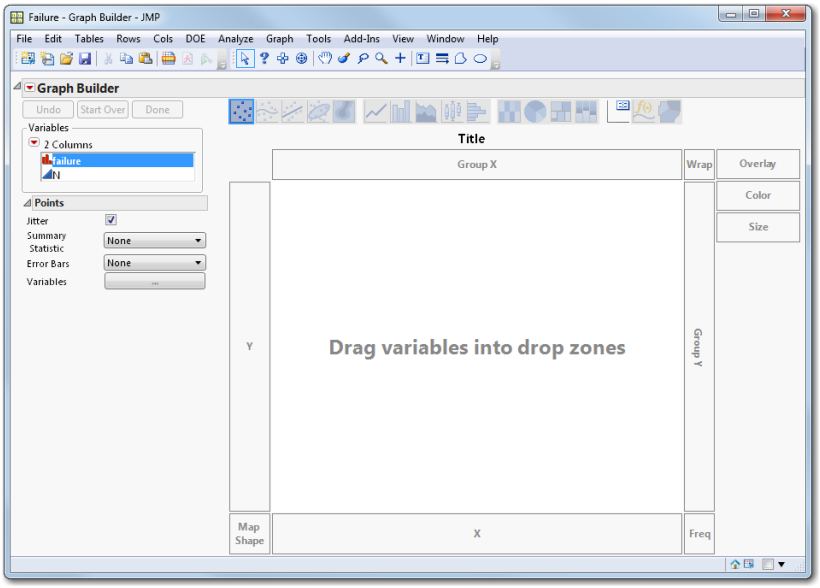
The Graph Builder platform is accessed from the Graph menu (see Figure 3.1).

Figure 3.1 Graph Builder Menu



It produces appropriate graphs for each continuous nominal or ordinal column that you drag and release into a drop zone (see Figure 3.2).

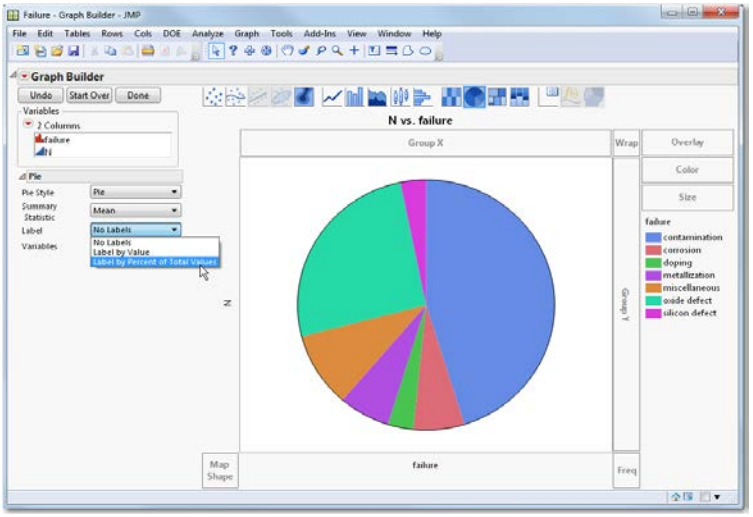
Figure 3.2 Graph Builder Drop Zones



You can also learn more about using the Graph Builder platform to assist you in problem-solving in Chapters 5 and 6.

Graphs appear in the center area framed by X and Y drop zones. When columns are dragged to the drop zones, graphs instantly appear, and additional options become enabled (see Figure 3.3).

Figure 3.3 Graph Builder Pie Chart

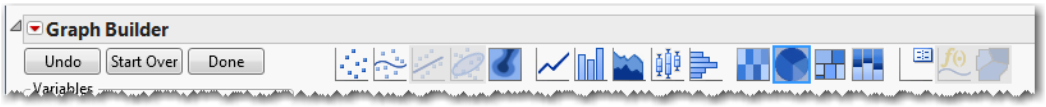


If column modeling types like continuous and nominal and ordinal are unfamiliar to you, see Section 2.3.

By default, a point chart is generated for any column type when dragged to the X or Y drop zones.

Charting choices are revealed in the element icon palette at the top of the Graph Builder window (see Figure 3.4).

Figure 3.4 Element Icon Palette

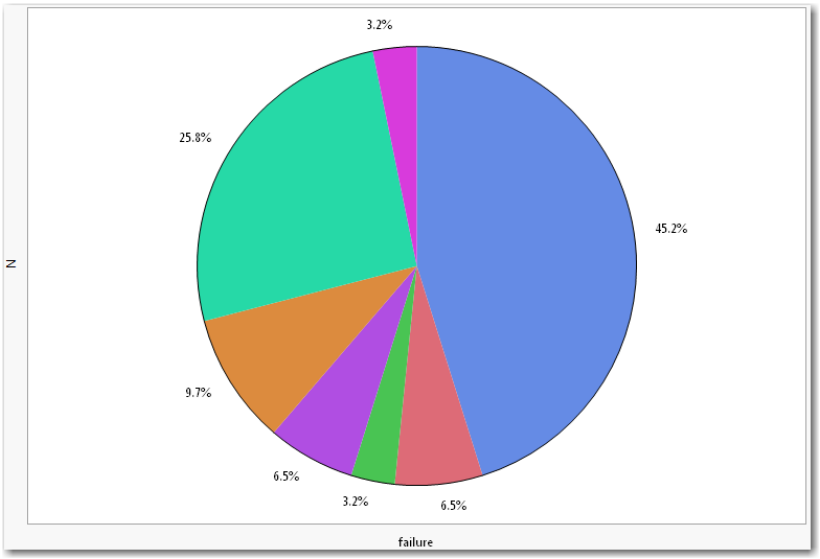


As columns are dragged to drop zones, graphing options become enabled and their icons are highlighted in the palette. Unavailable graphing options for the column combinations selected are automatically disabled and appear to be grayed out. Graphing options respond to the modeling types of the columns and the drop zones selected. You can use the SHIFT key to apply multiple elements at once. If you are unfamiliar with the graphs depicted in the element type icon palette, experiment by clicking on them when they are enabled. You can also click **Undo** if you don't like the result.

Pie Chart

A circular chart divided into areas proportional to the percentages of the whole or total.

Figure 3.5 Pie Chart



Data Table Access

To access the data table used for the above pie chart example, follow **Help ▶ Sample Data ▶ Control Charts ▶ Failure.**

Usage

Used when representing proportions, percentages, or fractions of any measured quantity. Some examples are market share, customer preferences, and percent of any kind of category or group. As shown, pie charts can display percentages of seven different failure types (see Figure 3.5).

Required

One ordinal or nominal column for labels and a continuous column to define the size of the pie chart sections.

Other column combinations are supported. See **Help ▶ Search**. Type Pie Chart in the search field.

Using the Graph Builder method, select **Graph ▶ Graph Builder**. Drag a continuous column with count data to the Y drop zone and drag a nominal variable with the group identifier to the X drop zone. From the element palette, choose the pie element. To add percentages to the pie slices

as depicted in Figure 3.5, choose **Label by Percent to Total Values** from the **Element Properties Panel** on the left.

Using the legacy Chart platform, select **Graph ▶ Chart.** Then select **Pie Chart** from the **Options** drop-down menu. Select a continuous column, and then from the **Statistics** drop-down menu, select a summary such as **Sum**. Select a nominal column, move it to the **Categories, X, Levels** role, and click **OK**. To add percentages to the pie slices as depicted, select the red triangle, and then select **Label Options ▶ Label by Percent of Total Values**.

Bar Chart, Line Chart, and Scatter Chart

Charts with bars, lines, and points showing lengths or positions proportional to quantities.

Figure 3.6a Bar Chart with Values

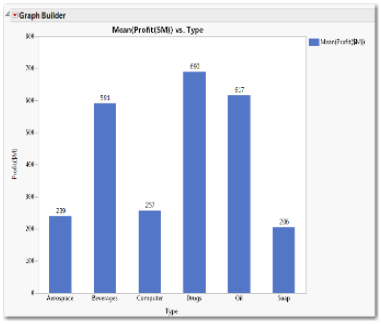


Figure 3.6b Line Chart

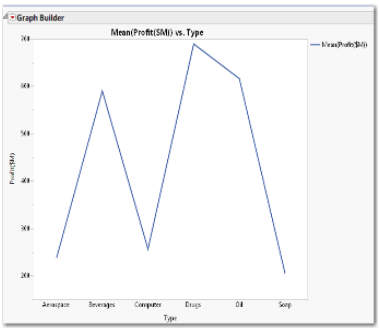


Figure 3.6c Point Chart

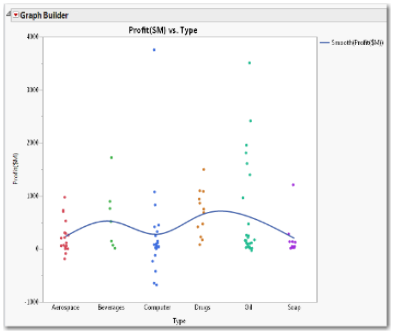
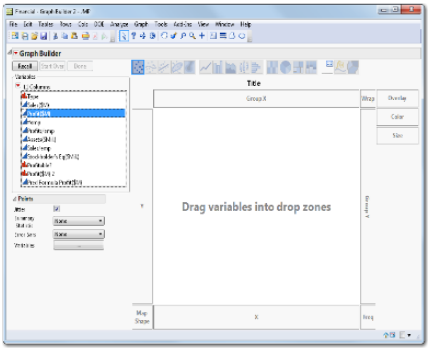


Figure 3.6d Graph Builder Pallette



Data Table Access

To access the data table used for the above chart examples, follow **Help ▶ Sample Data ▶ Business and Demographic ▶ Financial.**

Usage

Similar to pie charts representing individual values, proportions, percentages, or depictions of any measured quantity. Some examples are market share proportions and customer preferences expressed as percentages of any kind of category or group. As shown, these charts display profits for six company types in three different chart styles (see Figures 3.6a, 3.6b and 3.6c).

Required

One continuous column for the Y drop zone and one nominal or ordinal column for the X drop zone.

Using the Graph Builder method, select **Graph ▶ Graph Builder**. Then drag a continuous column to the Y drop zone and drag a nominal or ordinal column to X drop zone (see Figure 3.6d).

For a bar chart, choose the bar element . To add value totals to the bars as depicted, choose **Label ▶ Value** from the Bar Element Properties Panel on the left. Additional elements like Summary Statistic, Error Bars, and labeling are available from the Bar Element Properties Panel on the left. To generate a line chart, choose the line element . To generate a scatter chart with a smoother, choose both elements (scatter chart and smoother) . Many other charts are supported. Experiment!

Using the legacy Chart platform, select **Graph ▶ Chart**. Then select **Bar Chart, Line Chart, Needle Chart,** or **Point Chart** from the **Options** drop-down menu for the chart listed. Select a continuous column and then from the Statistics drop-down menu, select a summary such as **Sum** or **Mean**. Optionally select a nominal or ordinal column for the **Categories, X, Levels** role. To add percentages as shown, select the red triangle, and then select **Label Options ▶ Label by Percent of Total Values.**

3.2 Thematic Maps

The Graph Builder platform can be used to create interactive maps with boundaries such as states, provinces, or county boundaries, and in JMP 11, street level mapping. These mapping tools are included in JMP and stored as shape files, background maps, or as a link to a street map server. Other sources of shape files (for example, ESRI) or map shapes you create yourself can be utilized in JMP. See the JMP.com website for add-in utilities to make your own map shapes.

To create thematic maps such as those in Figure 3.7a or 3.7b, your data needs to contain boundary names or abbreviations in a column that match those that appear in the shape file e.g., "California", "CA" or "Calif". That column is dragged to the **Map Shape** drop zone (Figure 3.7d).

Figure 3.7a SAT Score Math Map

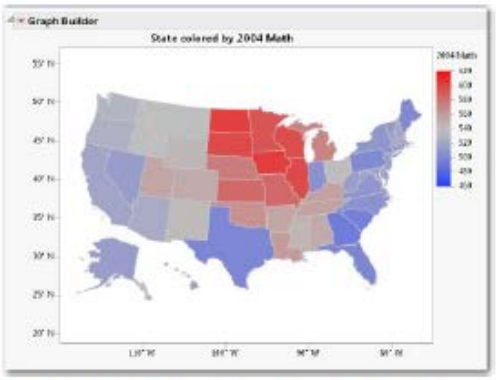


Figure 3.7b SAT Math Percent Taking Overlay

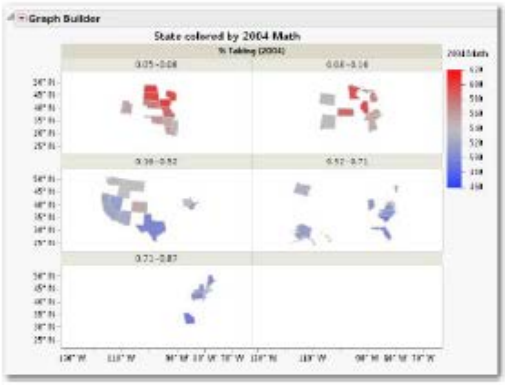
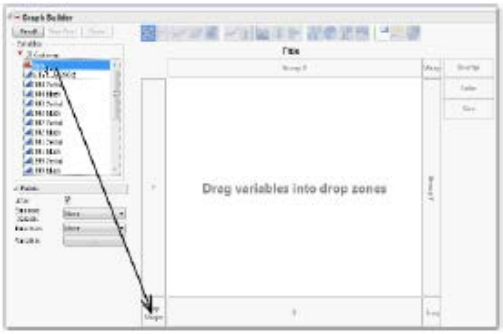


Figure 3.7c San Francisco Crime Map

Figure 3.7d Graph Builder Map Shape Drop Zone with Arrow



Data Table Access

To access the data table used using a map shape file, select **Help ▶ Sample Data ▶ Graph Builder ▶ SAT**.

To access the data table used using latitude and longitude, select **Help ▶ Sample Data ▶ Graph Builder ▶ San Francisco Crime**.

If you have latitude and longitude columns, you may plot points on a background map as those in Figure 3.7c. (see section 2.7 for more information about these file types).

Usage

A method to graphically visualize data in a spatial system. Some examples are socio-economic indicators overlaid on political boundaries, crime incidents overlaid on street maps, or where boundaries can be defined in two dimensions.

Required for thematic maps (with a shape file)

One column recognized as a map shape to provide map boundaries and one continuous column to provide a color range.

Optional

A continuous, nominal, or ordinal column for the additional visualizations. See Appendix B for descriptions of these terms. As shown in Figures 3.7a and 3.7b, the maps display US State boundaries for 2004 SAT Scores for Math and percentage of students taking the SAT in that year.

Select **Graph ▶ Graph Builder**. Then, drag a map shape column from the **Variables** list (for example, State) to the map shape drop zone (Figure 3.7d). The states can be whole names or two-letter abbreviations.

Required for thematic maps (with latitude and longitude)

One column with latitude coordinates in the **Y** drop zone and one column with longitude coordinates in the **X** drop zone. **Optional:** A continuous, nominal, or ordinal column for the additional visualizations. As shown in Figure 3.7c, San Francisco crime incidents are displayed by type overlaid upon a street level map.

Select **Graph ▶ Graph Builder**. Then, drag the longitude column from the **Variables** list to the **Y** drop zone. Then, drag the latitude column from the **Variables** list to the **X** drop zone. Right click in the map and select **Graph ▶ Background Map ▶ Street Map Service ▶ OK**.

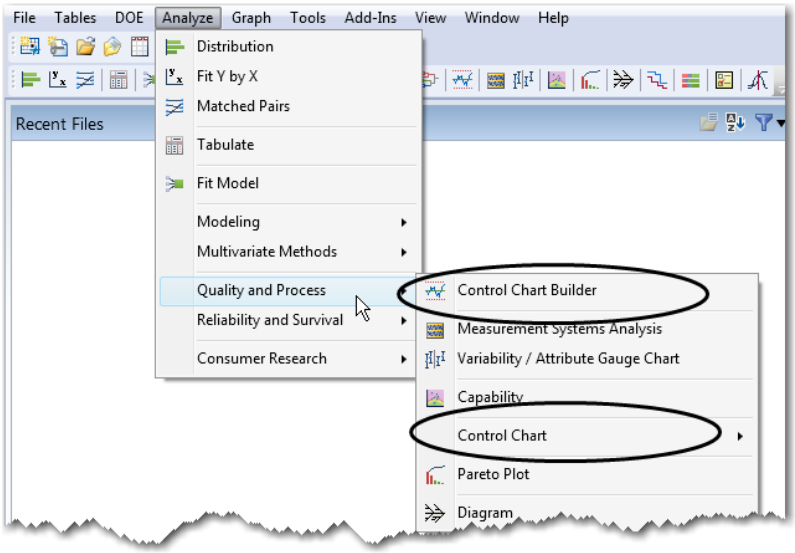
3.3 Control Charts, Pareto, Variability and Overlay Plots

This section groups typical graphs used to measure product or service quality. The graphs are common in quality improvement scenarios.

Control charts are graphical and analytic tools for deciding whether a process is in a state of statistical control and for monitoring an in-control process. Control charts help determine if variations in measurement of a product are caused by small, normal variations that cannot be controlled or by some larger, special cause that can be controlled. The type of chart to use is based on the nature of the data.

Control charts are broadly classified into control charts for variables and control charts for attributes. Control charts for variables and control charts for attributes come in several varieties with names or letters attached to them. Control charts can be generated from either the **Control Chart Builder** or the legacy Control Chart menu under **Analyze ▶ Quality and Process** (see Figure 3.8).

Figure 3.8 Quality and Process Control Chart Builder and Control Chart



Important Note: In a few of the graph instructions that follow, we use the term "numeric" -which refers to the nature of the data- in place of modeling type (continuous, nominal, ordinal). We use this term as some of the legacy platforms, like Control Chart, allow you to use either continuous or nominal/ordinal columns provided that the data is numeric. While newer platforms and their options in JMP are tied more closely to modeling type, you will find that your numeric data is likely to be continuous most of the time.

Run Chart

Displays a column of data as a connected series of points (see Figure 3.9a).

Figure 3.9a Control Chart Builder Individual Moving Range Chart

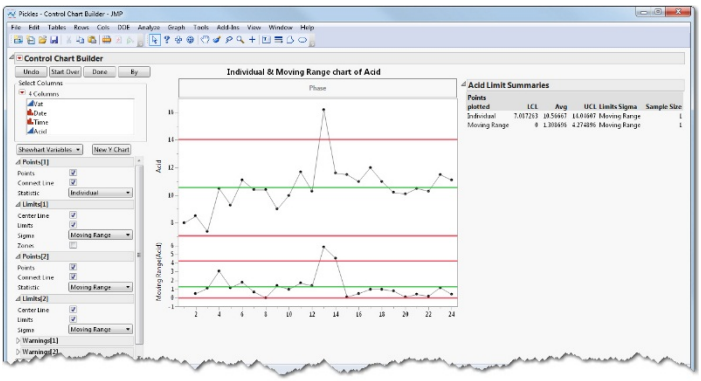
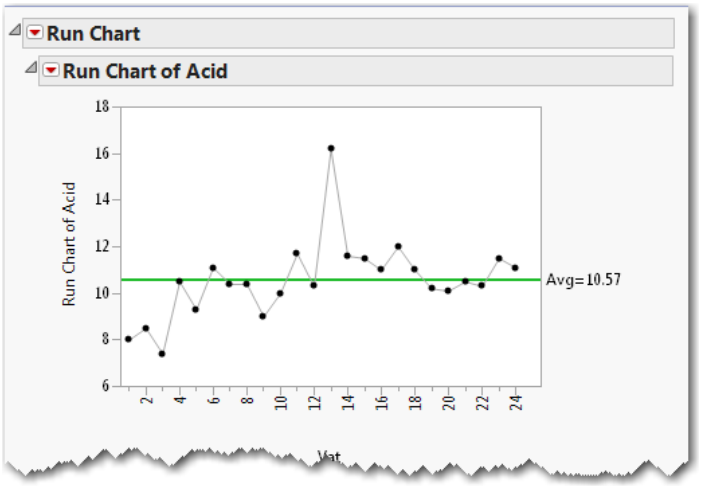


Figure 3.9b Classic Control Chart Run Chart



Data Table Access

To access the data table used for this example, follow **Help ▶ Sample Data ▶ Control Charts ▶ Pickles.**

Usage

Displays data from a column. Frequently used as a first visualization of quality data and to assess ranges of variability. Examples include delay times, counts of rejected items, or measurement of variability in a product or service. As shown, the run chart displays Acid measurements over 24 Pickle Vats with an average shown as a green center line.

Required

One or more continuous columns.

Optional

Nominal, ordinal, or continuous columns for the **Sample Label** and **By** and **X** roles.

If data in a column is sorted by ascending values of time, then the x-axis will display in the time-sorted order.

Using the Control Chart Builder method, select **Analyze ▶ Quality and Process ▶ Control Chart Builder**. Drag a continuous column to the **Y** drop zone. Use the options panel on the left to add or remove chart elements (see Figure 3.9a).

Using the legacy Control Chart platform, select **Analyze ▶ Quality and Process ▶ Control Chart ▶ Run Chart** (see Figure 3.9b).

Individual & Moving Range Chart

A process behavior chart that displays individual measurements. Individual measurement charts are appropriate when only one measurement is available for each subgroup sample.

The accompanying moving range chart displays moving ranges of two or more successive measurements. Moving ranges are computed using the number of consecutive measurements entered in the Range Span box. The default range span is 2.

Figure 3.10a Control Chart Builder IR Chart with Phases

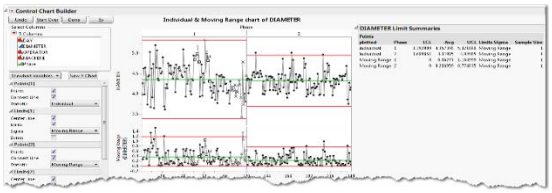
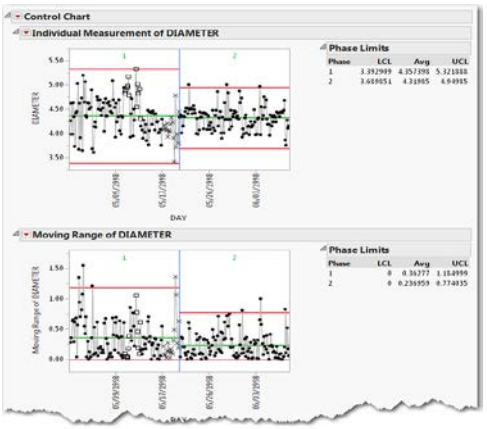


Figure 3.10b Classic Control Chart IR Chart with Phases



Data Table Access

To access the data table used for the above chart examples, follow **Help ▶ Sample Data ▶ Control Charts ▶ Diameter.**

Usage

Used when only one measurement is available for each subgroup. An example would include samples of part diameters from various lots. Individual & Moving Range charts are efficient at detecting relatively large shifts in the process average. As shown, the charts display the contrasting control limits and process performance before and after a quality improvement as phases 1 and 2 (see Figure 3.10a and Figure 3.10b).

Required

One or more numeric columns for the **Process** or **Y** role.

Optional

Continuous, nominal, or ordinal columns for the Sample Label, Phase, and By roles.

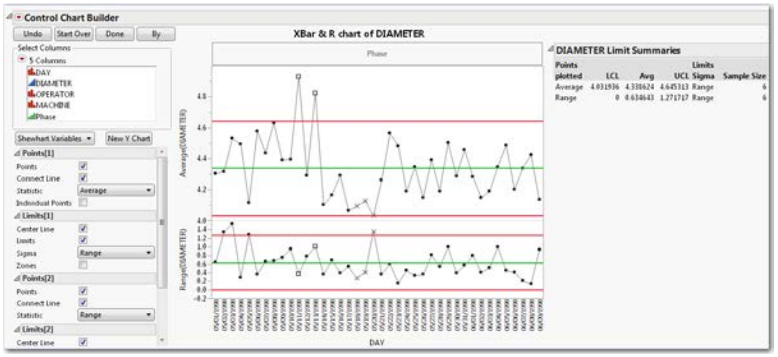
Using the Control Chart Builder method, select **Analyze ▶ Quality and Process ▶ Control Chart Builder**. Drag a continuous column to the **Y** drop zone. Drag a nominal or ordinal column to the **Phase** drop zone. Use the options panel on the left to add or remove chart elements (see Figure 3.10a). To add phases as shown, select a nominal or ordinal column for the **Phase** drop zone near the chart top.

Using the legacy Control Chart platform, select **Analyze ▶ Quality and Process ▶ Control Chart ▶ IR** (see Figure 3.10b). To add phases as shown, select a nominal or ordinal column for the **Phase** role in the dialog box.

X-Bar/R-Charts

Displays quality characteristics measured on a continuous scale. A typical analysis shows both the process mean and its variability, aligned above a corresponding range or standard deviation chart, respectively.

Figure 3.11a Control Chart Builder XBar and R chart



Data Table Access

To access the data table used for these chart examples, follow **Help ▶ Sample Data ▶ Control Charts ▶ Diameter**.

Usage

Normally used for numerical data that is recorded in subgroups in some logical manner (for example, three production parts measured every hour). A special cause, such as a broken tool, will then appear as an abnormal pattern of points on the chart. As shown, the chart displays several part diameters outside of the control limits (see Figure 3.11a).

Required

One or more numeric columns for the **Process** role.

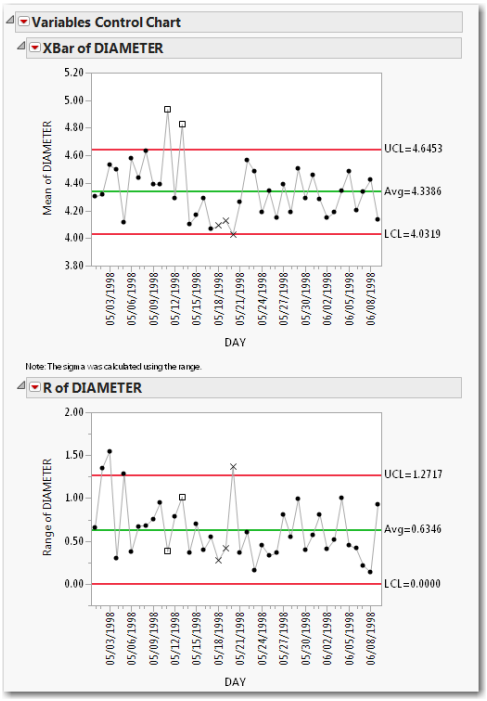
Optional

Nominal, ordinal, or continuous columns for the **Sample Label** and **By** roles.

Using the Control Chart Builder method, select **Analyze ▶ Quality and Process ▶ Control Chart Builder**. Drag a continuous column to the **Y** drop zone. Drag a nominal column to the **X** drop zone (see Figure 3.11a).

Using the legacy Control Chart method, select **Analyze ▶ Quality and Process ▶ Control Chart ▶ XBar** (see Figure 3.11b).

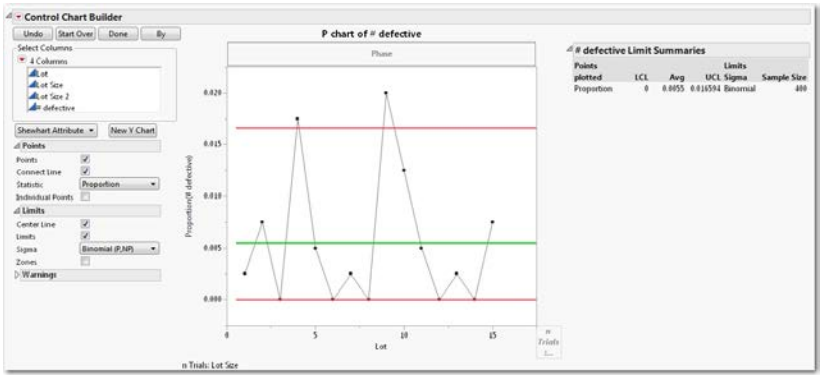
Figure 3.11b Classic Control Chart XBar and R



P Chart

An attribute chart that displays the proportion of nonconforming (defective) items in subgroup samples, which can vary in size. Because each subgroup for a P chart consists of N items, and an item is judged as either conforming or nonconforming, the maximum number of nonconforming items in a subgroup is N. The P Chart assumes a binomial distribution.

Figure 3.12a Control Chart Builder P chart of defective



Data Table Access

To access the data table used for these chart examples, follow **Help ▶ Sample Data ▶ Control Charts ▶ Washers.**

Usage

Used when one or more errors might propagate within the same sample (for example, loan transactions where more than one error may occur or where one or more errors may be present in the same unit such as on the surface of a DVD). As shown, the chart displays the proportion of defective washers across many sub-groups of washers within and outside the control limits (see Figure 3.12a).

Required

One or more numeric columns for the **Process** role.

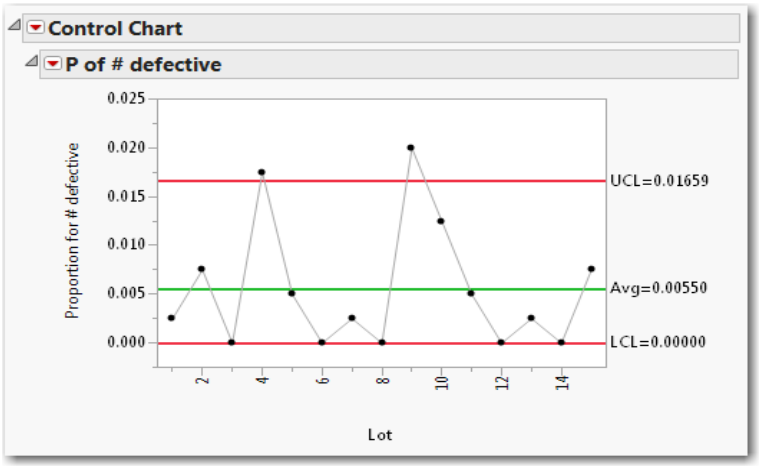
Optional

Continuous, nominal, or ordinal columns for the **Sample Label**, **Phase**, and **By** roles. Sample Size must be a numeric column. A constant or variable sample size can be specified and must be numeric.

Using the Control Chart Builder method, select **Analyze ▶ Quality and Process ▶ Control Chart Builder**. Select **Shewhart Attribute** from the chart type drop-down. Drag a column to the **Y** drop zone. Drag a column to **X** drop zone to represent subgroups. Drag a continuous column to the **n Trial** drop zone. From the Points Control, select **Statistic ▶ Proportion** (see Figure 3.12a).

Using the legacy Control Chart method, select **Analyze ▶ Quality and Process ▶ Control Chart ▶ P** (see Figure 3.12b).

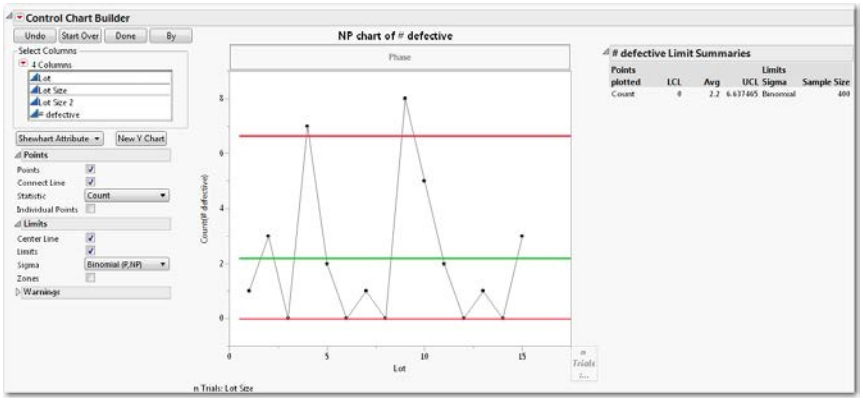
Figure 3.12b Classic Control Chart P Chart



NP Chart

An attribute chart that displays the number of nonconforming (defective) items in fixed-sized subgroup samples. Because each subgroup for an NP chart consists of N_i items, and an item is judged as either conforming or nonconforming, the maximum number of nonconforming items in subgroup i is N_i. The NP Chart assumes a Binomial distribution.

Figure 3.13a Control Chart Builder NP Chart of Defective



Data Table Access

To access the data table used for the above chart examples, follow **Help ▶ Sample Data ▶ Control Charts ▶ Washers.**

Usage

A fixed sample is taken from an established number of transactions or manufactured items each month. From this sample, the number of transactions or items that had one or more errors is counted. The control chart then tracks the number of errors per group or lot. As shown, the chart displays the number of defective washers across many lots of washers within and outside the control limits (see Figure 3.13a).

Required

One or more numeric columns for the **Process** role.

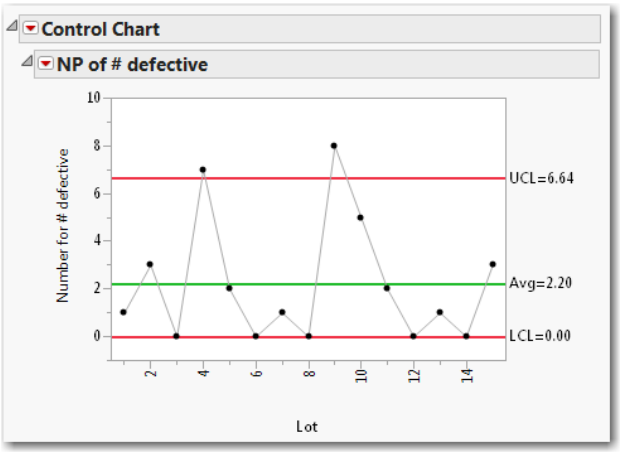
Optional

Continuous, nominal, or ordinal columns for the **Sample Label** role, **Phase**, and **By** roles. **Sample Size** must be a numeric column. A constant or variable sample size can be specified and must be numeric.

Using the Control Chart Builder method, select **Analyze ▶ Quality and Process ▶ Control Chart Builder**. From the drop-down menu, select **Shewhart Attribute**. Drag a continuous column to the **Y** drop zone. Drag a column representing subgroups to the **X** drop zone. Drag a column representing lot size to the **n Trials** drop zone. From the **Limits** menu, choose **Sigma, Binomial (P, NP).**

Using the legacy Control Chart method, select **Analyze ▶ Quality and Process ▶ Control Chart ▶ NP** (see Figure 3.13b).

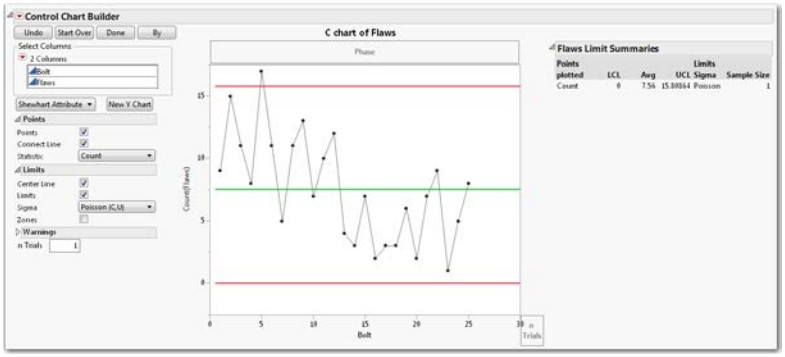
Figure 3.13b Classic Control Chart NP of Defective



C Chart

An attribute chart that displays the number of nonconformities (defects) in a subgroup. The C Chart assumes a Poisson distribution.

Figure 3.14a Control Chart Builder C Chart of Flaws



Data Table Access

To access the data table used for the above chart examples, follow: **Help ▶ Sample Data ▶ Control Charts ▶ Fabric.**

Usage

The values of C computed from each subgroup are plotted on the vertical axis and can then be used to control the quality of the subgroup. As shown, the chart displays the number of flaws in each bolt of fabric within and outside the control limits (see Figure 3.14a).

Required

One or more numeric columns for the **Process** role.

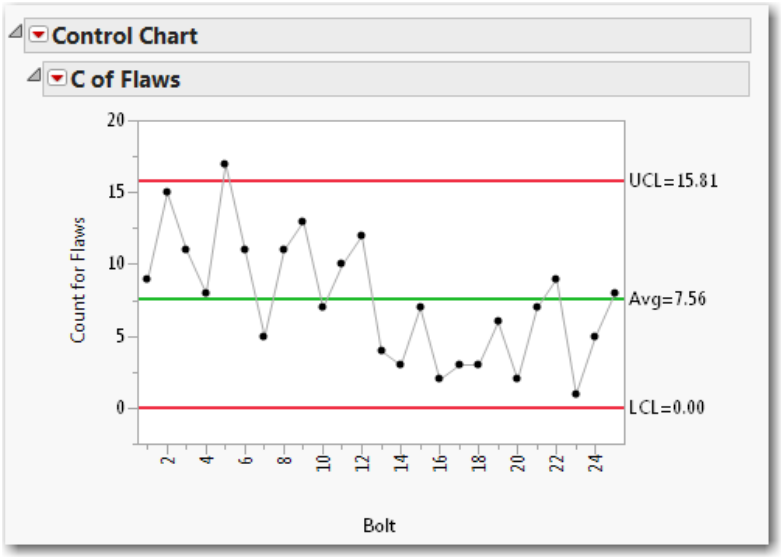
Optional

Continuous, nominal, or ordinal columns for the **Sample Label**, **Phase**, and **By** roles. **Sample Size** must be a numeric column. A constant or variable sample size can be specified and must be numeric.

Using the Control Chart Builder method, select **Analyze ▶ Quality and Process ▶ Control Chart Builder**. From the drop-down menu, select **Shewhart Attribute**. Drag a continuous column representing defects to the **Y** drop zone. Drag a column representing the subgroups to the **X** drop zone.

Using the legacy Control Chart method, select **Analyze ▶ Quality and Process ▶ Control Chart ▶ C** (see Figure 3.14b).

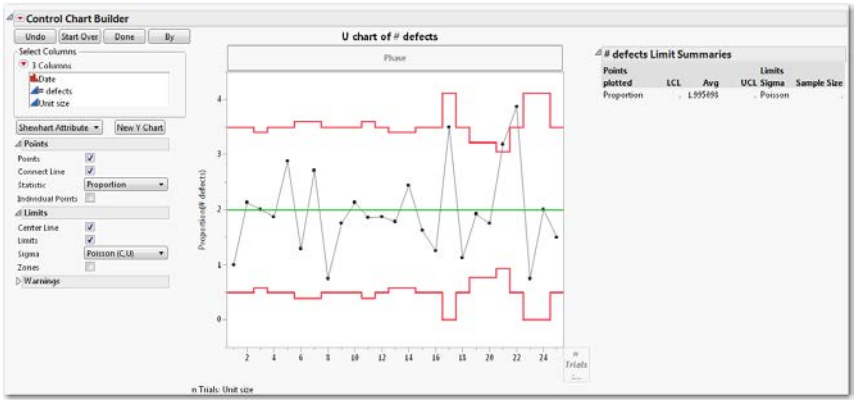
Figure 3.14b Classic Control Chart C of Flaws



U Chart

An attribute chart that displays the number of nonconformities (defects) per unit in subgroup samples that can have a varying number of inspection units. The U Chart assumes the Poisson distribution.

Figure 3.15a Control Chart Builder U Chart Defects



Data Table Access

To access the data table used for the above chart examples, follow **Help ▶ Sample Data ▶ Control Charts ▶ Braces.**

Usage

To count the number of defective units in subgroups of varying numbers. As shown, the chart counts the number of defective braces in groups of braces of varying size on specific dates and indicates if the count is within or outside of the control limit (see Figure 3.15a).

Required

One or more numeric columns for the **Process** role.

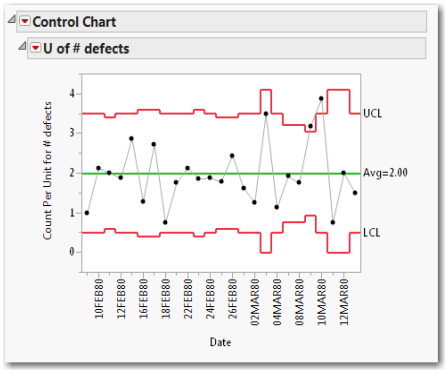
Optional

Continuous, nominal, or ordinal columns for the Sample Label, Phase, and By roles. Sample Size must be a numeric column.

Using the Control Chart Builder method, select **Analyze ▶ Quality and Process ▶ Control Chart Builder**. From the drop-down menu, select **Shewhart Attribute**. Drag a continuous column representing defects to the **Y** drop zone. Drag a continuous column representing the number of units to the **n Trials** drop zone. From the **Points** menu, choose the **Statistic** drop-down, then choose **Proportion**.

Using the legacy Control Chart method, select **Analyze ▶ Quality and Process ▶ Control Chart ▶ U** (see Figure 3.15b).

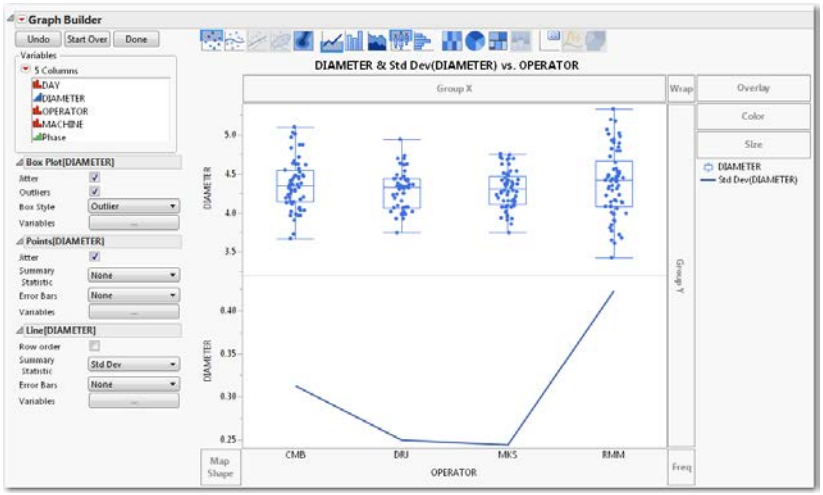
Figure 3.15b Classic Control Chart U Defects



Variability Chart

A chart that illustrates how numeric values vary across a hierarchy of categories. Along with the data, you can view the mean, range, and standard deviation of the data in each category. The analysis options assume that the primary interest is in how the mean, range, and variance change across the categories.

Figure 3.16a Control Chart Builder Variability Chart of Diameter



Data Table Access

To access the data table used for these chart examples, follow **Help ▶ Sample Data ▶ Control Charts ▶ Diameter.**

Usage

For viewing the ranges, standard deviation, and means of a measured column across groups and subgroups. As shown, part diameter variability is displayed across different operators (see Figure 3.16a).

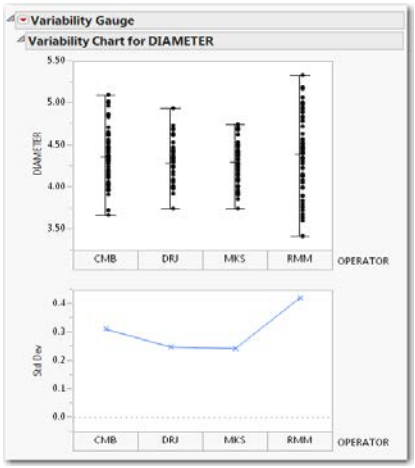
Required

At least one numeric column for the **Y, Response** role and at least one nominal, ordinal, or continuous column for the **X, Grouping** role.

Using the Graph Builder method, **select Graph ▶ Graph Builder.** Drag a continuous column to the **Y** drop zone. Drag one or more nominal or ordinal columns to the **X** drop zone for groups nested in hierarchies. Use selections on the element palette and line and point controls to modify the graph.

Using the legacy approach, select **Analyze ▶ Quality and Process ▶ Variability / Attribute Gauge Chart** (see Figure 3.16b).

Figure 3.16b Classic Variability Chart

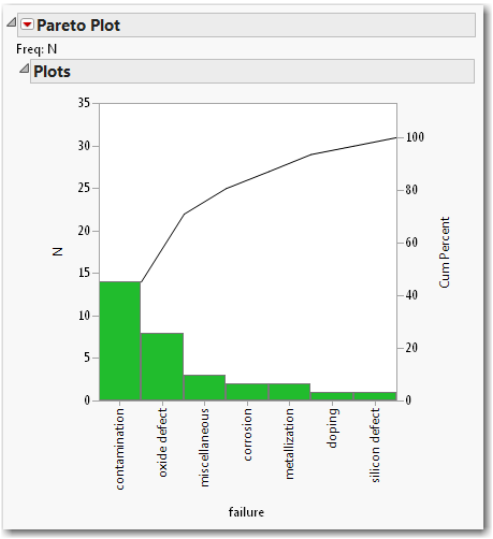


Select one column for the **Y, Response** role and one column for the **X, Grouping** role. An optional nominal column for the **X, Grouping** role produces horizontally nested results by the subgroup, overlaid. Click the **Help** button for many additional options.

Pareto Plot

A chart often included as a quality metric for processes and products. The Pareto Plot produces charts to display the relative frequency of problems in a quality-related process or operation. A *Pareto plot* is a bar chart that displays the classification of problem occurrences, arranged in decreasing order. The column with values that are the cause of a problem is displayed as **X** in the plot. An optional column with values assigning the frequencies is assigned as **Freq**. An optional column whose value holds a weighting value is assigned as **Weight**.

Figure 3.17 Pareto Plot of Failure



Data Table Access

To access the data table used for the above chart example, follow **Help ▶ Sample Data ▶ Control Charts ▶ Failure.**

Usage

For counts of defects by occurrence of defect causes. The plot can be used to target improvement efforts toward those failures that are most serious or common. As shown, the chart displays defect counts and cumulative percents of seven types of semiconductor defects (see Figure 3.17).

Required

At least one continuous or nominal column for the **Cause** role. Additional options for **Grouping**, **Frequency** and **Weight** roles.

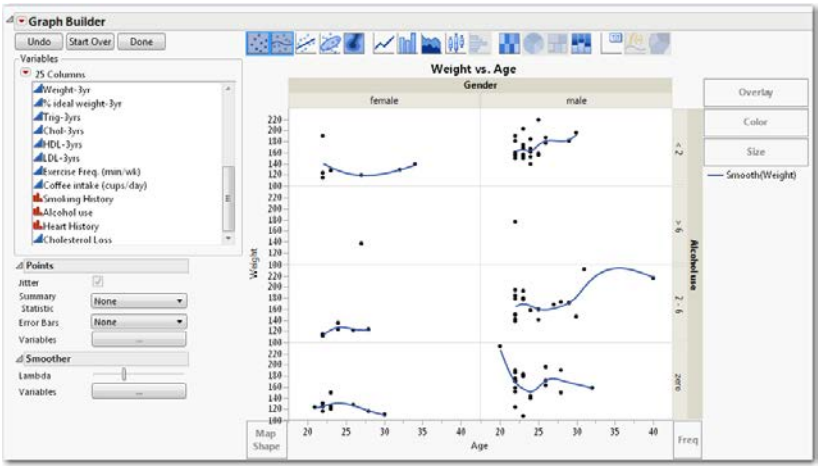
Select **Help ▶ Search ▶ Pareto Plot** for more information.

To generate the plot, select **Analyze ▶ Quality and Process ▶ Pareto Plot**.

Overlay Plots

Like the variability chart presented earlier, the overlay plot allows you to visualize values over any specified times or groups. A key difference, however, is that it allows you to specify multiple Y columns and group those values in a meaningful way. An example of an overlay plot with more than one Y value is shown for reference here (see Figure 3.18a).

Figure 3.18a Overlay of Lipid Data Weight by Age by Gender by Alcohol Use



Data Table Access

To access the data table used for the above chart example, follow **Help ▶ Sample Data ▶ Medical Studies ▶ Lipid Data.**

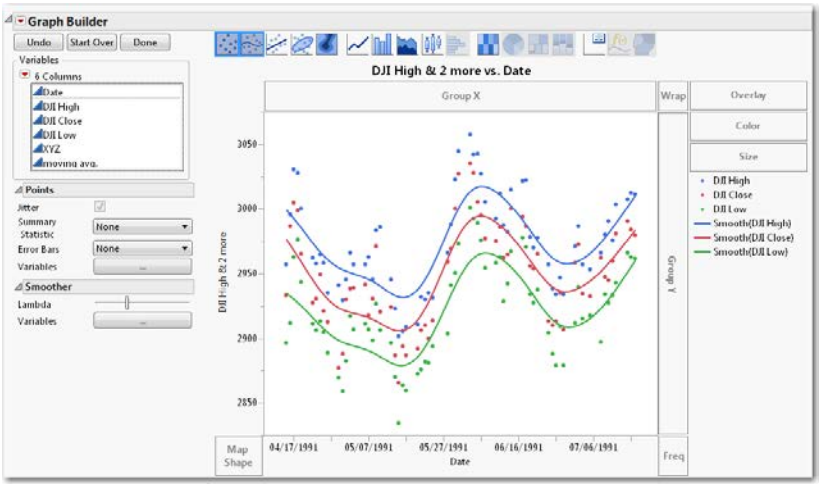
Many overlay plots can be generated with Graph Builder. Some overlay plots are supported in the legacy Chart Platform. A couple of examples of overlay plots with Graph Builder follow.

To generate an overlay plot like the one pictured (see Figure 3.18a), select **Graph ▶ Graph Builder**. Drag a continuous column to the **Y** drop zone. Drag a continuous column to the **X** drop zone. Drag a nominal column to the **Group X** drop zone. Drag a nominal column to the **Group Y** drop zone.

Alternate Overlay Plot

The overlay plot produces overlays of columns or groups on a single bivariate plot (see Figure 3.18b).

Figure 3.18b Alternate Overlay DJI High Close Low with Smoother by Date



Data Table Access

To access the data table used for the above chart examples, follow **Help ▶ Sample Data ▶ Business and Demographic ▶ XYZ Stock Averages (Plots)**.

Usage

You want to overlay multiple groups of data into one graph when there is a single x axis. The chart shows the Dow Jones High, Low, and Close index for a period of time with contrasting colors for High, Low, and Close (see Figure 3.18b).

Required

At least two numeric columns for the **Y** role. **Optional:** Continuous, nominal, or ordinal columns for the **X** and **Grouping** roles.

Select **Graph ▶ Graph Builder**. Select at least two numeric columns simultaneously and drag them to the **Y** drop zone and drag at least one continuous column to the **X** drop zone.

Note: When the overlay chart appears, additional customizations are available on the element icon palette and by a right mouse click inside the chart frame.

3.4 Graphs of One Column

Unlike the charts introduced so far, graphs in this section are accompanied by statistical results. These graphs depict the distribution of values for one column of data (a.k.a. univariate) and provide appropriate tools to assess their properties.

These graphs help you understand the nature of a column, such as how widely the values vary or whether there are any curious qualities to the data.

Most of these graphs are found within the Distribution platform from the **Analyze** menu. We also briefly cover time series in this section.

Many JMP graphs can be saved as interactive HTML and retain their interactivity when opened in a web browser.

Note: You can choose more than one column with these graphs, but each column will be graphed and independently analyzed side-by-side. When you are looking at more than one column, know that the graphs are linked, which allows you to click on any part of the graph to see and explore those values represented in the graphs of other selected columns. See Chapter 2 for more information on how graphs and data are linked.

Distribution Plot

Examines properties of a continuous, nominal, or ordinal column individually or in a univariate fashion.

Figure 3.19a Distribution of Profit($M)

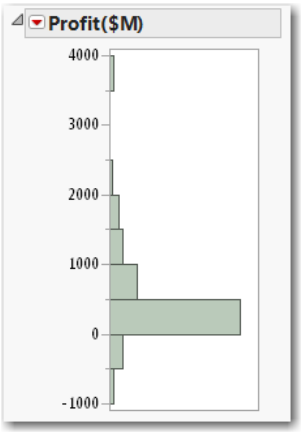
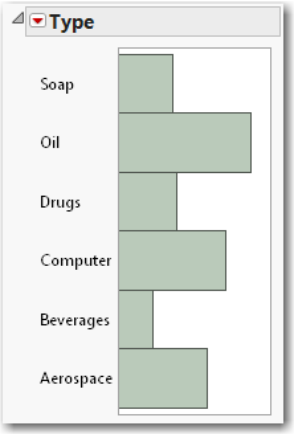


Figure 3.19b Distribution of Company Type



Data Table Access

To access the data table used for the above distribution chart examples, follow **Help ▶ Sample Data ▶ Business and Demographic ▶ Financial**.

Continuous Usage

To view the properties of a continuous distribution such as shape, range, and data density. As shown in Figure 3.19a, the chart displays the profits (or losses) of a selection of technology companies from the late 1990s.

Continuous Distribution Requires: One or more continuous columns for the Y, Columns role.

Nominal, Ordinal Usage

The Distribution platform is similar to a bar chart and allows you to view the properties of a frequency distribution such as the relative counts or percentages of fixed groups. As shown in Figure 3.19b, the chart displays the frequency of company types as bars for a selection of technology companies from the late 1990s. The nominal and ordinal distribution plots are related to the mosaic plot in the next section.

Frequency Distribution Requires

One or more nominal or ordinal columns for the **Y, Columns** role.

Select **Analyze ▶ Distribution**. Select a column and place it in the **Y, Columns** role, and click **OK**.

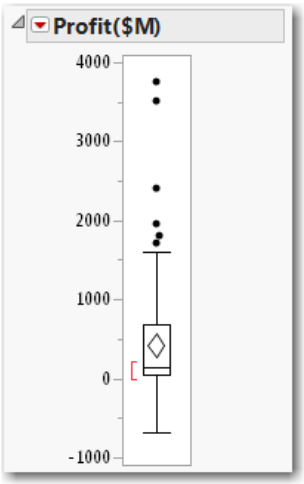
Optional

To generate a distribution plot like the ones pictured using **Graph Builder** select **Graph ▶ Graph Builder**. Drag a continuous or nominal column to the **Y** drop zone. Select the histogram element from the elements palette.

Outlier Box Plot

A chart for detecting extreme values and properties of a distribution, sometimes called a *Tukey Box Plot.* See Appendix B for a description of this term.

Figure 3.20 Tukey box plot of Profits



Data Table Access

To access the data table used for the above chart example, follow **Help ▶ Sample Data ▶ Business and Demographic ▶ Financial.**

Usage

To view the properties of a continuous distribution such as quartiles, moments, and outliers. See Appendix B for a description of these terms.

As shown, the plot displays a few very profitable companies as points that are well beyond the main body of companies, the middle half of which are contained in the box (see Figure 3.20).

Required

One or more continuous columns for the **Y, Columns** role.

Select **Analyze ▶ Distribution**. Select a continuous column and place it in the **Y, Columns** role, and click **OK**. Then click the red triangle and select **Outlier Box Plot**. (Program preferences might automatically generate the plot when running the Distribution platform.)

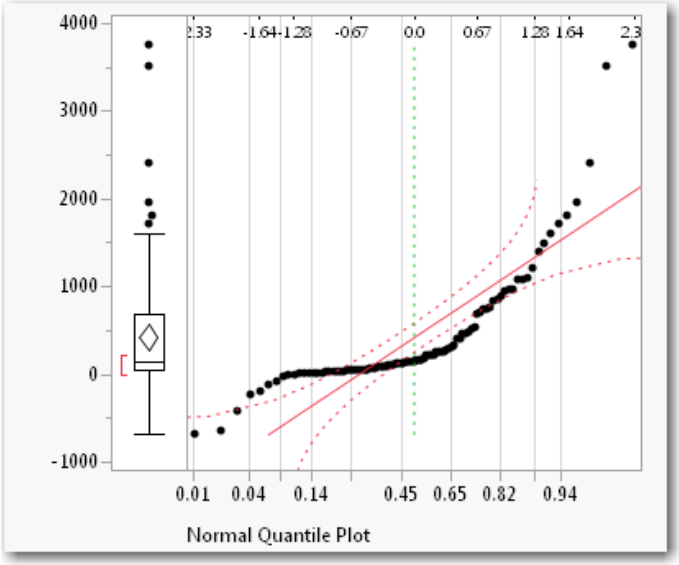
Optional

To generate an outlier box plot like the one pictured using **Graph Builder** select **Graph ▶ Graph Builder**. Drag a continuous column to the **Y** drop zone. Select the Box Plot element from the elements palette.

Normal Quantile Plot

A chart for visualizing the extent to which a column is consistent with a normal distribution. In a symmetrical distribution, the points would fall about the solid red line in the display and not beyond the confidence curves.

Figure 3.21 Normal Quantile Plot of Profits



Data Table Access

To access the data table used for the above chart example, follow **Help ▶ Sample Data ▶ Business and Demographic ▶ Financial.**

Usage

To view the properties and visually assess the extent to which the data is normally distributed (see Figure 3.21).

In this example, the plot displays the profits from a sample of companies. The data do not follow the solid red line; some fall beyond the dotted red confidence bands. The data are not consistent with a normal distribution; they are more consistent with a skewed distribution.

Required

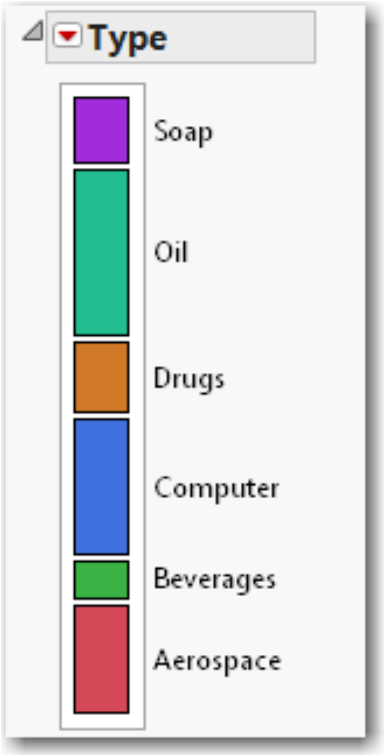
One or more continuous columns for the **Y, Columns** role.

Select **Analyze ▶ Distribution.** Drag a continuous column into the **Y, Columns** role, and click **OK**. Click the red triangle, and select **Normal Quantile Plot**.

Mosaic Plot

A stacked bar chart where each segment is proportional to its group's frequency count.

Figure 3.22 Mosaic Plot of Type



Data Table Access

To access the data table used for the above chart examples, follow **Help ▶ Sample Data ▶ Business and Demographic ▶ Financial.**

Usage

To view the properties of a nominal or ordinal distribution, or to visually assess the proportions of data that fall within each group. These charts are especially useful when comparing groups. Dynamic linking allows you to select a group and see if the proportion of the selection in each level remains even. If not, you have evidence of an association between the variables. As shown, the chart displays the proportions or counts of each type of company from a stock portfolio (see Figure 3.22).

Required

One or more nominal or ordinal columns for the **Y,Columns** role.

Select **Analyze ▶ Distribution.** Drag a nominal or ordinal column into the **Y, Columns** role, and click **OK**. Click the red triangle, and select **Mosaic Plot**.

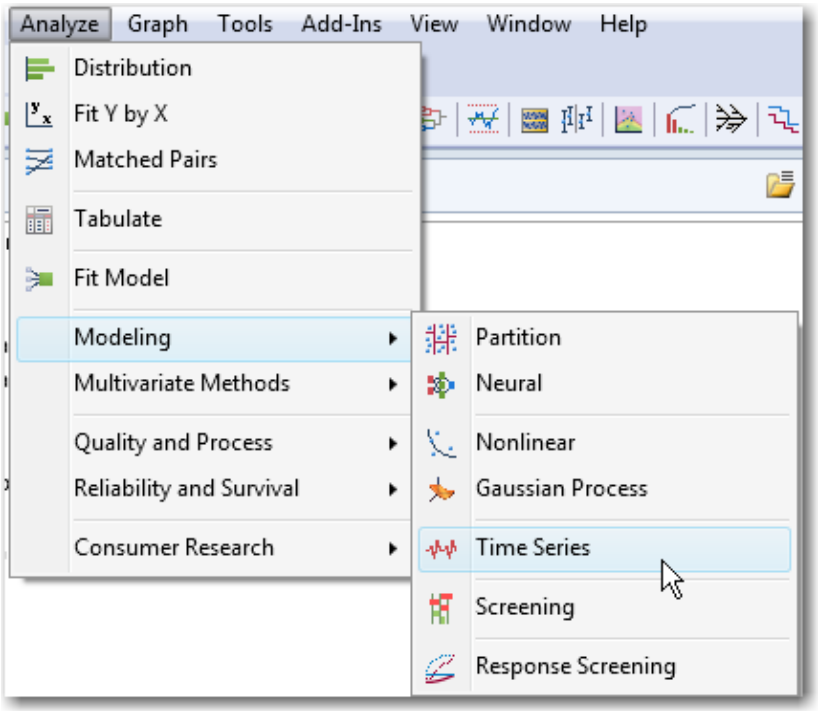
Optional

To generate a mosaic plot like the one pictured using **Graph Builder,** select **Graph ▶ Graph Builder**. Drag a nominal column to the **Y** drop zone. Select the Mosaic element from the elements palette.

Time Series

Time Series is a separate platform that generates a graph of a numeric value over time. It also serves as a platform to employ forecasting techniques and produces statistical results. For more information on these techniques, see the *JMP Statistics and Graphics Guide* (**Help > Books > Specialized Models)**. The Time Series platform is available from the **Analyze** menu and the **Modeling** submenu (see Figure 3.23).

Figure 3.23 Analyze Modeling Time Series



Data Table Access

To access the data table used for these chart examples, follow **Help > Sample Data > Time Series > Raleigh Temps.**

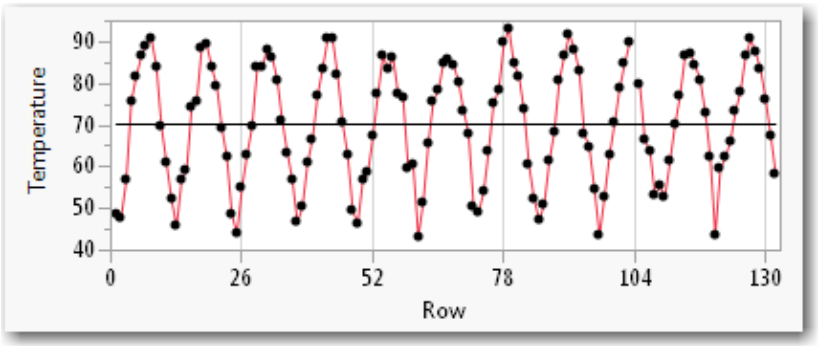
Time Series Plot

A graph of numeric values, Y, over a time order, X.

Usage

To view and fit the variability and potential seasonality of a measured value over time. For example, the chart displays the average monthly temperatures with a clear seasonal trend in Raleigh, North Carolina, over a 130-month period (see Figure 3.24).

Figure 3.24 Time Series



Required

One numeric column for the **Y, Time Series** role. Options include a numeric Time column (**X, Time ID**) with corresponding values and an input column. If an **X, Time ID** column is not specified, JMP orders the data over the rows sequentially. It is assumed that the time interval is constant between every pair of time points.

Select **Analyze** ▶ **Modeling** ▶ **Time Series**, drag a continuous measured column to the **Y, Time Series** role, and click **OK**. Select a numeric column for the **Time, ID** role.

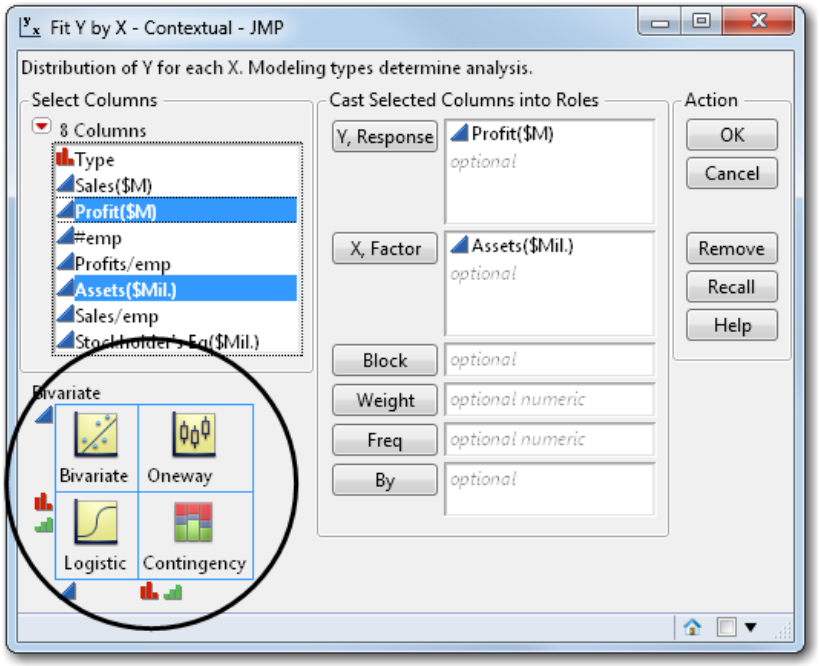
3.5 Graphs Comparing Two Columns

The **Fit Y by X** command studies the relationship of two columns (a.k.a. bivariate). This command is available from the **Analyze** menu and shows graphs with statistical results for each pair of x and y columns. The type of graph generated by JMP is determined by the modeling types (continuous, nominal, or ordinal) of the columns that are cast into the **X** and **Y** roles. JMP always creates the right graphs based on the modeling type for any function from the Analyze menu. In important ways, **Fit Y by X** is four sets of graphs and analyses in one!

The matrix circled in the **Fit Y by X** window (see Figure 3.25) provides a visual preview of the graphs that will be generated depending on the modeling type of the Y (the vertical axis) and the X (the horizontal axis), which can be altered to obtain the desired analysis or plot.

Note: If the column modeling types nominal, ordinal, and continuous are unfamiliar to you, see Section 2.3.

Figure 3.25 Fit Y by X Contextual



Scatterplot

A graph of the continuous-by-continuous personality within the **Fit Y by X** command. The analysis begins as a scatterplot of points to which you can interactively add a linear fit and confidence curves.

Figure 3.26a Bivariate Profit by Assets

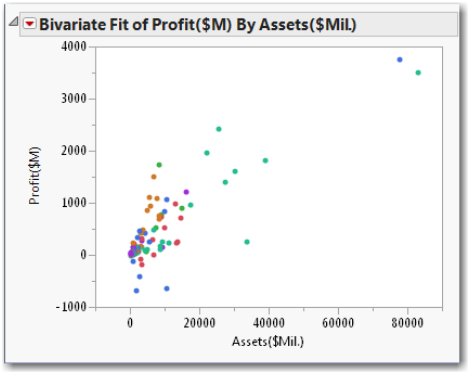


Figure 3.26b Bivariate Profit by Assets Fit Line

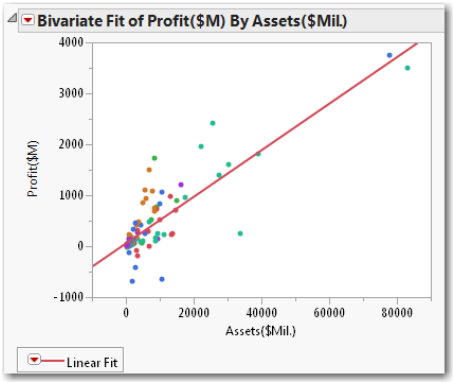
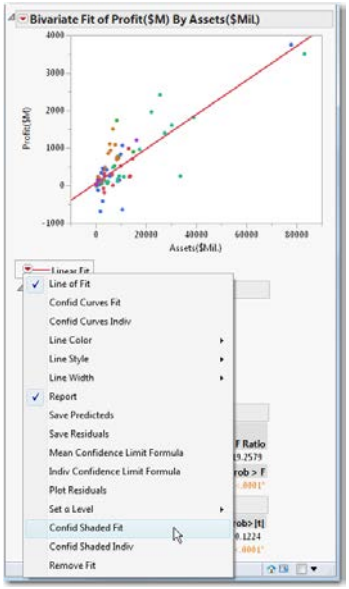


Figure 3.26c Bivariate Profit by Assets Fit Line Conf Shaded Fit



Data Table Access

To access the data table used for the above chart examples, follow **Help ▶ Sample Data ▶ Business and Demographic ▶ Financial.**

Usage

To view the relationship of a continuous column to another continuous column. An example might be graphing the relationship of profits to assets for a selection of Fortune 500 companies and then fitting a regression line with 95% confidence curves, as shown (see Figures 3.26a, 3.26b, and 3.26c).

Required

One continuous column for the **Y, Response** role and one continuous column for the **X, Factor** role.

Select **Analyze** ▶ **Fit Y by X,** select a continuous **Y, Response** column and a continuous **X, Factor** column, and click **OK** (see Figure 3.26a).

To add the simple linear least squares fit: From the red triangle next to **Bivariate Fit**, select **Fit Line** (see Figure 3.26b).

To add the confidence shaded curves to the fit: From the red triangle on the **Linear Fit** item, select **Confid Shaded Fit** (see Figure 3.26c).

Optional

Using the Graph Builder method, select **Graph** ▶ **Graph Builder** and drag a continuous column to the Y drop zone and a continuous column to the X drop zone. Select the Line of Fit

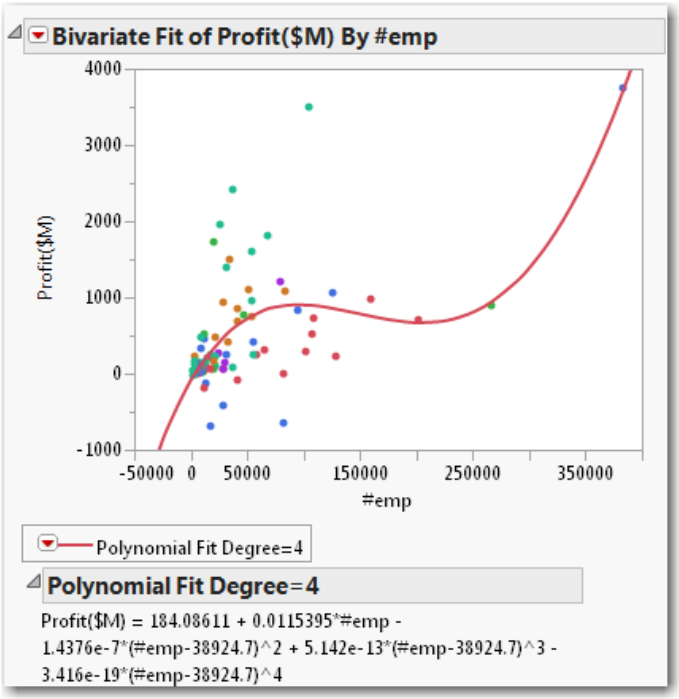
element from the elements palette.

Note: Bivariate and scatter plots with fitting are also supported in the Graph Builder Platform. These graphs show colored markers. For more information on how to color or mark rows, see Section 2.6.

Scatterplot (with Polynomial Fit)

A graph that fits a polynomial curve to the degree you select from the **Fit Polynomial** submenu. After you select the polynomial degree, the curve is fit to the data points using least squares regression.

Figure 3.27 Bivariate 4th Order Fit



Data Access Table

To access the data table used for the above chart example, follow **Help ▶ Sample Data ▶ Business and Demographic ▶ Financial.**

Usage

To view the relationship of a continuous column to another continuous column using a linear polynomial fit where curves produce the best fit of the data. The chart displays the fourth-order polynomial fit showing the relationship of profits to number of employees for a selection of companies (see Figure 3.27).

Required

One continuous column for the **Y, Response** column and one continuous column for the **X, Factor** role.

Select **Analyze** ▶ **Fit Y by X,** select a continuous **Y, Response** column and a continuous **X, Factor** column, and click **OK**. From the red triangle, select **Fit Polynomial** and select a degree number from the submenu.

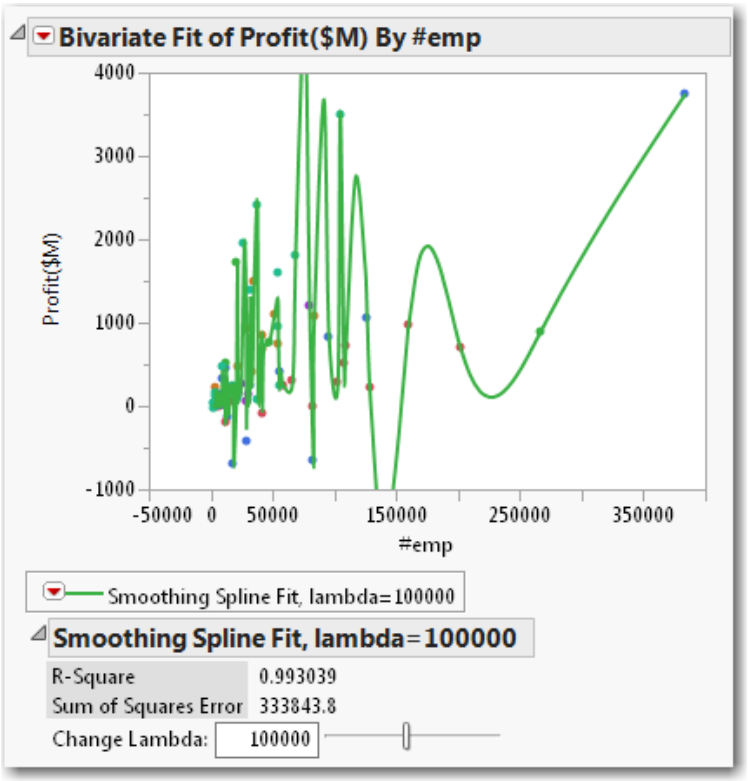
Optional

Using the Graph Builder method, select **Graph** ▶ **Graph Builder** and drag a continuous column to the Y drop zone and a continuous column to the X drop zone. Select the Line Of Fit element from the elements palette, then adjust the degree of fit in the **Line Of Fit** panel.

Scatterplot (with Spline Fit)

A chart that fits a smoothing spline that varies in smoothness (or flexibility) according to a tuning parameter in the spline formula. Splines contrast the polynomial fit using least squares regression. You can use a spline of varying smoothness to highlight the overall trends in the data without using a linear function to describe the relationship.

Figure 3.28 Bivariate Spline Fit



Data Table Access

To access the data table used for the above chart example, follow **Help ▶ Sample Data ▶ Business and Demographic ▶ Financial.**

Usage

To view the relationship of a continuous column to another continuous column. For example, the chart illustrates that limited profit variation is present in companies with lower numbers of employees. The plot also shows that fewer companies have higher numbers of employees and higher profit variation (see Figure 3.28).

Required

One continuous column for the **Y, Response** role and one continuous column for the **X, Factor** role.

Select **Analyze ▶ Fit Y by X**. Select a continuous column and place it in the **Y, Response** role. Select a continuous column, place it in the **X, Factor** role, and click **OK**. From the red triangle, select **Fit Spline**, and from the submenu, select the degree of flexibility you want in the spline fit by changing the lambda value.

Optional

Using the Graph Builder method, select **Graph ▶ Graph Builder** and drag a continuous column to the Y drop zone and drag a continuous column to the X drop zone. Adjust lambda to the value desired.

Oneway Plots

The Oneway platform analyzes how the distribution of a continuous Y column differs across groups defined by a categorical X column. Group means, as well as other statistics and tests, can be calculated and tested. The Oneway platform is the continuous (placed as Y) by nominal/ordinal (placed as X) personality of the **Fit Y by X** command.

Figure 3.29a Oneway Profits emp by Type

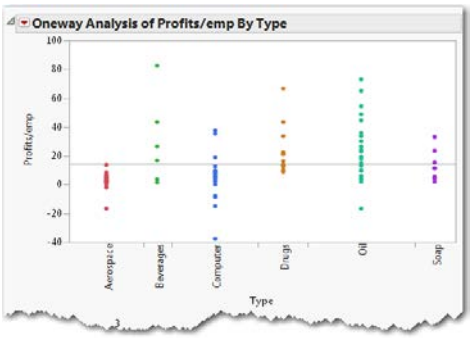


Figure 3.29b Oneway Profits emp by Type

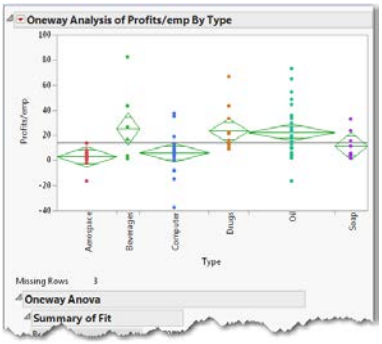
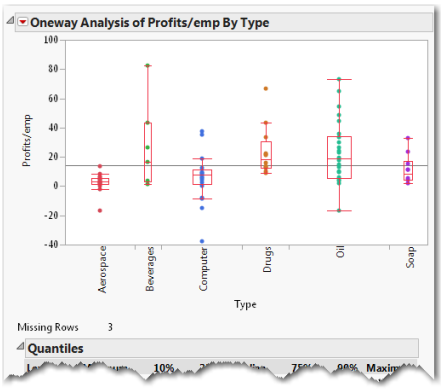


Figure 3.29c Oneway Profits emp by Type ANOVA



Data Table Access

To access the data table used for these chart examples, follow **Help ▶ Sample Data ▶ Business and Demographic ▶ Financial.**

Usage

To view the relationship of a continuous column across the groups (nominal/ordinal) in another column. For example, the chart displays the difference in means and variation in profits and employees across six company types (see Figures 3.29a, 3.29b, and 3.29c).

Required

One continuous column for the **Y, Response** column and one nominal or ordinal column for the **X, Factor** role.

Select **Analyze ▶ Fit Y by X.** Select a continuous column for the **Y, Response** role and a nominal or ordinal column for the **X, Factor** role, then click **OK** (see Figure 3.29a).

From the red triangle, select **Means/Anova** (see Figure 3.29b).

From the red triangle, select **Quantiles** (see Figure 3.29c).

Optional

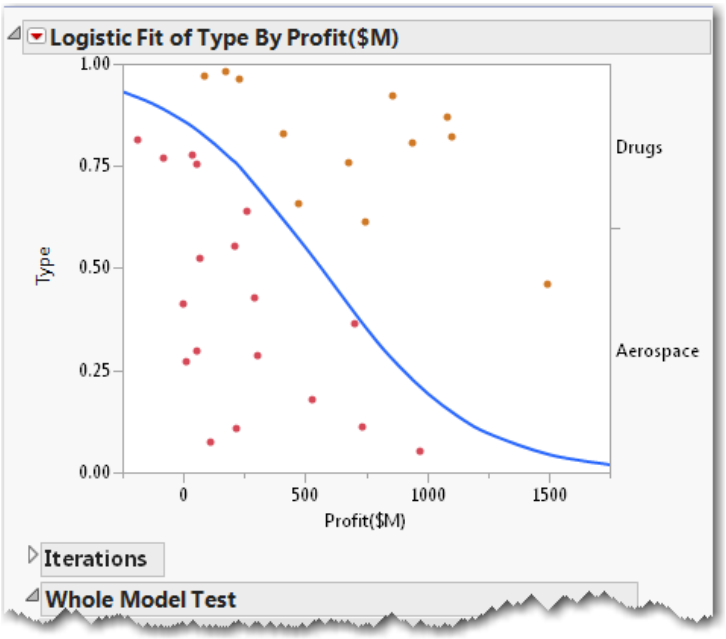
Using the Graph Builder method, select **Graph ▶ Graph Builder,** drag a continuous column to the **Y** drop zone and a nominal column to the **X** drop zone. Select the box plot element from the elements palette. Right-click in the graph frame to choose additional box plot options.

Logistic Fit

A chart that estimates the probability of choosing one of the Y response levels as a smooth function of the X factor. The fitted probabilities must be between 0 and 1 and must sum to 1 across the response levels for a given factor value.

In a logistic probability plot, the y-axis represents probability.

Figure 3.30 Fit Y by X Logistic



Data Table Access

To access the data table used for the above example, follow **Help ▶ Sample Data ▶ Business and Demographic ▶ Financial**.

Usage

To predict a group or groups by some continuous column. The chart displays a prediction separation (as a probability/percentage) of the type of company (Drugs or Aerospace) by its profits (see Figure 3.30).

Required

One nominal or ordinal column for the **Y, Response** column and one continuous column for the **X, Factor** role.

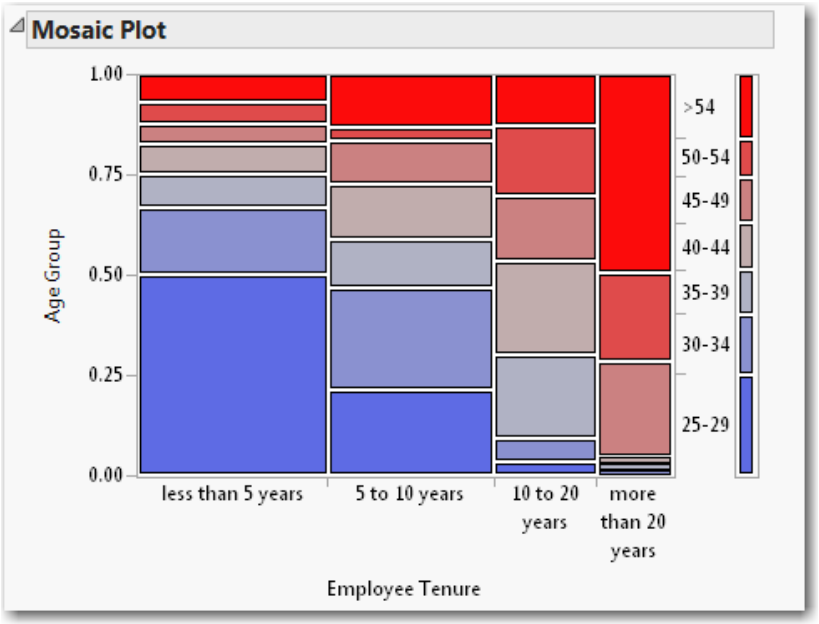
Select **Analyze ▶ Fit Y by X**. Select a nominal or ordinal column for the **Y, Response** role and a continuous column for the **X, Factor** role, then click **OK**.

Mosaic Plot

A chart that is divided into small rectangles such that the area of each rectangle is proportional to a frequency count of interest.

The Mosaic Plot appears in the Contingency Platform and is the personality of the **Fit Y by X** command when both the Y and X columns are nominal or ordinal. Mosaic examines the distribution of a categorical Y column by the values of a categorical X column.

Figure 3.31 Fit Y by X Contingency Mosaic Plot



Data Access Table

To access the data table used for the above chart example, follow **Help ▶ Sample Data ▶ Psychology and Social Science ▶ Consumer Preferences**.

Usage

Group-by-group counts are shown as proportional colored rectangles in a two-by-two arrangement. As shown, the graph displays a simple color chart of the proportion of seven age groups displayed by four employee tenure groups in a large company. Older employees are represented in red shades while younger employees are represented in blue shades (see Figure 3.31).

Required

One nominal or ordinal column for the **Y, Response** column and at least one nominal or ordinal column for the **X, Factor** role.

Select **Analyze ▶ Fit Y by X**. Select a nominal or ordinal column for the **Y, Response** role and a nominal or ordinal column for the **X, Factor** role, and click **OK**.

Optional

Using the Graph Builder method, select **Graph ▶ Graph Builder** and drag a nominal or ordinal column to the **Y** drop zone and a nominal or ordinal column to the **X** drop zone. Select the mosaic icon from the elements palette.

Parallel Plot

A plot that displays lines representing differences among specified row values. The strength of a parallel plot is that major differences show up easily as crossed lines. The columns often represent a related measured quantity across time or location.

Figure 3.32 Parallel Plot of Movies Domestic Worldwide

Data Access Tables

To access the data table used for this chart example, follow **Help ▶ Sample Data ▶ Business and Demographic ▶ Movies.**

Usage

To visually detect measurement differences across many rows of data. The chart displays which movies performed better in worldwide revenues relative to domestic revenues and vice-versa (see Figure 3.32).

Required

At least one column of any type; two are recommended.

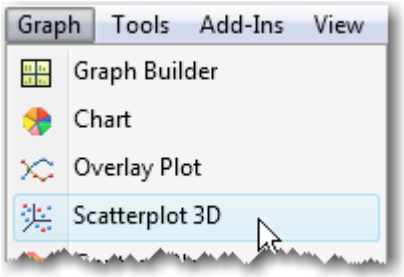
Select **Graph ▶ Parallel Plot**. Select at least two columns, drag them to the **Y, Response** role, and click **OK.**

To produce the row labels, add a label attribute to the Movie column, select the line in the plot and select **row label**. To produce the row colors, select **Type** and select **Rows ▶ Color or Mark by Column**. See Section 2.5 for details on row labels.

3.6 Graphs Displaying Multiple Columns

Sometimes it is valuable to see a problem in more than two dimensions. This section uses JMP to visualize three or more columns at once. With the exception of the profiler, which concludes this section, these graphs appear under the **Graph** menu and contain only a few built-in analytic procedures (see Figure 3.33).

Figure 3.33 Graph Scatterplot 3D

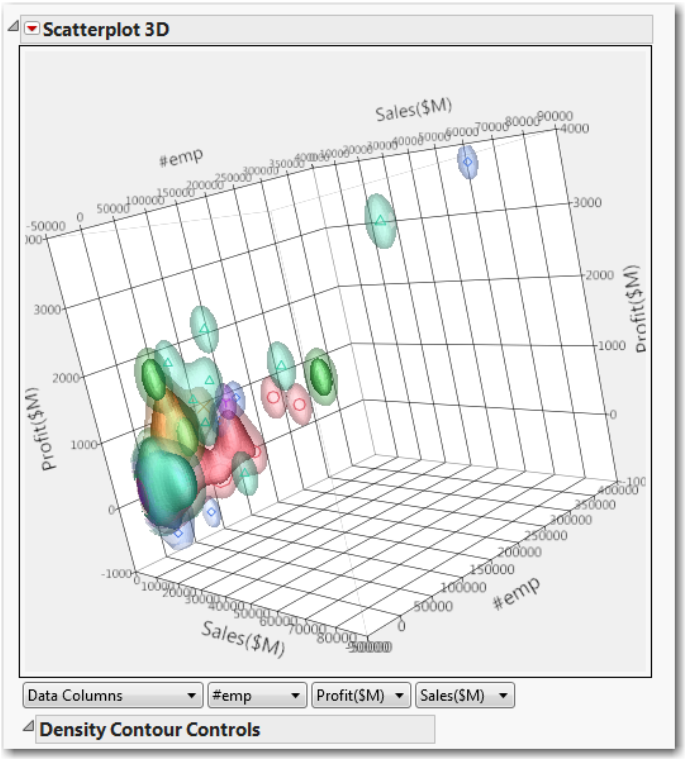


Like most JMP graphs, these multi-dimensional (or multi- column) graphs are interactive and allow you to select, rotate, and animate them. You can copy and paste these into other documents. Two of them, the bubble plot and the profiler, allow you to create exportable Flash files that retain their interactivity and animation. At this book's release, many graphs can be saved as interactive HTML and retain their interactivity when opened in a web browser.

Scatterplot 3D

Accessible from the **Graph** menu, this chart displays a three-dimensional scatterplot that can be rotated with your mouse. The Scatterplot 3D platform displays three columns at a time from the columns you select.

Figure 3.34 Scatterplot 3D



Data Access Table

To access the data table used for the above chart example, follow **Help ▶ Sample Data ▶ Business and Demographic ▶ Financial**.

Usage

To view patterns among any two or three columns of data. This plot is very useful for exploring data in three dimensions. The chart displays sales (y-axis) by number of employees (x-axis) by profits (z-axis) with colored 3D density contours by company type (see Figure 3.34). The eye can detect possible differences among company types across the three columns. This is an interactive plot and can be rotated on any axis. To rotate the graph, click and hold the graph and move the mouse.

Required

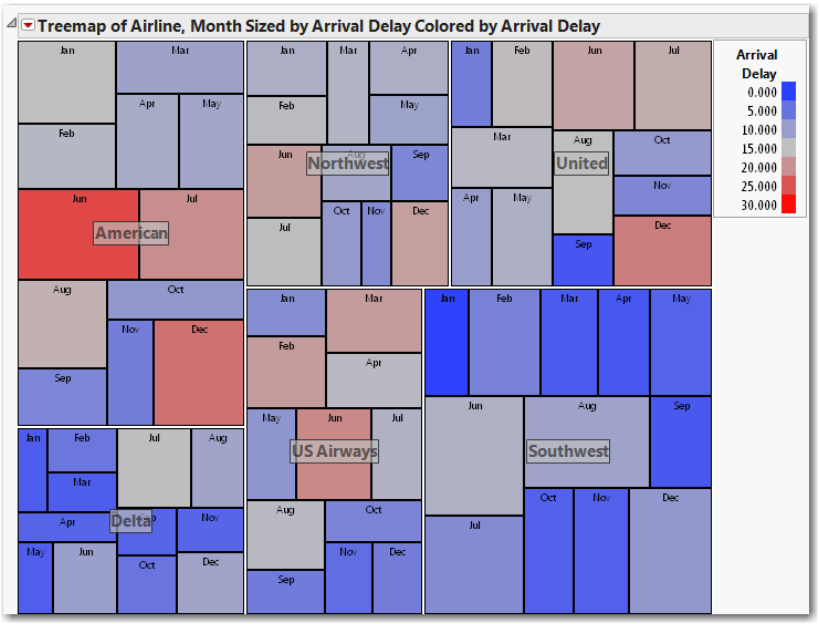
Two or more columns of any modeling type (can be continuous, nominal, or ordinal). Three columns are required for a three-dimensional plot.

Select **Graph ▶ Scatterplot 3D**. Select at least two columns (three are recommended), place them in the **Y, Columns** role, and click **OK**. To include surfaces as displayed, from the red triangle, select **Nonpar Density Contour**.

Tree Map

A graphical technique of observing patterns among groups that have many levels. Tree maps are especially useful in cases where histograms are ineffective because there are so many bars.

Figure 3.35 Tree Map Airline Delays



Data Table Access

To access the data table used for the above chart example, follow **Help ▶ Sample Data ▶ Open Sample Data Directory ▶ Airline Delays.**

Usage

For example, the chart displays airline arrival delays by month, where hot colors represent longer delays and cold colors represent shorter delays by airline. Larger squares are longer delays as well (see Figure 3.35). These maps produce convenient visual rankings or groups within groups.

Required

At least one continuous, nominal, or ordinal column for the **Categories** role.

Using the legacy method, select **Graph ▶ Tree Map**.

Select one nominal or ordinal column for the **Categories** role and click **OK**. Optionally, select numeric columns and place them in the **Sizes** and **Ordering** roles. Also optionally, select a column and place it in the **Coloring** role, and click **OK**.

To generate a legend as shown, a Coloring column must be set. Then from the red triangle, select **Legend**.

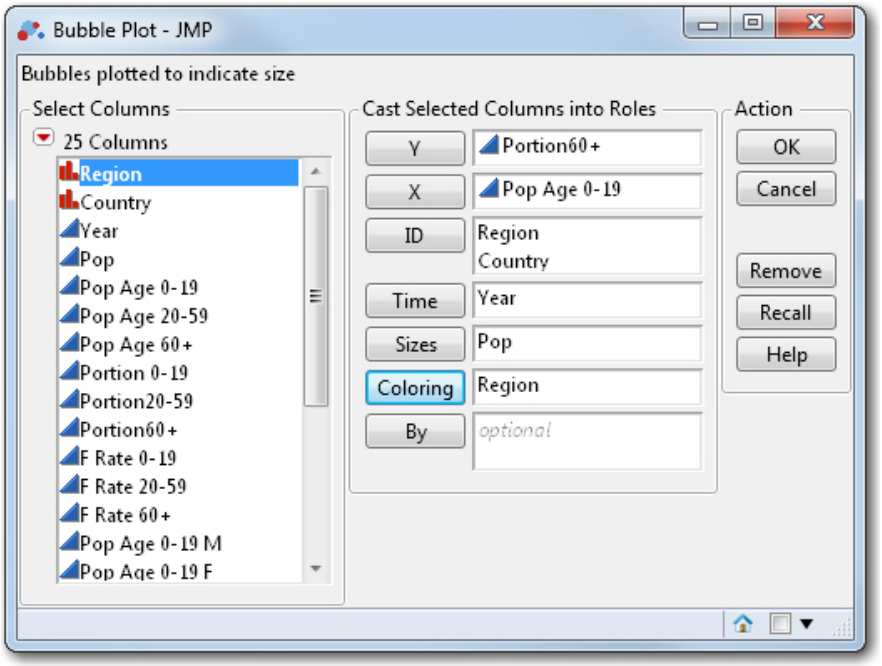
Option

Using the Graph Builder method, select **Graph ▶ Graph Builder** and drag a nominal or ordinal column to the **Y** drop zone and a nominal or ordinal column to the **X** drop zone. Select the tree map icon from the elements palette. Drag additional columns to drop zones. Experiment!

Bubble Plot

An interactive scatterplot that represents its points as circles (bubbles). Optionally, the bubbles can be sized according to another column, colored by yet another column, and dynamically indexed by a time column. With the opportunity to see up to five dimensions at once (x position, y position, size, color, and time), bubble plots can produce dramatic animated visualizations and are effective at communicating complex relationships.

Figure 3.36 Bubble Plot Window with Vars



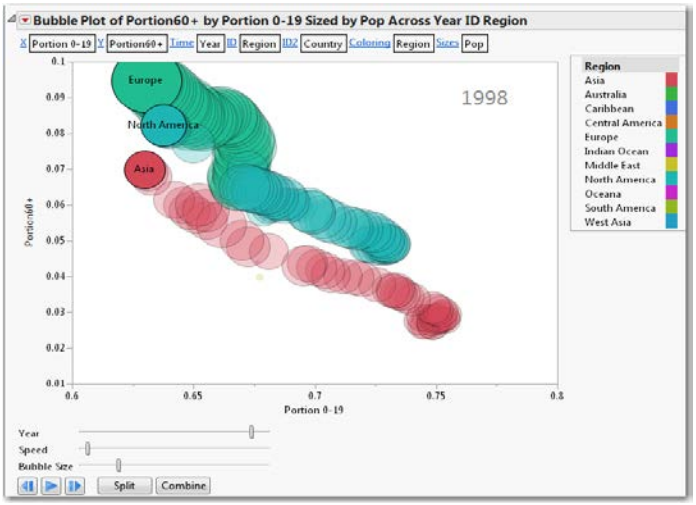
Data Table Access

To access the data table used for these chart examples, follow **Help ▶ Sample Data ▶ Open the Sample Data Directory ▶ PopAgeGroup**.

Usage

Summarizing multi-column data in an interactive two-dimensional display. Frequently used where time is one of the columns. The bubble plot in this example displays the relationship of the portion of the population over 60 years old related to the portion of the population between 0 and 19 years old, colored by continent over a 55-year time span (see Figure 3.36). Sizes of bubbles show population size (see Figure 3.37).

Figure 3.37 Bubble Plot Portion 60plus vs Portion 0 to 19



Required

One **X** column and one **Y** column of any type (continuous, nominal, or ordinal). For time animation, select a column for the **Time** role.

Select **Graph ▶ Bubble Plot**. Select one column and place it in the **Y** role. Select one column and place it in the **X** role. For an animated time series plot, specify a **Time** column. The **ID** column produces a label for each bubble. The **Sizes** and **Coloring** columns can be specified to increase information density (see Figure 3.36).

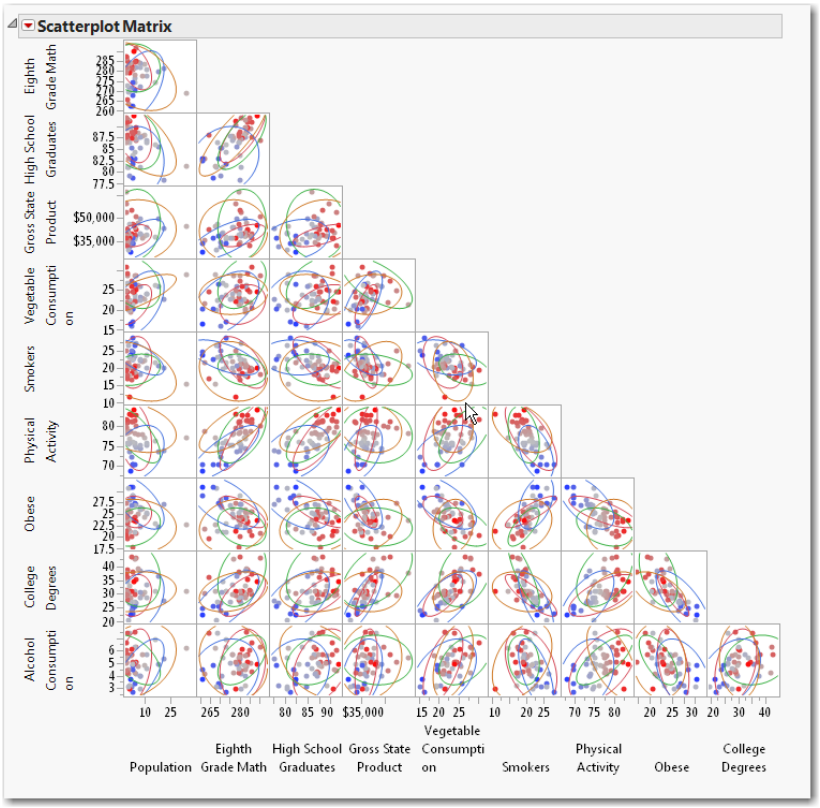
To generate trail lines (as shown), select the **Trail Lines** and **Trail Bubbles** options from the red triangle. To save a Flash file of this plot for viewing in a Web browser or PowerPoint. From the red triangle, select **Save As Flash (SWF)** and specify a location for the Flash file.

To turn on the legend, select the red triangle, then select **Legend**.

Scatterplot Matrix

A chart that provides quick and orderly production of many bivariate graphs. These are assembled so that comparisons among many columns can be conducted visually and that correlation and data pattern can be easily detected. In addition, the plots can be customized and employ advanced features (such as density ellipses) to provide for further analysis.

Figure 3.38 Scatterplot Matrix with Density Elipses



Data Access Table

To access the data table used for the above chart example, follow **Help ▶ Sample Data ▶ Open the Sample Data Directory ▶ US Demographics**.

Usage

In this example, the scatterplot matrix provides every bivariate combination of nine columns of US States demographic data (see Figure 3.38). This chart quickly produces many correlation plots of all variables specified for easy identification of interesting groups and patterns.

Required

Two or more columns of any type (continuous, nominal, or ordinal) for the **Y, Columns** role. More than two columns are recommended.

Select **Graph ▶ Scatterplot Matrix**.

Select at least one column for the **Y, Columns** role. Select multiple Y and X columns for a matrix of graphs. Optionally, select a column and place it in the **X** role. Select a nominal or an ordinal column and place it in the **Group** role.

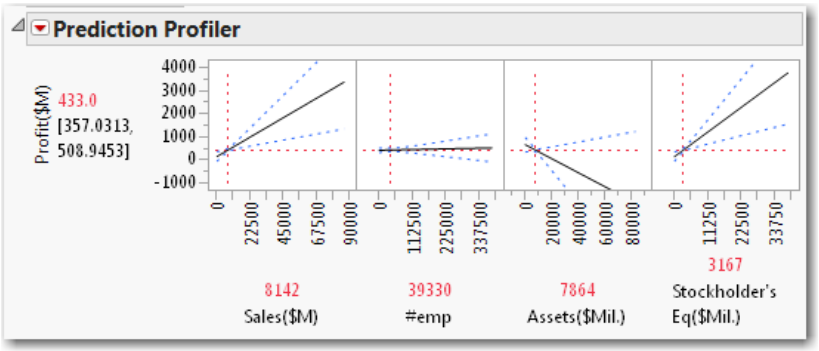
To include grouped ellipses as shown, include a column for the **Group** role in the window. Then for the plot, select **Density Ellipses** from the red triangle.

Profiler

An interactive graph that provides a simple way to view complex relationships within a model. It lets you visualize what-if scenarios quickly and easily by allowing you to see the effect that changes in one column have on the remaining columns. This tool is especially useful when describing multiple variable models by demonstrating the sensitivity of changes in one or more X columns on the predicted Y.

The profiler displays profile traces for each X column. A profile trace is the predicted response as one column is changed (by dragging the vertical red dotted line in the graphs) while the others are held constant at the current values. The profiler recalculates the predicted responses (in real time) as you vary the value of an X column.

Figure 3.39 Prediction Profiler



Data Table Access

To access the data table used for the above example, follow **Help ▶ Sample Data ▶ Business and Demographic ▶ Financial.**

Usage

In this example, the chart displays the relationship between a continuous column Profits($M) and four predictors: sales [Sales($M)], number of employees (#emp), assets [Assets($Mil.)], and stockholder's equity [Stockholder's Eq($Mil.)] (see Figure 3.39).

The profiler shows that as sales increase, predicted profits increase, and as assets increase, profits decrease. To generate interactive, real-time predictions for profits, drag the vertical red trace lines in the profiler in JMP (or in the saved Flash file).

Required

A formula column. Can be created in one of two ways: from Fit Model (which generates a formula), or from a formula column either saved from Fit Model or entered in a data table by hand.

To create a formula in a column manually, see Section 2.4.

Create a Profiler from Fit Model

Select **Analyze ▶ Fit Model**, and select one or more columns for the **Y** role and one or more columns for the **Construct Model Effects**. Select the **Emphasis** pop-down menu, select **Effect Screening,** and click **Run Model**. The profiler appears at the bottom of the report.

Create a Profiler for Use Outside of JMP (Flash File)

From the Fit Model report window from the previous step, select the red triangle and select **Save Columns ▶ Prediction Formula**. This action saves a prediction formula to a JMP data table column as the last column in the table.

Once you have a prediction formula in a column, select **Graph ▶ Profiler** from the top menu. Select the prediction formula column that has appeared in the data table, place it in the **Y, Prediction Formula role**, and click **OK**. This implementation allows you to save the profiler as a Flash file by selecting **Save as Flash (swf)** from the red triangle in the profiler report.

3.7 Summary

This chapter presented a series of the most frequently used graphs and their step-by-step recipes. They are presented in a cookbook style so that each can be recognized by its picture, name, or definition and easily replicated with simple steps. These charts represent the first tier of commonly used graphs. You can find additional graphs and instructions on how to create them in the JMP documentation, including *The Essential Graphing*. Go to **Help ▶ JMP ▶ Books ▶ Essential Graphing** for more information.

Chapter 4 Finding the Right Graph or Summary

4.1 Using Graph Builder to Produce Graphs of Data

4.2 Using Graph Builder to Produce Maps

4.3 Using Tabulate

4.4 Summary

In the last chapter, we presented an overview of the commonly used graphs in JMP with accompanying recipes. This works well when you know what you want your graph to look like. However, at other times, and particularly when you are exploring your data for the first time, it makes sense to take a different approach for several important reasons. First and foremost, you probably don't know what your data is going to tell you. Second, you can learn a great deal about your data when you explore it both visually and numerically. Reviewing your data in both ways can lead you to find the best way to display information.

In this chapter, we take advantage of a key feature of JMP: its ability to create graphs, maps, and summaries interactively and intuitively. While all platforms in JMP promote exploration, two features in particular—Graph Builder and Tabulate—are especially designed for this purpose. Because the former is geared toward interactively creating maps and graphs and the latter at numerical summaries, we cover them separately. Let us begin with finding the right picture of your data.

4.1 Using Graph Builder to Produce Graphs of Data

Graph Builder is a great tool to quickly see your data expressed in many different forms. In many ways, Graph Builder is like a blank canvas waiting for your artistic direction. With Graph Builder, it is helpful to begin thinking about your problem and the columns that are central to your questions at hand. Graph Builder is found under the **Graphs** menu when you select **Graphs ► Graph Builder** (see Figure 4.2).

Graph Builder works in a drag-and-drop manner. Simply click a column in the **Select Columns** box and drag it (without letting go) to one of the zones located around the canvas. You can hold the column in a zone (without letting go) to get a preview of the graph, then move it to another zone to see an alternative display. Once you release the mouse button, Graph Builder keeps that selection and allows you to tailor the display or select another column to begin exploring relationships. You can repeat this process by adding more columns to view more complex displays.

Example 4.1: TechStock

You can find the data set at **Help ► Sample Data ► Business and Demographic ► TechStock**.

We'll be working with one of the sample data files to illustrate the steps in this section. The file we'll use, **TechStock.jmp**, contains closing price data from selected technology stocks. The columns include:

Date the date expressed as dd/mm/yyyy

Open the opening price of the stock

High the highest price of the stock on that date

Low the lowest price of the stock on that date

Close the closing price of the stock on that date

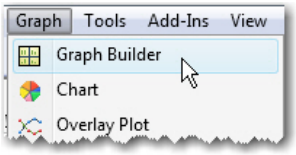
Volume the number of shares traded on that date

Adj. Close the adjusted price of the closing price

YearWeek the week of the year expressed as yyyy/week#

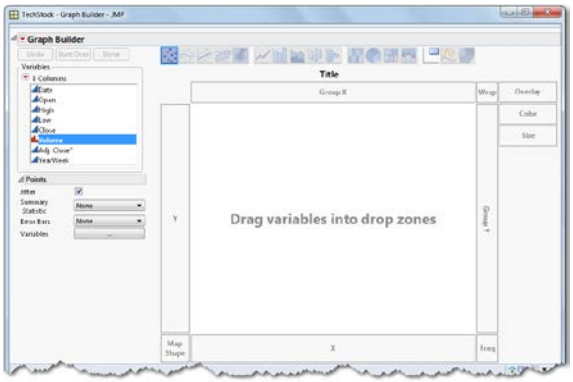
1. First, open the sample data table **TechStock**, as indicated previously.
2. Open Graph Builder. Select **Graph ► Graph Builder** (see Figure 4.1).

Figure 4.1 Open Graph Builder



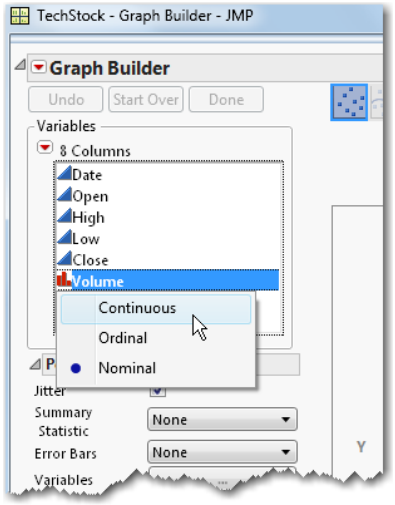
3. Notice the columns in your data table appear on the left in the **Variables** box (see Figure 4.2).

Figure 4.2 Graph Builder Variables



4. In the **Variables** box, change the **Volume** column from **Nominal** to **Continuous** by clicking on the icon in the **Variables** box on the left (as in Figure 4.3).

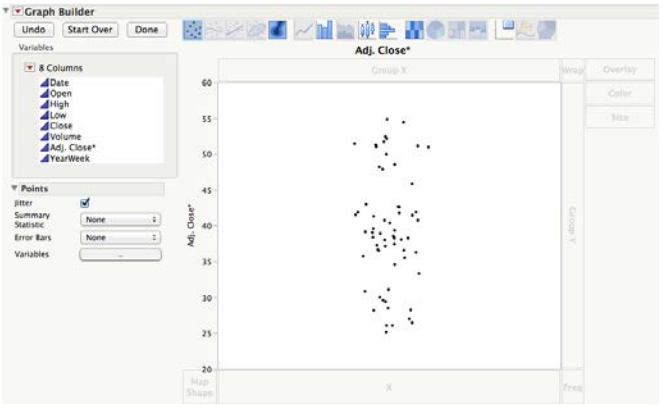
Figure 4.3 Change Volume from Nominal to Continuous



5. Drag the **Adj. Close*** column from the **Variables** box to one of the labeled zones located around the graph canvas. You will notice that as soon as the column is dragged into the zone, a graph of that data is immediately displayed. Keeping the mouse button depressed, experiment by dragging the column into different drop zones to see what happens. If you let go of the mouse button, you can always click the Undo or **Start Over** button to begin again.

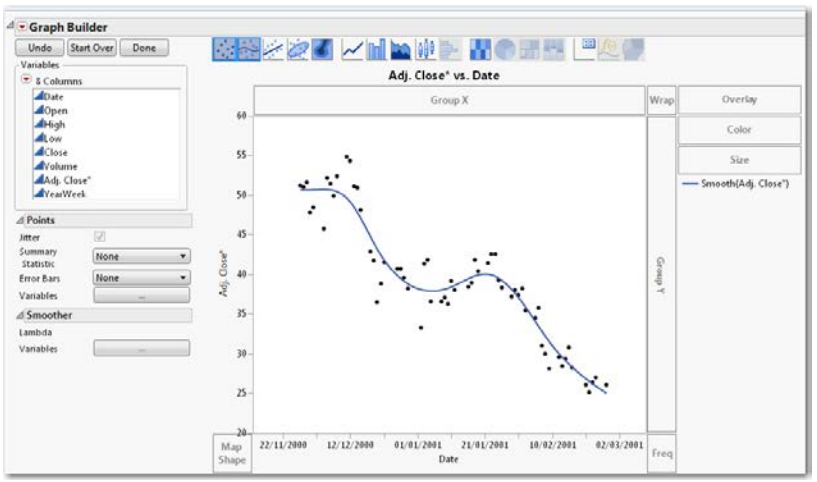
6. Click, hold, and Drag the **Adj. Close*** column to the **Y** drop zone and release the mouse button. You should see a point graph (see Figure 4.4).

Figure 4.4 Drag Adjusted Close to Y Drop Zone



7. Now click and drag the **Date** column to the **X** drop zone. This action immediately produces a scatter plot with dates on the x-axis (see Figure 4.5). Notice that the trend for the close prices is downward and a smoothing line is fit to the data.

Figure 4.5 Drag Date to X Drop Zone



8. Right-click in the graph region to change the look of the graph (see Figure 4.6). A menu appears, containing options to make graph changes. Make no changes at this time. Additional visualization changes can be accomplished by choosing from the palette of icons across the top of the graph (see Figure 4.7). These are the Element Type Icons.

Figure 4.6 Right Mouse Click in the Graph Region

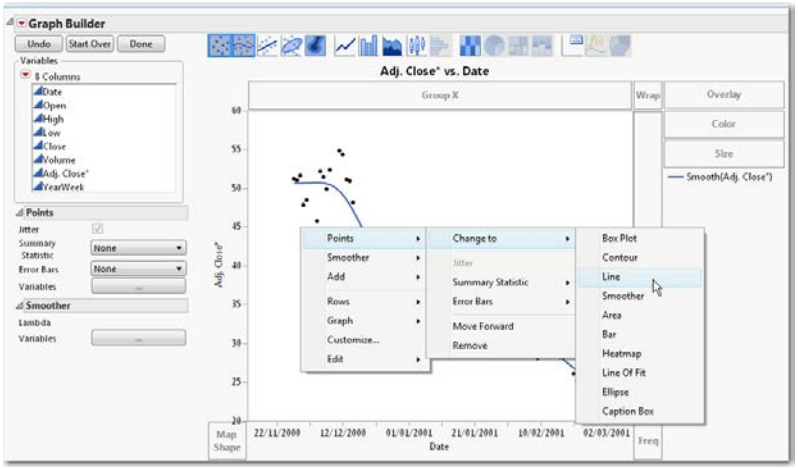
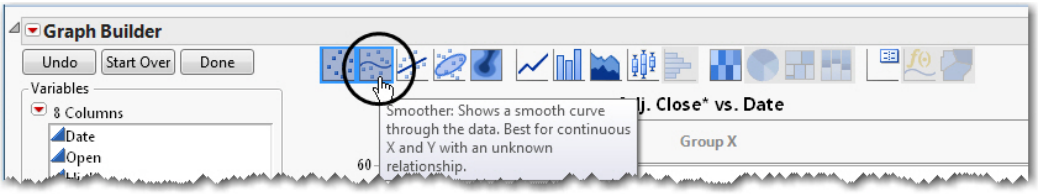


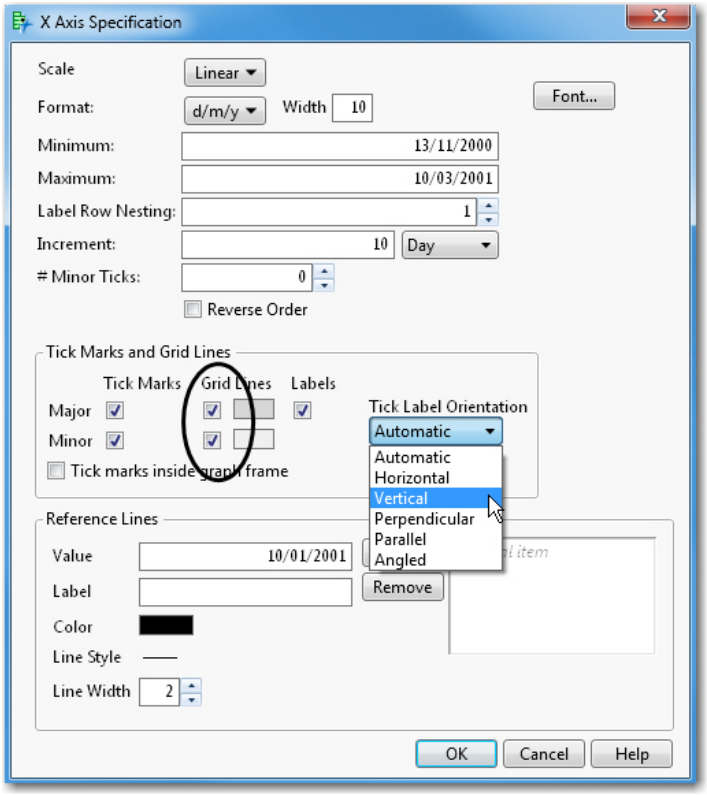
Figure 4.7 Element Type Icon Smoother



9. Select the icon that is second from the left on the palette of icons at the top (circled in Figure 4.7). This toggles the smooth curve on and off. Highlighted choices are enabled. Grayed out choices are disabled. Leave the smooth curve enabled.

10. Now double-click on the **Date** axis. You see a window box of choices (see Figure 4.8). Change the **Tick Label Orientation** to **Vertical**. In **Grid Lines**, add a check for grid lines for the **Major** and **Minor** axes and click **OK**.

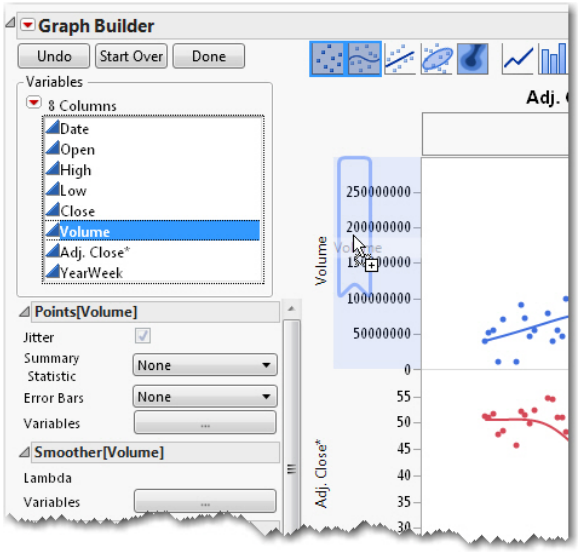
Figure 4.8 Change Tick Orientation and Gridlines



.

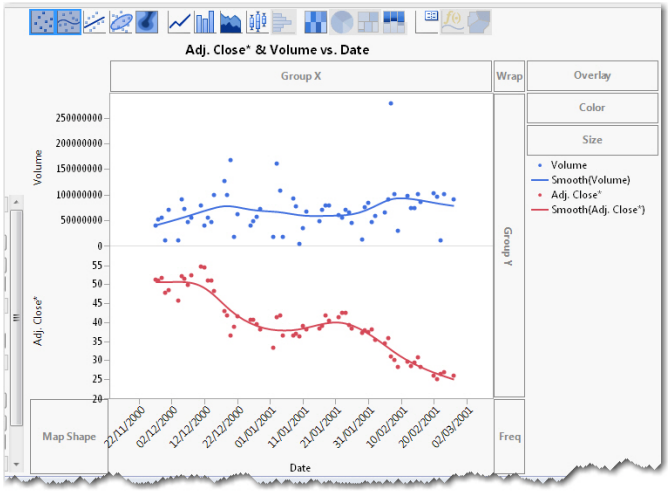
11. You should now see a graph with vertical dates and gridlines at the major tick date intervals. Now drag the **Volume** column to the area just above the *y*-axis label for **Adj. Close*** (see Figure 4.9).

Figure 4.9 Drag Volume to Y Drop Zone



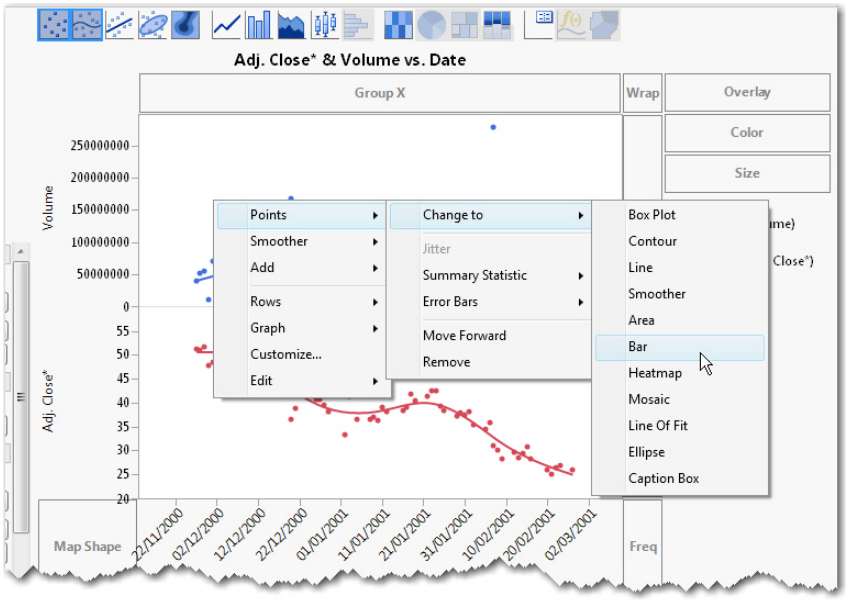
12. You will see a graph rendered (see Figure 4.10).

Figure 4.10 Graph Rendered



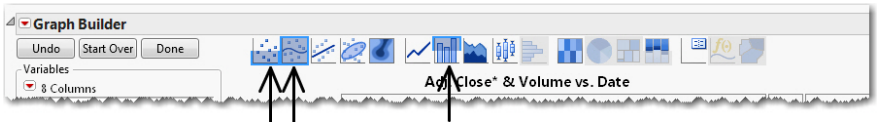
13. Right-click on the **Volume** graph points (see Figure 4.11), and select **Points ► Change to ► Bar**.

Figure 4.11 Change Points to Bars



Note that the Element Type Icons at the top of the chart give an iconographic indication of what parts of the graph elements are currently enabled (see Figure 4.12).

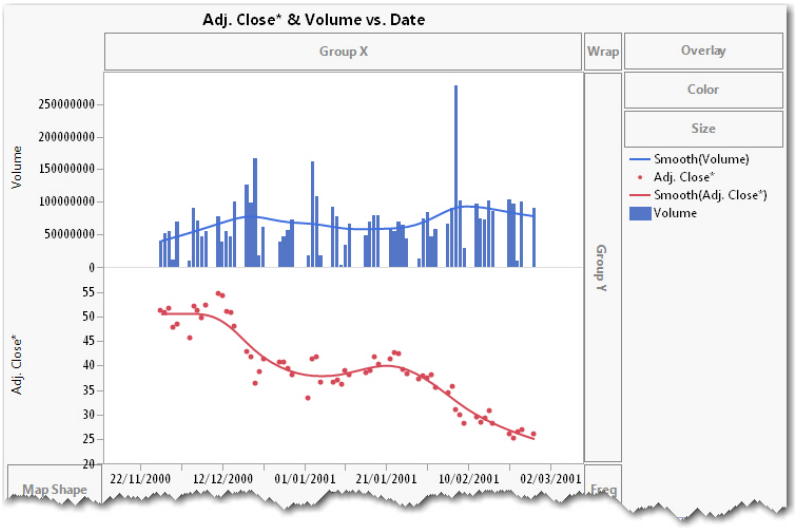
Figure 4.12 Element Type Icons



Elements of the graph's rendering can be controlled from this panel. Reading the icons for the current graph from left to right, points are enabled for the **Adj. Close*** ***y*-axis** only (indicated by the lower half of the icon being highlighted and corresponding to the lower half of the graph in Figure 4.13). Smoother is enabled for both **Adj. Close*** and for **Volume**. Bars are enabled for **Volume** only (the upper half of the graph).

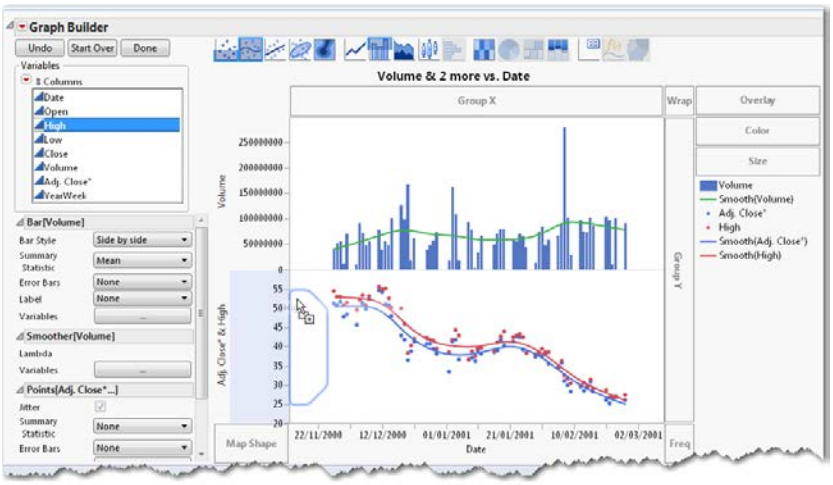
14. This renders a graph showing trading volume and adjusted closing prices for the time period (see Figure 4.13).

Figure 4.13 Enable Smoother for Volume and Adjusted Close



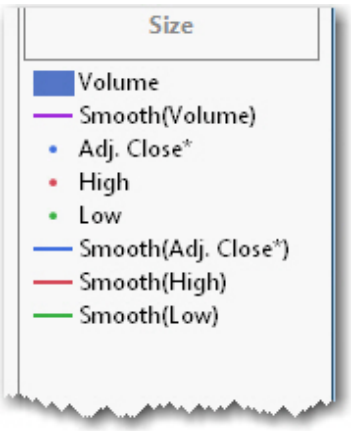
15. Now let's include the daily high and low prices in the graph. Drag the **High** column to the location just to the right of the *y*-axis for **Adj. Close*** (see Figure 4.14).

Figure 4.14 Include Daily High and Low Prices



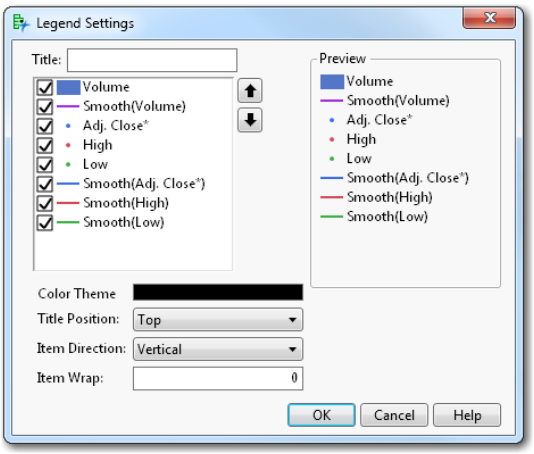
16. A red **High** line is now rendered that represents the maximum price of the stock portfolio by day. If you don't get the picture to match Figure 4.14 the first time, you can always click undo and try again. Now do the same with the **Low** column by clicking and dragging the **Low** column to the same location. A green **Low** line is now rendered that represents the minimum price for each day.
17. To complete the graph, edit the last few items. Double-click in the inside of the legend (see Figure 4.15).

Figure 4.15 Edit the Legend



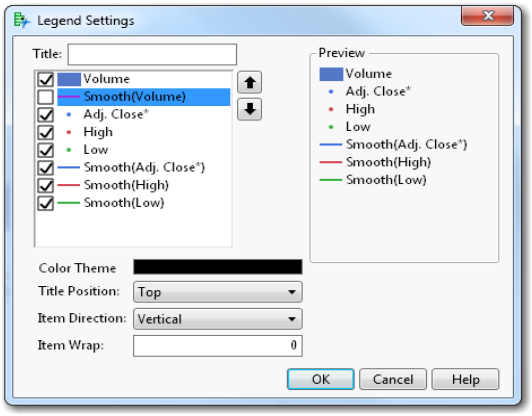
18. A Legend Settings window appears (see Figure 4.16).

Figure 4.16 Legend Settings Window



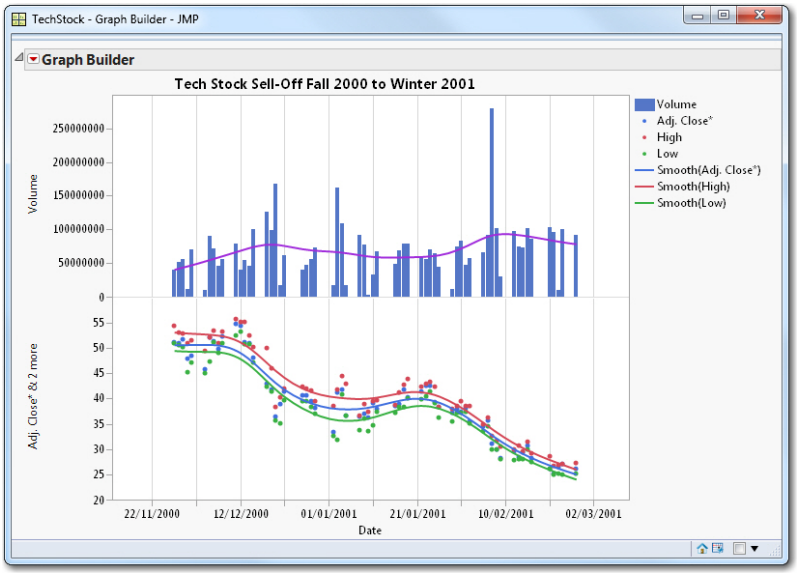
19. Click on the purple **Volume** line and remove the check mark for that item. Then click in the **Preview** area of the dialog box. The dialog box should look like Figure 4.17. Click **OK**.

Figure 4.17 Volume Line and Preview



20. Now let's add a few last details. We'll change the graph title. Click on the graph title and change it to **Tech Stock Sell-Off Fall 2000 to Winter 2001**. Then click **Done**. You should see a graph like that in Figure 4.18.

Figure 4.18 Final Graph



4.2 Using Graph Builder to Produce Maps

The Graph Builder platform also enables you to visualize data on a map, even at the street level. In JMP, maps can be created two ways: using shape files (which are data tables containing the boundaries and names of the shapes such as states or counties of the USA) or background maps (which are background images that recognize latitude and longitude and allow you to plot points accordingly).

Today, mapping data is abundant; JMP provides an easy and flexible environment in which to convey your data on maps. The example below will illustrate how to create a thematic map using the background map service called OpenStreetMaps. An Internet connection is required to use OpenStreetMaps, and you must be using JMP 11 or later.

To make maps in Graph Builder, simply use the same drag and drop methods you practiced in the previous section. The key step is dragging a shape column to the Map Shape drop zone (to utilize a Shape file) or in using latitude and longitude data columns in the X and Y drop zones respectively (to utilize a background map). This action enables maps to be visualized with your columns of interest data.

If you are unfamiliar with the special conditions of how map data is handled in JMP see Section 2.7.

Example 4.2: San Francisco Crime

We'll be working with one of the sample data files to illustrate the steps in this section. The file we'll use, **San Francisco Crime.jmp**, which contains data on individual crime incidents in San Francisco from April 2012. You can find the data set at **Help ► Sample Data ► Graph Builder ► San Francisco Crime.**

The columns include:

Incident Number identifier represented as an integer

Date the date expressed as dd/mm/yyyy

Time hour and minute expressed as HH:MM: AM/PM

Day of Week day of the week expressed as a word

Category 32 offense categories, expressed as words

Incident Description 336 standardized sub categories of the Category column, expressed as words

Traffic Incident 2 categories standardized for indicators of traffic incident, expressed as words

Resolution 16 categories of standardized indicators of incident resolution, expressed as words

Police District 10 categories of geographic police district names in the city of San Francisco

Incident Address nearest cross-street

Longitude longitude expressed in degrees and directions east and west

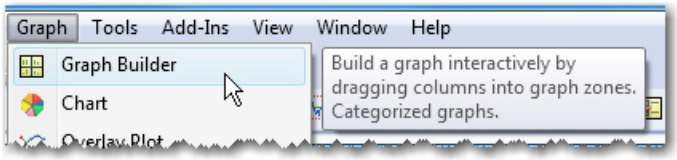
Latitude latitude expressed in degrees and directions north and south

Reference a compact set of 4 hidden columns that automatically performs recoding

Police Department Color a row state column indicating a color marker indexed to each police district

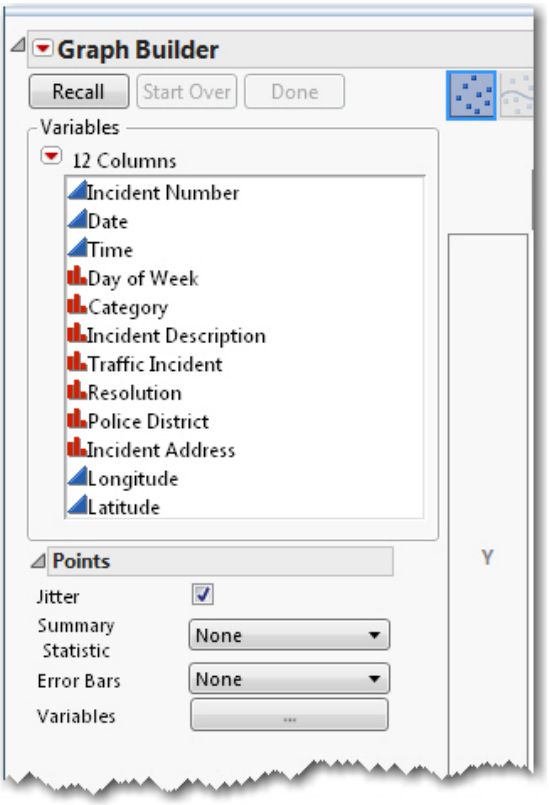
1. First, open the sample data table by selecting **Help ► Sample Data ► Graph Builder ► San Francisco Crime.**
2. Open Graph Builder. Select **Graph ► Graph Builder** (see Figure 4.19).

Figure 4.19 Open Graph Builder



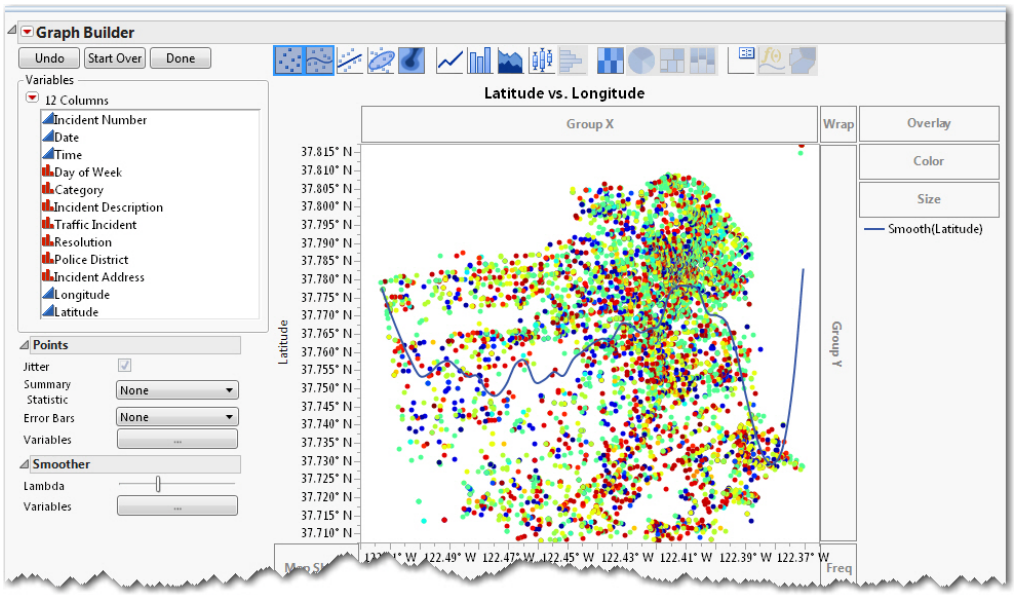
3. Notice the columns in your data table appear on the left in the **Variables** list (see Figure 4.20).

Figure 4.20 San Francisco Crime Variables



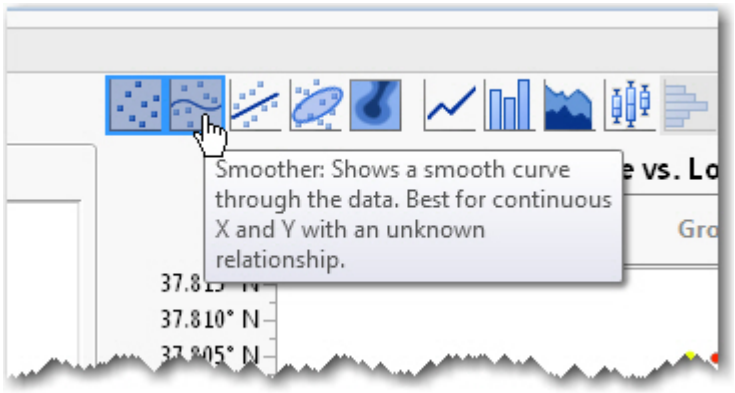
4. Now drag the **Latitude** column from the **Variables** list to the **Y** drop zone located in the graph canvas. Then drag the **Longitude** column from the **Variables** list to the **X** drop zone located in the graph canvas. You should see a point graph with a smoother line (see Figure 4.21).

Figure 4.21 Latitude and Longitude Point Graph



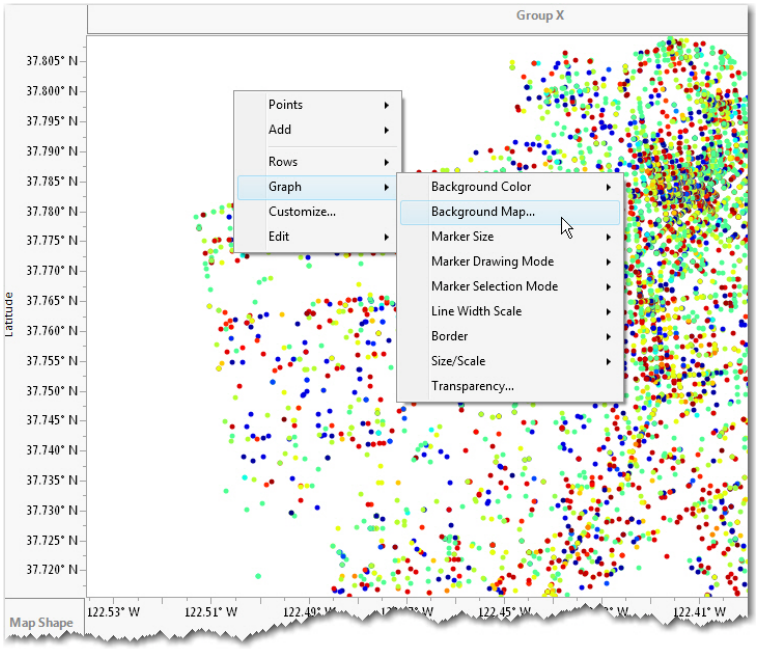
5. Let's de-select the second icon on the element type palette by clicking on the icon to remove the smoother line from the graph (Figure 4.22).

Figure 4.22 Remove Smoother Line



6. Now right-click in the middle of the graph area and choose **Graphs ► Background Map** from the submenu (see Figure 4.23).

Figure 4.23 Graph Builder Background Map

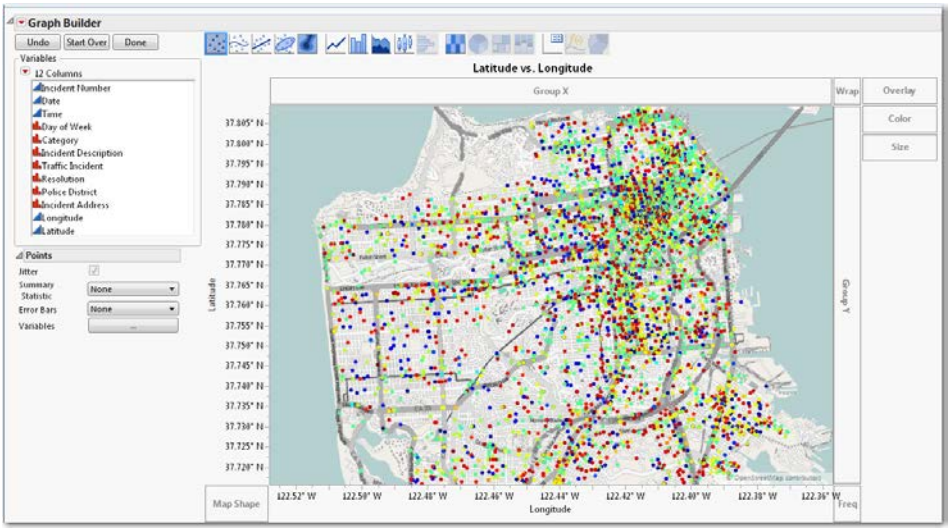


7. Select **Street Map Service** and click **OK** (see Figure 4.24).

Figure 4.24 Graph Builder Street Map Service

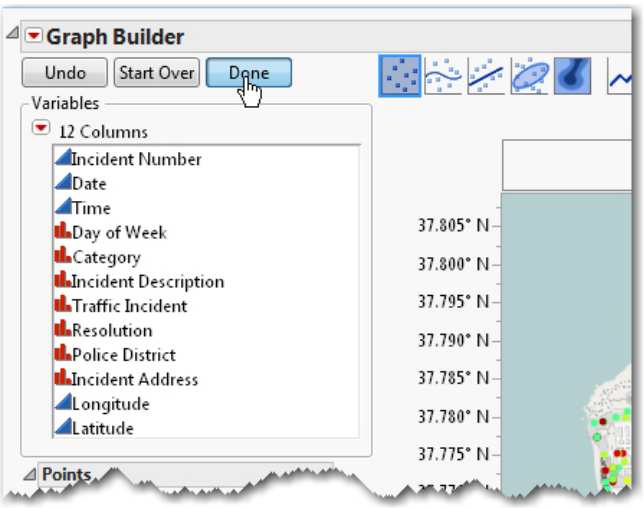
8. The addition of San Francisco streets now appear as a background to the color-coded crime locations (see Figure 4.25).

Figure 4.25 Background Street Map Overlay—Color-Coded Crimes



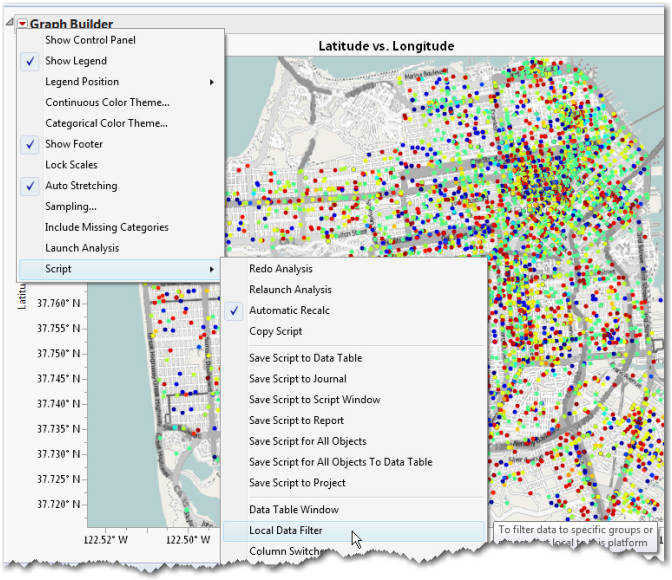
9. You are now finished with the Variables control panel. Click the **Done** button from the control panel to close the panel (see Figure 4.26).

Figure 4.26 Close the Variables Control Panel



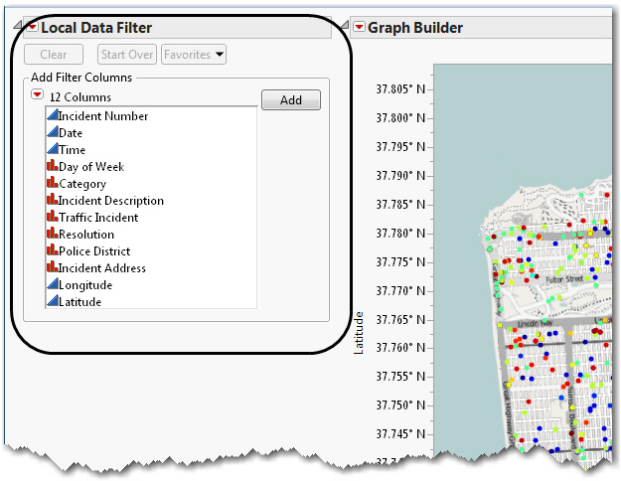
10. Now let's focus on assaults in one particular police district. To do this, we will use a powerful feature called the **Local Data Filter**. From the red triangle for Graph Builder, choose **Script ► Local Data Filter** (see Figure 4.27).

Figure 4.27 Select Local Data Filter



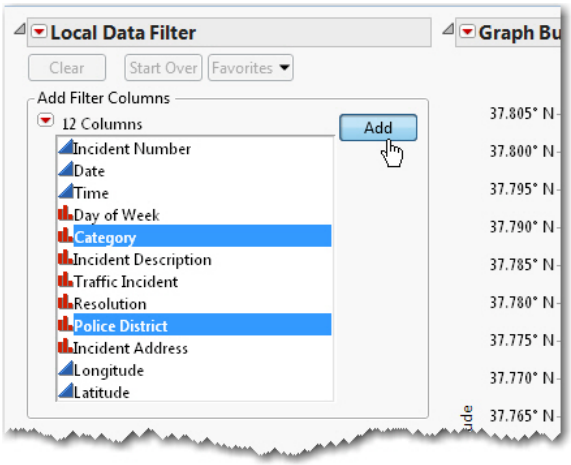
11. A **Local Data Filter** panel appears to the left and inside the Graph Builder window (see Figure 4.28).

Figure 4.28 Local Data Filter Panel



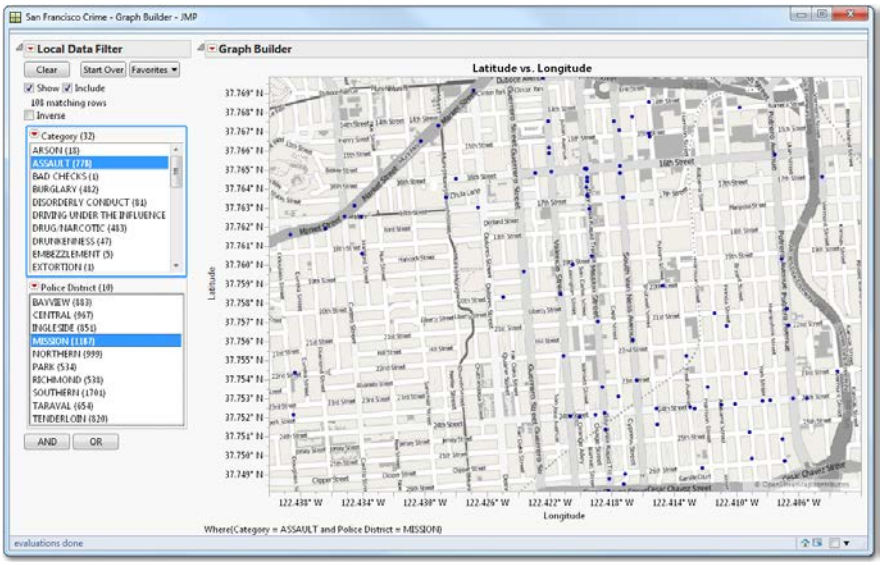
12. Hold down the **Ctrl** key, select **Category** and **Police District** and then click the **Add** button (see Figure 4.29).

Figure 4.29 Data Filter Category and Police District



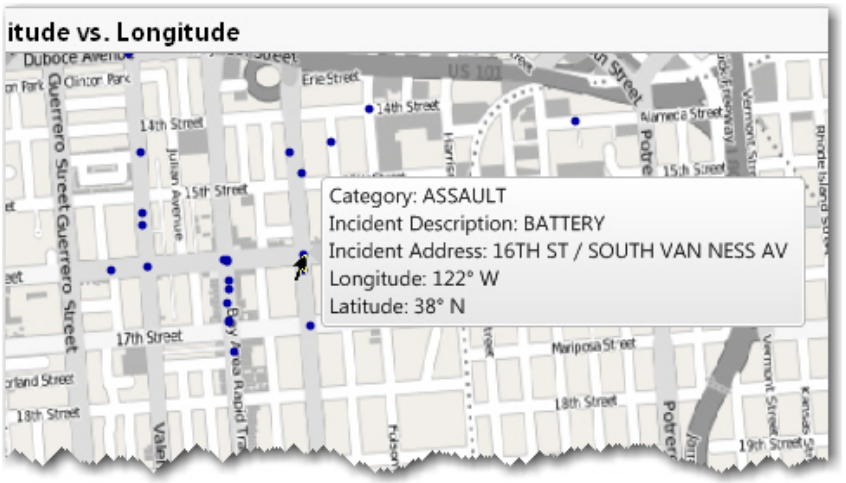
13. A list of crimes and police districts appears in the local data filter. Select **ASSAULT** from the **Category** list, hold down the CTRL key and select **MISSION** from the **Police District** list. The map automatically zooms to the Mission District and maps assault crimes as blue dots (see Figure 4.30).

Figure 4.30 Data Filter Assault and Mission



14. Now move your mouse pointer and hover over any of the blue dots. Labeling information for the crime is automatically displayed (see Figure 4.31).

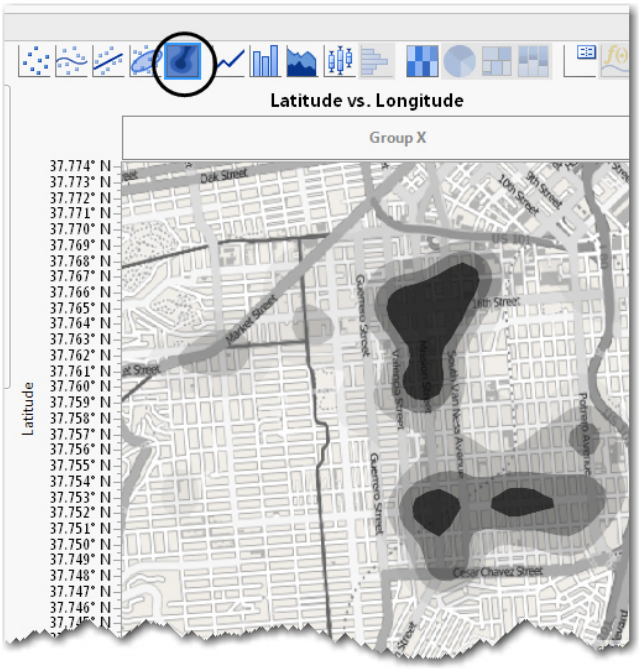
Figure 4.31 Display Label Information



Note: Column information may be used as labels by turning on the label property in the columns list in the data table. If labeling columns is unfamiliar to you, see "Adding Labels to Data" in Chapter 2.

15. Now, let's add contours to this map. Contours will help to plot the density of assaults for the Mission District in San Francisco during this time period. From the red triangle for Graph Builder, select **Graph Builder ► Control Panel**. Hold the **Shift** key and select the contour icon (circled in Figure 4.32).

Figure 4.32 Add Contours for Density of Assaults



16. Many additional visualizations are possible. Feel free to experiment. You can always click the **Undo** button if you want to go back. Click the **Done** button when you are finished.

4.3 Using Tabulate

Now that you are familiar with using drop zones to generate a graph, we introduce the Tabulate platform. Tabulate is used in a similar fashion to Graph Builder, but instead of creating graphs, your objective is to create numerical summaries for their own sake or for the development of new data tables based on summarized data for further analysis.

Example 4.3: Movies

We'll use **Movies.jmp** to demonstrate Tabulate. This data table is a listing of popular US movies that were released between 1937 through 2003. The movies are categorized by type, rating, and director and contain earned gross dollar amounts for both the US domestic market and the worldwide market. **Note:** Dollar amounts are expressed in millions of dollars for both domestic and worldwide markets.

You can find the data set at **Help ► Sample Data ► Business and Demographic ► Movies**.

The columns include:

Movie name of movie

Director director of movie

Type genre/category of movie (for example: comedy, family)

Rating US movie rating system (for example: general audience [G], adult [R])

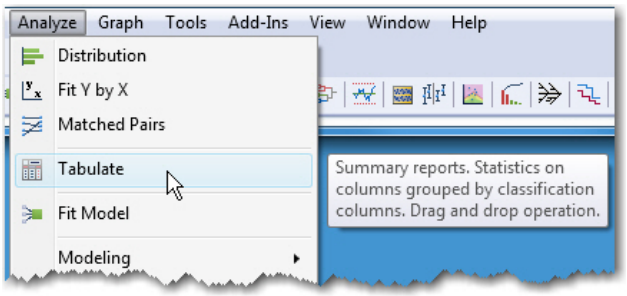
Year year of movie release (for example: 1937)

Domestic $ US domestic revenue in $ earned by the movie in that year

Worldwide $ Worldwide revenue in $ earned by the movie in that year

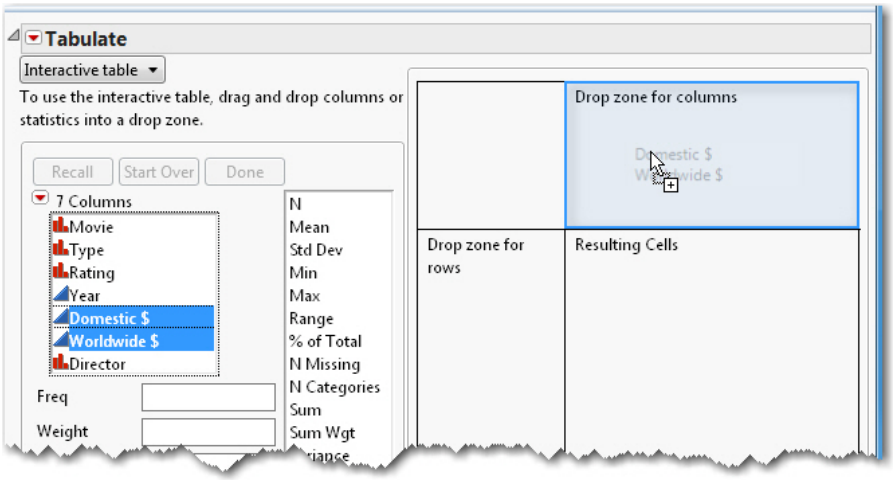
1. After opening the data table, open **Tabulate** (select **Analyze ► Tabulate**) (see Figure 4.33).

Figure 4.33 Open Tabulate Platform



2. The Tabulate platform presents a control panel with your columns, a statistics list, and drop zones for rows and columns (see Figure 4.34)

Figure 4.34 Tabulate with Domestic and Worldwide Columns



3. Click on both the **Domestic $** and **Worldwide $** columns and drag them to the part of the table labeled **Drop zone for columns** (see Figure 4.34).

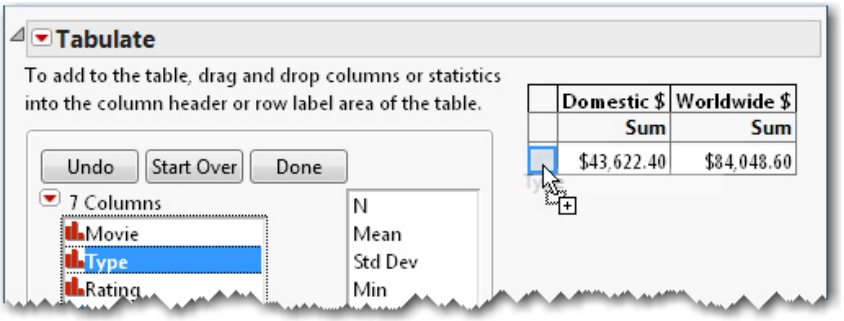
4. A sum of movie revenue by domestic $ and worldwide $ appears (see Figure 4.35).

Figure 4.35 Sum of Domestic and Worldwide Movie Revenue

	Domestic $	Worldwide $
	Sum	Sum
	$43,622.40	$84,048.60

5. Now drag the **Type** column to the row area (see Figure 4.36).

Figure 4.36 Drag Type Column to Row Area



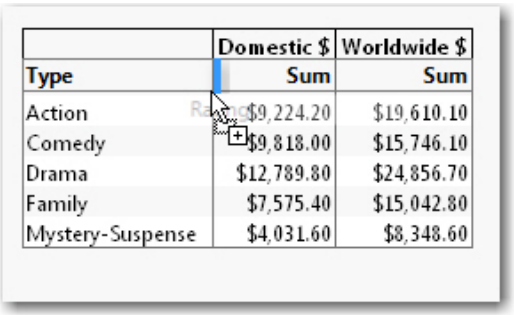
6. Release the mouse button; the table now sums the revenue by type for both domestic and worldwide (see Figure 4.37).

Figure 4.37

	Domestic $	Worldwide $
Type	Sum	Sum
Action	$9,224.20	$19,610.10
Comedy	$9,818.00	$15,746.10
Drama	$12,789.80	$24,856.70
Family	$7,575.40	$15,042.80
Mystery-Suspense	$4,031.60	$8,348.60

7. We would like to add a Rating subgroup within the **Type** column. Drag the **Rating** column to the area just to the left of the **Domestic Sum** column (see Figure 4.38).

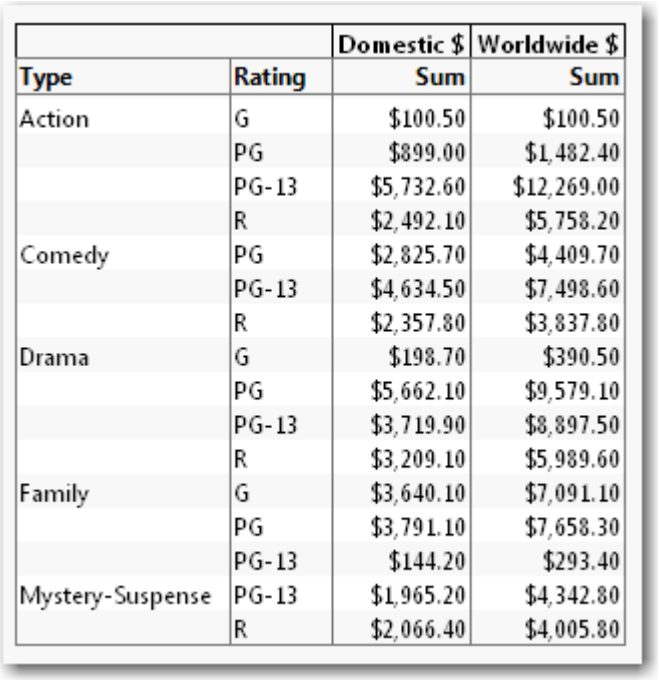
Figure 4.38 Add Rating Subgroup Within Type Column



	Domestic $	Worldwide $
Type	Sum	Sum
Action	$9,224.20	$19,610.10
Comedy	$9,818.00	$15,746.10
Drama	$12,789.80	$24,856.70
Family	$7,575.40	$15,042.80
Mystery-Suspense	$4,031.60	$8,348.60

8. Release the mouse button and a **Rating** column appears (see Figure 4.39).

Figure 4.39 Table with Rating Column Added

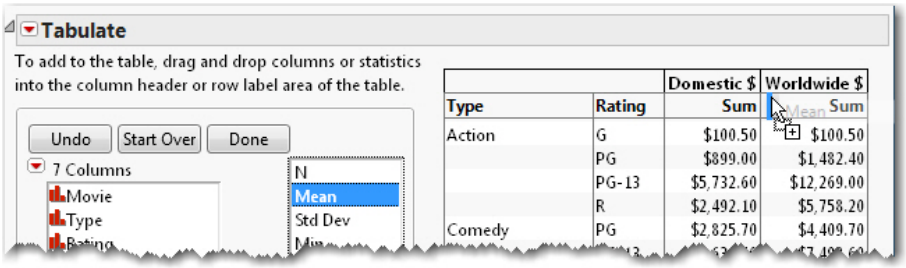


		Domestic $	Worldwide $
Type	Rating	Sum	Sum
Action	G	$100.50	$100.50
	PG	$899.00	$1,482.40
	PG-13	$5,732.60	$12,269.00
	R	$2,492.10	$5,758.20
Comedy	PG	$2,825.70	$4,409.70
	PG-13	$4,634.50	$7,498.60
	R	$2,357.80	$3,837.80
Drama	G	$198.70	$390.50
	PG	$5,662.10	$9,579.10
	PG-13	$3,719.90	$8,897.50
	R	$3,209.10	$5,989.60
Family	G	$3,640.10	$7,091.10
	PG	$3,791.10	$7,658.30
	PG-13	$144.20	$293.40
Mystery-Suspense	PG-13	$1,965.20	$4,342.80
	R	$2,066.40	$4,005.80

Note: You can click the **Undo** button at any time to undo the last action, or you can click **Start Over** to start over again.

9. Drag the Mean from the **Statistics** column and drop it on one of the **Sum** column headings (see Figure 4.40).

Figure 4.40 Drag Mean Column to Sum Column



10. Then, release the mouse button. The **Sum** columns have been transformed to **Mean** columns (see Figure 4.41).

Figure 4.41 Sum Columns Changed to Mean Columns

		Domestic $	Worldwide $
Type	**Rating**	**Mean**	**Mean**
Action	G	$100.500	$100.500
	PG	$149.833	$247.067
	PG-13	$184.923	$395.774
	R	$138.450	$319.900
Comedy	PG	$141.285	$220.485
	PG-13	$144.828	$234.331
	R	$138.694	$225.753
Drama	G	$198.700	$390.500
	PG	$161.774	$273.689
	PG-13	$232.494	$556.094
	R	$128.364	$239.584
Family	G	$158.265	$308.309
	PG	$180.529	$364.681
	PG-13	$144.200	$293.400
Mystery-Suspense	PG-13	$151.169	$334.062
	R	$129.150	$250.363

11. Now drag the **Sum** statistic to the right of the **Domestic $**, **Mean** column (see Figure 4.42). Then, release the mouse button. **Sum** columns then reappear in the table (see Figure 4.43).

Figure 4.42 Drag Sum Column to the Right of Domestic $ Column

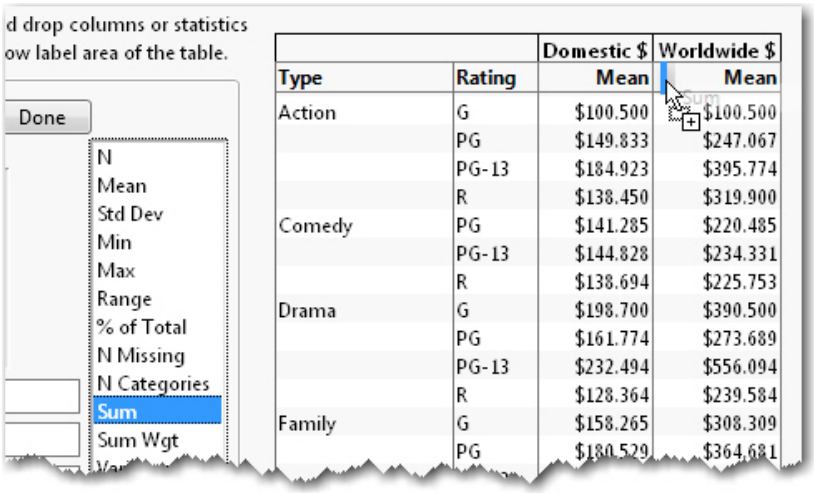
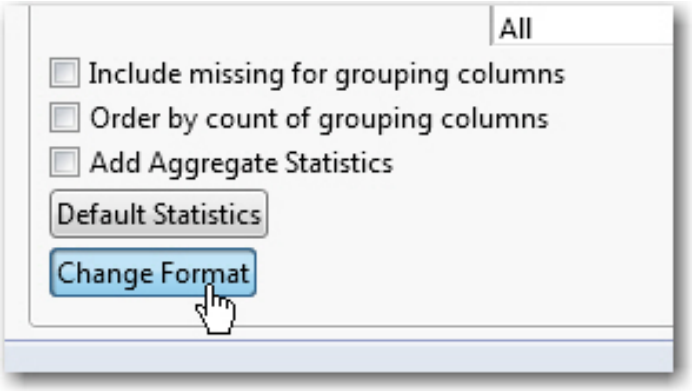


Figure 4.43 Sum Column Added

		Domestic $		Worldwide $	
Type	**Rating**	**Sum**	**Mean**	**Sum**	**Mean**
Action	G	$100.50	$100.500	$100.50	$100.500
	PG	$899.00	$149.833	$1,482.40	$247.067
	PG-13	$5,732.60	$184.923	$12,269.00	$395.774
	R	$2,492.10	$138.450	$5,758.20	$319.900
Comedy	PG	$2,825.70	$141.285	$4,409.70	$220.485
	PG-13	$4,634.50	$144.828	$7,498.60	$234.331
	R	$2,357.80	$138.694	$3,837.80	$225.753
Drama	G	$198.70	$198.700	$390.50	$390.500
	PG	$5,662.10	$161.774	$9,579.10	$273.689
	PG-13	$3,719.90	$232.494	$8,897.50	$556.094
	R	$3,209.10	$128.364	$5,989.60	$239.584
Family	G	$3,640.10	$158.265	$7,091.10	$308.309
	PG	$3,791.10	$180.529	$7,658.30	$364.681
	PG-13	$144.20	$144.200	$293.40	$293.400
Mystery-Suspense	PG-13	$1,965.20	$151.169	$4,342.80	$334.062
	R	$2,066.40	$129.150	$4,005.80	$250.363

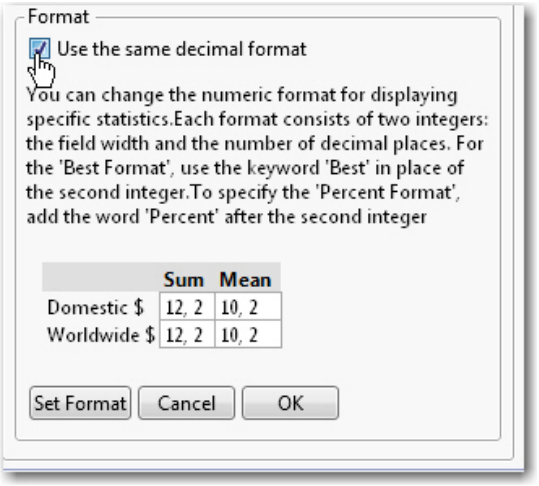
12. The decimal and display formats can be changed. In the previous example, there are three decimals displayed to the right of the decimal point for the **Mean** column. We would like to display only two decimal places for values in the table. To adjust the display of decimals, click **Change Format** (see Figure 4.44a).

4.44a Change Format



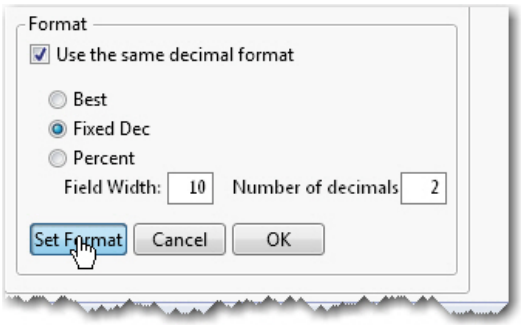
13. An additional format panel appears. In the format pane, enter a 2 for the **Mean** column for both **Domestic $** and **Worldwide $** so the panel looks as it is shown (see Figure 4.44b).

Figure 4.44b Change Decimal Display



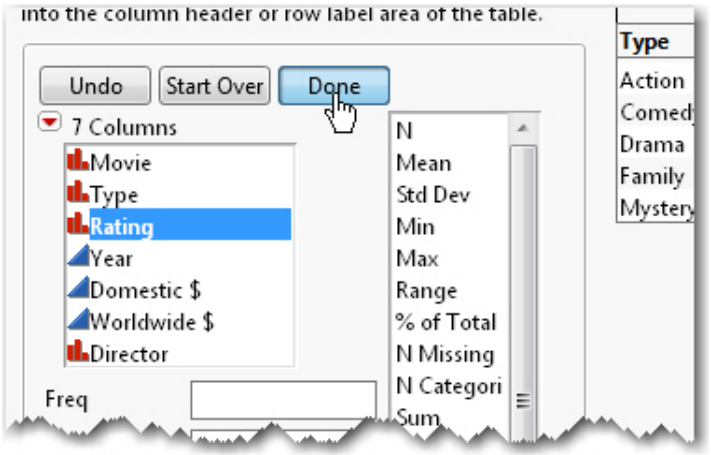
Click the **Set Format** button. Select the check box for **Use Same Decimal format for all**. The window changes (see Figure 4.44c). Click **Fixed Dec** and enter a 2 for **Number of Decimals**. Then, click **Set Format** and **OK**.

Figure 4.44c Specify Fixed Decimal Width



14. The table is now complete. Click **Done** (see Figure 4.45).

Figure 4.45 Finalize Changes



15. Recall that additional options are available on the red triangle. Click the red triangle next to Tabulate and select **Make Into Data Table** (see Figure 4.46), which will produce a JMP data table of the Tabulate results (see Figure 4.47). **Make Into Data Table** will accomplish two tasks simultaneously; you will have a summary table for reporting and a data table of summarized data ready for further analysis or visualization.

    Tabulate, therefore, is a useful tool for reshaping and reorganizing data for further analysis or graphing in addition to producing tabular reports.

Figure 4.46 Make Data Table from Tabulate Results

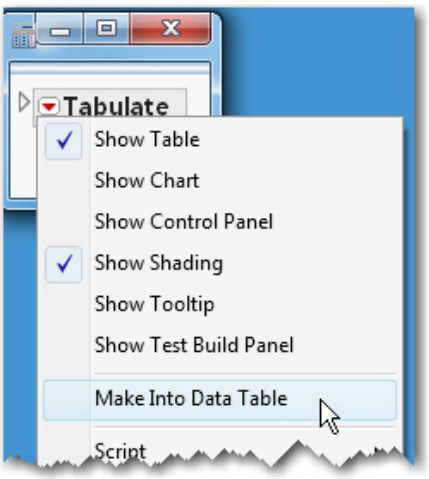
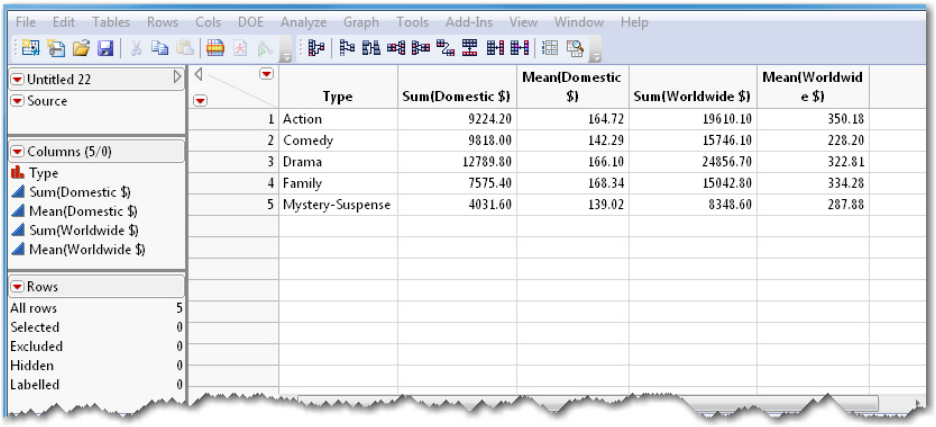


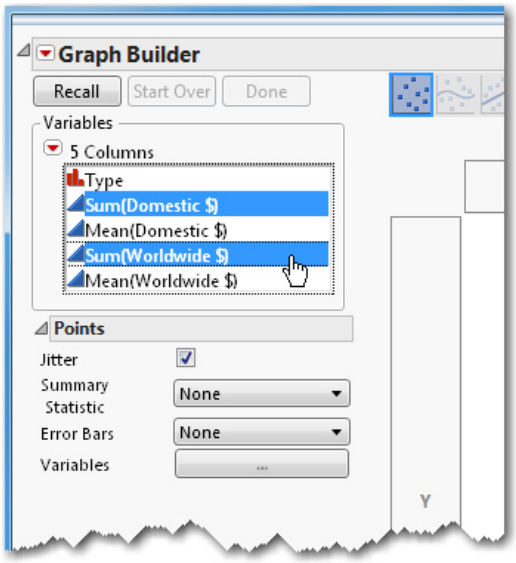
Figure 4.47 Newly Created Data Table



What do you think you can do with the new table in Figure 4.47? Try putting your new skills to work. Use Graph Builder to make a graph out of the data table you just produced (select **Graph ► Graph Builder**).

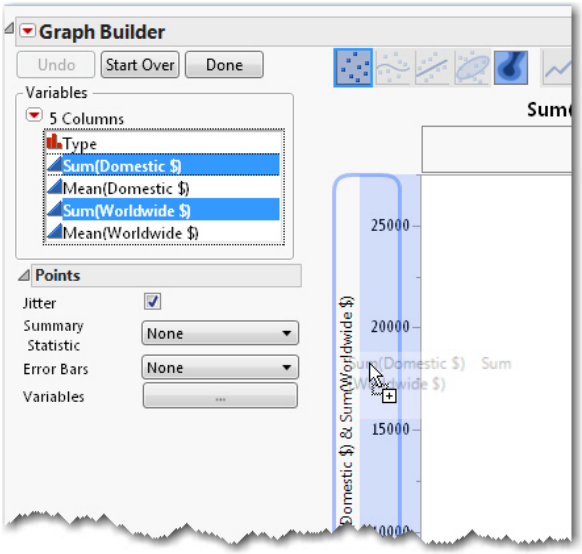
16. Now, press **CTRL** and select the columns **Sum (Domestic $)** and **Sum (Worldwide $)** so that they are both highlighted (see Figure 4.48).

Figure 4.48 Select Sum Columns



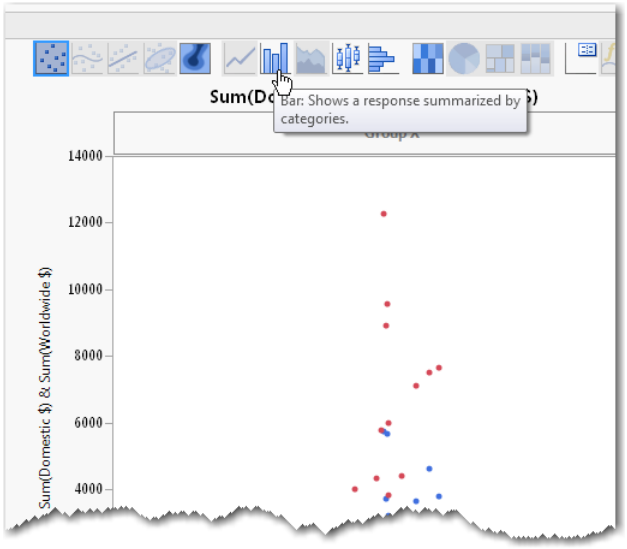
17. Drag the two selected columns to the **Y** drop zone (see Figure 4.49).

Figure 4.49 Drag Sum Columns to Y Drop Zone



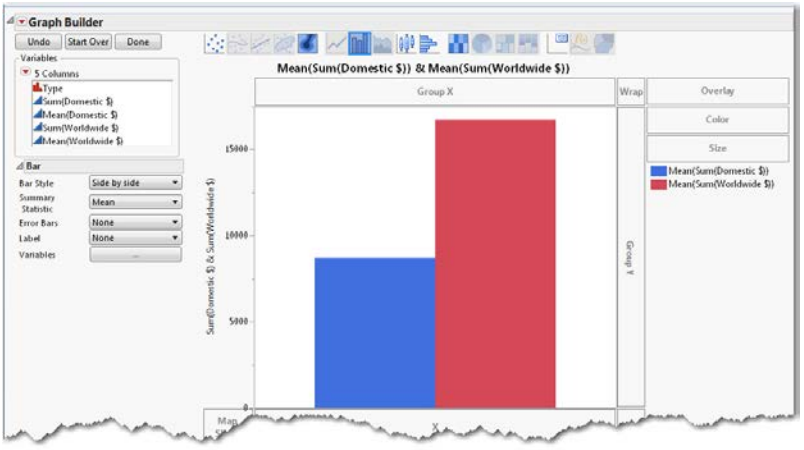
18. A point chart initially appears with blue and red dots. Select the Bar element type icon (see Figure 4.50). **Note:** Of course, you could right click in the graph as you learned earlier in the chapter.

Figure 4.50 Convert Point Chart to Bar Chart



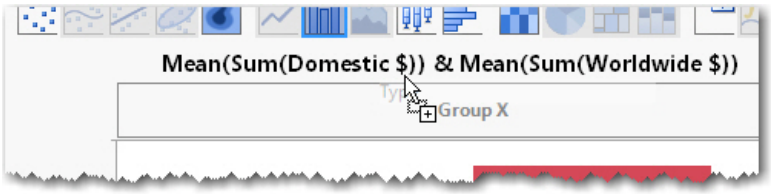
19. A bar chart appears, showing the worldwide and domestic revenue comparisons (see Figure 4.51).

Figure 4.51 Bar Chart



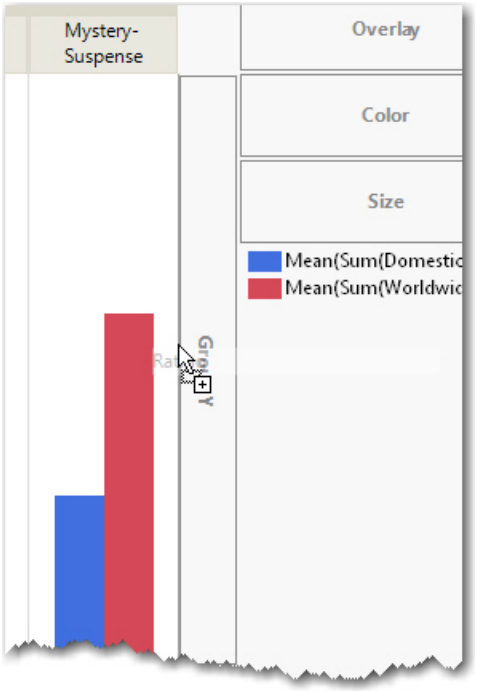
20. Now, drag **Type** to the **Group X** drop zone (see Figure 4.52).

Figure 4.52 Drag Type to Group X Drop Zone



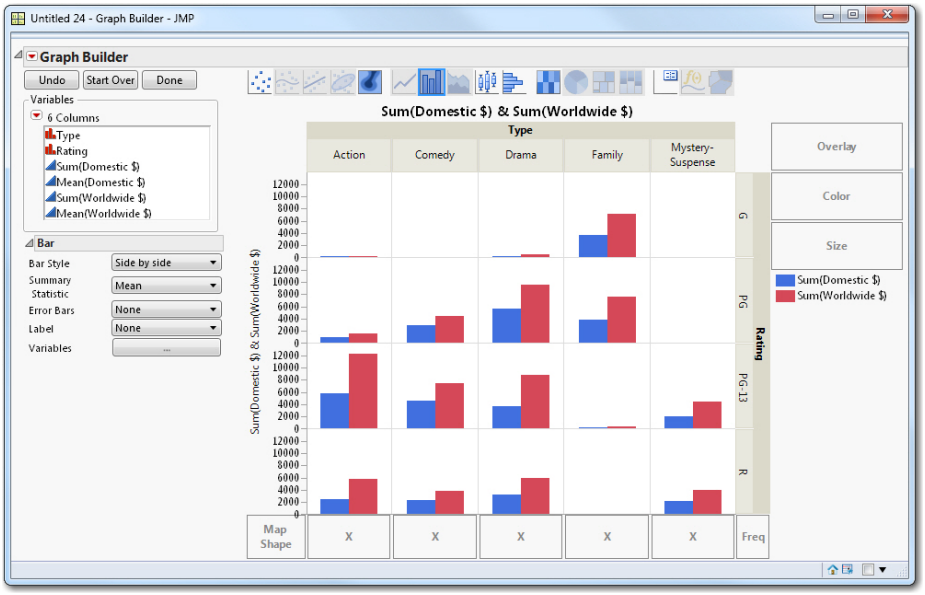
21. Now, drag **Rating** to the **Group Y** drop zone (see Figure 4.53).

Figure 4.53 Drag Rating to Group Y Drop Zone



22. The nearly finished graph appears (see Figure 4.54).

Figure 4.54 Nearly Finished Graph



23. Click **Done,** and the finished graph appears (see Figures 4.55 and 4.56).

Figure 4.55 Click Done to Fnalize

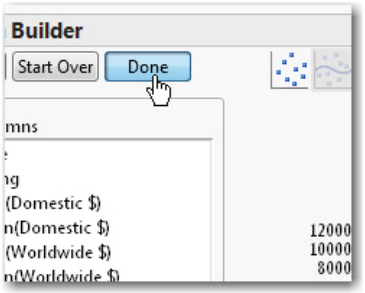
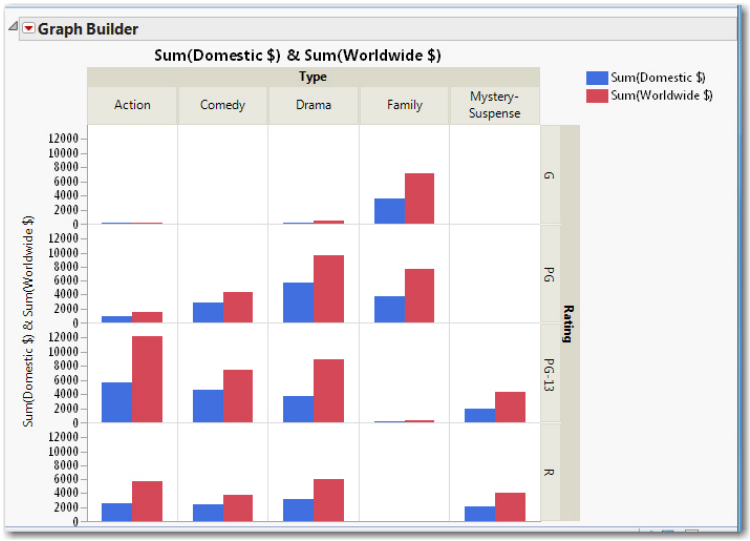


Figure 4.56 Final Graph



4.4 Summary

In this chapter, we examined a key feature of JMP, which is its ability to create graphs and summaries interactively and intuitively in an environment that enables quick exploration and discovery. This chapter introduced you to two tools, Graph Builder and Tabulate.

Graph Builder shows you quick previews of graphs and maps as you drag columns to drop zones, enabling you to flip through choices until you find just the right one to tell the graphical story in your data. With a local data filter, Graph Builder allows you to drill down to the right level in your data. Contrast using Graph Builder to the methods employed in Chapter 3, where it is assumed that you have a specific graph in mind and need directions to reproduce it.

Tabulate shows you quick previews of numerical table summaries as you drag columns to drop zones. Tabulate enables you to flip through choices until you find the right numerical table summary of your data.

Chapter 5 Problem Solving with One and Two Columns

5.1 Introduction

5.2 Analyzing One Column

5.3 Comparing One Column to Another

5.4 Summary

In contrast to earlier chapters that have focused on describing data and producing graphs and maps, this chapter is about getting answers to your questions and making sense of your data in a condensed and rapid way. This problem-solving activity is often a process of trial and error, and it does not lend itself to brief descriptive steps. It takes thought and practice to do this well, but JMP is the perfect companion on this journey. This chapter helps you develop some appreciation and basic JMP skills in this problem-solving process.

Just as we discussed in the first chapter, JMP provides a navigation framework that is designed around the workflow of the problem solver. So what do we mean by, "the workflow of the problem solver?" First, we are talking about the class of problems that are measurable or countable or that already have data that is written down. If you need assistance importing or accessing your data, see Chapter 2. By workflow, we are referring to the process by which you analyze data to arrive at some understanding or insight. This involves thinking about the questions you may have of the data and recognizing how the questions translate to the menu items. Users who learn this find JMP to be a very intuitive partner in that problem-solving process.

In the problem solving process, the answer to one question often prompts other questions. JMP is designed to help you answer these follow-up questions quickly, resulting in more rapid learning. Throughout this chapter, we use simple examples with scenarios that prompt questions you might want to answer. We show you how JMP's menus translate to these questions and how the results help you answer them. Just as many real-world problems start with basic questions and understanding and evolve into more complex ones, we start off with the basics here as well.

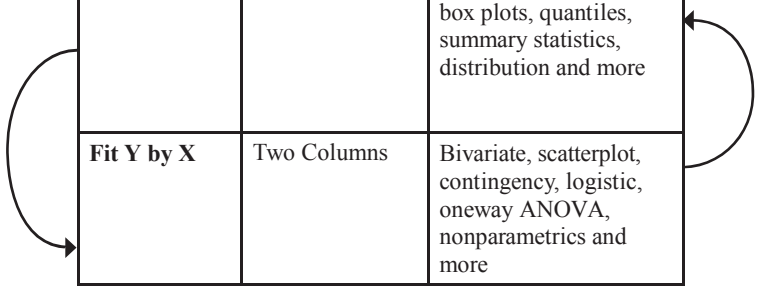
5.1 Introduction

The transformation of data into information is a process that involves a few basic JMP platforms that we introduce in this chapter, including **Distribution** and **Fit Y by X**. Distribution and Fit X by Y correspond to analyzing the characteristics of one and two columns, respectively, which is the scope of this chapter. The tools are found in the Analyze menu. Table 5.1 outlines the tool name, how many columns it supports, and its statistical terms. Don't worry if you don't recognize the statistical terms and acronyms; we'll present the basic ideas as we go. Appendix B describes the statistical terms and concepts in this chapter.

The organization of the items on the **Analyze** menu is the same framework discussed in Section 1.3. Within this framework, we cover just a few menu items but in the process give you access to over 100 statistical methods.

Table 5.1 Tool Name, Columns Supported, and Statistical Terms

Analyze Menu	How Many Columns	Statistical Terminology
Distribution	Single Columns	One sample univariate methods, histograms, box plots, quantiles, summary statistics, distribution and more
Fit Y by X	Two Columns	Bivariate, scatterplot, contingency, logistic, oneway ANOVA, nonparametrics and more



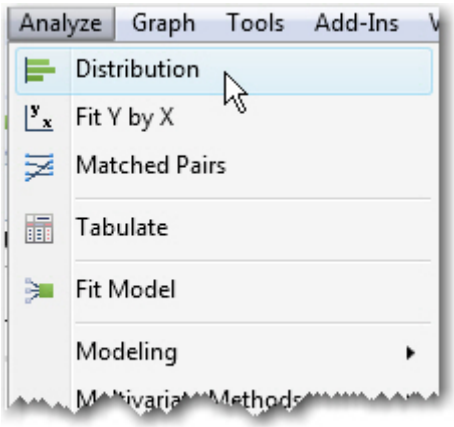
Note: If there is magic to JMP, this is it. The arrows above indicate how the **Analyze** menu works. First and simple questions are answered with the Distribution platform and then further questions are often addressed with items further down the menu like **Fit Y by X** until you get the answers you need to solve your problem. Thus, you proceed from the simple to just the level of complexity you need to answer your questions as you work your way down the **Analyze** menu. Sometimes a discovery is made with a more complex method farther down the **Analyze** menu that leads you to confirm it with a simpler method back up the **Analyze** menu.

5.2 Analyzing One Column

JMP's goal in using this menu framework is to expose you to powerful methods in a logical order that allows you to learn about your data progressively and rapidly. The order is built into the menu structure.

Figure 5.1 displays the first item on the Analyze menu. It is the Distribution platform. It's first on the menu because this platform answers many basic questions that you should ask at the beginning of analysis of data.

Figure 5.1 Analyze Distribution



Another way to remember Distribution as a starting point is that the platform allows you to look at one column at a time and produces results for those individual columns. In fact, a group of statistics is called one sample univariate, meaning analysis of the properties of one variable represented as a sample. (In JMP variables are referred to as columns as we've mentioned in earlier chapters.) There are dozens of one sample univariate statistics, and most are conveniently arranged in the Distribution platform.

Example 5.1: Financial

We will be using the **Financial.jmp** data file to illustrate the steps in this chapter. The data are from the Fortune 500 selected from the April 23, 1990 Fortune magazine issue. You can find the data set at Help ► Sample Data ► Business & Demographic ► Financial.

This data file includes columns for:

Type type of company

Sales($M) yearly sales in millions of dollars

Profit($M) yearly profits in millions of dollars

#emp number of employees at time of measurement

Profits/emp profits per employee in thousands of dollars

Assets($Mil.) assets in millions of dollars

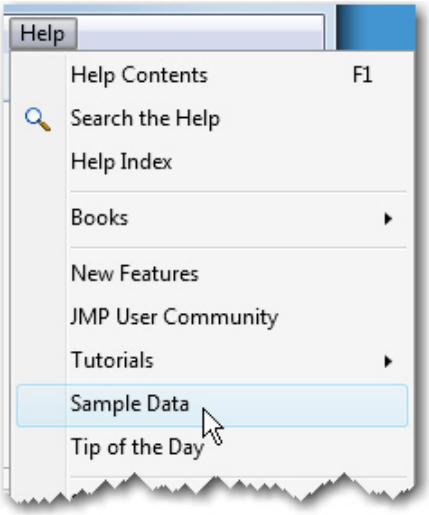
Sales/emp sales per employee in thousands of dollars

Stockholder's Eq($Mil.) stockholder's equity in millions of dollars

Let's start working our way through an analysis using this process. We'll use this exercise to practice asking the early questions of unknown data using the Distribution platform in JMP. Let's open a data table on which to base some analysis:

1. Select **Help ► Sample Data** (see Figure 5.2).

Figure 5.2 Help Sample Data



2. Then select **Business and Demographic ► Financial** (see Figure 5.3). Some financial performance data are displayed (see Figure 5.4).

Figure 5.3 Select Business and Demographic, then Financial

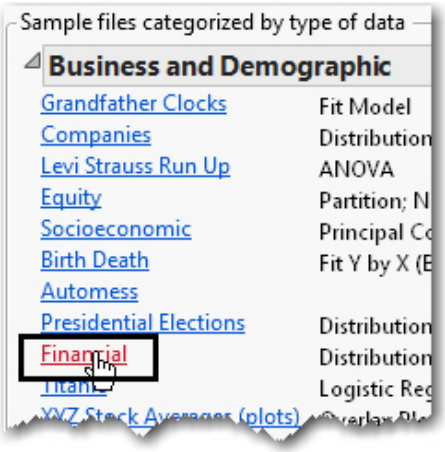
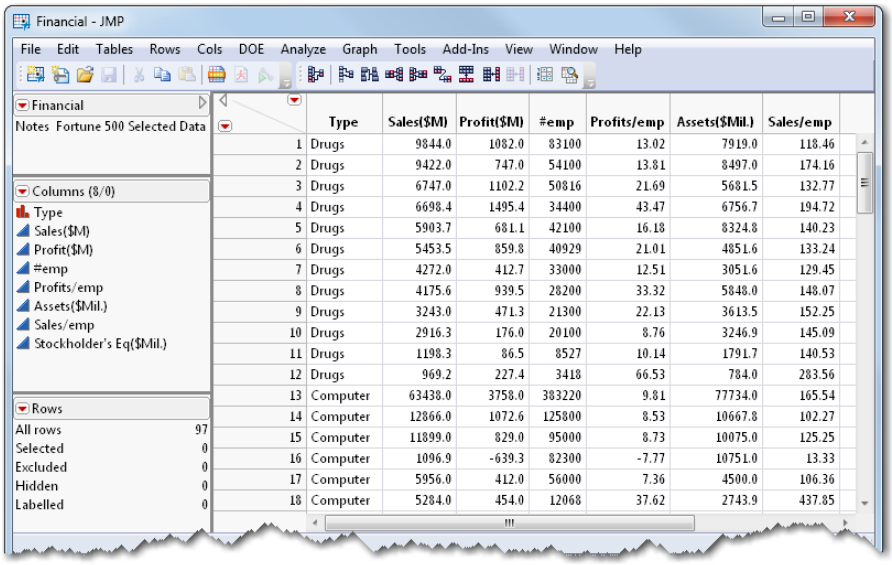


Figure 5.4 Financial Performance Data



	Type	Sales($M)	Profit($M)	#emp	Profits/emp	Assets($Mil.)	Sales/emp
1	Drugs	9844.0	1082.0	83100	13.02	7919.0	118.46
2	Drugs	9422.0	747.0	54100	13.81	8497.0	174.16
3	Drugs	6747.0	1102.2	50816	21.69	5681.5	132.77
4	Drugs	6698.4	1495.4	34400	43.47	6756.7	194.72
5	Drugs	5903.7	681.1	42100	16.18	8324.8	140.23
6	Drugs	5453.5	859.8	40929	21.01	4851.6	133.24
7	Drugs	4272.0	412.7	33000	12.51	3051.6	129.45
8	Drugs	4175.6	939.5	28200	33.32	5848.0	148.07
9	Drugs	3243.0	471.3	21300	22.13	3613.5	152.25
10	Drugs	2916.3	176.0	20100	8.76	3246.9	145.09
11	Drugs	1198.3	86.5	8527	10.14	1791.7	140.53
12	Drugs	969.2	227.4	3418	66.53	784.0	283.56
13	Computer	63438.0	3758.0	383220	9.81	77734.0	165.54
14	Computer	12866.0	1072.6	125800	8.53	10667.8	102.27
15	Computer	11899.0	829.0	95000	8.73	10075.0	125.25
16	Computer	1096.9	-639.3	82300	-7.77	10751.0	13.33
17	Computer	5956.0	412.0	56000	7.36	4500.0	106.36
18	Computer	5284.0	454.0	12068	37.62	2743.9	437.85

Let's assume that your objective is to use this financial data to help you select company stocks to add to your portfolio. Your goal is to select stocks that will maximize the likelihood of positive returns and to use those returns to fund your favorite charity. We assume that:

- The most profitable companies will also tend to have the highest positive returns.
- You have sufficient data to make a reasonable prediction.
- Each row represents a company stock. You will need to select about 10 company stocks to sufficiently diversify your portfolio.
- Market conditions for the next 6 months will remain mostly the same for all sectors.

What kinds of questions will you ask in order to pick company stocks?

Note: Time is an important variable to consider with financial data in particular. For the purposes of illustration, however, we have omitted this variable.

Questions That Involve One Column

In this example, you might ask what range of company profits has existed for these stocks. Or, what has been the average (the mean) profit for the companies? How much variability (standard deviation) has there been? Are there some companies whose profits or losses are very extreme (outliers) relative to others? Are these extremes genuine or did someone make a mistake when entering the data (data quality)?

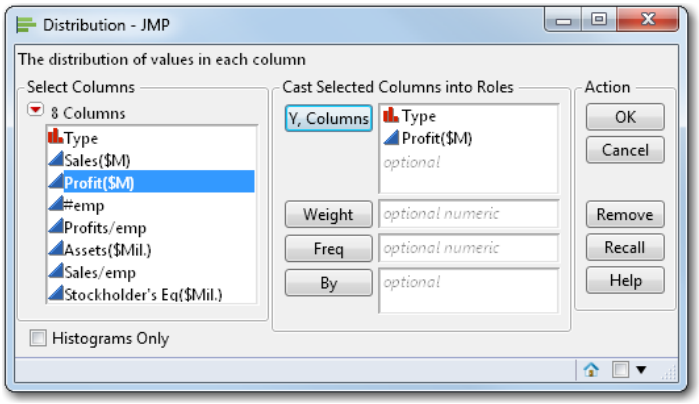
These are the types of questions you might ask of any set of data. These initial questions and many more are all answered with the Distribution platform from the Analyze Menu.

Using Distribution to Understand a Column of Data

Let's perform a distribution analysis to answer these early questions for company performance.

1. Select **Analyze ► Distribution**.
2. Select **Type** and **Profits($M)** and click **Y, Columns**. A fully populated window now appears (see Figure 5.5).

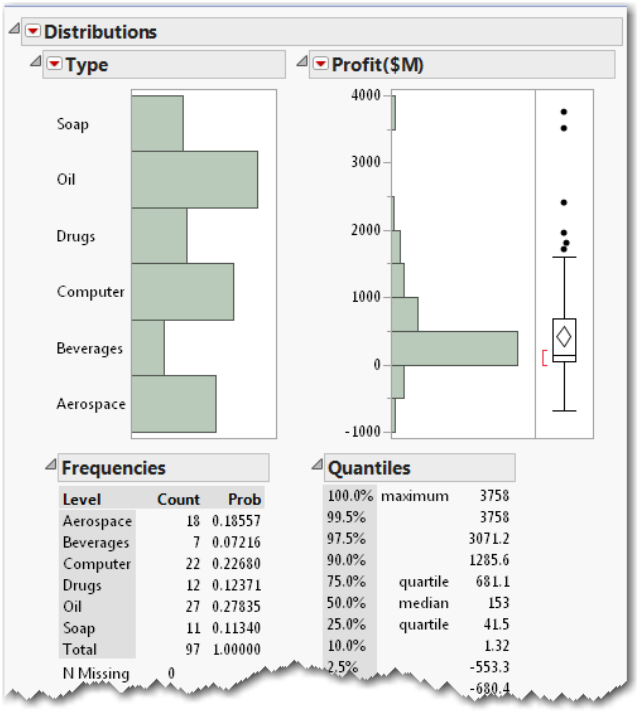
Figure 5.5 Distribution Window



3. Click **OK**.

   You are presented with a result (see Figure 5.6). In this example, you can see several things that might capture your attention. There are six black dots at the high end of the range. There are the highly profitable companies.

Figure 5.6 Distribution Report



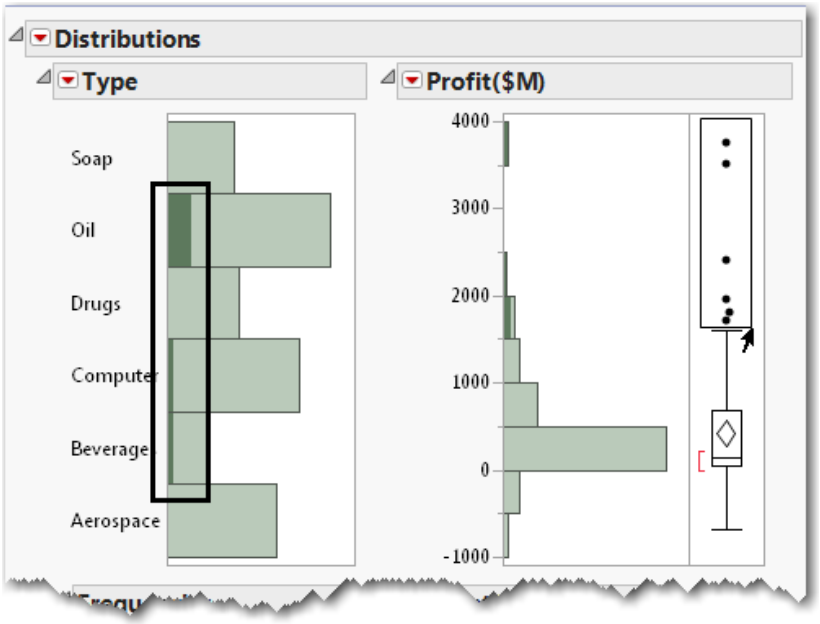
You might also notice that some company types in the portfolio are more heavily represented than others. The size of the bars in the Type distribution graph indicates that Oil has the most companies and Beverages the least.

Note: Did you know that graphs and tables of numerical results appear together in report windows by design? People can learn faster when both graphs and numerical results appear together in the same context. This is a guiding principle of all JMP reports.

Now, let's try something new to help us answer some important questions:

4. Draw a box with your pointer around some of the highest profit entries (see Figure 5.7). Just take the pointer, left-click and drag down over the dots in the profit graph, and a rectangle appears.

Figure 5.7 Draw a Box to Show Highest Profits



5. Notice that as you draw the box, the round dots become dark.
6. Notice also the graph bars on the left for company type, where certain types of company bars have started to turn dark green, including Oil, Computer, and Beverages. This means that companies with the greatest profits are also those mostly associated with the oil sector. Notice the tiny slivers of Computer and Beverages are turning dark green, which indicates a small number of highly profitable companies are in these two business sectors.

7. Now select the **Window** menu from the top and select **Financial**.
   This brings the data table to the foreground. You can also select the data table icon from the lower right corner in the associated report window to bring the data table to the foreground (see Figure 5.8).

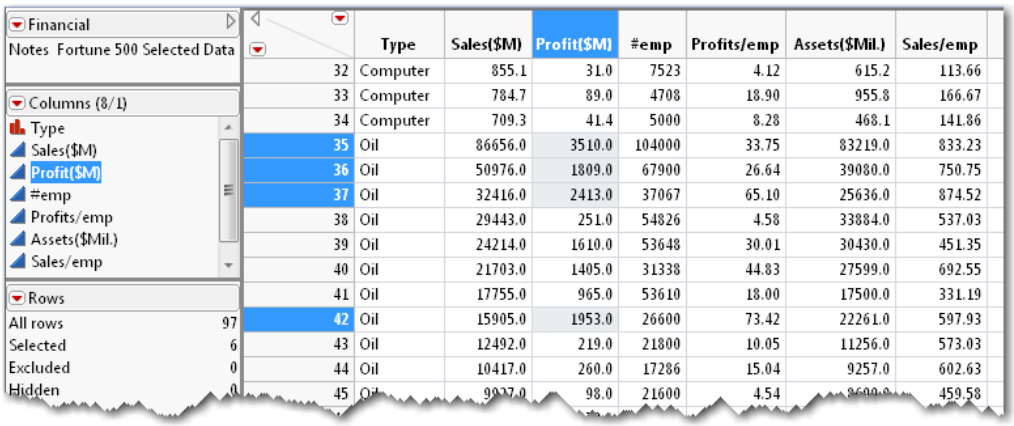
Figure 5.8 Select the Report Icon to Bring Data Table to the Foreground

Note: Color references used in this chapter indicate what you will see on your screen using the standard default settings in JMP.

Key Concept: Dynamic Linking of Graphs and Data. Nearly all graphs that appear in report windows in JMP are tied directly to any other displayed graphs AND to rows in the corresponding data table. By selecting graphical attributes, you see those corresponding values represented in other graphs AND highlighted in the data table.

8. Now scroll down to around row 35 (see Figure 5.9).

Figure 5.9 Rows Selected



	Type	Sales($M)	Profit($M)	#emp	Profits/emp	Assets($Mil.)	Sales/emp
32	Computer	855.1	31.0	7523	4.12	615.2	113.66
33	Computer	784.7	89.0	4708	18.90	955.8	166.67
34	Computer	709.3	41.4	5000	8.28	468.1	141.86
35	Oil	86656.0	3510.0	104000	33.75	83219.0	833.23
36	Oil	50976.0	1809.0	67900	26.64	39080.0	750.75
37	Oil	32416.0	2413.0	37067	65.10	25636.0	874.52
38	Oil	29443.0	251.0	54826	4.58	33884.0	537.03
39	Oil	24214.0	1610.0	53648	30.01	30430.0	451.35
40	Oil	21703.0	1405.0	31338	44.83	27599.0	692.55
41	Oil	17755.0	965.0	53610	18.00	17500.0	331.19
42	Oil	15905.0	1953.0	26600	73.42	22261.0	597.93
43	Oil	12492.0	219.0	21800	10.05	11256.0	573.03
44	Oil	10417.0	260.0	17286	15.04	9257.0	602.63
45	Oil	9027.0	98.0	21600	4.54	9600.0	459.58

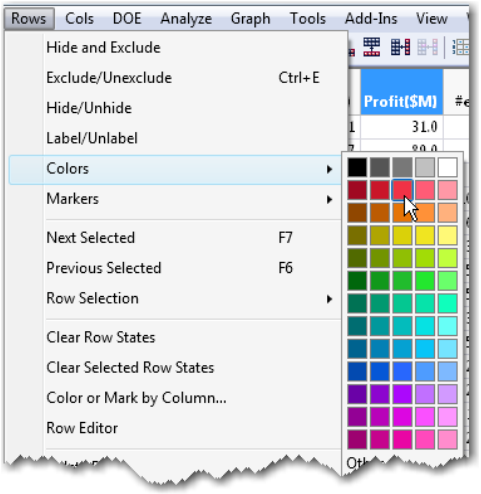
Do you see how the highlighted rows match those same points you highlighted by drawing the box in the Distribution window?

Note: The highlighting is what we refer to as JMP's dynamic linking, which allows you to link graphs to data, data to graphs, and graphs to other graphs.

Because these highlighted rows show high profits, let's mark them so we can always find them.

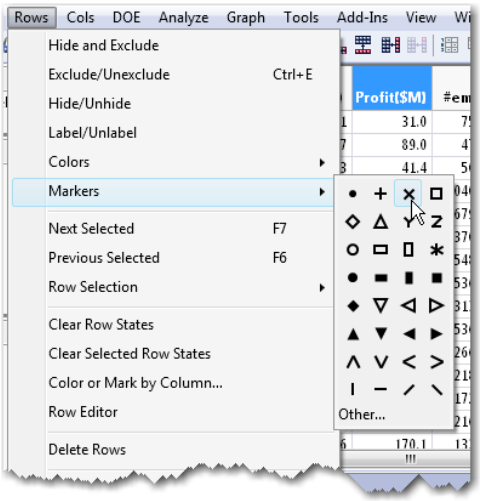
9. Select **Rows ► Colors**, and then select the color red from the color palette (see Figure 5.10).

Figure 5.10 Select Red for Row Color



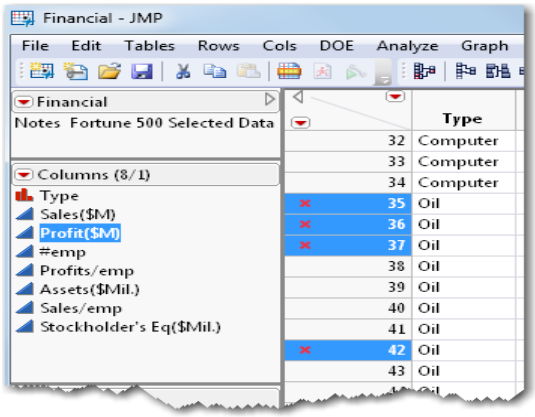
10. Select **Rows ► Markers**, and select the marker type **X** (see Figure 5.11).

Figure 5.11 Select Marker Type X for Rows



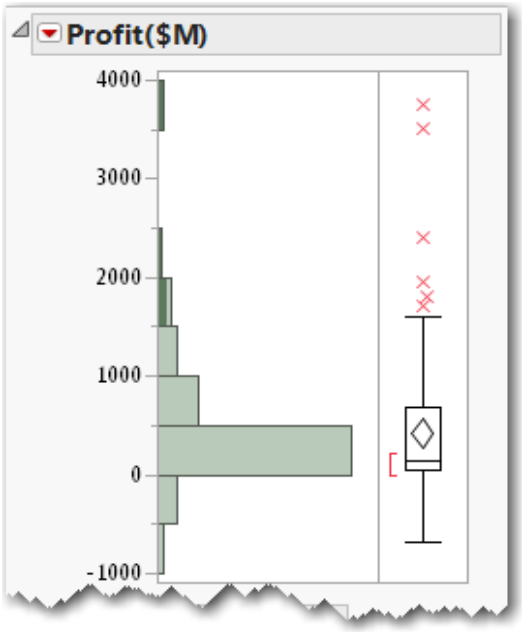
These rows in the data table are now marked red and appear with a marker type of **X**, as shown in the row number column of the data table (see Figure 5.12).

Figure 5.12 Rows Colored Red and Marked with X



11. Now select **Window** and bring the **Financial – Distributions of Profit($M), Type** result to the foreground. Focus on the Distributions result for **Profit($M)** (see Figure 5.13).

Figure 5.13 Top Profit Marked with Red Xs

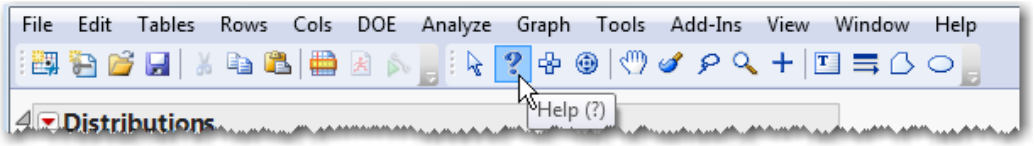


The distribution graph for profit also shows the red X markers as well as a box plot. The top X represents a company with over $3.7 billion in profits (see Figure 5.13). On the bottom of the range is a very unprofitable company at approximately -$1 billion (Figure 5.13).

If a box plot or any graphical or numerical result is unfamiliar to you, use the question mark tool (?) from the **Tools** menu to identify the item in question and locate the corresponding documentation about the item.

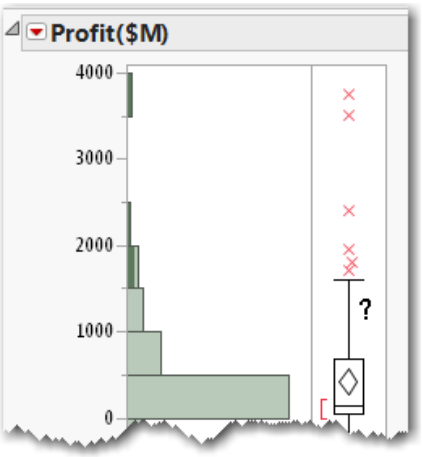
12. Select the question mark (**?**) from the toolbar. The pointer changes to a question mark (see Figure 5.14).

Figure 5.14 Question Mark Tool from Tool Bar



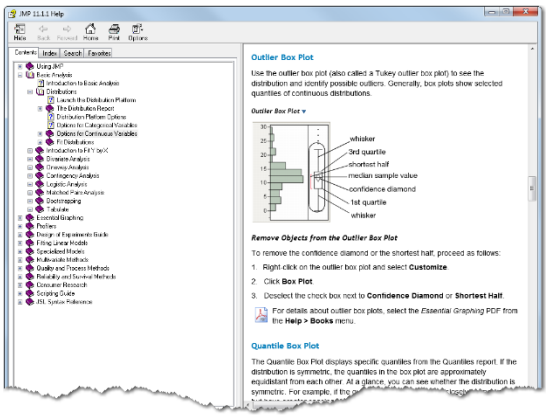
13. Then, move the question mark on top of the item you're unfamiliar with and click on that item (see Figure 5.15). In this example, it is the item next to the distribution graph.

Figure 5.15 Question Mark on Distribution



The section of the documentation associated with the outlier box plot appears automatically (see Figure 5.16). Don't forget to scroll down, because sometimes the topic of interest is a little below where you landed in the documentation.

Figure 5.16 Help for Box Plots



Key Concept: Unfamiliar graphics, geometric shapes, and results are explained using the context-sensitive question mark tool. Just select the question mark tool (**?**) from the **Tools** menu and click on the part of the result you want to learn about. You can learn statistics while you explore your data!

Summary of 5.2: Answers the Distribution Platform Provides

By observing the distribution, you found that the range of profits for the companies was mostly on the profitable side, topping out with a highly profitable company at over $3.7 billion. On the bottom of the range was a very unprofitable company at approximately -$1 billion (Figure 5.13).

You may have noticed that the average, or mean, profits for all companies was $426 million.

The distribution results indicated that the variability, or standard deviation, of profits was about $717 million. Most of the profitable companies fall within this range around the mean.

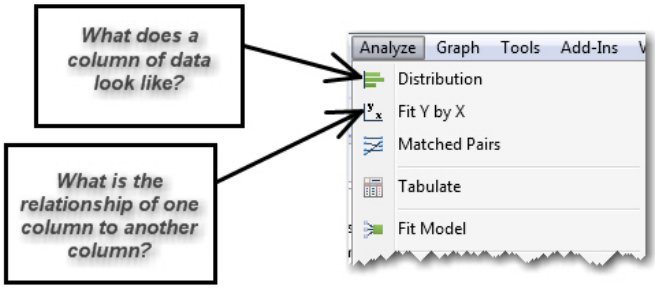
You found that most of the extreme values were on the high side. By identifying them, you determined that most of these companies came from the oil sector and a couple of other industry types.

5.3 Comparing One Column to Another

Building upon the Distribution platform, the answers to questions you asked about one column have motivated you to ask further questions about the relationship between two columns. For example, what might be the relationship of profits to other columns in the data table? In fact, you have already identified an interesting relationship between the company type and its profits using dynamic linking as some types of companies are associated with higher profits. This visual relationship implies something is happening with profits that includes more than just one column. This implies that there may be a relationship between at least two columns that we'll explore in this section.

The second item under the **Analyze** menu is **Fit Y by X**. It is designed to explore relationships between one column and another column. These are sometimes called bivariate relationships (meaning two variables). You might have noticed a pattern developing for items under the Analyze menu. The first item is **Distribution** and is useful for looking at one column at a time. The second item is **Fit Y by X** for looking at the relationship between two columns. Figure 5.17 provides a description of each menu item.

Figure 5.17 Analyze Menu Item Descriptions

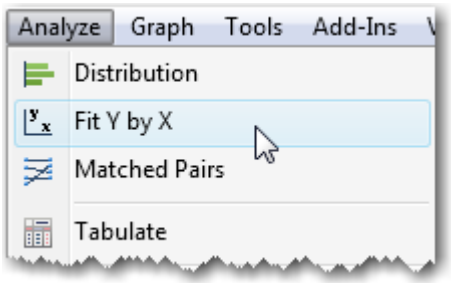


Let's continue with our example and perform a Fit Y by X analysis. We already suspect that there are interesting profit differences among the different company types, and we've discovered a few by just looking at one column at a time using the Distribution platform and dynamic linking. We'll now formalize that inference.

The relationship we want to explore includes the **Profit($M)** column and the **Type** column:

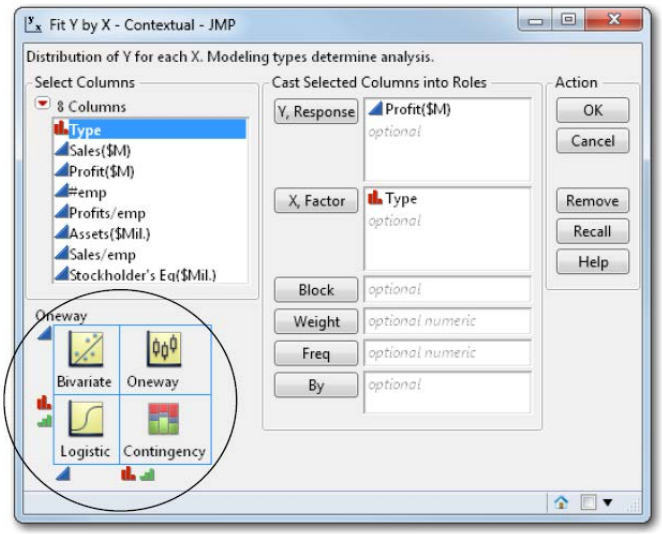
1. Select **Analyze ► Fit Y by X** (see Figure 5.18).

Figure 5.18 Select Fit Y by X



2. In the window (see Figure 5.19), select **Profit($M)** and click **Y, Response.**

Figure 5.19 Fit Y by X Profit by Type Oneway Analysis



3. Select **Type** and click **X, Factor.**
4. Click **OK**.

Before we continue with the example, let's examine that last Fit Y by X window. We'll focus on the preview or circled area (see Figure 5.19).

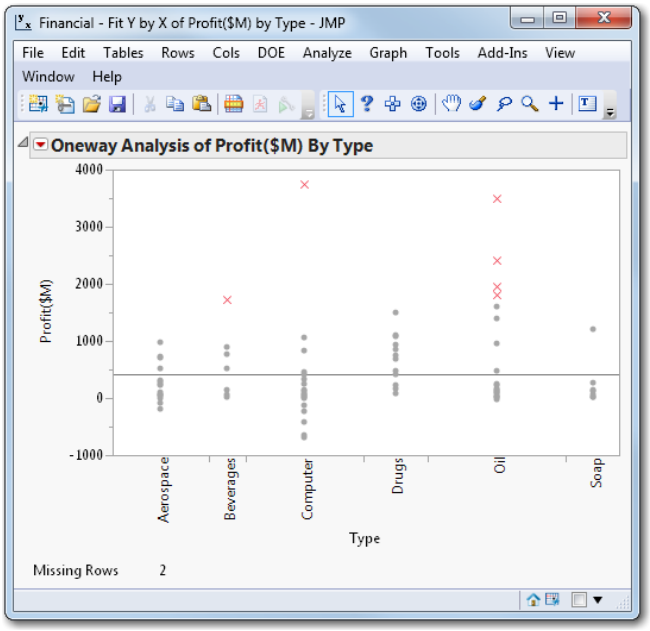
Notice that each modeling type has its corresponding icon (continuous, nominal, ordinal which are described in section 2.3). The modeling type of the column that you cast into a role determines the type of analysis that is produced (see Figure 5.20). In this case, we have selected **Profit($M)** for the **Y, Response** (the vertical axis), which is continuous, and **Type** for the **X, Factor** (the horizontal axis), which is nominal. The arrows in Figure 5.20 show these selections and where they intersect on the matrix indicates the type of analysis that will be produced; in this case, a Oneway.

Figure 5.20 Oneway Analysis Illustration

For now, don't worry about terms like *Oneway* in the preview. If you want to learn about the result, you can place the question mark tool on the result after generating it. Just note that the picture previews are there so you can get an idea of what kinds of analyses will be produced when you cast certain types of columns into Y and X roles within the platform.

When you click **OK**, a *Oneway* analysis appears (see Figure 5.21).

Figure 5.21 Oneway Analysis Graph



The Fit Y by X (two-column) analysis confirms what we started to observe earlier. We see that the highest profits have come from a mix of mostly oil companies, one computer company, and one beverage company. These are conveniently marked by X's from a previous step.

Note: Your selection of the most profitable companies might differ slightly from the selection shown here. This is ok.

Based on the observed best performers among the company types, we might start asking more complex questions like these:

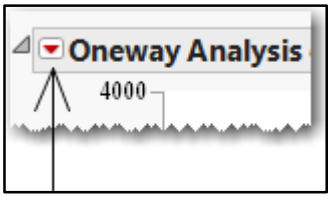
Questions Answered with Two Column Analysis

- Are there differences in profits among the company types?
- How big and in what direction (negative or positive) are these differences?
- Should I choose more companies from one particular company type than from another type?

These questions and others can be answered using Fit Y by X because they involve two columns. How? As you make selections from the choices in the **Analyze** menu, the types of questions you start asking at each step are anticipated for you.

You might notice the red triangle associated with the results you have generated (see Figure 5.22).

Figure 5.22 Red Triangle



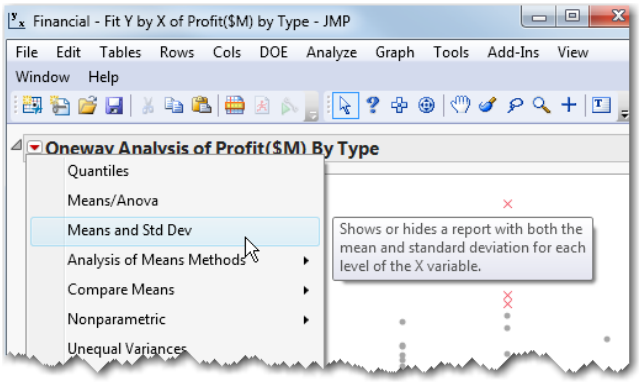
As we introduced earlier, *red triangles* anticipate questions you might have at any stage of analysis and have been carefully placed on the report in the context of your analysis.

Let's use the choices on the menu for the oneway analysis to find out if there are differences in profits among company types.

You can already see some of these differences in the graph. Let's further quantify the differences in profits by company type.

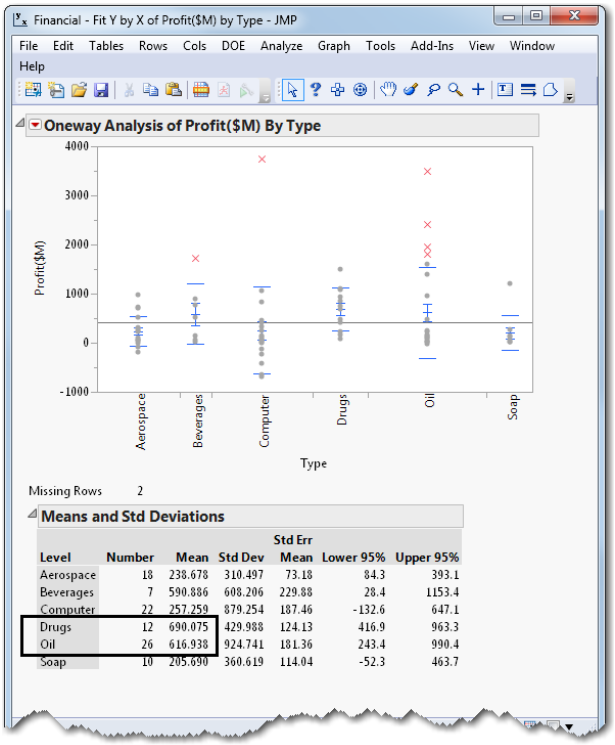
From the red triangle, select **Means and Std Dev** (see Figure 5.23).

Figure 5.23 Select Means and Std Dev



An additional table appears below your graph and blue mean error bars and standard deviation lines appear on your graph. The blue lines on the graph correspond to Mean Lines, Mean Error Bars, and Std Dev Lines. These values also appear in the table as numbers (see Figure 5.24).

Figure 5.24 Means for Drugs and Oil



Using the table, we can now see that the highest average profits (mean) were obtained from the 12 companies represented in the drugs category. The second highest average profits were obtained from the oil category, and there were 26 companies represented.

Let's review what we have learned so far. In Distribution, we learned that companies with the highest profits are likely to be in the oil category. From Fit Y by X, we learned that the highest average profits are found in the drug category and the next highest average profits are found in the oil category. Remember, you are trying to accumulate just enough knowledge to make your decisions. If you are keeping score, Oil is showing up as a good performer by at least two measures. The most extreme profits are coming from some of these oil companies and overall averages are also high for Oil.

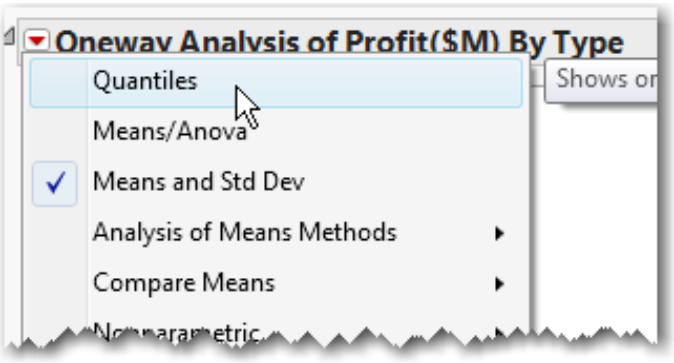
Did you also notice the number of oil companies represented? This is another vote for Oil to be represented in our selection of company stocks because many highly profitable stocks are from this category. The risk associated with one oil company going down is hedged by having many profitable ones in the oil category. **Note:** This is true when the companies in the category are not highly correlated on the fundamentals. Seek professional advice when making investment decisions. If you are still keeping score, Oil looks better by three measures now.

We don't really have numerical measurements of range yet to help answer the question: "How big and in what direction (negative and positive) do these differences go?"

Where might we go to answer this question? Yes, it's in the red triangle for the oneway result.

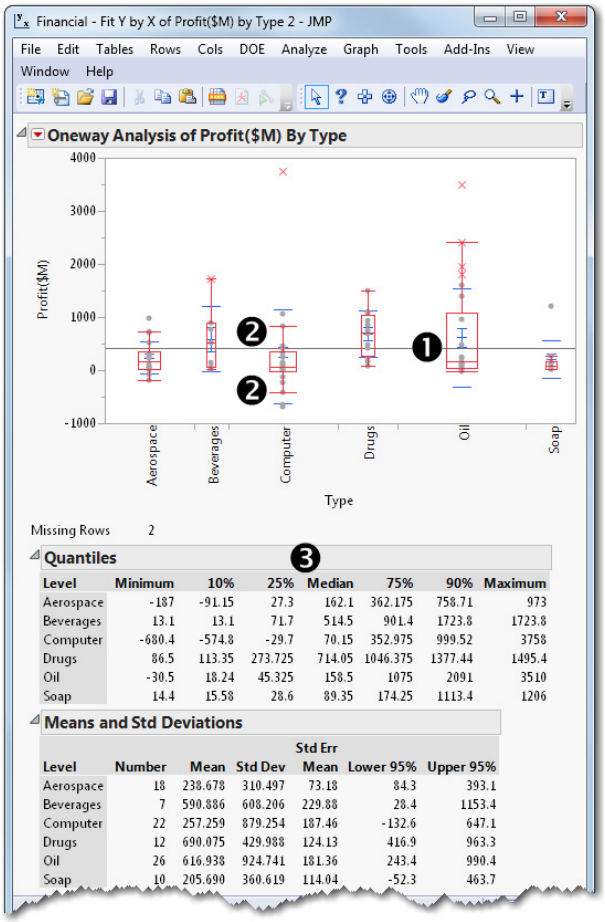
1. Select the red triangle next to **Oneway**.
2. Select **Quantiles** (see Figure 5.25). **Quantiles** are values that divide an order set of continuous data (from smallest to largest) into equal proportions. See "Quantiles" in Appendix B for more information.

Figure 5.25 Select Quantiles



A Quantiles table appears. This table appears in the context of your initial analysis, and box plots are added to your graph illustrating the quantiles (see Figure 5.26).

Figure 5.26 Oneway Means and Quantiles



❶ The box encompasses the 25th through 75th percentiles.

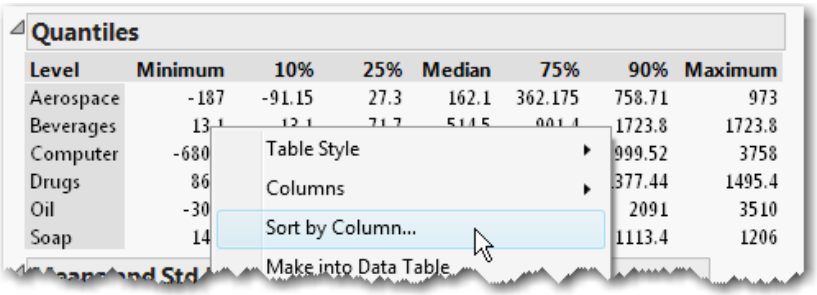
❷ The whisker lines stretch out to the 10th and 90th percentiles on the corresponding sides of the box.

❸ What type of company has the worst profit by examining the **Minimum** column? What type of company has the best profit in the **Maximum** column? Which has the lowest value for the **Maximum** column?

You can make the differences easier to see. Try this:

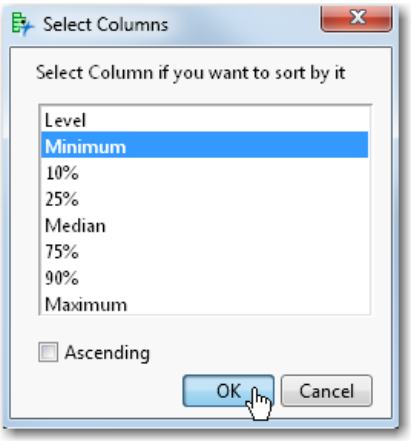
1. Right-click on the **Quantiles** table.
2. From the submenu, select **Sort by Column** (see Figure 5.27).

Figure 5.27 Sort by Column



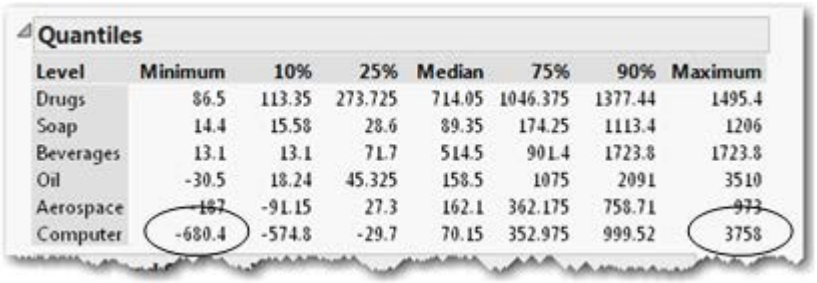
3. Select **Minimum** and click **OK** (see Figure 5.28).

Figure 5.28 Sort Column by Minimum



4. The Profit table now appears sorted by the **Minimum** column (see Figure 5.29).

Figure 5.29 Quantiles Sorted by Minimum



Quantiles

Level	Minimum	10%	25%	Median	75%	90%	Maximum
Drugs	86.5	113.35	273.725	714.05	1046.375	1377.44	1495.4
Soap	14.4	15.58	28.6	89.35	174.25	1113.4	1206
Beverages	13.1	13.1	71.7	514.5	901.4	1723.8	1723.8
Oil	-30.5	18.24	45.325	158.5	1075	2091	3510
Aerospace	-187	-91.15	27.3	162.1	362.175	758.71	973
Computer	-680.4	-574.8	-29.7	70.15	352.975	999.52	3758

You can now see that the worst loss (or least profitable) is in the Computer category at -680.4. Can you find the biggest profits in the **Maximum** column? Yes, it's also in the Computer category at 3758. They are circled for you (see Figure 5.29).

Now you can start asking yourself about your risk tolerance. Review the oneway graph you produced, the mean and standard dev, and the Quantiles table. If you are conservative, what types of companies would you pick to assure profits? Wouldn't you choose only companies that show no negative profits?

Which ones are those? They are Drugs, Soap, and Beverages.

Higher profits appear in other categories, but more of those companies show higher variability and losses and, therefore, present more risk.

5. Finally, save the data table **Finance.jmp**. You will need this data table for the next chapter.

Note: The "Soap" company type above includes a group of diversified consumer products companies.

Summary of 5.3: What You Learned by Comparing One Column to Another

You may have noticed that the highest mean (average) profits are found overall in the 12 companies in the drug category and the next highest mean profits are found in the 26 companies in the oil category.

You found that the biggest losers were in the computer category as well as the most profitable ones.

You found that picking companies depends on risk tolerance. If you are conservative, you might pick Drugs, Soap, and Beverages. Higher profits might be achieved elsewhere, but they are in company types where there is higher variability or risk.

5.4 Summary

This chapter has presented an approach to problem-solving that is unique to JMP. The approach underscores the progressive nature of problem solving that tends to build from simple descriptions of one column of data to relationships between two columns of data, leading to a better understanding of the data.

Learning about data using JMP tends to start slowly, increase rapidly, and then reach understanding. Also, the process does not go simply in one direction as you move from one column to two columns. As discoveries are made between two columns, confirmation and further analysis can be made with simple one-column tools like Distribution from which our journey began.

Marking rows with colors and markers helps certain groups stand out in subsequent analyses. Visual identification with markers and colors speeds discovery and the effective communication of results.

Because many real-world problems involve multiple variables or columns, you will learn in the next chapter that JMP easily handles this increased complexity. We will build upon our example and the basic analyses introduced in this chapter to explore multivariable relationships. In the next chapter, we introduce several tools including the Partition platform, the Data Filter, and the Prediction Profiler to help you explore and discover deeper insights in your data.

Chapter 6 Problem-Solving with Multiple Columns

6.1 Introduction

6.2 Comparing Multiple Columns

6.3 Filtering Data for Insight

6.3 Model Fitting, Visualization, and What-If Analysis

6.5 Summary

The previous chapter focused on problem-solving when you have one or two columns of interest. This chapter builds on the previous chapter's framework for problem solving and introduces a method to understand multiple column and group relationships.

We introduce advanced analytics including Partition, Fit Model, and Prediction Profiler to help us understand the multi-column/variable relationships in the data. Most real-world problems involve more than one- or two-column relationships so this approach is warranted.

We also introduce the Data Filter for real-time slicing of the data. The Data Filter enables easy inference testing and hypothesis development by excluding, hiding, and marking selected observations across any number of column ranges or groups.

The Fit Model platform includes several modeling personalities, including Multiple Regression. Unlike the two- column analysis (one Y and one X) we introduced in the last chapter, Multiple Regression allows you to employ additional X columns. We briefly introduce this topic at the end of the chapter, along with a very powerful model visualization tool, the Prediction Profiler.

6.1 Introduction

As we explained in the previous chapter, the transformation of data into information is a process that involves a few basic JMP platforms including Distribution and Fit Y by X, which are found in the **Analyze** menu. Table 6.1 provides a review of the tool name, how many columns it supports, and its common statistical identity. The process of discovery starts with univariate/one-column analysis using the Distribution platform to explore the single-column/variable properties and proceeds to bivariate/two-column relationships using the **Fit Y by X** platform. As in life itself,

these analyses can often lead to more complex questions involving more than two columns, which is the subject of this chapter.

In Table 6.1, we add Fit Model and Modeling to the items on the Analyze menu column. Partition and Multiple Regression are associated with these menus and are added to the table framework. These methods support multivariable or multi-column relationships.

Table 6.1 Tool Name, Columns Supported, and Statistical Terms

Analyze Menu	How Many Columns	Statistical Terminology
Distribution	Single column	Univariate methods, histograms, box plots, quantiles, and descriptive statistics, and more
Fit Y by X	Two columns	Bivariate, contingency, logistic, oneway ANOVA, nonparametrics, linear regression, and more
Fit Model	Multi-Column	Partition, Data Mining, Multiple Regression, Logistic regression, Stepwise, and more
Modeling	Multi-Column	Partition, Data Mining, Time Series, and more

Remember, JMP's goal in using this menu framework is to introduce you to methods in a logical order, allowing you to learn about your data progressively without forcing you to first understand statistical jargon or the conditions required to implement them. The process of digging deeper into your data is built into the menu structure.

6.2 Comparing Multiple Columns

What remains to be answered is whether there are important multiple-column relationships (looking at more than two columns) that might be present in the data. Complex products or systems often have these relationships. While these multiple-column relationships are not always present in the questions you are trying to answer, knowing that you have access to methods to identify them should help. In fact, sometimes to see the forest for the trees, you need these methods so that you can focus on only those columns that drive your learning from the data.

We do not need to change the profit column to a nominal type to use Partition. Output columns like profit can be either continuous, nominal, or ordinal in the Partition platform as well as any input columns. We made this change only to simplify the interpretation of our analysis for illustration purposes.

Example 6.2: Financial

We will be using the **Financial.jmp** data file to illustrate the steps in this chapter. The data is from companies in the Fortune 500, selected from the April 23, 1990 *Fortune* magazine issue. This data includes columns for:

Type type of company

Sales($M) yearly sales in millions of dollars

Profit($M) yearly profits in millions of dollars

#emp number of employees at time of measurement

Profits/emp profits per employee in thousands of dollars

Assets($Mil.) assets in millions of dollars

Sales/emp sales per employee in thousands of dollars

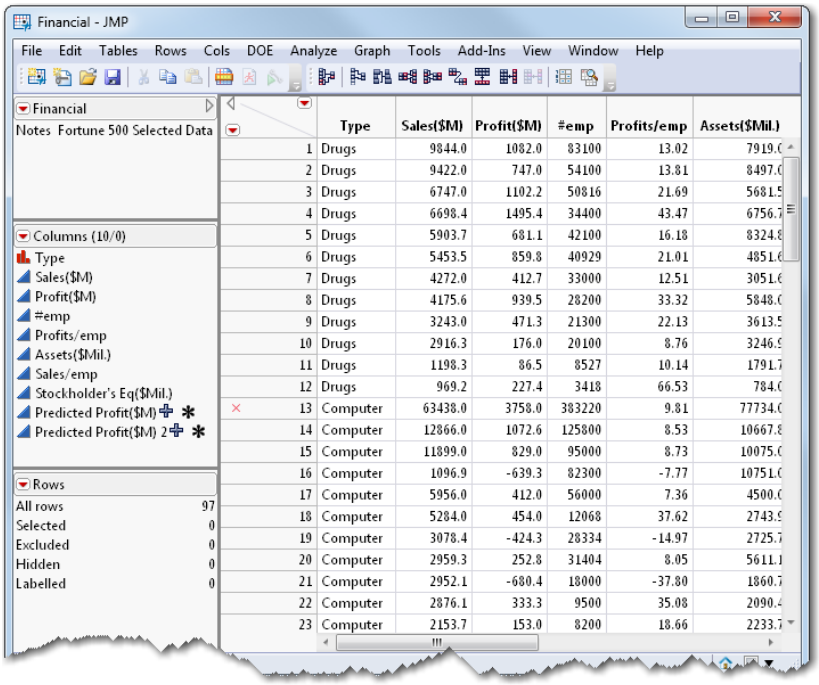
Stockholder's Eq($Mil.) stockholder's equity in millions of dollars

You can access this data at **Help ▶ Sample Data ▶ Business & Demographic ▶ Financial**.

Note: Statistical modeling in practice is both art and science and simply adding more columns to your model is not always better. While real world models often include more than two columns, adding too many columns will undermine the usefulness of the model on new data. There are tools built into JMP to help you decide the right number of columns to use in a model. JMP Pro includes many state of the art platforms and tools to validate your models.

You should have the Financial.jmp data table open from the previous chapter. If you don't, you can open it at: **Help ▶ Sample Data ▶ Business & Demographic ▶ Financial** (see Figure 6.1).

Figure 6.1 Financial Data Table



Preparing Data Using Recode

Before we perform an analysis, let's create a new column that simply identifies each company as being profitable or not (we will, in effect, be making a new profit column that indicates whether it meets a certain profit threshold and will be treated as nominal). Our intention is to identify profitable companies based on all the other columns. We want to pick companies whose profits will be sufficient to provide us with solid returns, so we want to identify companies whose profits are above $10 million and store them in a new column. The $10 million mark is an arbitrary break point (it could be any value), but it is based on the idea that companies with higher profits in general will probably be larger and less volatile. For purposes of illustration, we will use $10 million as a reasonable and conservative strategy at this point.

Note: We do not need to change the profit column to a nominal type to use Partition. Output columns like profit can be either continuous, nominal, or ordinal in the Partition platform as well as any input columns. We made this change only to simplify the interpretation of our analysis for illustration purposes.

Let's make our new Profit column by deriving what we need from the existing **Profits($M) column**:

1. Click the **Rows** menu.
2. Select Row Selection.
3. Click **Select Where…** (see Figure 6.2). This generates the Select rows window (see Figure 6.3).

Figure 6.2 Select Where to Generate the Select Rows Window

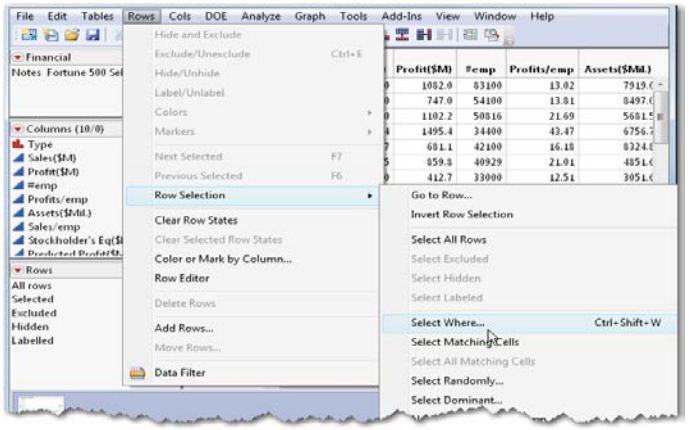
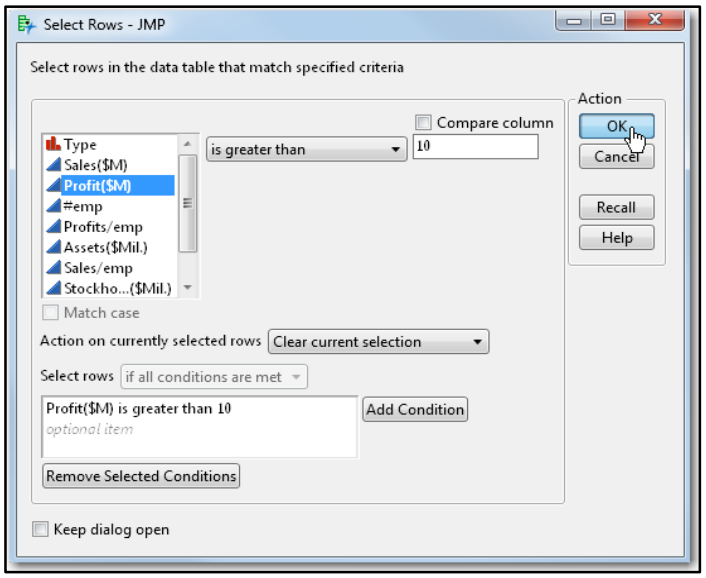


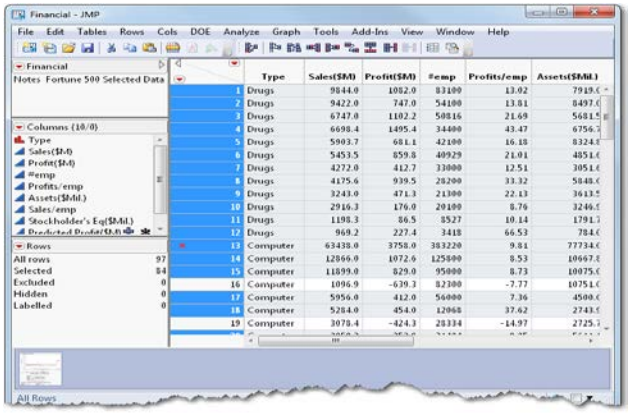
Figure 6.3 Select Where Profit is Greater Than $10M



4. Select **Profits($M)**.
5. Select is **greater than**.
6. Enter **10**. (The data is expressed in millions.)
7. Click **Add Condition**.
8. Click **OK**.

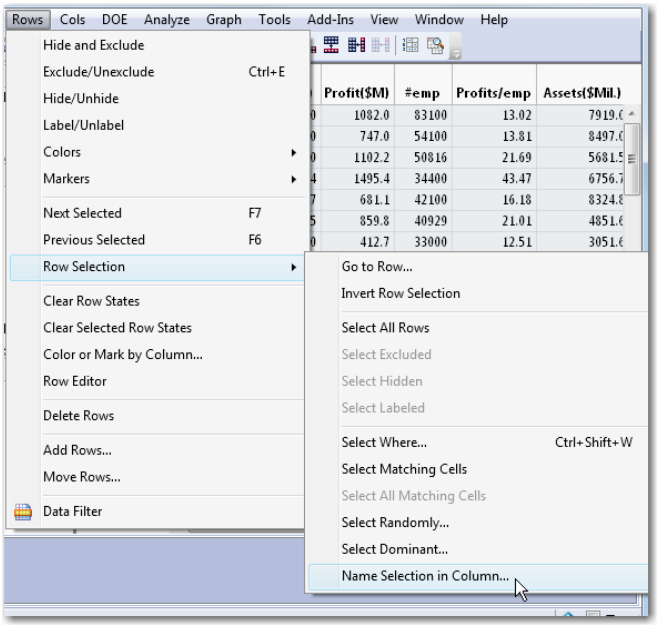
   Rows that meet the condition of Profits being greater than $10 million are now highlighted in the data (see Figure 6.4).

Figure 6.4 Selected Rows are Highlighted



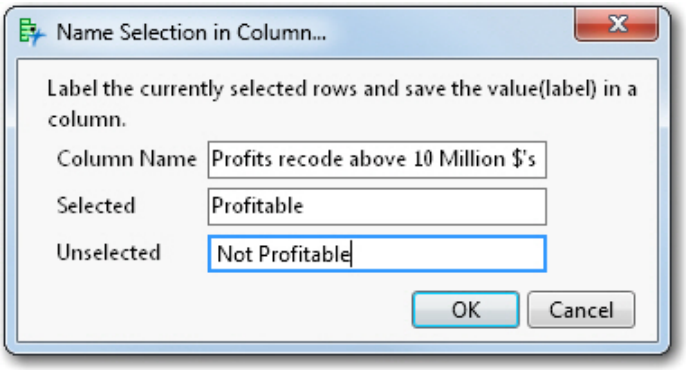
9. Now select **Rows ▶ Row Selection ▶ Name Selection in Column…** (see Figure 6.5).

Figure 6.5 Select Name Selection in Column



10. Type the new column name and labels for the **Selected** and the **Unselected** rows exactly as shown (see Figure 6.6) and click **OK**.

Figure 6.6 Name Selection in Column

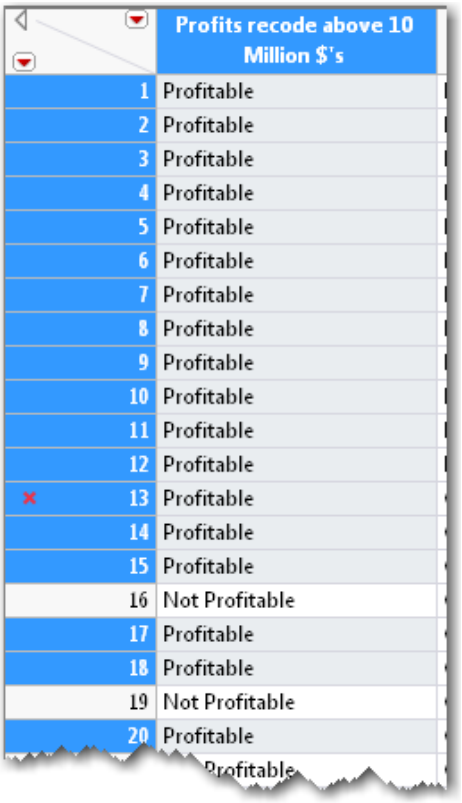


Note that we will hereafter refer to this new column as "Profits recode".

A new column is created that identifies profitable and not profitable companies for each row in the table, with a dividing point of $10 million and higher for those we deem to be

sufficiently profitable (see Figure 6.7). This is called recoding the data. Not having to write any code to do this is very handy. To learn about another method to accomplish data recoding using the Recode command on the Cols menu, see section 2.4.

Figure 6.7 New Recoded Profits Column

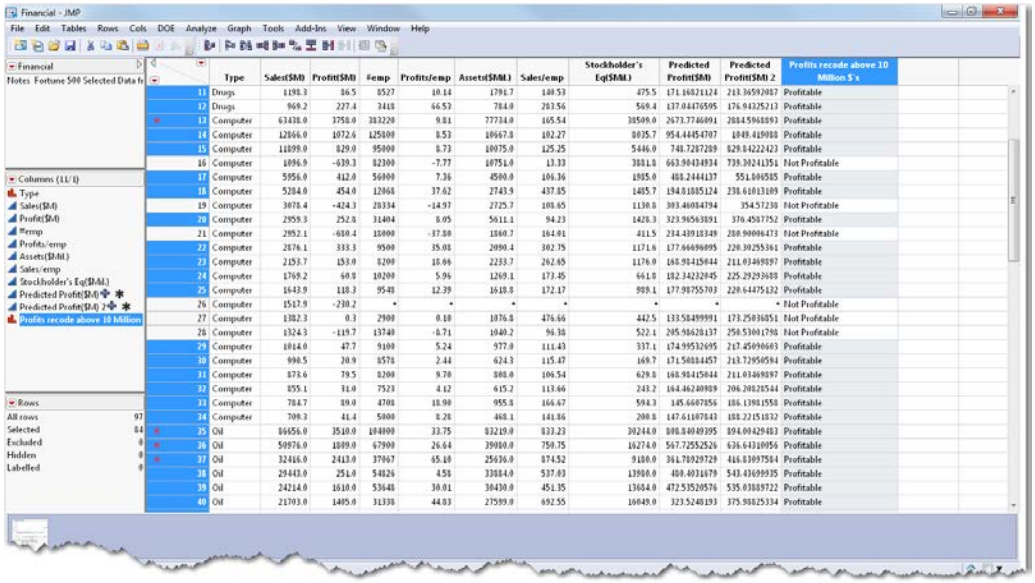


	Profits recode above 10 Million $'s
1	Profitable
2	Profitable
3	Profitable
4	Profitable
5	Profitable
6	Profitable
7	Profitable
8	Profitable
9	Profitable
10	Profitable
11	Profitable
12	Profitable
13	Profitable
14	Profitable
15	Profitable
16	Not Profitable
17	Profitable
18	Profitable
19	Not Profitable
20	Profitable
	Profitable

Let's move on with the analysis.

Consider the data table in Figure 6.8. More than one column might be related to the **Profits recode** column: for example, **Sales($M)**, **#emp**, and **Assets($M)** may explain or influence a company's profitability. With more than two columns to investigate we need to move further down the Analyze menu. There are many methods to investigate this kind of relationship. Let's look at one of these with a tool called Partition.

Figure 6.8 Recoded Profits



Note: You might find these multi-way relationships by simply (but laboriously) producing distributions or examining output from Fit Y by X. We use Partition because it is a powerful and quick method for finding these relationships when you have more than two potential relationships in your data. Partition results are usually easier to interpret as well.

Mining Data Using Partition

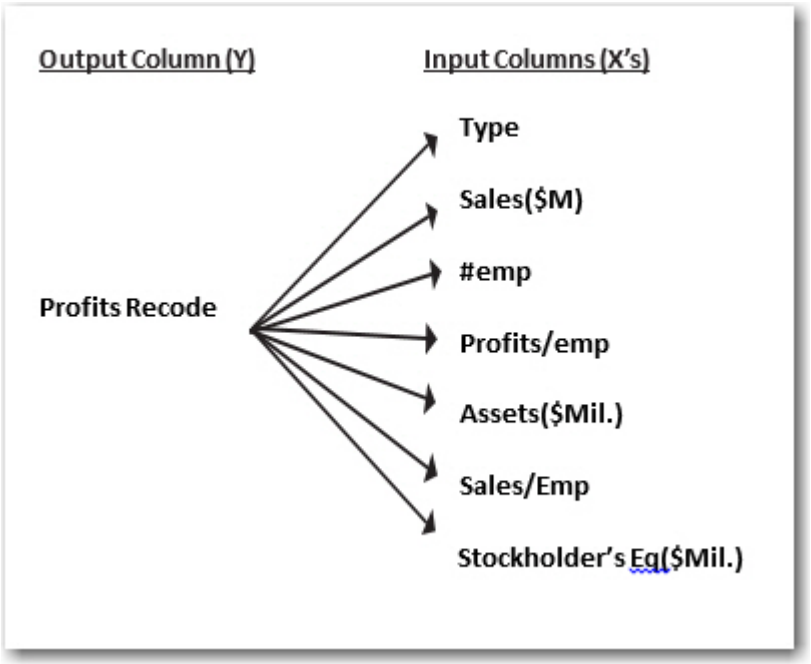
The Partition platform allows you to determine which columns in a data table most influence or predict the outcome of another column (profits, in our example). It achieves this by searching all of your selected X columns and finding a set of splits or subgroups within them whose values best predict your Y value. These splits (or partitions) of the data are done recursively (starting with the best predictor), forming a tree of decision rules until the desired fit is reached.

A decision tree is essentially a ranked set of predictors (such as **Type** or **Assets($Mil.),** in our example) placed in a hierarchical tree structure according to the strength of their relationship to the column of interest (**Profits** or Y, in our example). Items higher in the tree have more influence than items lower in the tree.

In our example, **Profit recode** is our column of interest (Y Column) and we want to determine which of the remaining columns (X Column) in our data table most influence or predict Profit. See Figure 6.9.

The Partition method introduced here is one of a set of methods that is sometimes termed data mining because you are mining relationships in your data to find which columns most relate to or predict profits.

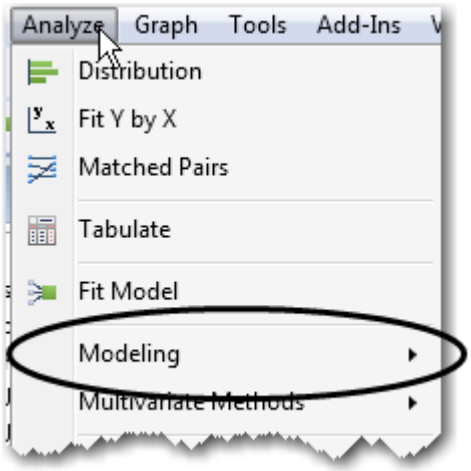
Figure 6.9 Determine Which Columns Affect Profit Recode (Outputs versus Inputs)



Note: The Partition technique is introduced here to enable quick exploration of your data. As an advanced analytical technique it is recommended that you familiarize yourself with the underlying concepts by using the question mark tool to call up the relevant chapters in the documentation. Because these tools often require greater sophistication in their interpretation, we recommend that you seek out experienced data analysts for assistance when necessary.

To use Partition we will move further down the **Analyze** menu to the **Modeling** menu (see Figure 6.10).

Figure 6.10 Analyze Modeling Menu



Because we have already explored profits compared with the type of company, we might now ask other questions.

These are just a few of the questions that can be answered using the Modeling platforms.

Framing Our Analysis

Questions That Involve Multiple Columns

What are the best predictors of profit?

Are there relationships between profits and other columns?

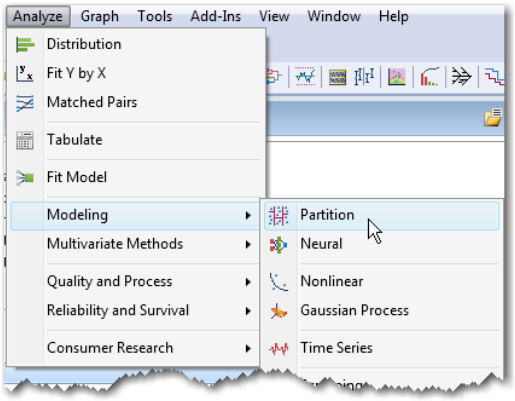
What or where are the biggest differences found between company profits?

Where are there no differences?

Is there some way to use multiple columns to pick the most profitable companies?

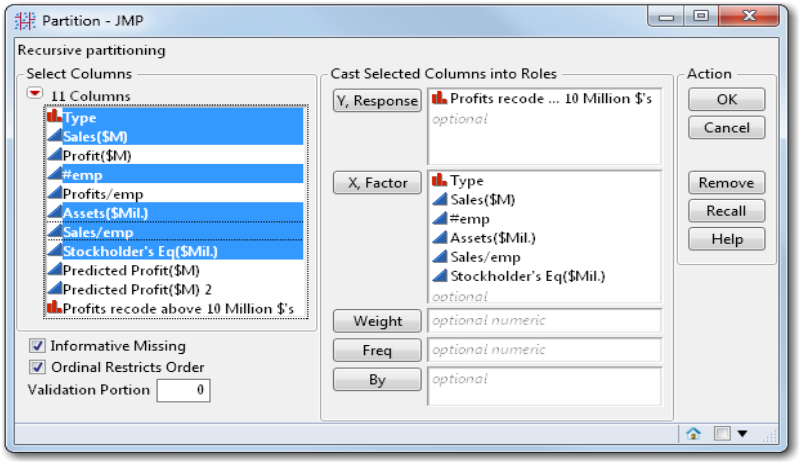
1. Select **Analyze ▶ Modeling ▶ Partition** (see Figure 6.11).

Figure 6.11 Analyze Modeling Partition



2. Select **Profits recode** for **Y, Response** (see Figure 6.12).

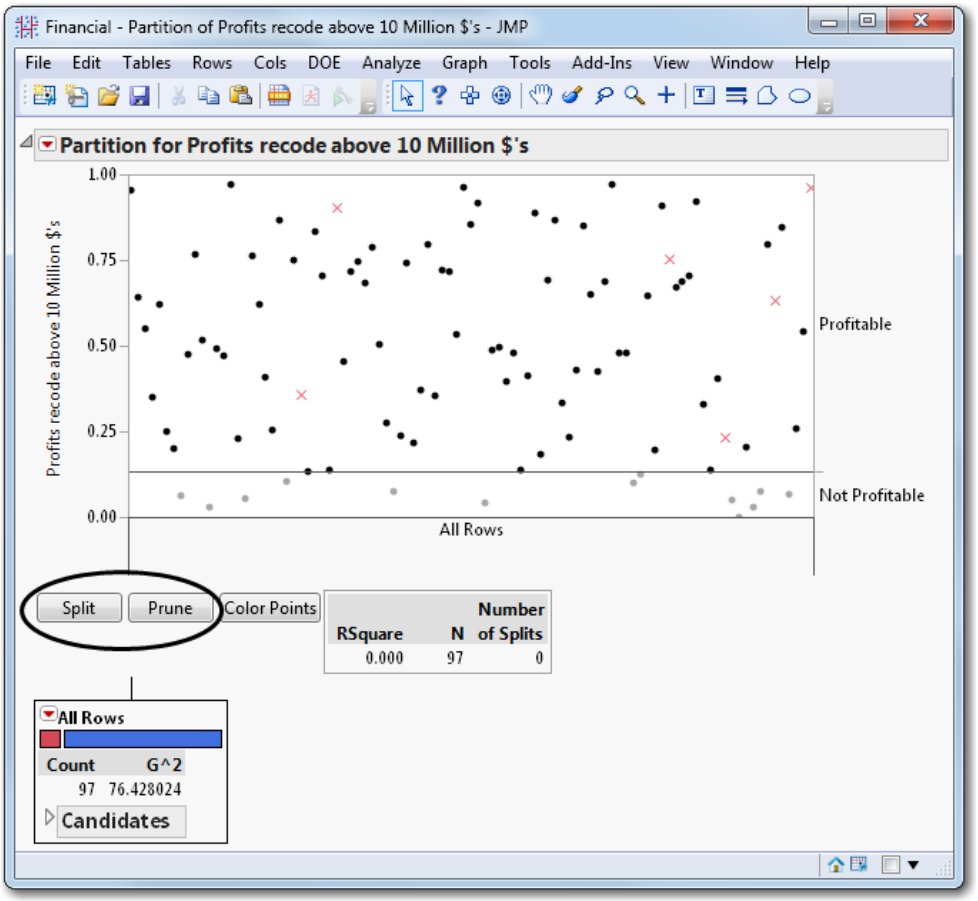
Figure 6.12 Partition Window Selections



3. And for **X, Factor**, select **Type, Sales, #emp, Assets($Mil.), Sales/emp**, and **Stockholder's Eq** (see Figure 6.12).
4. Click **OK**.

You now see profits for all the companies represented by a random pattern of dots from left to right in the partition graph (see Figure 6.13).

Figure 6.13 Profits Before Split



The horizontal black line that is perpendicular to the Y axis is the dividing point we set for companies that have at least $10 million in profit.

The companies that are above the line are those that we have deemed profitable ones and those below are not profitable enough and thus deemed the unprofitable ones. So, looking at the line on the scatterplot, about 13% of the companies are not profitable and about 87% are profitable. Your graph may look slightly different in that the dots representing the data points are randomly arranged or in statistical terms, jittered.

You can see those very high values (outliers) to which we assigned a red X marker from the previous chapter. We saved these in the table as row properties previously. If you don't see the red X's, it's ok. We can go forward without them.

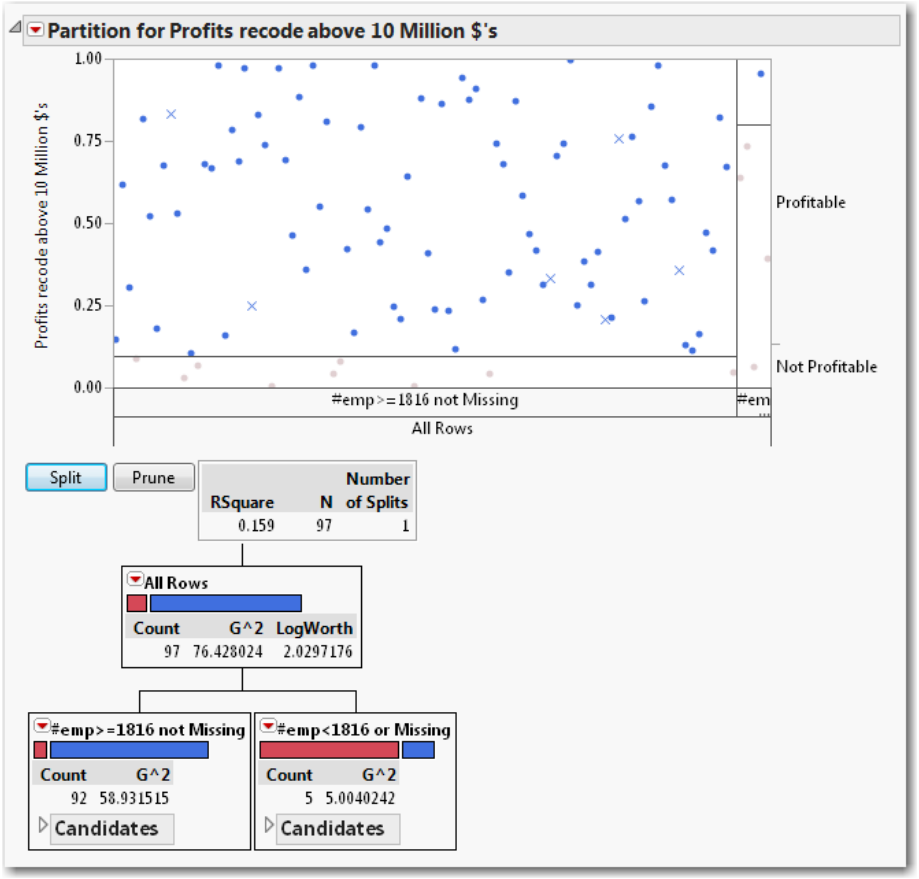
Now let's try something.

5. Click **Color Points**. It's the button next to **Split** and **Prune** (see Figure 6.13).

   Remember that the point of this analysis is to understand the multiple-column relationships between profits and the other columns in the data table.

6. Select **Split** and new output appears (see Figure 6.14).

Figure 6.14 First Split of Profits



What happened? Partition went through all the columns you selected as X's and found that the best predictor of **Profits Recode** was number of employees (**#emp**). Remember, you also included a

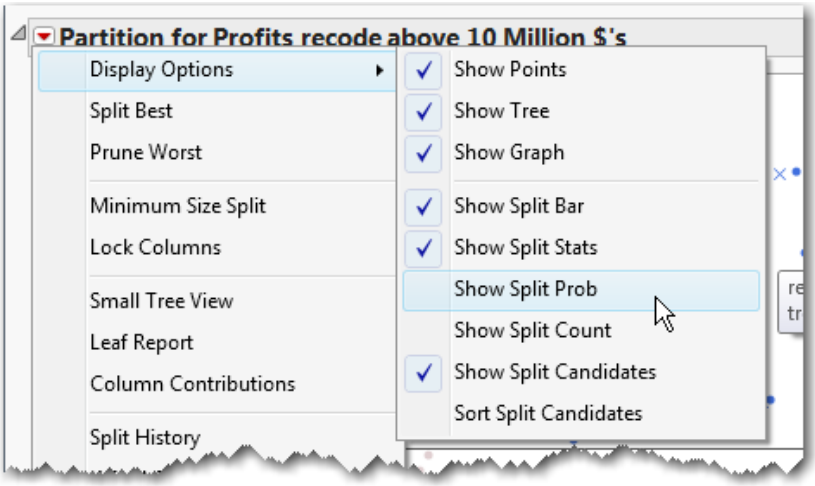
half-dozen other columns in your model, but the first to be selected was the #emp column. Also note that your high performers from the previous chapter's work show up as X's (that is, if you saved them in the previous chapter). These should show up as profitable as you discovered them using the Distribution platform. It's a good confirmation that Partition found them also!

This predictor is important and is expressed by finding the column whose differences in means (or categories, in the case of a nominal column) are most different as they relate to our Y, thus allowing us to predict our Y from our split X values. Want to know more about the analytical method used by Partition? Go to **Help ▶ Books ▶ Specialized Models ▶ Partition Models**.

Let's express the splits as probabilities, so go to the red triangle:

7. Select Display Options.
8. Select **Show Split Prob** (see Figure 6.15).

Figure 6.15 Select Show Split Prob

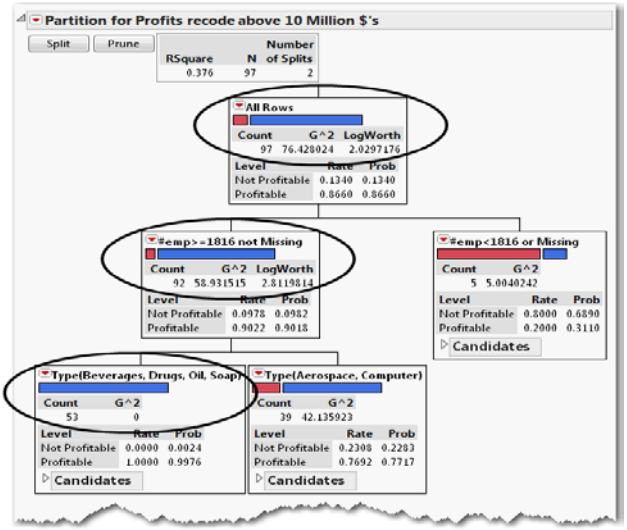


9. Select **Split** again. Another branch appears on the report (Figure 6.16).

Notes on Partition Trees: Partition trees are useful when you have large amounts of unexplored data. Partition trees are also flexible to your column modeling types. The output column (**Profits recode** in our example) can be either continuous, nominal, or ordinal AND the input columns (in our example, **#emp** and **Type** among others) can be either continuous, nominal, or ordinal. However, you should be cautious on drawing conclusions from partition trees when the data sets are small, sparse, or messy, especially as you continue to split. Starting in JMP 11, enhancements were made to many modeling methods, including Partition, to utilize the information value of missing data in improving model accuracy. Methods to measure the

usefulness of a partition model are also supported in JMP. To learn more about the Partition platform, go to **Help ▶ Books ▶ Specialized Models ▶ Partition Models**.

Figure 6.16 Probabilities for Each Split



The partition tree output now also shows the probabilities for each split (see Figure 6.16). Out of the total group of companies we found about 13% of them are not profitable and about 86% of them are profitable.

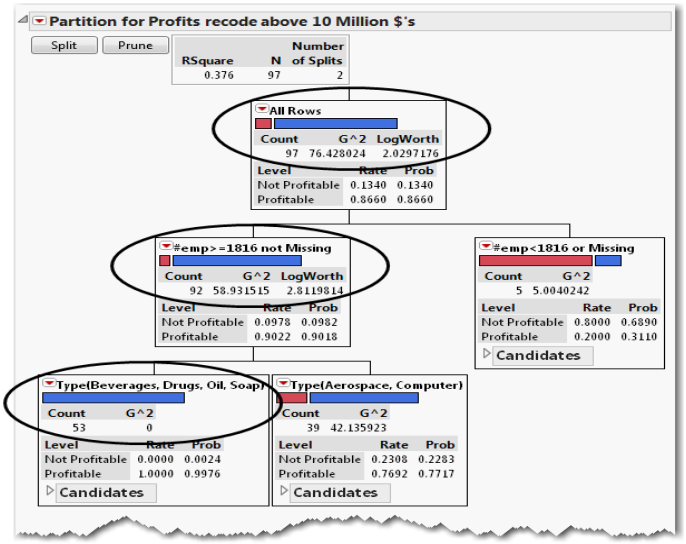
Now where is most of the blue? (Reminder: Blue is profitable!) It's mostly on the left-hand branch of the tree. These blue bars have been circled for you. It shows that 90% of the companies are profitable; about 10% are not profitable and come from companies with employees greater or equal to 1816 when the data is not missing. On the right-hand side, the method picked companies where the number of employees was less than 1816. You see more red on the right side of the tree. (Reminder: Red is unprofitable by our definition.) So, we conclude that the most significant predictor of profit is the **#emp** column where about 90% of the companies that have more than 1816 employees meet our criteria for profitable companies. In comparison, on the right side of the tree, we found about 68% of the companies are not profitable and about 31% are profitable where the number of employees is less than 1816 or the data are missing.

Note that in the output JMP has taken Missing or not Missing into account in the model. The information value of missing data can result in improved models. In JMP 11, a new feature called Informative Missing was added to improve modeling results and is reflected here. Go to **Help ▶ Books ▶ Specialized Models ▶ Partition Models** and the 'Informative Missing' section to learn more.

Based on what you just learned, what type of companies would you choose?

Where are the most profitable groupings after your last split (see Figure 6.17)? Hint: Follow the blue. On what side of the tree did the split happen? Can you find the combination where 99% of the companies are profitable? We find these where the Type of companies are Beverages, Drugs, Oil, and Soap, and the number of employees are greater than 1816. The best combinations in the tree are circled for you.

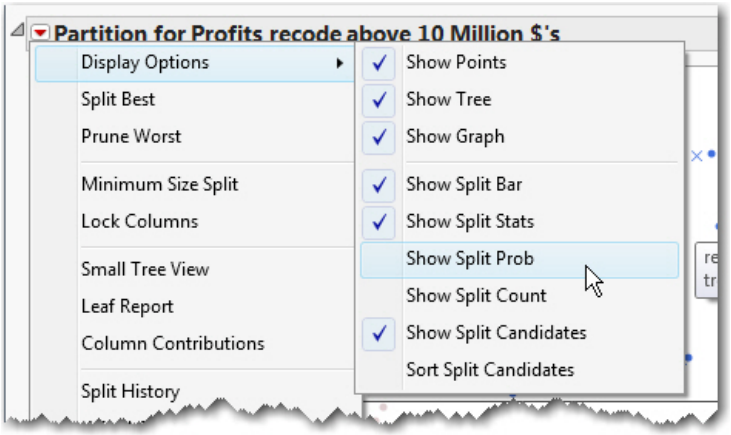
Figure 6.17 Most Profitable Groups after Last Split



You have now identified the most likely combination of predictors for profitable companies so far. Let's make that easier to see in the Leaf Report.

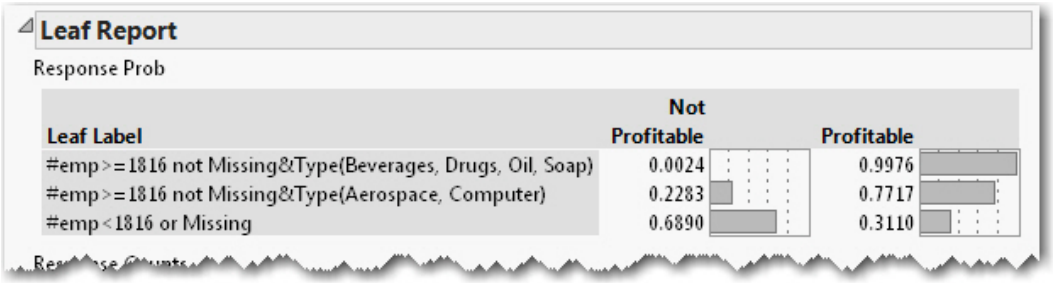
10. From the red triangle for partition, select **Leaf Report** (see Figure 6.18).

Figure 6.18 Select Partition Leaf Report



The Leaf report assembles your discoveries in a table format that mirrors the tree format we just covered (see Figure 6.19). Looking at the Leaf report and reading the **Leaf Label** from left to right, we can see which split combinations are most or least desirable. What combination looks the most risky so far? It appears to be where the number of employees is less than 1816. Some 68% of companies in that group do not achieve our definition of profitable.

Figure 6.19 Leaf Report



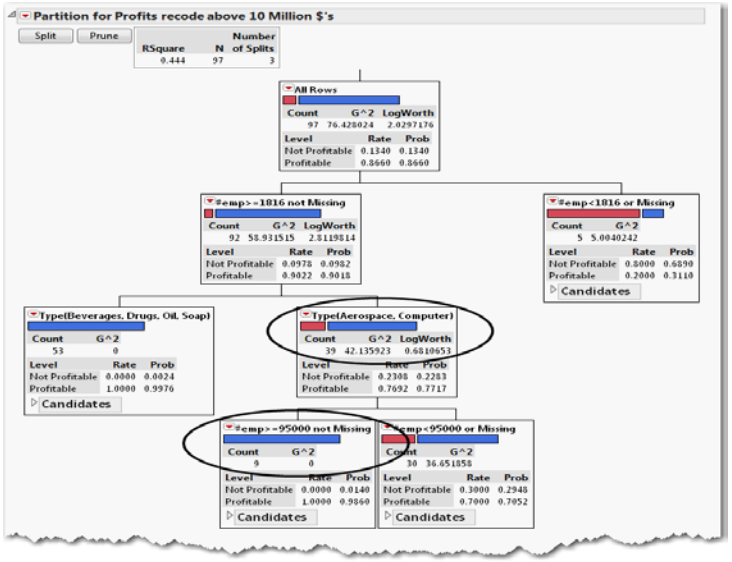
What group looks most profitable?

It appears to be where the number of employees is greater than or equal to 1816 and the type of companies are Beverages, Drugs, Oil, and Soap. In our example, 99% of companies in this group are profitable. Remember this group, as we'll be using it in the Using Data Filter section a bit later.

11. Back to our tree, click **Split** again.

    Where did the next split happen? (See Figure 6.20). We have found another possible combination where profits are predicted on the left-hand side of the tree.

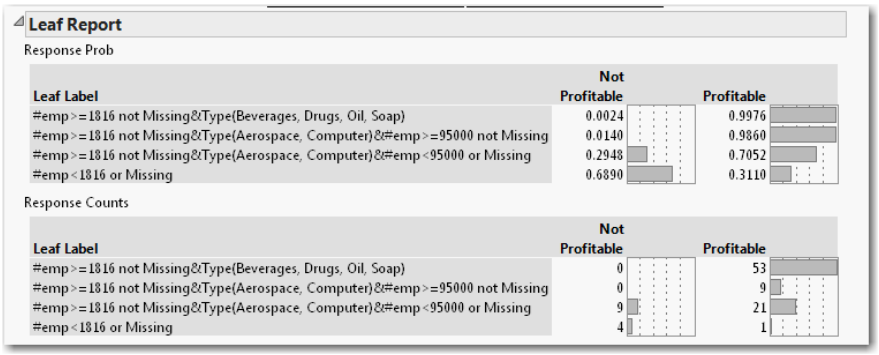
Figure 6.20 Results Showing Third Split



The combination of Aerospace and Computer and Number of Employees greater or equal to 95,000 has identified nine companies that are profitable. All of them are profitable in that combination. This profitable set has been circled for you (see Figure 6.20).

With each split, the columns in the model are sorted and ranked by the next most significant predictor. You can now see this last split in the updated Leaf report that appears in Figure 6.21.

Figure 6.21Third Split Leaf Report



Leaf Report

Response Prob

Leaf Label	Not Profitable	Profitable
#emp>=1816 not Missing&Type(Beverages, Drugs, Oil, Soap)	0.0024	0.9976
#emp>=1816 not Missing&Type(Aerospace, Computer)&#emp>=95000 not Missing	0.0140	0.9860
#emp>=1816 not Missing&Type(Aerospace, Computer)&#emp<95000 or Missing	0.2948	0.7052
#emp<1816 or Missing	0.6890	0.3110

Response Counts

Leaf Label	Not Profitable	Profitable
#emp>=1816 not Missing&Type(Beverages, Drugs, Oil, Soap)	0	53
#emp>=1816 not Missing&Type(Aerospace, Computer)&#emp>=95000 not Missing	0	9
#emp>=1816 not Missing&Type(Aerospace, Computer)&#emp<95000 or Missing	9	21
#emp<1816 or Missing	4	1

6.3 Filtering Data for Insight

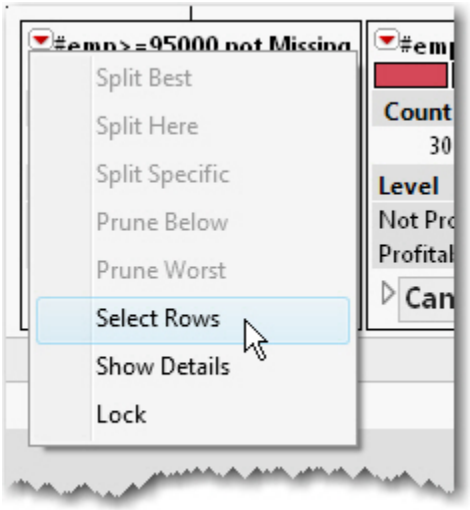
Drilling down or filtering your data is an important objective in data analysis and can be accomplished in several ways with JMP. In this section, we will show you three distinct ways this can be done with our example including commands from the Tables menu, the Data Filter platform, and Lasso tool. You will find that each filtering approach provides unique benefits and is suited to the context of the analysis you are performing.

Using a Table Command to Extract a Subset

Now let's identify those nine companies:

12. Select the red triangle within the tree node where **#emp>=95000 not Missing**
13. Select **Select Rows** (see Figure 6.22).

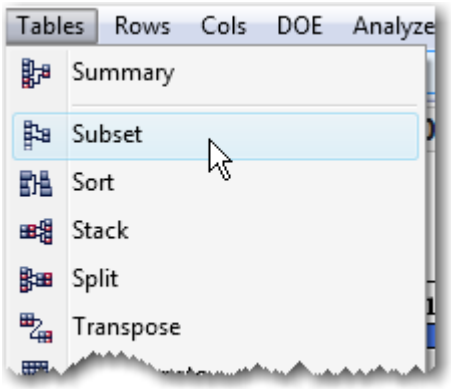
Figure 6.22 Select Rows on the Partition Node



14. Select the **Tables** menu.

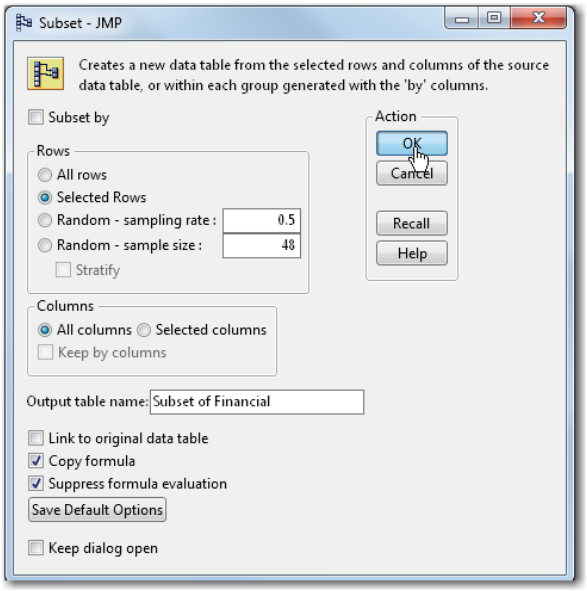
15. Select **Subset** (see Figure 6.23)

Figure 6.23 Tables Subset



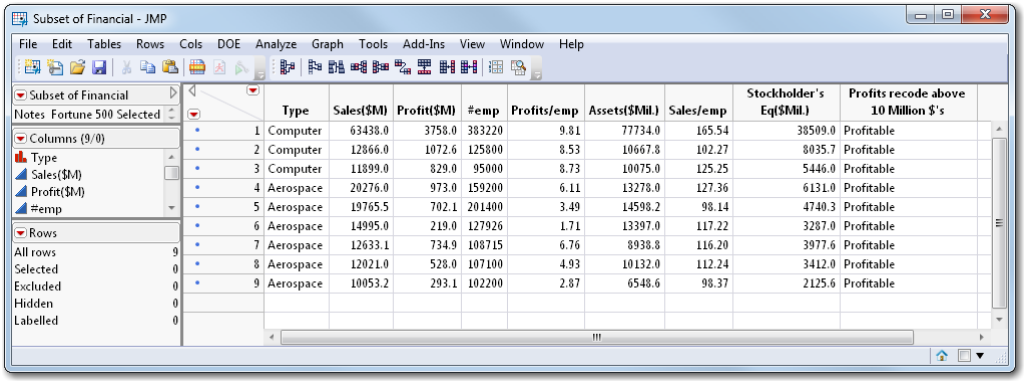
16. In the window, select **Selected Rows** and **All Columns** as shown, and click **OK** (see Figure 6.24).

Figure 6.24 Tables Subset Selected



A new data table appears (see Figure 6.25). You have identified a subset of desirable and profitable companies that are from the computer and aerospace categories and that have sales above $10 billion. These companies can now be submitted for further study and consideration for the investment portfolio. This subset step will save you time from finding them in the data table.

Figure 6.25 New Data Table Showing Profitable Subset



	Type	Sales($M)	Profit($M)	#emp	Profits/emp	Assets($Mil.)	Sales/emp	Stockholder's Eq($Mil.)	Profits recode above 10 Million $'s
1	Computer	63438.0	3758.0	383220	9.81	77734.0	165.54	38509.0	Profitable
2	Computer	12866.0	1072.6	125800	8.53	10667.8	102.27	8035.7	Profitable
3	Computer	11899.0	829.0	95000	8.73	10075.0	125.25	5446.0	Profitable
4	Aerospace	20276.0	973.0	159200	6.11	13278.0	127.36	6131.0	Profitable
5	Aerospace	19765.5	702.1	201400	3.49	14598.2	98.14	4740.3	Profitable
6	Aerospace	14995.0	219.0	127926	1.71	13397.0	117.22	3287.0	Profitable
7	Aerospace	12633.1	734.9	108715	6.76	8938.8	116.20	3977.6	Profitable
8	Aerospace	12021.0	528.0	107100	4.93	10132.0	112.24	3412.0	Profitable
9	Aerospace	10053.2	293.1	102200	2.87	6548.6	98.37	2125.6	Profitable

Using Data Filter

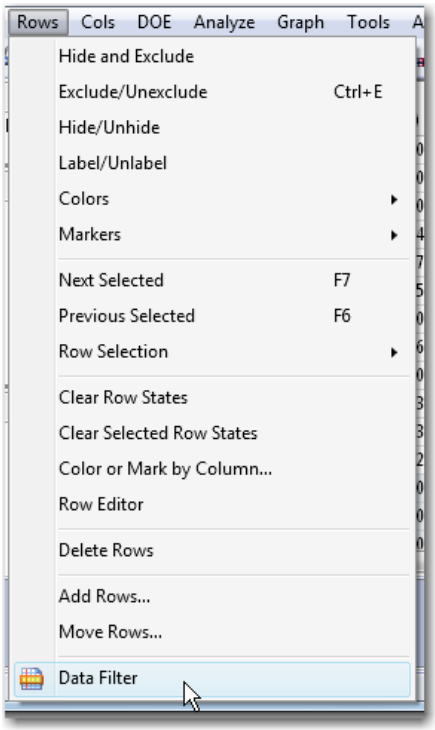
The Data Filter command from the Rows menu is a powerful means to visualize a subset of your data interactively. The Data Filter is especially useful when you are visualizing large data tables where graphs are packed with data points and it is difficult to see meaning in them. The Data Filter allows you to easily select rows (which can be ranges or categories) within any column of interest and, in effect, hide or exclude all other rows in your data table (see also section 2.5).

We'll use the Data Filter to identify the best of the best from among the groups you already identified using Partition.

Early in the last section, recall that a promising group of companies were from the Type column (Soap, Oil, Beverages, and Drugs) where the number of employees is greater than or equal to 1816 where the data are not missing. We need to restrict the next analysis only to this promising subset. Here's how:

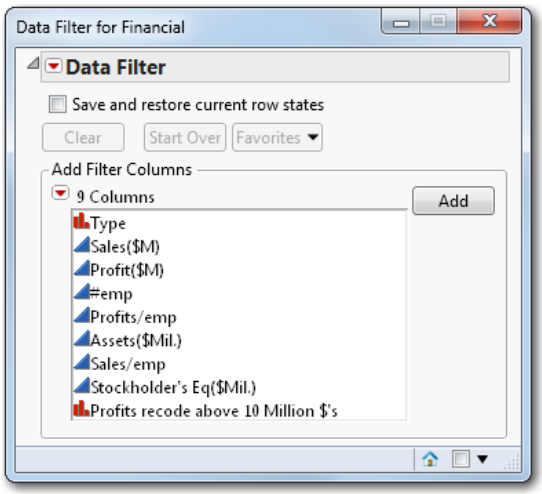
1. We need to bring the **Financial** data table to the front again. Select **Window ▶ Financial**.
2. From the Rows menu, select **Data Filter** (see Figure 6.26).

Figure 6.26 Select Rows Data Filter



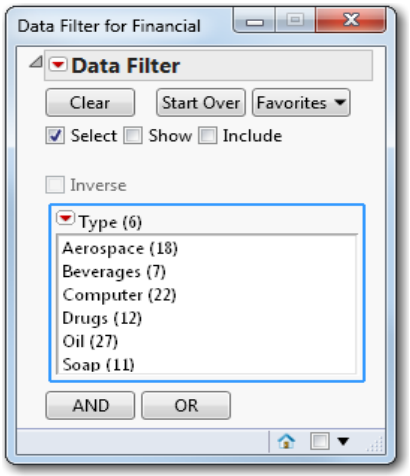
The Data Filter window appears (see Figure 6.27).

Figure 6.27 Data Filter Window



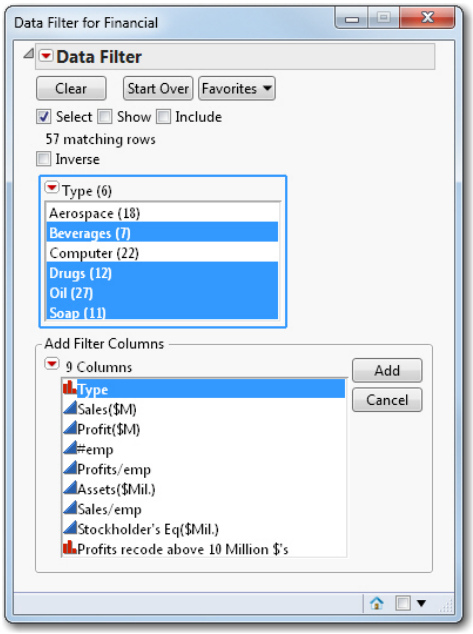
3. Select **Type** and click **Add**. A change to the Data Filter appears showing the different company types (see Figure 6.28).

Figure 6.28 Data Filter Type



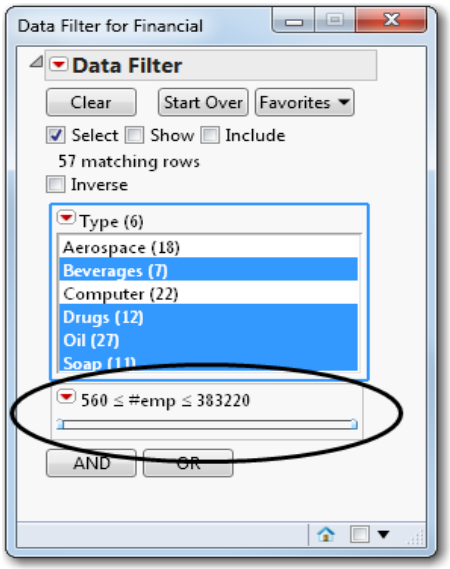
4. Now, press and hold the **CTRL** key and select **Beverages, Drugs, Oil,** and **Soap**. Click the **AND** button. (The results are shown in Figure 6.29).

Figure 6.29 Data Filter Beverages, Drugs, Oil, and Soap



5. Select the **#emp** column and click **Add.** (The results with **#emp** slider control are shown circled in Figure 6.30).

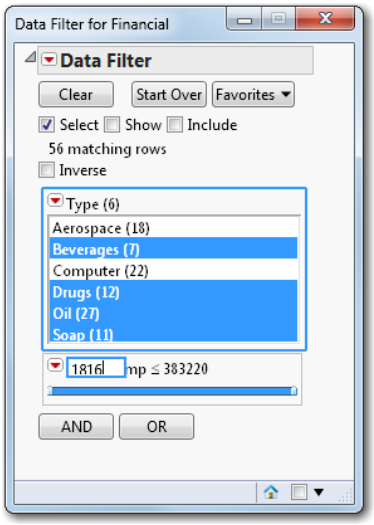
Figure 6.30 Select Number of Employees



A slider appears under the range, labeled **560 <= #emp <= 383220**. Recall that the dividing point discovered in the Partition platform earlier favored companies with a number of employees greater than or equal to 1,816 where the data was not missing.

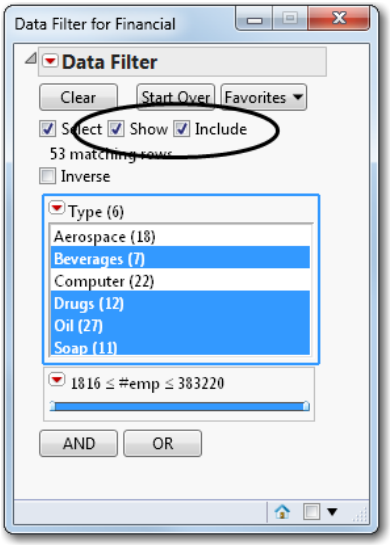
6. Click and enter **1816** where you see 560 on the left-hand side of the slider. On your keyboard, select the Tab key (see Figure 6.31)

Figure 6.31 Data Filter



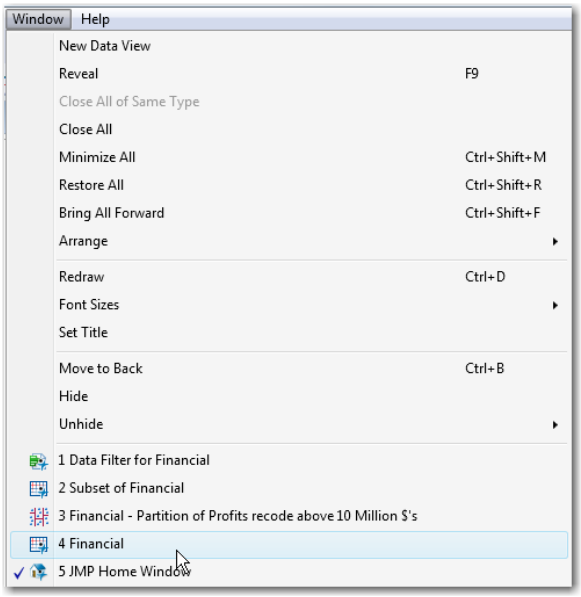
7. Now select the **Show** and **Include** check boxes (see the circled area in Figure 6.32). This will restrict the analysis to those same profitable companies identified in the partition and leaf report earlier in this chapter.

Figure 6.32 Select Show and Include



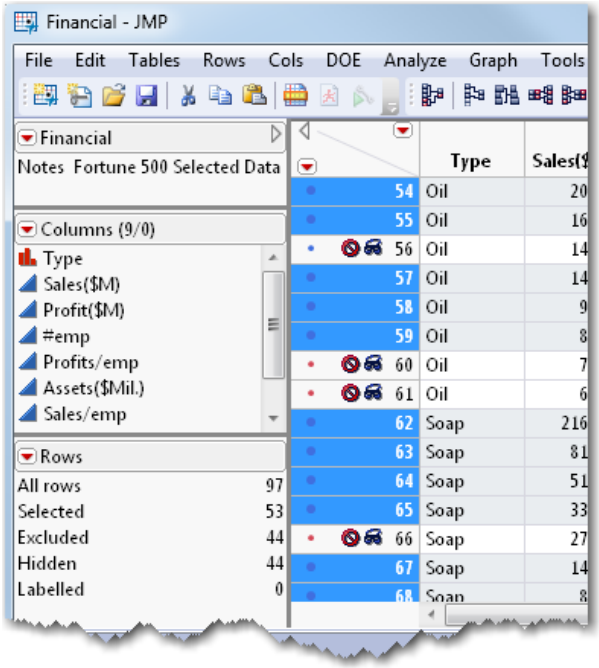
8. Let's take a look at the result of this exercise by returning to the data table, Select **Window ▶ Financial** (see Figure 6.33).

Figure 6.33 Bring Financial Window to Front



Take a look at the data table and notice what has happened. It now shows groups of rows that are selected and included for profitable rows, and hidden and excluded for every non-profitable company group (see Figure 6.34).

Figure 6.34 Data Table Showing Rows Hidden and Excluded



As we covered in section 2.5, the rows that are hidden (and that will not appear in graphs) feature a mask icon in the row, while rows that are excluded (not used in any analysis) feature a circle icon with a strike-through symbol.

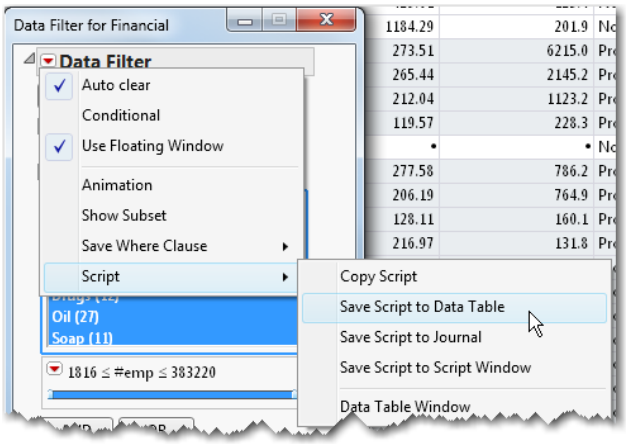
Rows that are still included and selected (highlighted) are those that meet the criteria for selection we established from our Partition example. They are now selected with the Data Filter.

The Data Filter is a terrific tool for exploring your data visually by allowing you to see subsets of your data in any JMP graph. These subsets can be derived from a prior analysis as demonstrated here or can simply be used to restrict your analysis to any subset you wish. As you will see in the next few pages, the Data Filter will be employed to further subset the most desirable companies in our example.

Now let's save the Data Filter for the future:

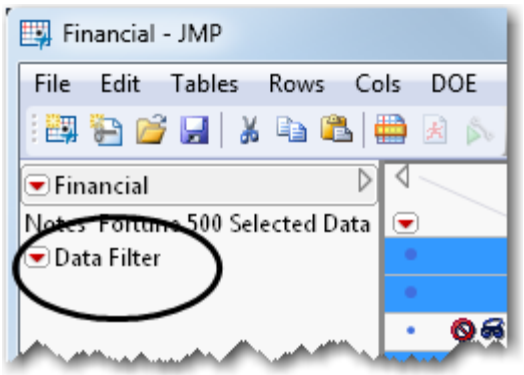
9. From the red triangle on the Data Filter, select **Script ▶ Save Script to Data Table** (see Figure 6.35).

Figure 6.35 Save Script to Data Table



In the upper left-hand panel of the data table, a new item appears named **Data Filter**. It's circled for you (see Figure 6.36). The **Data Filter** item has created a JMP script that stores the steps needed to reproduce the subset you selected.

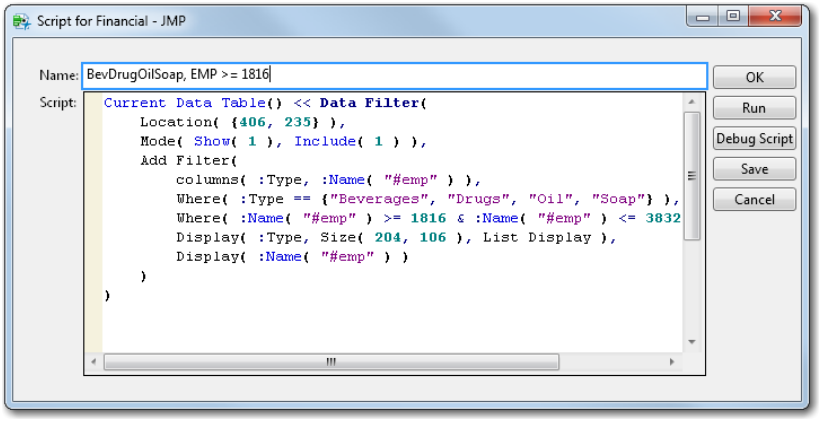
Figure 6.36 Data Filter Script in Data Table



Let's rename it something more descriptive so we can remember what it does.

10. Double-click on **Data Filter** and name it **BevDrugOilSoap, EMP >= 1816** in the window that appears (see Figure 6.37).

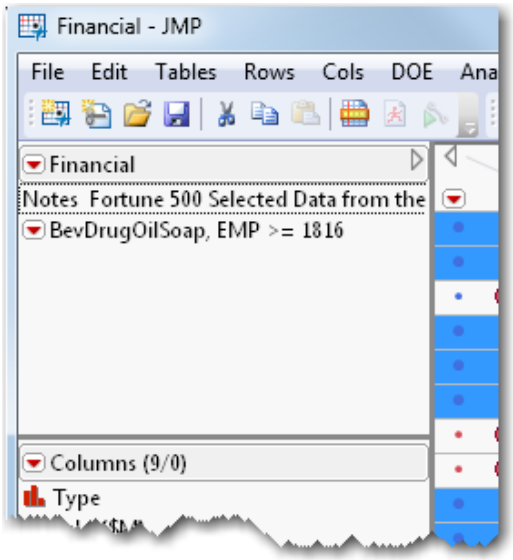
Figure 6.37 Data Filter Script Renamed



11. Click **OK**.

The Data Filter script is now renamed BevDrugOilSoap, EMP >= 1816 in the upper left-hand panel of the data table (see Figure 6.38).

Figure 6.38 Data Filter Renamed in Table



Now we can reuse the filter if we want to apply this criterion to the data. It's easy with the new script that applies the filter.

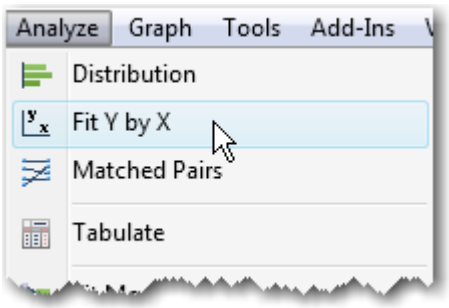
Note: The script also will act upon any new data you add or merge into the data table. If you add or change data in the data table, you may want to create a new subset. If that is true, use Partition again to analyze it and then use Data Filter again to apply it.

You have now filtered your data to only the best performers, so our task is to select only the very best from this group for our portfolio. There are many methods to finding high performers from among the groups we selected. We will go back and use the Fit Y by X platform (that we introduced in the previous chapter) along with the Lasso tool to visually select individual points. We find that using Lasso in this context easily enables visual selection of points that meet our performance requirements. The goal is to select the most profitable companies by industry type. In this example we'll use the continuous representation of profits, **Profits($M)**, as it allows us to distinguish with more precision among this group of high performers.

Recall that we also identified profitable companies from the Aerospace and Computer types in the partition Analysis and we sued the subset command to extract those. Because these companies also had the requirement of having sales over $10B, we could have used a conditioning statement (such as an "and" or "or" statement) in the Data Filter or merged that group with the one illustrated above to create one data table. To simplify this illustration, we've omitted these steps. To learn about how to merge data tables, see **Help ▶ Books ▶ Using JMP ▶** section **"Reshape Data"** and the topic **Concatenate**.

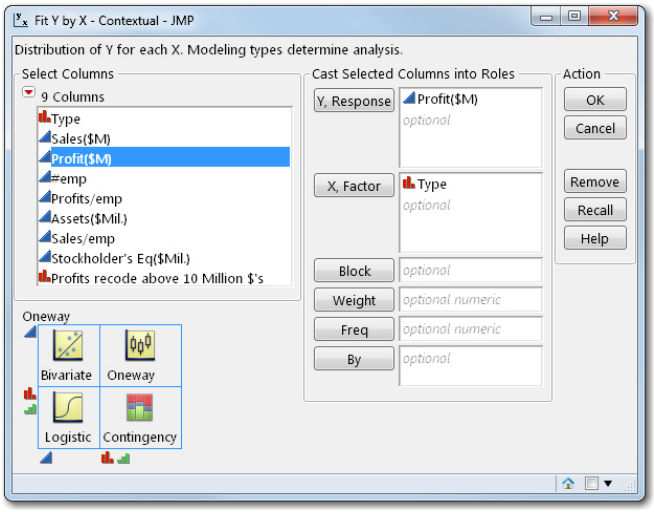
12. Select **Analyze ▶ Fit Y by X** (see Figure 6.39).

Figure 6.39 Analyze Fit Y by X



13. In the launch window, select **Profits($M)** as the **Y**, **Response** and select **Type** for the **X, Factor** (see Figure 6.40).

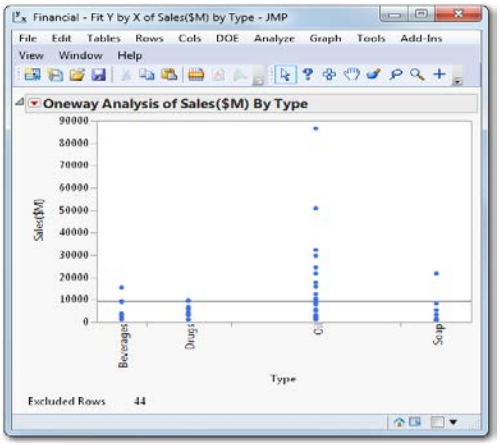
Figure 6.40 Fit Y by X Profit by Type



14. Click **OK**.

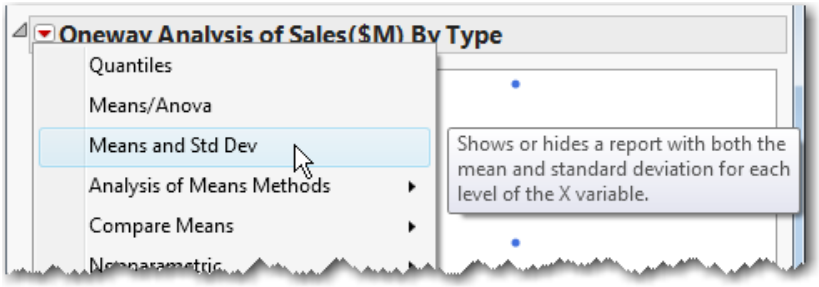
The Oneway graph (see Figure 6.41) shows profits from one of the best performing groups you identified in your partition analysis and selected using the Data Filter. All of the other companies have been excluded and hidden from this analysis. It is from these included companies that we want to identify the very best of these performers.

Figure 6.41 Oneway ANOVA Report



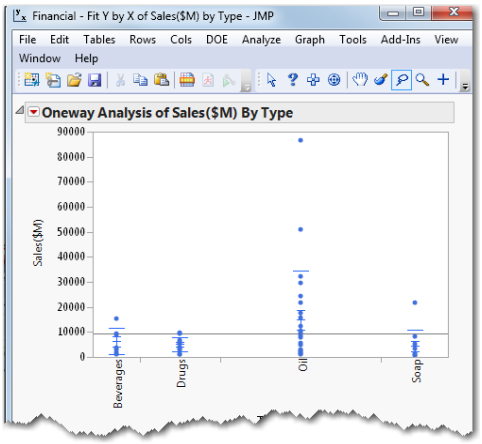
15. From the red triangle, select **Means and Std Dev** (see Figure 6.42).

Figure 6.42 Select Means and Std Dev from Red Triangle



The result appears in Figure 6.43a.

Figure 6.43a Oneway Means Std Dev Lines



We want to select companies that are exceptional performers and these are represented by those that are at least one standard deviation above the mean. Notice that the blue horizontal lines that appear in each type demark the standard deviations in the close-up in Figure 6.43b.

Figure 6.43b Mean Error Bars

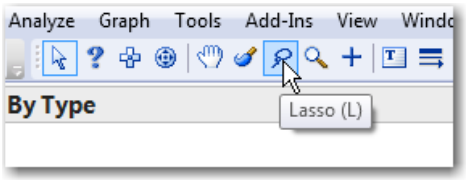
The smaller blue vertical lines with horizontal whiskers denote mean error bars (see Figure 6.43b). Use the question mark tool from the Tools menu and click on the Oneway plot to review standard deviations and other options.

Individual data points above or below horizontal error bars denote extremes. We are interested in those above the whiskers and above the blue horizontal lines because they are exceptional performers in terms of profit. Let's try a new method to select them.

Using Lasso to Select Individual Points

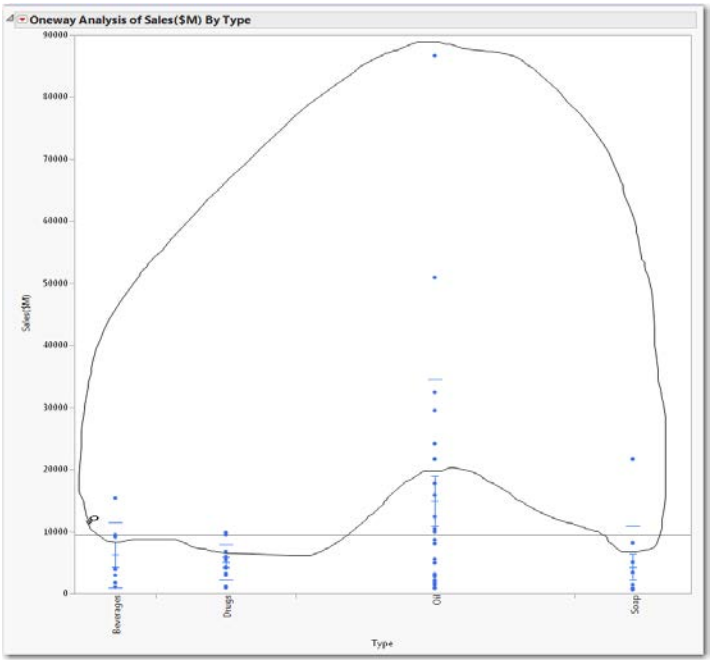
16. From the toolbar, select the Lasso tool (see Figure 6.44).

Figure 6.44 Select Lasso Tool



17. Now move the cursor (which should look like a little lasso) to the Oneway result. Left-click and draw around the points above the top whiskers of the error bars (see Figure 6.45), making sure that you close the circle around the points before releasing the button.

Figure 6.45 Lasso around Points

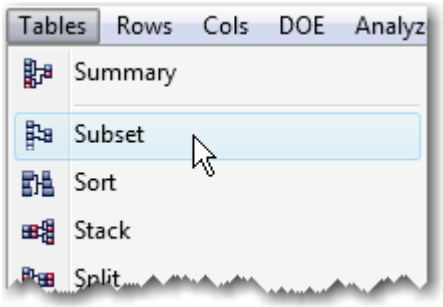


There should be about thirteen selected rows in your table. It's okay if the count is not exact. Now we just need to extract these very best historical performers from the rest.

Note: To help make these easier to see, the size of the graph has been increased by clicking and dragging the lower right corner to enlarge the graph.

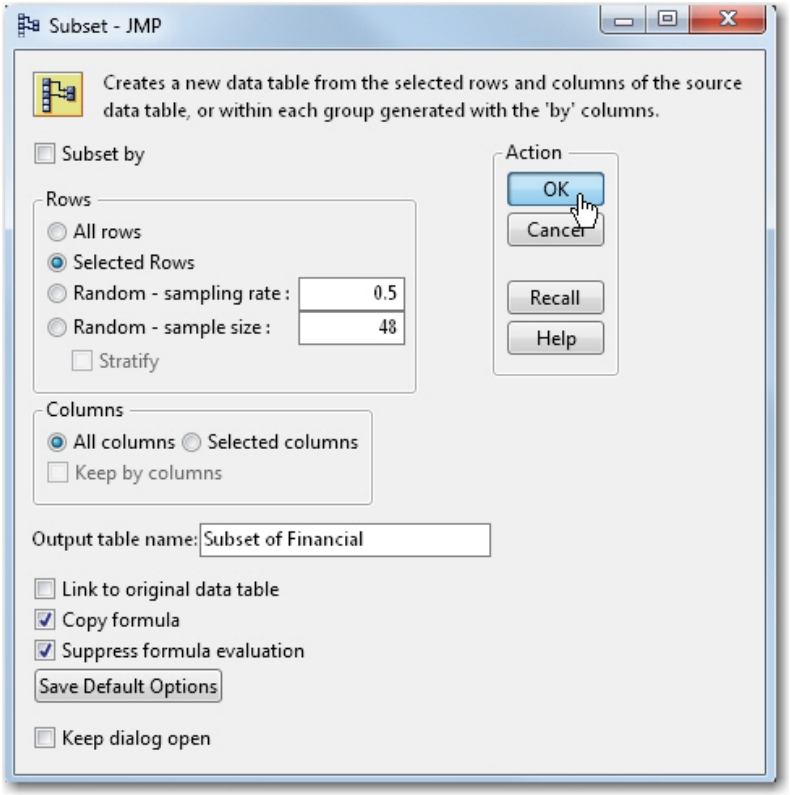
18. From the **Tables** menu, select **Subset** (see Figure 6.46).

Figure 6.46 Tables Subset



19. In the launch window, select **Selected Rows** and **All Columns** (see Figure 6.47).

Figure 6.47 Select Selected Rows and All Columns

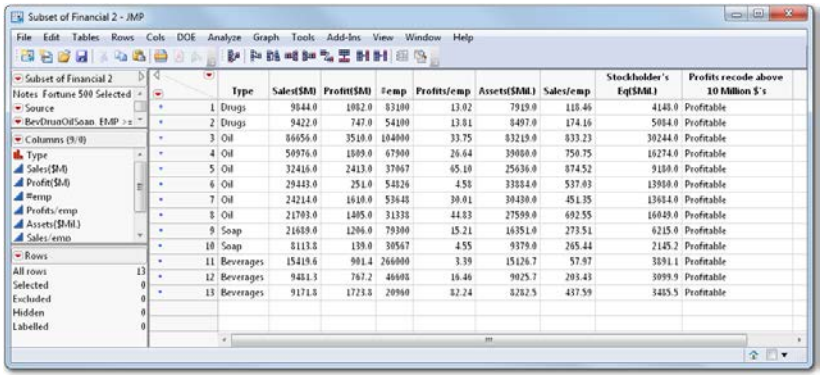


20. Click **OK**.

The resulting data table (see Figure 6.48) contains thirteen companies that have been sifted down from a group of 97 that represent the most profitable. The most profitable companies are mostly oil companies, though soap, drugs, and beverage companies each made a showing among the most profitable. We now have some potential portfolio selections for further study or investment.

Note: You may not get exactly 13. This is OK as some of the best performing companies might be challenging to grab using the lasso tool.

Figure 6.48 Data Table Showing Most Profitable Companies



	Type	Sales($M)	Profit($M)	#emp	Profits/emp	Assets($M)	Sales/emp	Stockholder's Eq($M)	Profits recode above 10 Million $'s
1	Drugs	9844.0	1082.0	83100	13.02	7919.0	118.46	4148.0	Profitable
2	Drugs	9422.0	747.0	54100	13.81	8497.0	174.16	5084.0	Profitable
3	Oil	86656.0	3510.0	104000	33.75	83219.0	833.23	30244.0	Profitable
4	Oil	50976.0	1589.0	67900	26.64	39060.0	750.75	16274.0	Profitable
5	Oil	32416.0	2413.0	37067	65.10	25636.0	874.52	9180.0	Profitable
6	Oil	29443.0	251.0	54826	4.58	33884.0	537.03	13910.0	Profitable
7	Oil	24214.0	1610.0	53648	30.01	30430.0	451.35	13684.0	Profitable
8	Oil	21703.0	1405.0	31338	44.83	27599.0	692.55	16049.0	Profitable
9	Soap	21689.0	1296.0	79300	15.21	16351.0	273.51	6215.0	Profitable
10	Soap	8113.8	139.0	30567	4.55	9379.0	265.44	2145.2	Profitable
11	Beverages	15419.6	901.4	266000	3.39	15126.7	57.97	3891.1	Profitable
12	Beverages	9481.3	767.2	46000	16.46	9025.7	203.43	3099.9	Profitable
13	Beverages	9171.8	1723.8	20960	82.24	8282.5	437.59	3455.5	Profitable

6.3 Model Fitting, Visualization, and What-If Analysis

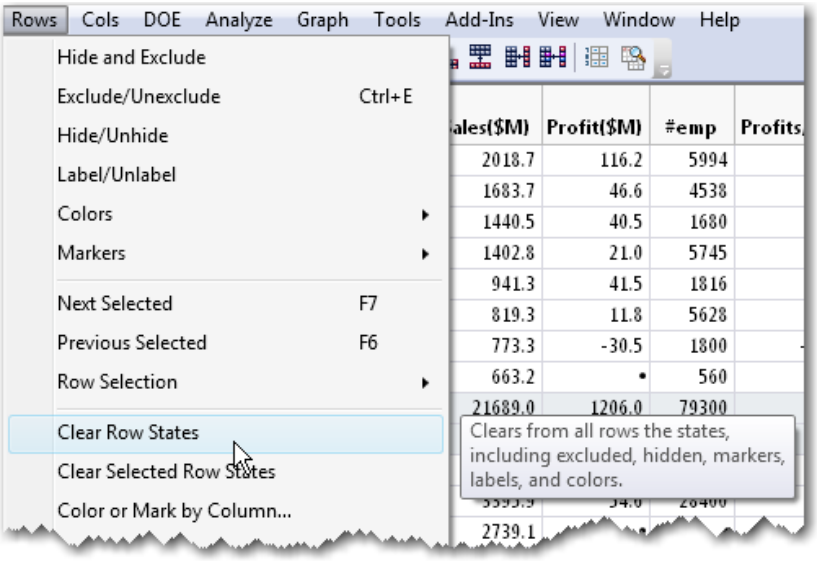
In this section, we will extend many of the JMP skills and concepts you've learned thus far into the insight you'll need to articulate your data effectively. We will now combine our models with the Data Filter in order to better understand the dynamics of our data. We will also introduce the Prediction Profiler for real time visual what-if analysis.

Consider for a moment a thought experiment. Some old radios have an analog tuner dial on them. Imagine you turn on the radio and hear noisy static. You slowly turn the tuner dial through the frequencies until you hear a transmission. Sometimes you might tune past the station, but you slowly tune back to obtain the optimal signal. Your ear heard the difference between the signal and the background noise. The Profiler and Data Filter (as described in the previous section) can be used as just such a tuner, but instead of using your ears, you use your eyes. The next section shows you how.

Let's test the idea that a positive relationship exists between profits and the number of employees in all of our companies. To do this, we need to use two things you already learned: Fit Y by X and the Data Filter.

1. Select **Rows ▶ Clear Row States** (see Figure 6.49)

Figure 6.49 Clear Row States

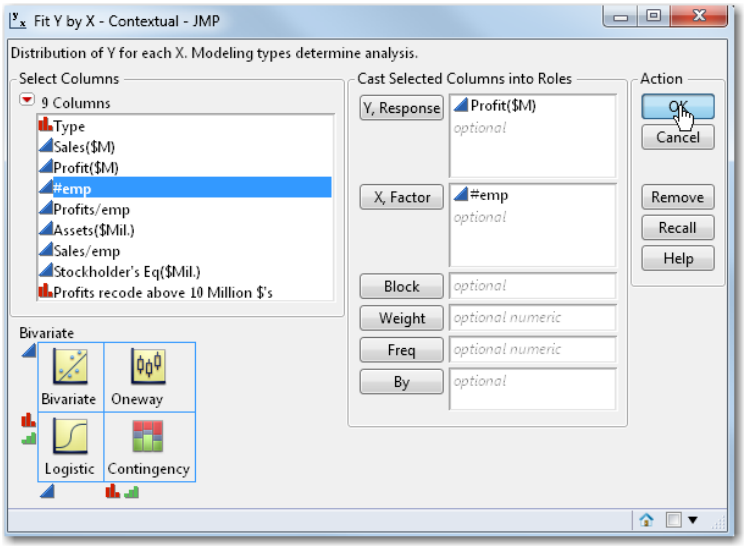


This clears the row selections, the markers, and the hidden and excluded rows from the last section.

Now let's build a simple model of the relationship between profits and number of employees.

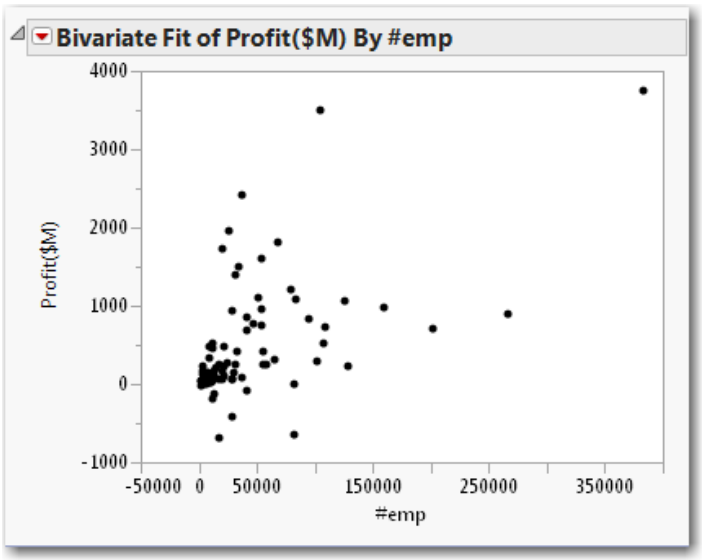
2. Select **Analyze ▶ Fit Y by X**. Select **Profit($M)** as the **Y, Response** role and **#emp** as the **X, Factor** role (see Figure 6.50).

Figure 6.50 Fix Y by X Profit Employees



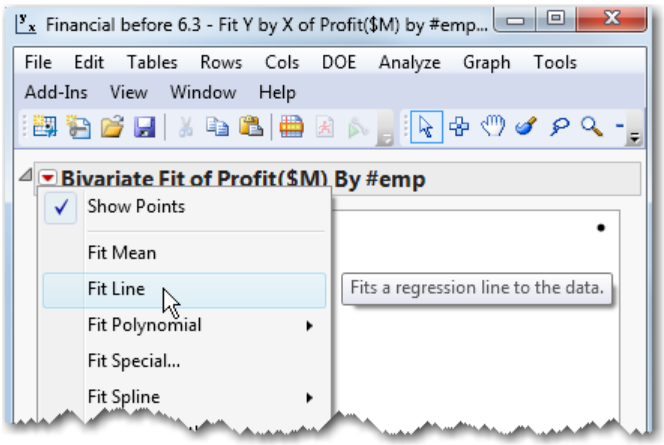
3. Click **OK**. A bivariate fit appears (see Figure 6.51).

Figure 6.51 Bivariate Plot



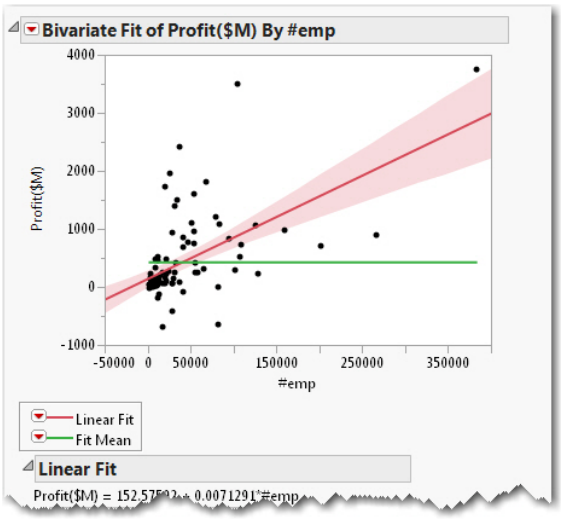
4. From the red triangle for the bivariate fit, select **Fit Line** and **Fit Mean** (see Figure 6.52).

Figure 6.52 Select Fit Line



A red line appears in the data that extends generally from the lower left of the graph to the upper right of the graph. A green line appears that represents the mean of the response or average profit (see Figure 6.53).

Figure 6.53 Bivariate Profits by Employee with Fit Line



This red line shows that, in general, as the number of employees increases, so do the profits. Notice how sparse the data is for companies above 150,000 employees. Because there are few companies or rows of data above 150,000 employees, it might suggest that the relationship is weaker for very

large companies. An unusually large and profitable company in the upper right is influencing the slope of the line as well.

Note: This is a simple or linear least squares regression fit, which fits a straight line through the points in a manner that balances the differences between those points above and below the line.

Let's see if graphics can help us interpret this relationship:

5. Select the red triangle item next to the line labeled Linear Fit and select **Confid Shaded Fit** (see Figure 6.54). A red shaded boundary around the line appears (see Figure 6.55).

Figure 6.54 Conf Shaded Fit

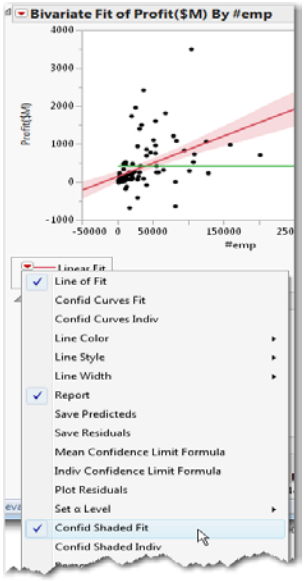
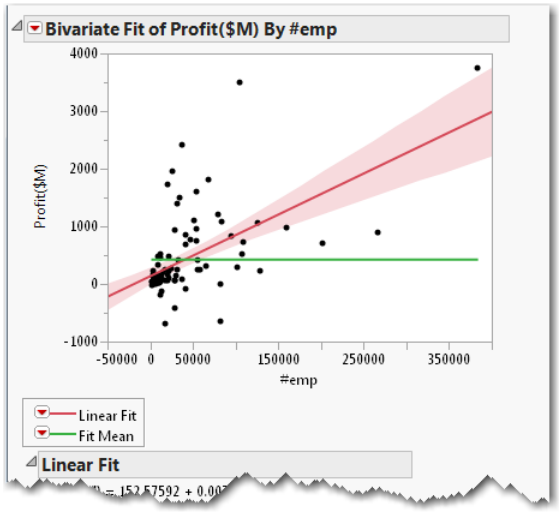


Figure 6.55 Conf Shaded Fit Displayed



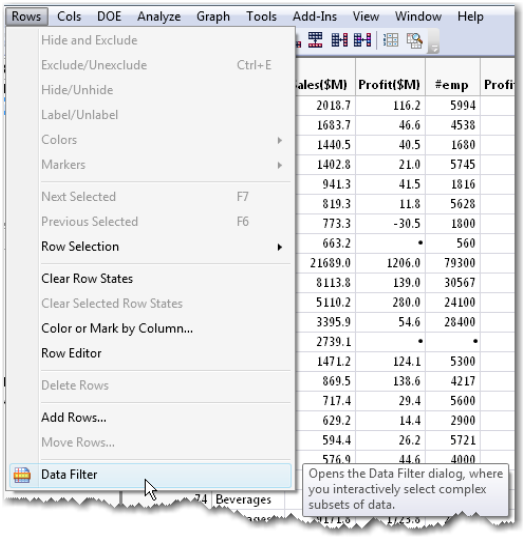
The curves appearing on the graph are 95% confidence curves. The shaded area around the line gets wider where data is sparser and the farther away from the mean of X. This is a visual indication that where data is sparse and far away from the mean of X, the estimates are less precise. The converse is also true: where data is denser and closer to the mean of X, the confidence curves are narrower and estimates are more precise. Thus, we can be more confident that the line better represents the relationship where the confidence curves are narrower than where they are wider. This might indicate that the relationship between profits and number of employees is strongest up to about 150,000 employees because the confidence curves start to widen there.

Notice, too, that a lot of the data points are pretty far away from the line. That's curious. It appears especially true for companies with more than 50,000 employees. Maybe even more than one pattern is present. How might we explore this?

The goal now is to see if the positive relationship between profits and number of employees holds across all company types. The Data Filter acts like a tuner as it toggles through the company types. Here's how.

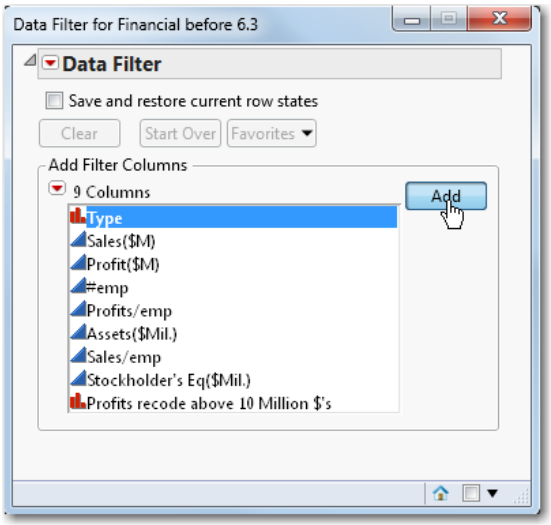
6. Select **Rows ▶ Data Filter** (see Figure 6.56).

Figure 6.56 Select Rows Data Filter



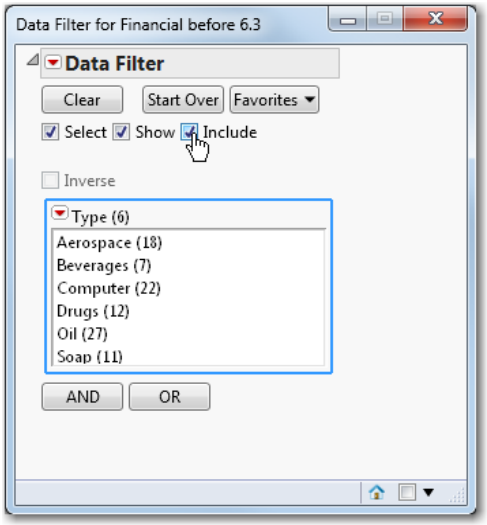
7. In the Data Filter window, select the **Type** column (see Figure 6.57) and click **Add**.

Figure 6.57 Select Data Filter Type



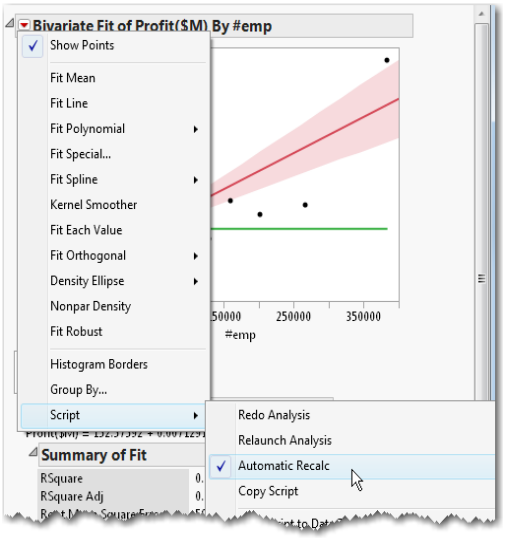
8. Put a check mark next to the **Select**, **Show** and **Include** check boxes (see Figure 6.58).

Figure 6.58 Data Filter with Select, Show, and Include Selected



9. Now return to your Fit Y by X results. From the red triangle, select **Script** ▶ **Automatic Recalc** (see Figure 6.59).

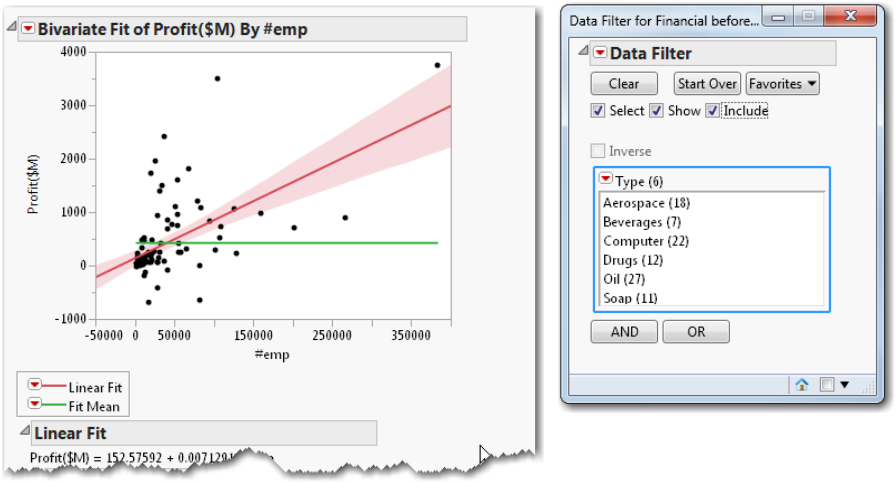
Figure 6.59 Automatic Recalc



Automatic Recalc updates the scatterplot and fit based on the new row states controlled by the **Data Filter**. Alternatively, you could leave **Automatic Recalc** off and re-fit the trend line interatively for multiple views but this takes more time to do.

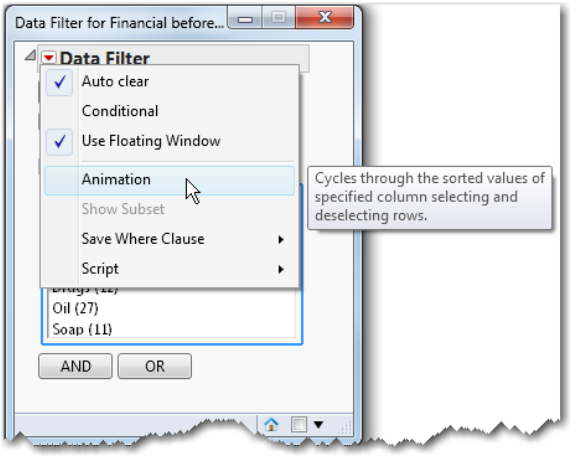
Now, you should see two floating windows (see Figure 6.60).

Figure 6.60 Bivariate Fit Y by X and Data Filter



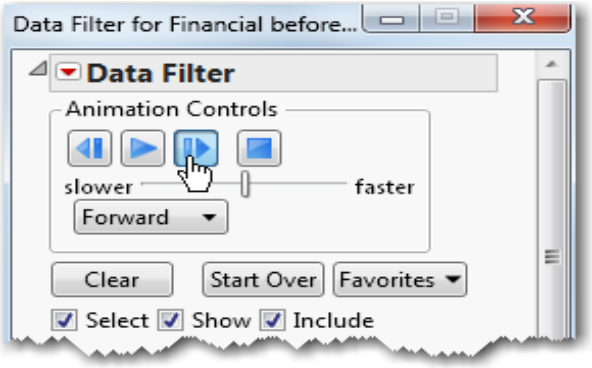
10. From the red triangle of the Data Filter, select **Animation** (see Figure 6.61). Animation controls will now appear in the Data Filter window that look like the controls on a DVD player.

Figure 6.61 Data Filter Animation



11. Click on (the step forward control) (see Figure 6.62).

Figure 6.62 The Step Forward Control

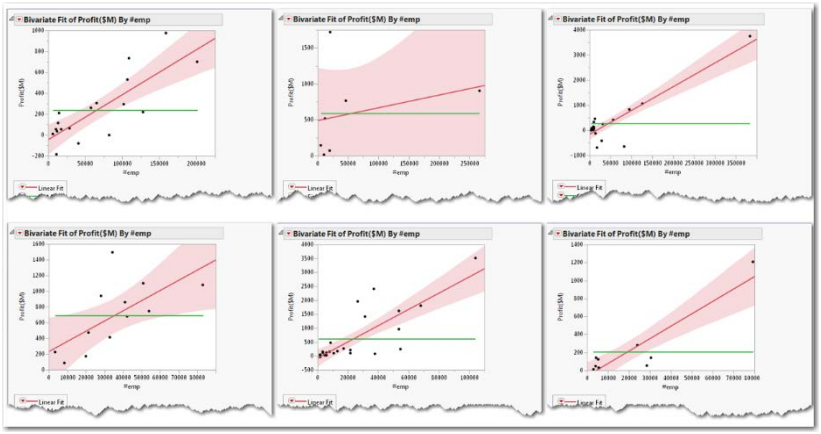


Each time you click the step forward control, you filter on just one of company groups within the **Type** column and you will see these highlighted in the Data Filter window.

Now click sequentially through the company types and notice what happens to the graph each time you click. Toggle through several times. Can you see the one type that is different from the rest?

Figure 6.63 includes the series of graphs you should be seeing as you toggle through each company type.

Figure 6.63 Results of Clicking Through Company Type



Which graph is most different from the rest? Which graph has a fit line that is almost flat? You might have noticed that Beverages shows a line that is much flatter than the others, which indicates that the number of employees or size of the company has less influence on the profitability. Notice that, in general, the other company types show a reasonably strong correlation between profits and number of employees.

Notice also that Beverages is the sparsest fit, with only seven data points widely dispersed so the confidence bands also flare out widely. Also notice that the green mean line for Beverages is completely inside the confidence curves. This is another indication that the correlation for Beverages is not strong. At least for the beverage company type, we can conclude that the greater number of employees does not predict greater profits in this sparse sample.

12. Select **Rows ▶ Clear Row States** for next section.

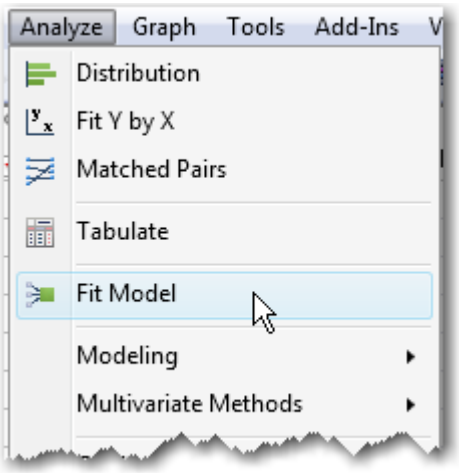
Conducting What If Analysis

The Prediction Profiler is a different type of graphical tuner for your data and provides a clear picture of your model. The Prediction Profiler is an interactive graph that produces estimates of your Y column of interest (profits, in our example) subject to your predictors, or X, columns, (such as number of employees from the last example). The interactive feature allows you to drag and change the settings of any column to see the estimated effect on the other columns.

The advantage of the Prediction Profiler is that it lets you try what-if scenarios dynamically and get immediate estimates on any column of interest. Very cool!

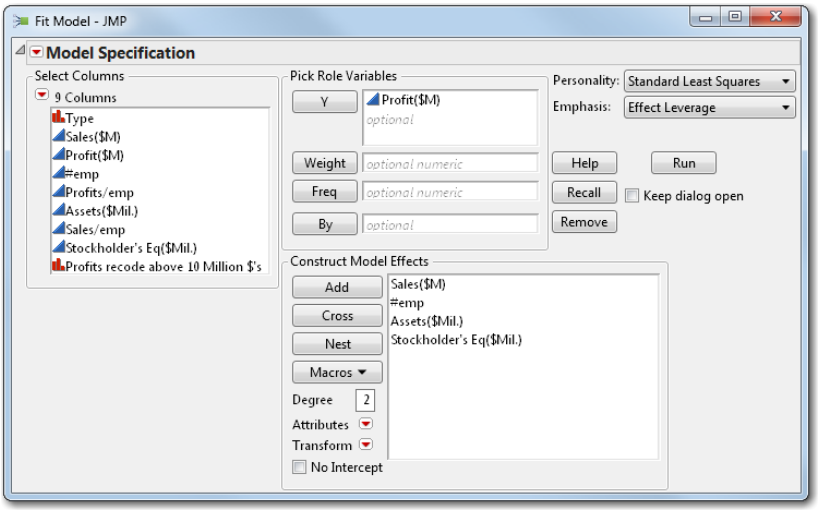
1. To create a profiler, you first need to create a model to describe the relationships in your data. Select **Analyze ▶ Fit Model** (see Figure 6.64).

Figure 6.64 Analyze Fit Model



2. In the Fit Model window, select **Profit($M)** as the Y column and then select **Sales($M), #emp**, **Assets($Mil.),** and **Stockholders Eq($Mil.).** Click Add to place them in the Construct Model Effects window (see Figure 6.65).

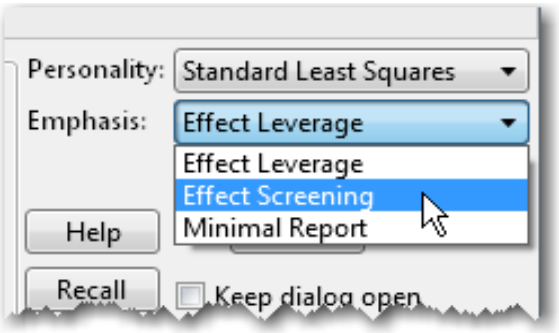
Figure 6.65 Fit Model Window



Caution: The Prediction Profiler appears in the context of a multiple regression model here. The technique is introduced here to enable quick exploration of your model. It is an advanced analytical method. You should familiarize yourself with the underlying analysis concepts by using the question mark tool or searching the relevant chapters in the documentation (**Help ▶ Search**).

3. Select the pop-down menu and change it from **Effect Leverage** to **Effect Screening** as shown. This just changes the output to include additional tools to interact with the output (see Figure 6.66).

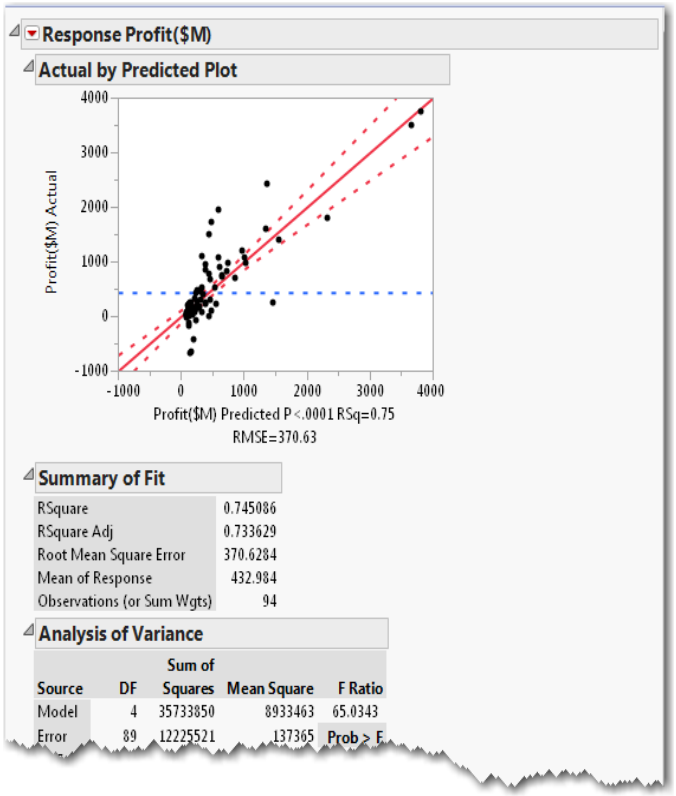
Figure 6.66 Fit Model with Emphasis Effect Screening



Now we are ready to run the model to test the relationships between profits and the other columns we've selected. This approach using the Fit Model platform employs a method called *multiple regression*. For now, we'll exclude the type of company from the model because we will be using it later to investigate differences.

4. Click **Run Model**. A report window appears (see Figure 6.67).

Figure 6.67 Fit Model Screening

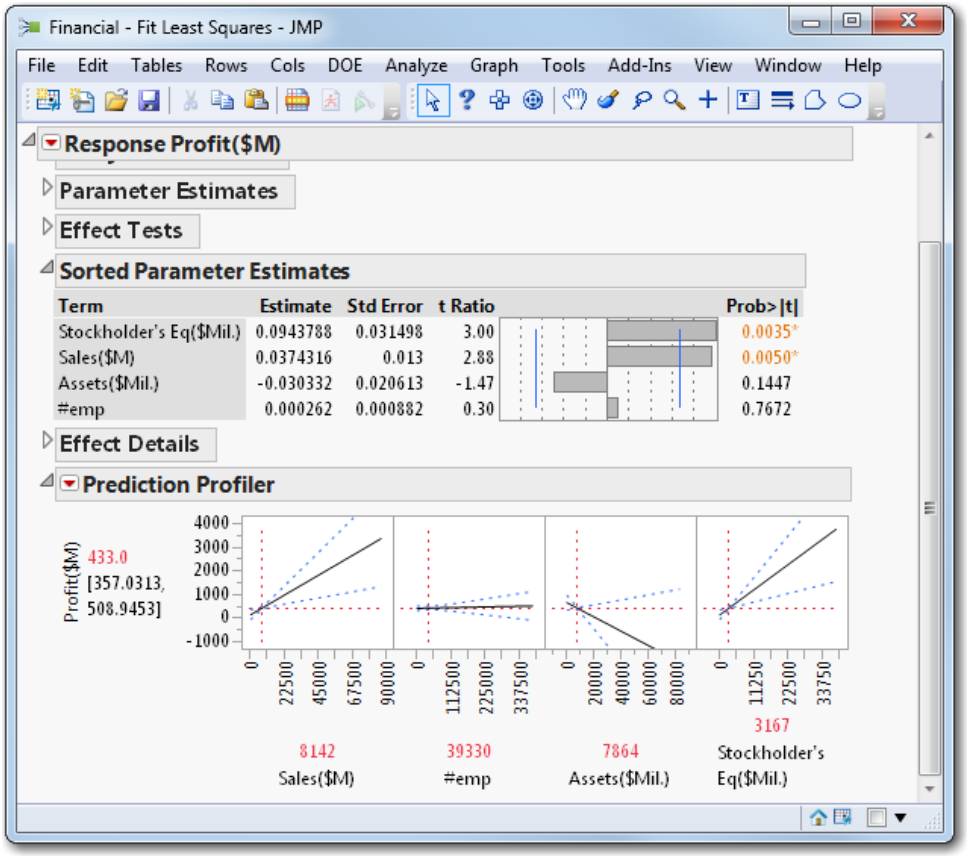


The Actual by Predicted Plot indicates there is a strong, positive relationship between profits and the columns you picked because of angle of the red fit line relative to the blue mean line. Just as in the earlier example, the blue mean line is not within the confidence bands, indicating a strong relationship. There are additional technical indicators of the quality of the relationships in the other report items. Use the question mark tool (**?**) to investigate the report items and learn more about the results.

5. Scroll down the window until you see the Prediction Profiler (see Figure 6.68).

   The Prediction Profiler enables you to see visual relationships between **Profits(#M)** (Y column) and sales, number of employees, assets, and stockholder equity (X columns) as depicted by the fitted solid black lines (see Figure 6.68).

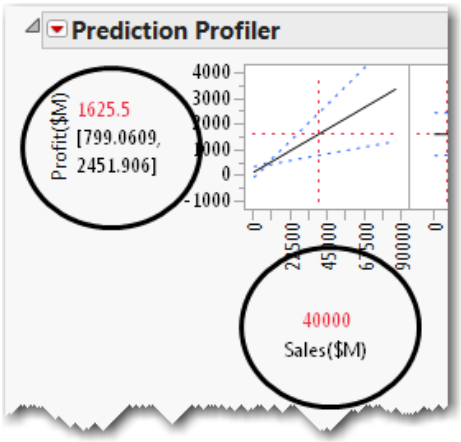
Figure 6.68 Prediction Profiler



Note: Each graph is bisected by a vertical red dotted line. These red dotted lines are interactive and provide us with a kind of X-ray vision into the relationships between the columns.

6. Drag the vertical dotted red line for sales from its current set point near 0 to a new set point near 40000 (see Figure 6.69).

Figure 6.69 Prediction Profiler Sliders

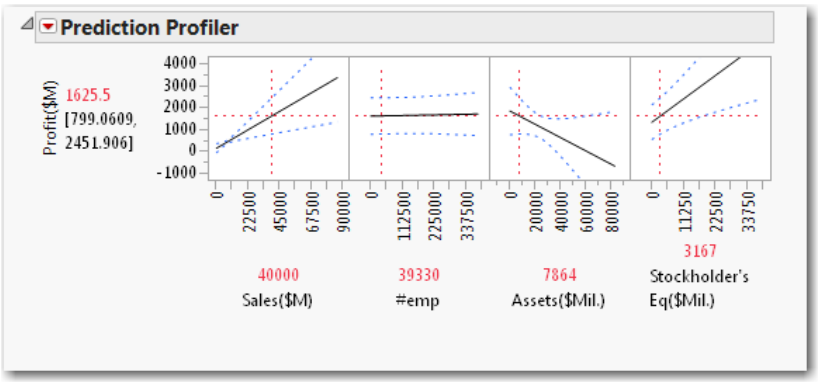


As you drag the red line for Sales($M), notice that the estimate for profits increases to around 1625. Dragging the line you may find it difficult to get it exactly. This is okay. You can also click on the red value and type in the new desired value, 40,000 in this case.

Each of the vertical and horizontal red dotted lines in the Prediction Profiler are interactive and work in this manner.

Look at the angle of the fitted line in the plots within the Prediction Profiler (see Figure 6.70). The angle of each of the fitted lines give us clues about their influence on Profit. A steeper line suggests that small changes in X (sales for example) will have significant changes in Y (profit).

Figure 6.70 Prediction Profiler

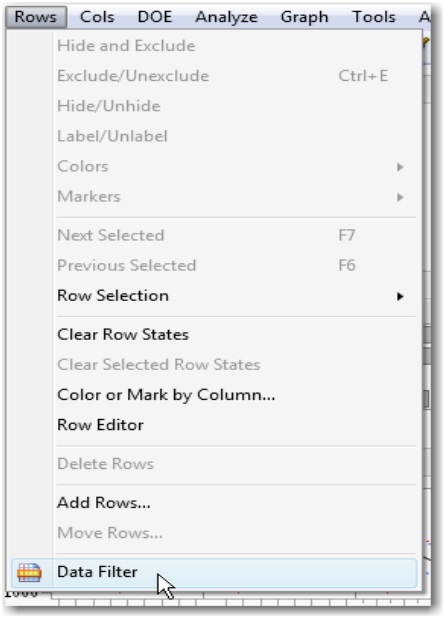


With sales, for example, this is an indication that our column of interest (profits) changes a lot when we move the red dotted vertical line associated with **Sales($M).** Thus, the steeper the line, the greater effect a change has on our Y column of interest. Lines that are flat, or nearly so (such as number of employees), mean that changes in these columns have little impact on our Y column (profits). With the Profiler, you can conduct dynamic, real-time what-if analyses.

We previously learned that profits varied among some company types. However, the model we just created did not include the **Type** column. So let's explore how each type of company might impact profits in this model. We again use the Data Filter to accomplish this:

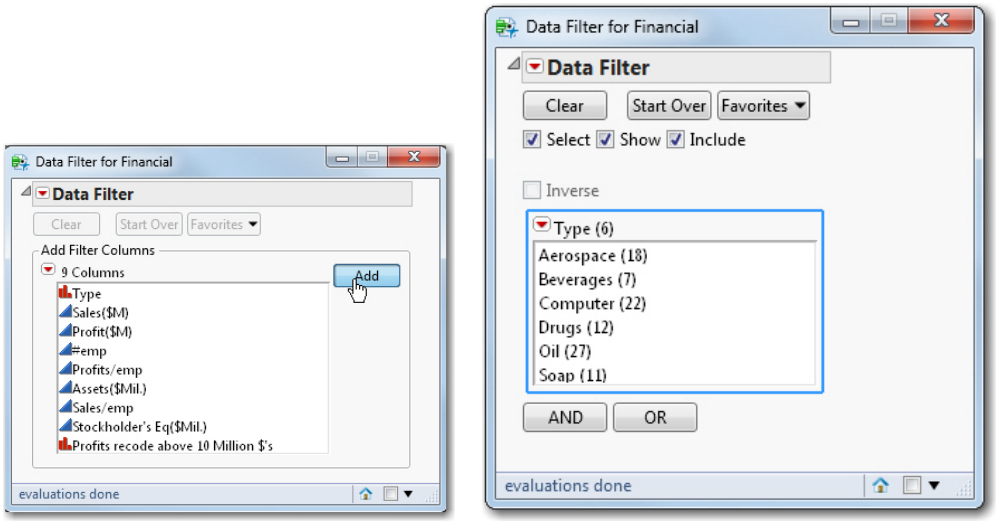
7. Select **Rows ▶ Data Filter** (see Figure 6.71).

Figure 6.71 Rows Data Filter



8. In the Data Filter window, select **Type** and click **Add**. Select **Show** and **Include** (see Figure 6.72).

Figure 6.72 Data Filter Type, and Show and Include

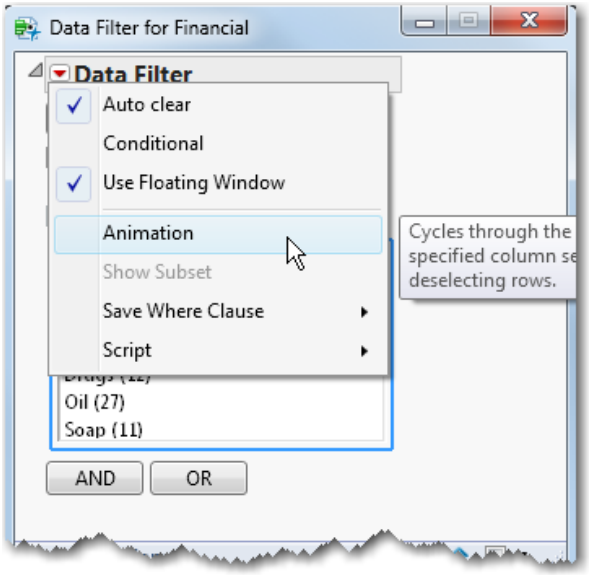


We need the report window to respond to the Data Filter dynamically using automatic recalculation. To do this, follow these steps:

9. From the red triangle (within the report window that contains the Prediction Profiler), select **Script ▶ Automatic Recalc**. The Prediction Profiler now responds to changes you make in the Data Filter.

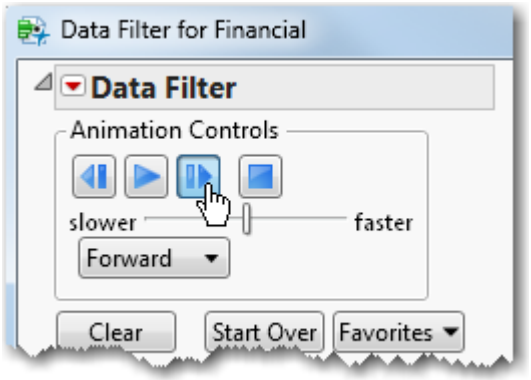
10. In the Data Filter window, click on the red triangle and select **Animation** (see Figure 6.73).

Figure 6.73 Data Filter Animation



11. Click on (the step forward control) (see Figure 6.74).

Figure 6.74 Data Filter Animation Step Control



Each time you click the step forward control, you are filtering the Fit Y by X analysis to just one of the company groups within the **Type** column and you will see these highlighted in the Data Filter window.

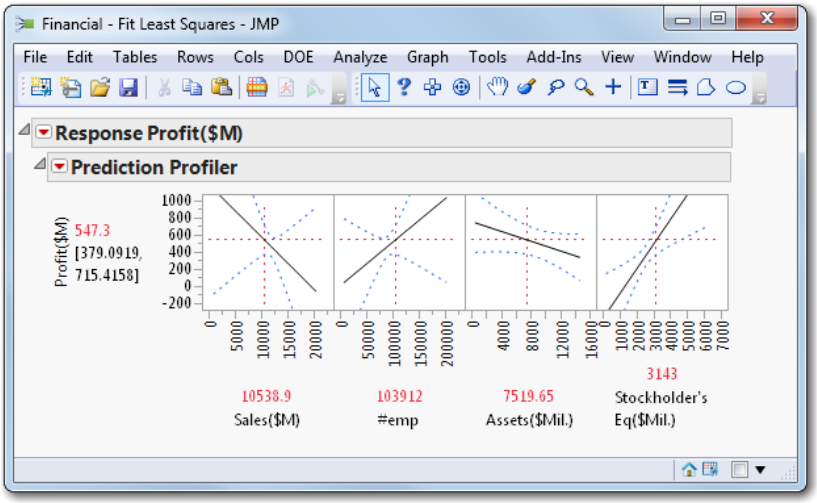
12. Now click the step forward control again (which toggles through the company types). Watch what happens to the Prediction Profiler with each click as it steps through the company types.

Have you noticed that some of the fitted lines in the profiler flip directions as the company type changes? How would you interpret these changes? We have summarized the first few of the profilers you see as you click through the company types. We will also provide a brief interpretation of each profile. Within each type, we can still drag the red dotted lines to investigate the sensitivity analysis or relationships within the model for that company type.

Aerospace

As sales increase, profits decrease (see Figure 6.75). As the number of employees increases, profits increase. As assets increase, profits decrease. As stockholder's equity increases, profits increase.

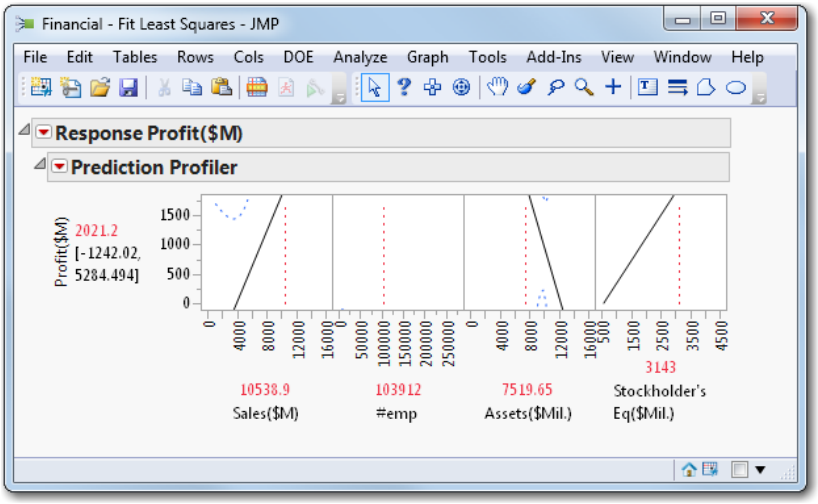
Figure 6.75 Prediction Profiler for Aerospace



Beverages

The confidence bands are so far away from the fit that we must be very cautious about trusting the profiler for Beverages (see Figure 6.76).

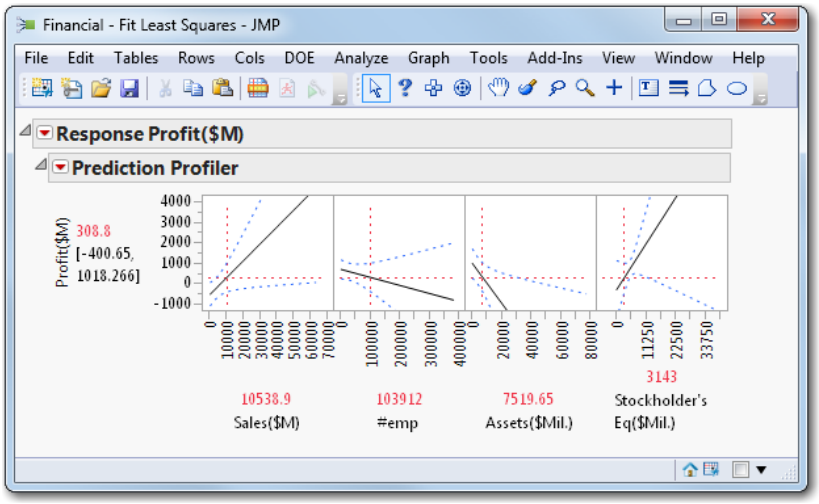
Figure 6.76 Prediction Profiler for Beverages



Computer

As sales increase, profits increase (see Figure 6.77). As the number of employees increases, profits decrease slightly. As assets increase, profits decrease steeply. As stockholder's equity increases, profits increase.

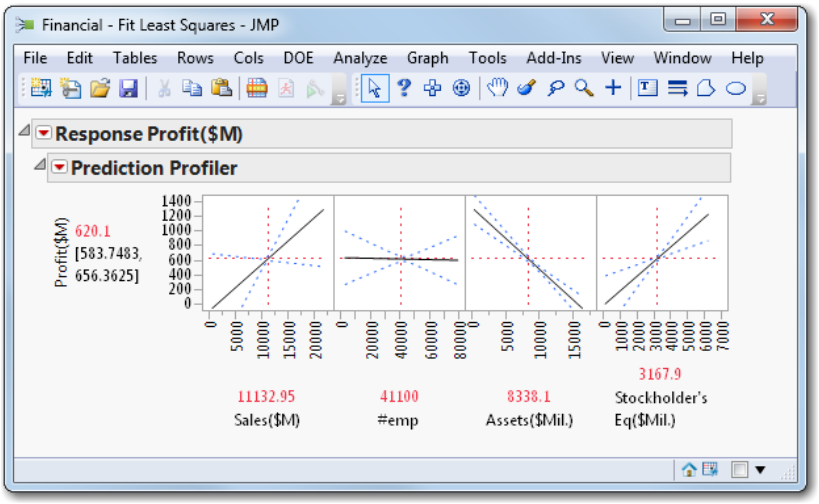
Figure 6.77 Prediction Profiler for Computer



Soap

As sales increase, profits increase (see Figure 6.78). As the number of employees increases, profits stay the same. As assets increase, profits decrease. As stockholder's equity increases, profits increase.

Figure 6.78 Prediction Profiler for Soap



Now it's your turn. On your computer screen, try reading the Prediction Profiler for Oil using what you have learned.

6.5 Summary

Most real-world problems are complex and involve multiple columns. The Partition platform is a flexible tool for solving many types of problems that involve multiple columns and rapidly identifies the key relationships in the data.

Tools like Data Filter and Lasso enable quick identification and extraction of interesting subsets of data. The Data Filter also provides an exploration method using animation that lets you tune in to just the slice of the data that best supports an inference or hunch.

The Profiler using the Fit Model platform offers a powerful way to understand the relationships among columns in your models. The ability to manipulate values (using the red dotted lines) within a column and immediately see their effect on other columns provides the means to conduct visual what-if analyses. With the Data Filter, you can drill down to subcategories or ranges of columns to discover those nuggets of insight that are often hidden at first glance.

This chapter and the last have presented an approach to problem-solving that is unique to JMP. The approach underscores the progressive nature of discovery that tends to build from simple descriptions of one column of data to complex relationships among many columns. This problem-solving process leads to a better understanding of your data and, in turn, to the insights and answers you seek. This process might not only go in one direction from simple to complex. As discoveries are made with the more advanced multi-column tools, confirmation and further analysis can be made with the simpler ones, Distribution and Fit Y by X, from where our journey began.

Chapter 7 Sharing Graphs

7.1 Presenting Graphs Effectively
7.2 Customizing Graphs for Presentation
7.3 Placing Graphs into PowerPoint or Word
7.4 Sharing Dynamic Graphs with HTML 5
7.5 Creating and Sharing Animated Graphs
7.6 Placing Animated Graphs into PowerPoint
7.7 Creating Dynamic Dashboards
7.8 Using the Layout Option to Share Results
7.9 Using Scripts to Save or Share Work
7.10 Summary

This chapter focuses on common ways you can customize and share JMP graphs with others whether you are presenting those findings yourself, placing them in a document, or sending a file. We also provide some advice for effectively doing so.

One of the most important things a JMP user can do is communicate results (data, statistics, and graphs) clearly and accurately. Most people recognize that it can be easy to manipulate results or create a graph that doesn't tell the whole story. The first section of this chapter is devoted to some principles of effective communication with graphs.

If you want to annotate a graph with comments or change colors or other settings before moving it into another document, JMP provides several convenient tools introduced in Section 7.2.

We'll work with three types of graphical output in this chapter:

1. Fixed or static graphs (sections 7.3, 7.8).
2. Dynamic or interactive graphs, which allow you to manually click on points to highlight (sections 7.4 and 7.7).
3. Animated graphs, which perform like movies. Simply press go and watch (sections 7.5, 7.6).

Moving *static* graphs into other applications, such as Microsoft Word and PowerPoint, is a straightforward operation in JMP. We illustrate some options in Section 7.3 that allow you to edit the results in other applications.

Moving *dynamic and animated* graphs onto the web or into other applications while retaining many of the interactive capabilities can be a little more involved, which we walk through in Sections 7.4 through 7.7.

Further, you might want to combine different graphs into an interactive dashboard by combining windows or using the JMP Layout tool to annotate and combine several graphs into a single static report. We cover the essentials of these features in Sections 7.7 and 7.8.

7.1 Presenting Graphs Effectively

Presenting statistical graphics in a report or presentation is a common task for many JMP users. As such, it is critical that you do this well. By *well*, we mean that you rely upon the data to tell the story, be succinct, and, most importantly, present an accurate and valid interpretation (visual, numeric, verbal). It helps to keep the following in mind when presenting statistical graphics:

- Understand your data and its limitations. Time spent with your data before you present it prepares you for the questions that may emerge.
- Focus on the important factors. What columns are core to the problem or question at hand?

  Note: How do you determine which variables or columns are most important? The methods described in Chapter 6 can help you identify important columns of interest. For example, the Partition platform (Section 6.1) can help you find out which columns are the key factors affecting another variable.
- Provide the simplest expression of the data to convey the most complete description of it.
- Present graphs accurately. Do not reduce ranges in graphs to magnify changes or increase ranges to hide trends.
- Let graphs speak for themselves and avoid using unnecessary background colors or dimensions that distract from the core information you are conveying. Make sure the graph can be seen by your audience.

An important feature of JMP is that it helps you adhere to these principles and commonly accepted standards. JMP can provide many different types of appropriate graphs of your data, so how do you know which is the most effective way to communicate your information? You will find that using common sense goes hand-in-hand with effective presentation of data and graphical integrity.

In recent years, there have been a number of new books on the subject of presenting quantitative graphs and visualizing data effectively. Perhaps the best-known of these are books by Edward Tufte, whose beautifully crafted works have won wide acclaim and illustrate the core principles of presenting quantitative information effectively. These books and those by Stephen Few (see the Bibliography) are highly recommended if you use graphs extensively.

7.2 Customizing Graphs for Presentation

There are many features in JMP that provide you with the ability to customize graphs for presentation. Sometimes you'd like to annotate a graph by pointing out a key attribute of the graph. Or, you might want to use color or markers to enhance other columns and attributes, as described in Section 2.6.

This section introduces the common tools needed to prepare your graphs and results before moving them into another application or document.

Most of JMP's formatting options appear by simply right-clicking on the area (for example, the graph or axis) you'd like to customize. When customizing your graphs, be sure to make changes within JMP before pasting them into another application. The first step, however, is to look at a powerful set of tools called the JMP Toolbar that we begin with in this section.

Example 7.1 SAT by Year

We will be using the SATByYear.jmp data table to illustrate the concepts in this chapter. SAT by Year is data representing SAT (Scholastic Aptitude Test) and ACT (Achievement Test) test performance by college-bound students in the United States covering the years 1992 to 2004. The tests serve as an important admission metric for colleges and universities.

The data consists of 17 columns and 408 rows of data and includes the following columns:

State Students' state of residence

Expenditures (1997) State expenditures per pupil

Student/Faculty ratio (1997) Ratio of students to faculty

Salary (1997) Mean salary of teachers (in thousands)

%Taking (2004) Percent taking the SAT exam in 2004

Population Population of the state

% Taking (1997) Percent of students taking the SAT in 1997

ACT Score (2004) Mean test scores by state in 2004

ACT % Taking (2004) Percent taking ACT in 2004

ACT Score (1997) Mean test scores by state in 1997

ACT %Taking (1997) Percent taking ACT in 1997

Year Year of the tests

SAT Verbal Mean SAT Verbal test score by state

SAT Math Mean SAT Math test score by state

Region Region in which state resides

You can access this data at **Help ► Sample Data ► Open the Sample Data Directory ► SATByYear.jmp.**

Using the JMP Toolbar

When JMP graphs are generated, the Toolbar contains several tools that help you copy, annotate, and share them (Figure 7.1).

Figure 7.1 JMP Toolbar

Once a tool is selected, your mouse is transformed into its corresponding function. If these do not appear on your home window, select **View ► Toolbars ► Tools**. These tools are useful in the JMP Layout platform (see Section 7.8), where you might construct a report. Many of these tools have familiar functions:

The **Selection** tool allows you to select graphs or reports (or portions of them) to be copied and pasted into Word, PowerPoint, or other applications.

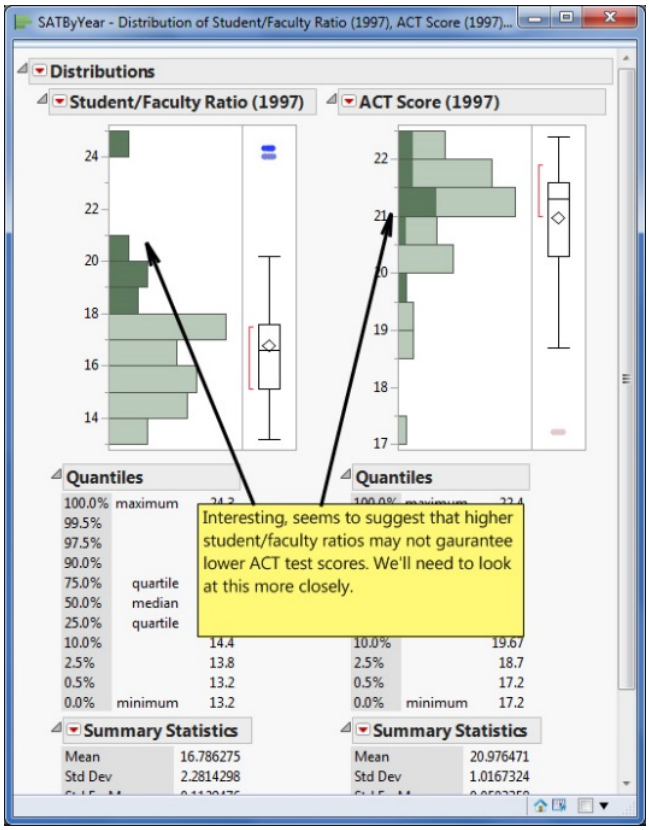
Within the Toolbar, the following functions allow you to add comments or highlight areas of your graphs:

- **Annotate** allows you to add a textbox or comment in your graphs.
- **Lines** allow you to draw lines on your graphs. For example, an arrow drawn from a textbox to a point on a graph.
- **Polygon** and **Simple Shape** allow you to draw shapes in your graphs.

Note: The Toolbar features are also available from the Tools menu that appears at the top of the JMP window.

Figure 7.2 is an example of what these tools can do for you. You can summarize key aspects of your data with the Annotate tool and create pointers using the Lines tool to call out this information. Interpretation of this kind can provide valuable direction when you are not there to guide the report recipient. To generate these graphs in the **SATByYear** data table, select **Analyze ► Distribution ► Student/Faculty Ratio** and **ACT Score, Y, Columns ► OK**.

Figure 7.2 JMP Results Window with Annotations



To create these features in JMP, do the following:

1. Select the **Annotate tool** from the JMP Toolbar.
   Click on your report where you want to add an annotation. Type your comment and then left-click outside the annotation box. You can then move or resize the box as you would any textbox or change the color of the background by right-clicking on the box.

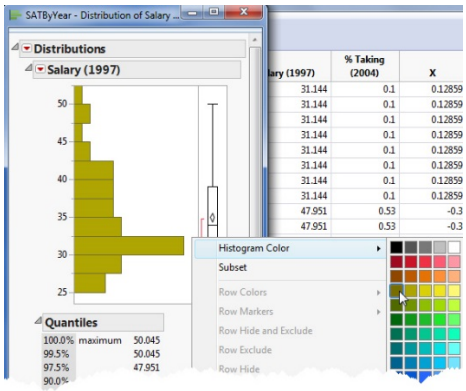
2. To add lines or pointers, select the **Lines** icon from the Toolbar and draw the line. Once the line has been drawn, right-click to access the options for pointers, thickness, style, and color.

Note: When you'd like to annotate more than one graph window and show these together, use the Layout tool (discussed in Section 7.6).

Using Color

Changing the colors of your basic graphs is simple. Just right- click within the graph area (see Figure 7.3). In the Distribution platform, select **Histogram Color** and select a color. The method of accessing color options varies slightly by platform but generally can be found by right-clicking on the graph or through the red triangle. If you want your graphs to always use the same color, you can change the JMP defaults by selecting **Save Color Preference**. For more information on setting preferences, see Chapter 1.

Figure 7.3 Changing Histogram Color

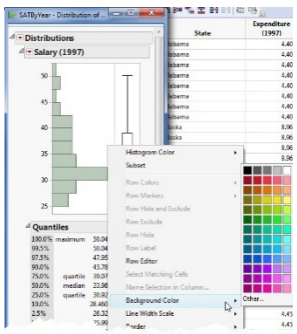


Coloring a graph in this manner does not affect graphs where individual rows are color-coded by a column value within the data table. (See Section 2.6 for more information on how to use color to denote a category or range within a graph.)

Background Color

You can change the background color of any graph by right- clicking in a graph area and selecting **Background Color** (see Figure 7.4). When using background colors, be mindful not to distort, distract, or obscure the information in the data.

Figure 7.4 Changing Background Color



Horizontal Layout

Within the Distribution platform, the default display places the histograms in a vertical (side-by-side) format. Clicking on any bin (or bar) within one histogram highlights those values in the remaining histograms. This is especially useful when you are exploring relationships among two or more columns and don't want to scroll down.

Sometimes, however, you'd like to see the histogram in the more traditional horizontal format. Rotate the display by simply selecting **Display Options** under the red triangle after you've created a histogram and then selecting **Horizontal Layout** (see Figures 7.5 and 7.6).

Figure 7.5 Specifying Horizontal Layout

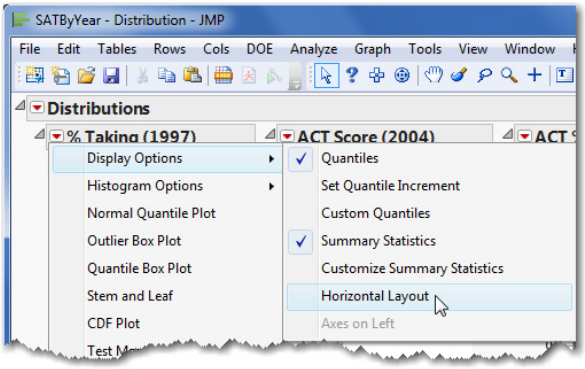
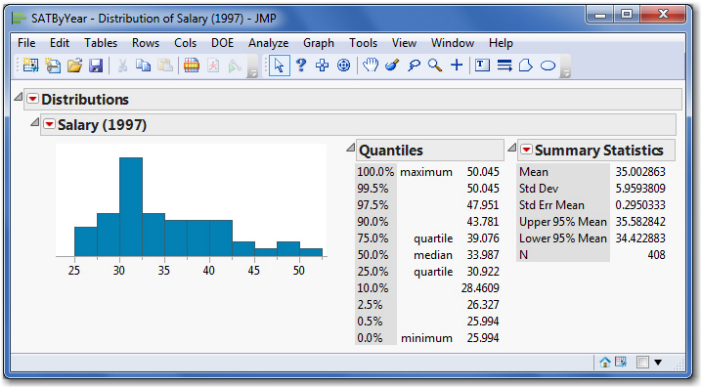


Figure 7.6 Horizontal Layout Result

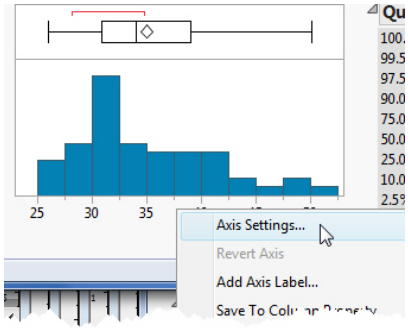


Note: If you want to stack multiple histograms horizontally, one on top of the other, you can use the Stack command under the Distributions red triangle (immediately above Salary's red triangle illustrated in this example).

Axis

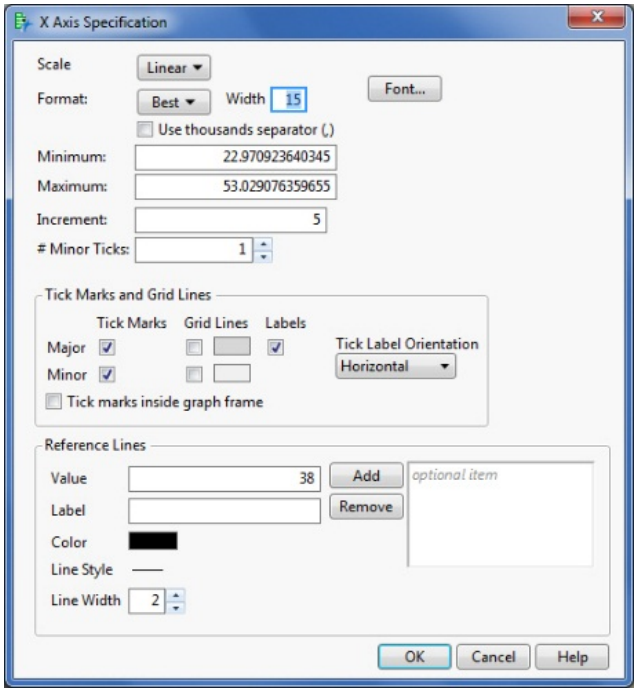
JMP allows you to easily change the axis of any graph by simply grabbing the axis and moving it. Alternatively, you can right-click on the axis to access the following menu (see Figure 7.7).

Figure 7.7 Axis Settings



By selecting **Axis settings**, you access the Y Axis Specification window (see Figure 7.8). In it, you can customize the range, increment, and other axis options.

Figure 7.8 Axis Settings Dialog Window



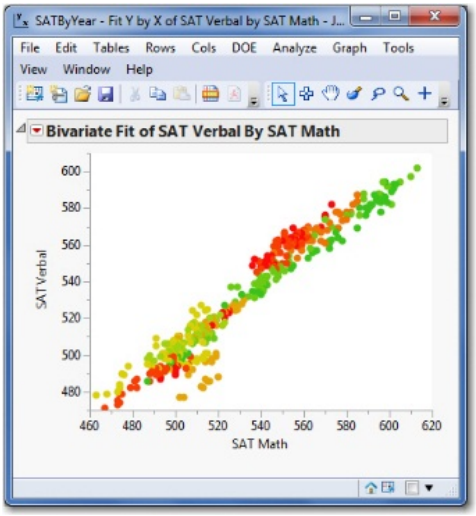
7.3 Placing Graphs into PowerPoint or Word

Copying a fixed or static JMP graph into Word or PowerPoint is simple. You just need to use the Selection tool to select what you want and then copy and paste the graph into your document.

Note: If you are using JMP 12, this process is made even easier by simply selecting **File ► Save**, select **Save as type ► PowerPoint Presentation** (or **Microsoft Word 2000+**) from the desired results window.

Let's illustrate this with the following example. Say you want to move the graph in Figure 7.9 from JMP into PowerPoint.

Figure 7.9 Moving a Graph

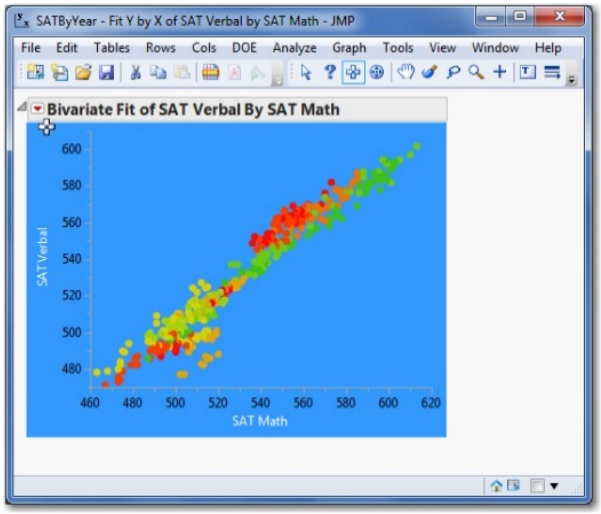


1. Select the Selection tool (see Figure 7.10).

Figure 7.10 Use the Selection Tool

2. Click on the upper left corner of the graph, as shown, to capture the entire picture (see Figure 7.11).

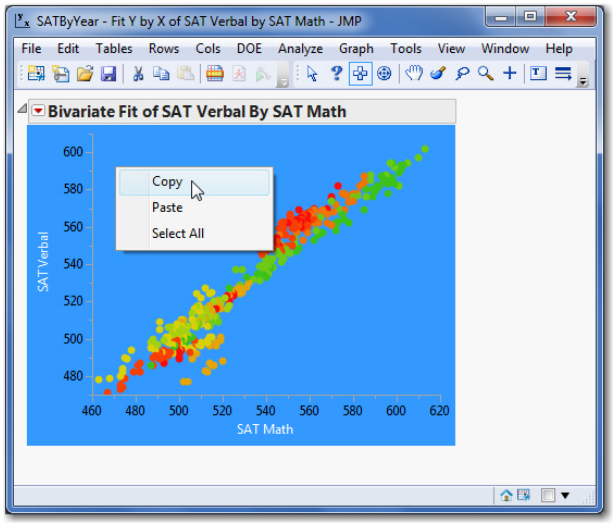
Figure 7.11 Selecting Graph Components



Alternatively, you might want to copy only selected elements of the graph, in which case you click and drag the cursor to select the elements you'd like to copy.

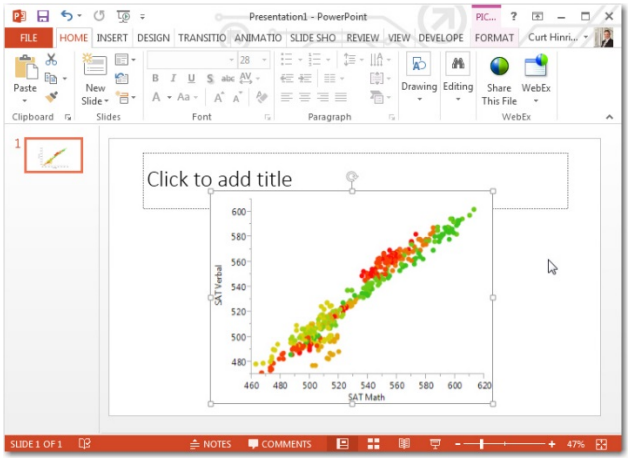
3. Right-click and select **Copy** (see Figure 7.12).

Figure 7.12 Copy the Graph



4. Open the desired application (PowerPoint in this case), open the presentation and create a slide with the desired layout, right-click, and select **Paste** (see Figure 7.13).

Figure 7.13 Pasting the Graph in PowerPoint



Note: When using JMP's default paste function, the object can be re-sized in the new application, but the contents are fixed.

Special Paste Functions

JMP also provides additional paste formats and functions. Sometimes you might want to edit the results or graph, or you need to provide the graph in a specific format. JMP has some options for you:

- Use **Paste Special**. This allows you to paste a .bmp or metafile (see Figures 7.14 and 7.15). **Metafile** is a vector art graphics format that can be edited in detail by Word and PowerPoint on Windows.

Figure 7.14 Paste Special

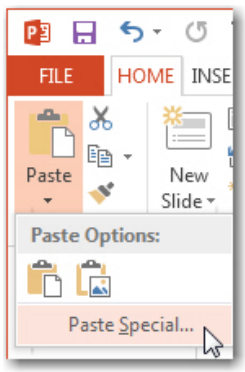
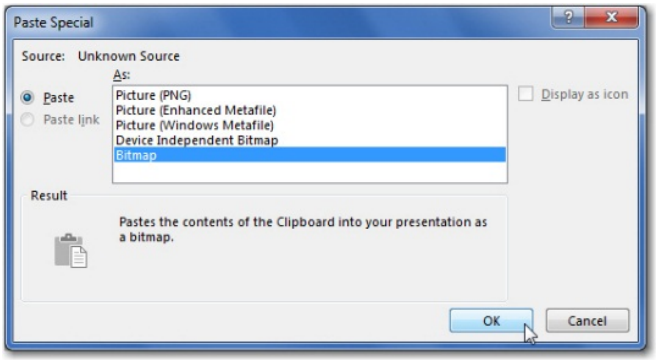


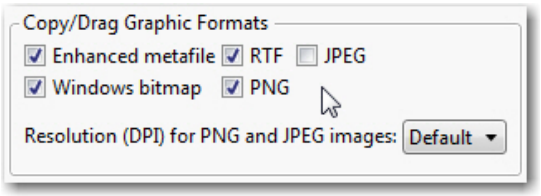
Figure 7.15 Paste Special Options



- If you want to change the formatting options of the graphs you want to paste, change your preferences to reflect this:

1. Select **File ► Preferences**.
2. Select **Windows Specific** (or **MAC OS**).
3. Under Copy/Drag Graphic formats, check the formats you want as options or remove options by unchecking them. (see Figure 7.16).

Figure 7.16 Changing Formatting Options



Note: if you simply wish to save your results in a certain file type, such as a JPEG, EPS, PDF or HTML, you can simply select **Save As ► type** from the File menu. As an example, we will describe how to save an interactive HTML file in the next section.

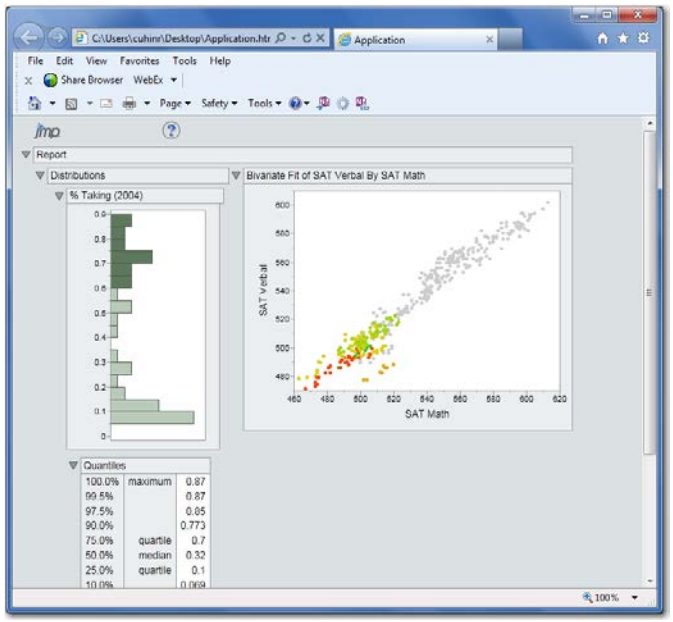
7.4 Sharing Dynamic Graphs with HTML 5

Because JMP's graphs are dynamic and allow you to interact with them to see interesting things, it is reasonable that you'd want to share these insights with others –including those who may not have a copy of JMP. JMP has introduced an important new way to share your discoveries with others by letting you save your reports as interactive HTML 5. HTML 5 is a standard format recognized by all browsers and enables you to save your results which maintain much of the same interactivity you get in JMP.

To save a report as HTML 5 simply generate the results you would like to share, go to **File ► Save**, select **Interactive HTML with Data** (under **Save as Type**), rename and locate your file as desired and select **Save**.

You may simply send this file, which can then be opened with any browser (Figure 7.17).

Figure 7.17 Interactive HTML5 Output in a Browser



Note: At the time of publication, Interactive HTML has been implemented in many, but not, all platforms in JMP.

Important Note about HTML 5 in JMP: When a file is saved in this file type, the data accompanies the graph. Although it may not be obvious, the data can be extracted from the file and the data cannot be encrypted. Thus, we caution users to be careful with sensitive or confidential data. Internal presentations may not be at issue, but sending Interactive HTML files externally may compromise confidentially.

7.5 Creating and Sharing Animated Graphs

Animated graphs are used to express trends among column/variable relationships. JMP provides the means to communicate a clear impression from a complex model and, with the Bubble Plot and Profiler, you can export these to other documents through Adobe Flash (or HTML5 in later releases). Unlike other graphs that are static or require direct interaction, this class of graphs brings an entirely new dimension to your presentation by allowing you to view changes in data relationships over time, range, or category– very much like a movie, except now, you can also interact with them.

In this section, we show you how to create a Flash file of a Bubble Plot and, in the following section, how to access and use these files in a PowerPoint presentation. We use the Bubble Plot to illustrate this section with time series data (steps for the Profiler are essentially the same; see Chapter 3). Animation of time series data is particularly effective because changes can be reflected over time.

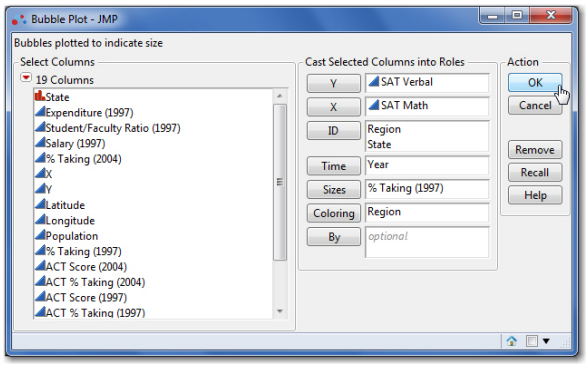
Creating Exportable Animated Graphs

The Bubble Plot is found in the **Graph** menu. The window is shown in Figure 7.16. As you see, the Bubble Plot allows you to visualize up to six variables at once, including x- and y-axes, time, size of bubble, color gradient, and By. JMP allows you to also select a second subgroup for your points. For example, if you have sales data that contains both individual territories and regional information, you might want to toggle between **Region** and **Territory**. This feature allows you to see the graph for the region and to split that region into its territories on the fly to see their individual performance.

Note: If you have JMP 12, Interactive HTML 5 is an alternative to Flash for sharing a Bubble Plot with others. However at the time of publication, HTML 5 output is not compatible with PowerPoint. Since we will illustrate how to integrate an animated Bubble plot with PowerPoint in the next section, we will use Flash in this section. Check the book's website for important updates concerning this and other sections.

1. Open your data file and select **Graph ► Bubble Plot**. Select columns to match desired roles (described in the following box). Only Y and X are required, but you will want to do more with this platform (see Figure 7.18).

Figure 7.18 Bubble Plot Launch Window

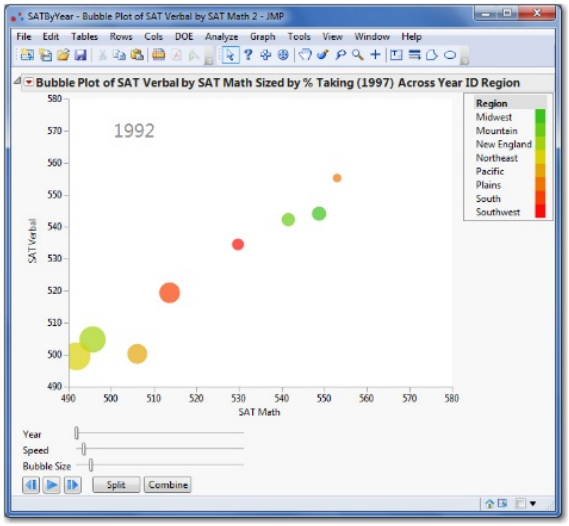


2. Using our chapter example SAT by Year, assign these columns to their corresponding roles:

Column		Role
SAT Verbal	>	Y
SAT Math	>	X
Region	>	ID
State	>	ID
Year	>	Time
% Taking	7) >	Sizes
Region	>	Coloring

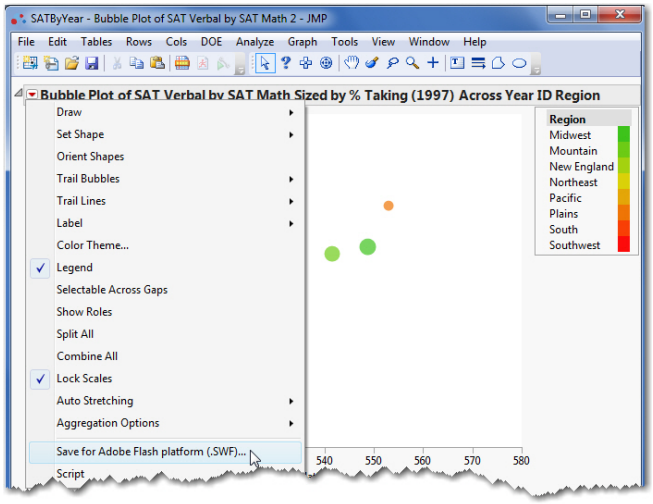
3. After selecting **OK** (see Figure 7.19), locate the red triangle in the top left corner.

Figure 7.19 Bubble Plot Results



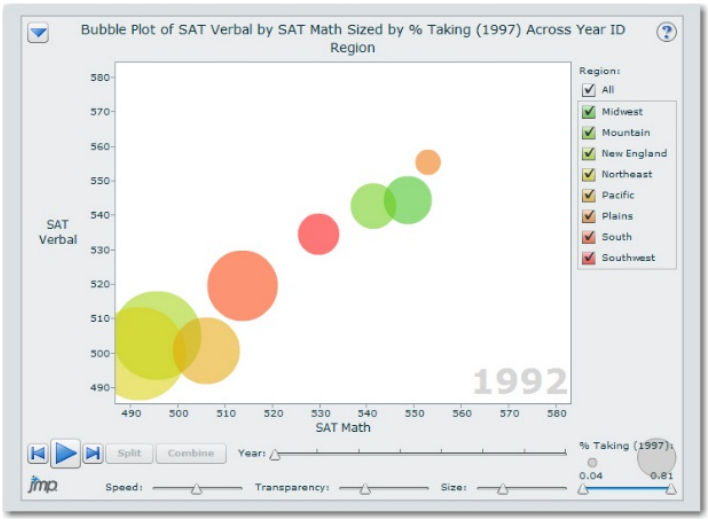
4. Under the red triangle, there is a menu option, **Save for Adobe Flash platform (.SWF)** (see Figure 7.20). Selecting this option saves the file (as a .swf file), along with a companion .htm file. Name your file and select **Save**.

Figure 7.20 Saving as a Flash File



5. It takes a few seconds, but you will see that a Flash version of that Bubble Plot now automatically appears in your browser on your desktop (See Figure 7.21). You can post this to a website.

Figure 7.21 Flash File Seen Through Browser

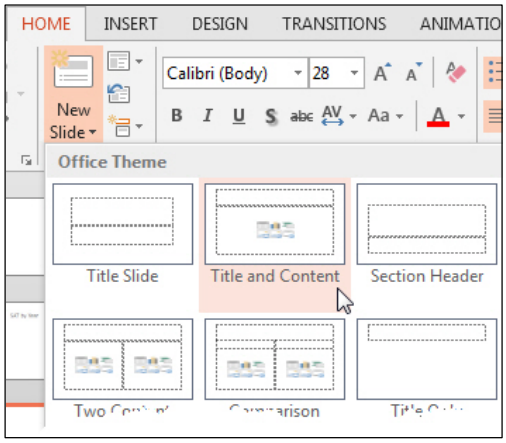


7.6 Placing Animated Graphs into PowerPoint

Before moving these graphs into PowerPoint, first save your graph as a Flash file as described in the previous section. Once you have done this, you will point to the .swf file from inside PowerPoint. This process is a little more involved, but we walk you through the steps here:

1. Save your Shockwave Flash (".swf") file. Note the file name and location on your computer.
2. Create a PowerPoint slide with a layout format that provides ample room to display the animation, such as Title and Content (see Figure 7.22).

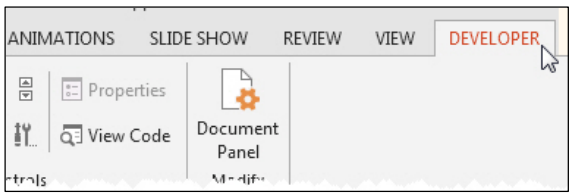
Figure 7.22 Specifying Theme of a New Slide



3. Once you have created the PowerPoint slide, select the **Developer** tab (see Figure 7.23). If you don't see the Developer tab, go to **File ► Options**, select **Customize Ribbon**, select **Developer**, then click **OK.**

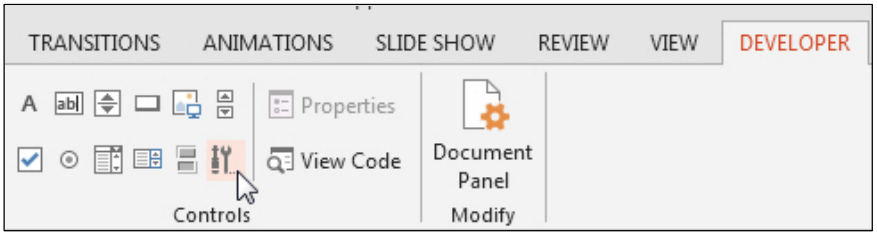
   Note: We are illustrating this example with PowerPoint 2013 and Flash 13. If you use other versions of PowerPoint or Flash, the steps may differ. If needed, consult PowerPoint Help or www.jmp.com/support/swfhelp/en/.

Figure 7.23 The Developer Tab



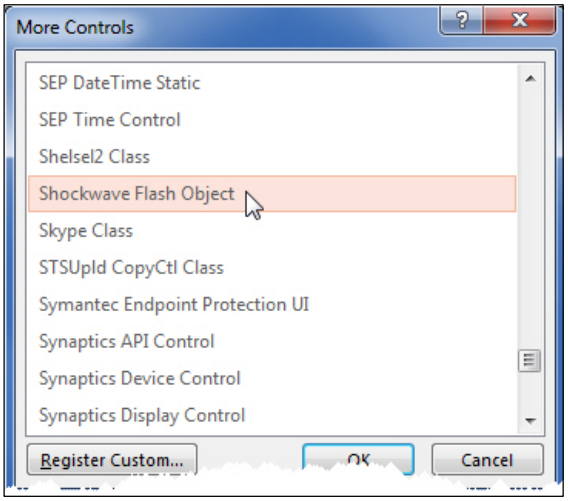
4. Select (the More Controls icon) to view more controls (see Figure 7.24).

Figure 7.24 More Controls Icon



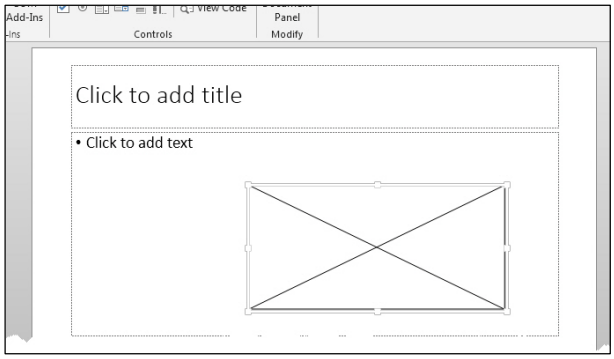
5. Select **Shockwave Flash Object**, and click **OK** (see Figure 7.25).

Figure 7.25 More Controls Dialog Window



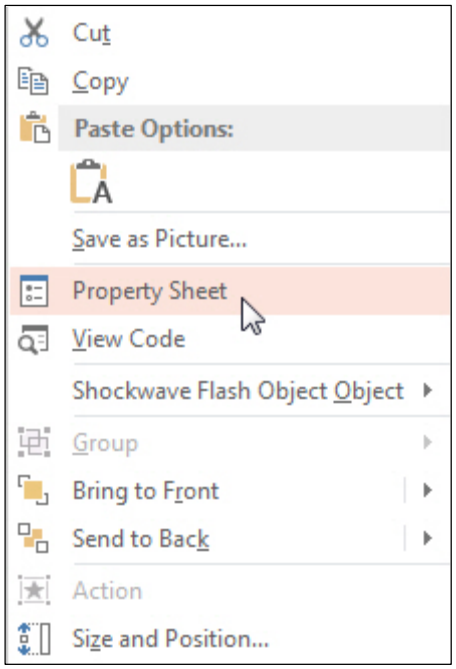
6. On your PowerPoint slide, adjust the box to the right size to accommodate the Flash-animated graph. For this example, enlarge this small box (the one with the lines bisecting each corner or what looks like an X) to fill the entire lower panel of the slide (see Figure 7.26).

Figure 7.26 Specify the Size of a Flash Object



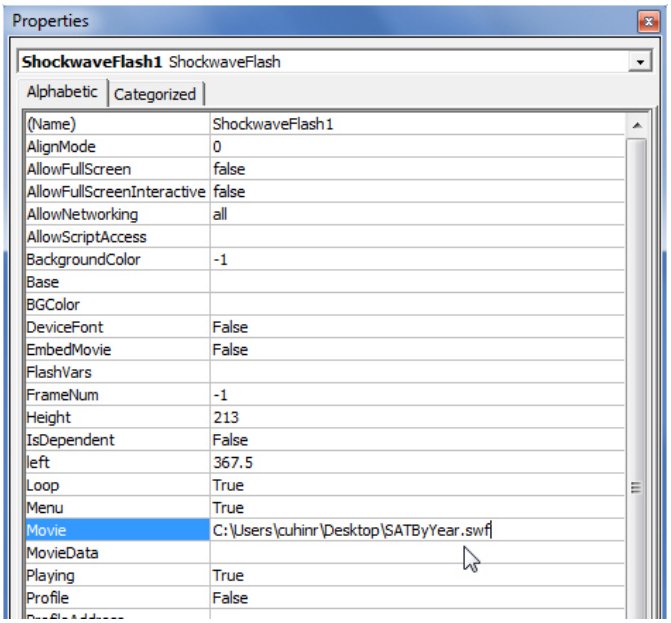
7. Right-click inside the box and select **Property Sheet** (see Figure 7.27).

Figure 7.27 Select the Property Sheet



8. Under **Movie**, specify the path to your .swf file (see Figure 7.28).

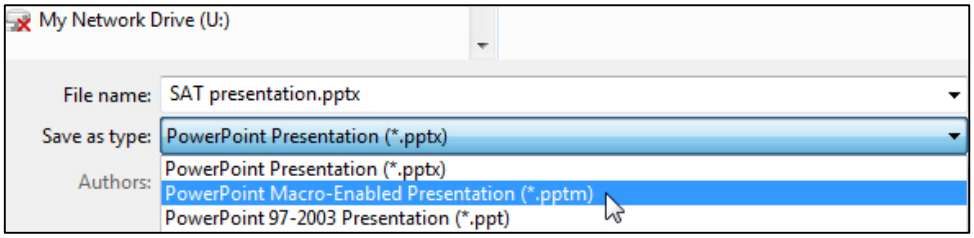
Figure 7.28 Specify the Path to the Animated File



9. When you are finished, close the window. Save your PowerPoint file.

   When saving this file, be sure to identify it as PowerPoint Macro-Enabled Presentation in the **Save as type** drop-down menu, as shown (see Figure 7.29).

Figure 7.29 Save as a Macro-Enabled PowerPoint File



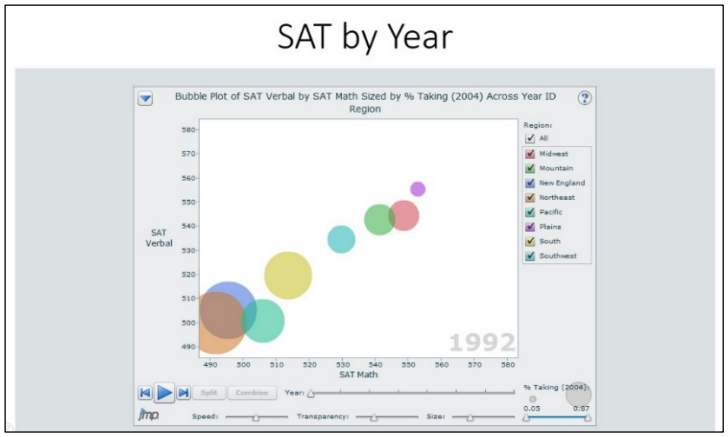
10. Select the Slide Show or Presentation icon (see Figure 7.30) and view the slide.

Figure 7.30 Select Slide Show Mode

Figure 7.31 Click Play

11. Click **Play** to see the animated graph in action (see Figure 7.31).

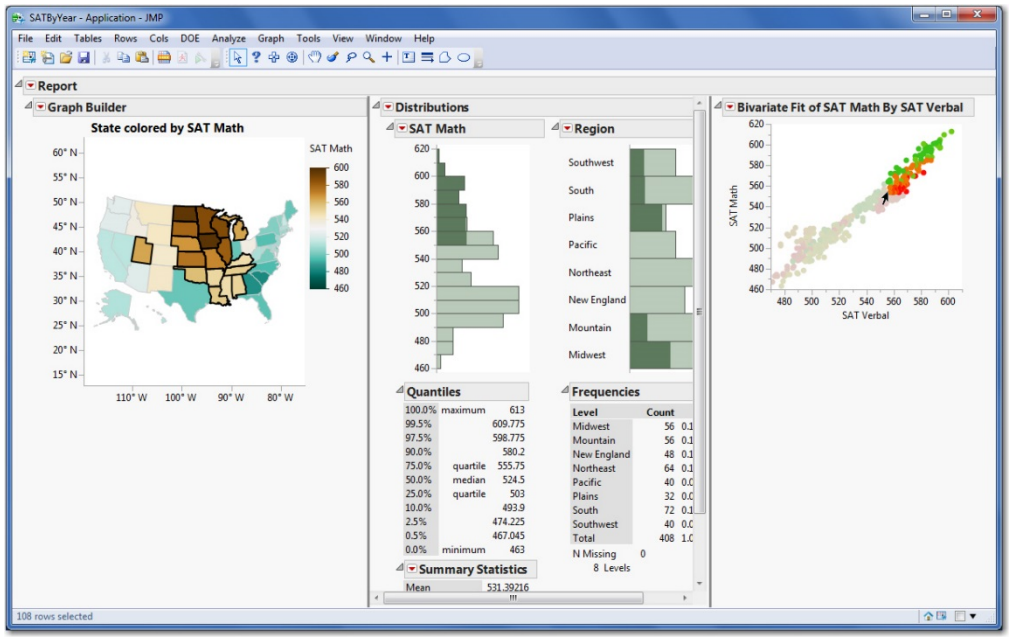
Figure 7.31a Flash Object Embedded in PowerPoint



7.7 Creating Dynamic Dashboards

One of the fundamental features of JMP is its ability to dynamically link multiple graphs and view selected rows or observations in any number of graphs. In the past, the challenge with this has been trying to understand relationships by viewing multiple graphs due to their separate platforms and results windows. A new feature allows you to combine several separate windows into a single window or dashboard (JMP documentation refers to these as Instant Applications, see Figure 7.32). You can save these dashboards as a script or application and, in most cases, as HTML 5 and share them with others. Let's revisit the SATbyYear data table. In this data set, we'd like to better understand SAT Math scores spatially in a map, but also in relationship with SAT Verbal scores.

Figure 7.32 Creating a Combined Window



The first step in creating a dashboard is to create the individual graphs.

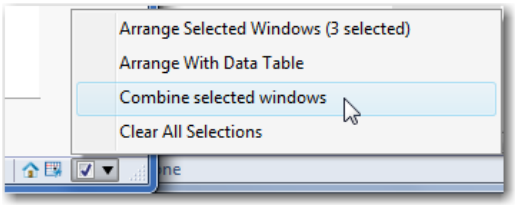
- In this example we created a Map in Graph Builder (**Graph ► Graph Builder**) by dragging the State column into the lower left box titled "Map/Shape" and then dragging SAT Math into the resulting US map.
- We created distributions of SAT Math and Region by using the Distribution platform (**Analyze ► Distribution**) and placing SAT Math and Region into the **Y Columns** role and selecting **OK**.
- We created the scatterplot of SAT Math and SAT Verbal by using Fit Y by X (**Analyze ► Fit Y by X**) and casting SAT Math into the **Y** role and SAT Verbal into the **X** role and selecting **OK**.

Once we have created the three graphs which will initially appear in separate windows, our next task is to combine them into a single results window*. To combine windows, locate the check box in the lower right hand corner of the results window (on Windows only), check the box (Figure 7.33), and repeat for the other windows you desire to include.

Figure 7.33 Select Windows to Combine

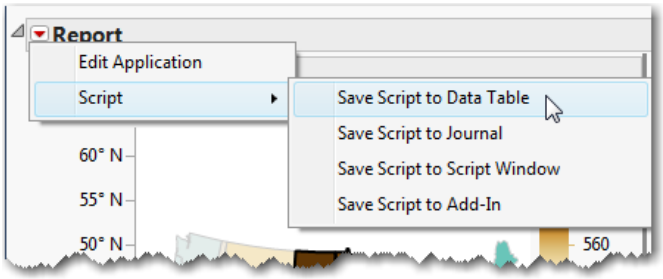
Next to the check box on one of the results windows, there is a down arrow button indicating a submenu. Click and Select, **Combine selected windows** (Figure 7.34).

Figure 7.34 Combine Selected Windows



This will create a single results window or dashboard that maintains the dynamic and interactive graphs you're used to in JMP (see Figure 7.32). To save this dashboard to the data table, select **Reports ► Script ► Save to Data Table** (Figure 7.35 and see also section 7.9).

Figure 7.35 Saving the Dashboard to the Data Table



Note: The order in which the results display in the combined window (from left to right) corresponds to where your original individual windows appeared on your screen (from left to right) before you combined them. To specify a different order to the graphs in the combined window, click and drag your original results windows so they appear in the desired sequence from left to right. Then repeat this process of combining selected windows.

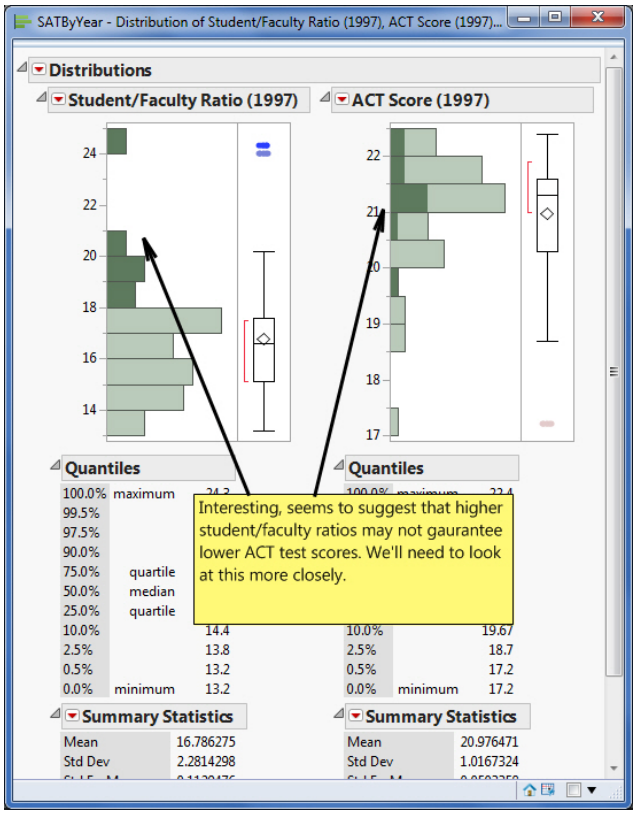
7.8 Using the Layout Option to Share Results

If you'd like to edit, annotate, or consolidate a series of static JMP graphs or results before printing, sharing, or moving them into another application, use the Layout platform. Layout lets you create a collection of graphics or output that you can print or paste elsewhere. The usual interactive JMP graphs are fixed after being placed into Layout.

Layout allows you to ungroup the elements of graphs and results and piece together only those components you need for your presentation. For example, if you'd like to change the title, remove some pieces of an analysis report, combine multiple graphs from different JMP platforms, or add annotations to elements of multiple graphs, Layout can help.

Let's illustrate this feature with our SAT example introduced earlier. We used the Distribution platform to create histograms of ACT Score (1997) and Student/Faculty Ratio, noting that there might be a positive relationship between these columns (as the Student/ Faculty ratio increases, so do the ACT scores) (see Figure 7.36).

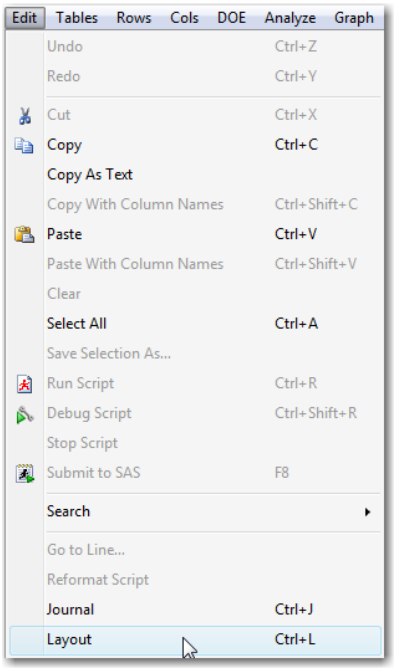
Figure 7.36 JMP Results Window with Annotations



So let's answer this question by adding the appropriate additional analyses to the mix. Our first step after annotating the distribution screens is to move these graphs into the Layout platform:

1. Choose **Edit ► Layout** (see Figure 7.37). A new window is created with the header Layout: Untitled. Think of this new window as a poster board that you can lay out and annotate any way you like. Now you can add other graphs to this window with ease.

Figure 7.37 Launching the Layout Platform

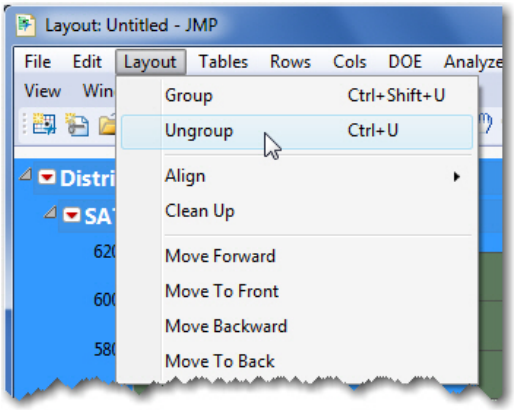


2. Return to the data table and select **Analyze ► Fit Y by X.** Then select for **ACT Score (1997)** for Y and **Student/Faculty Ratio (1997)** for X. Click **OK**. From the red triangle, select **Fit Line** and observe the graph depicted by the relationship. Does it appear from the graph that as one column increases, another also increases or decreases? We now want to combine this scatter plot with the histograms noted earlier.

With the Selection tool, highlight the graphs and output you want to combine with the distribution graphs. Right-click and select **Copy**.

3. Go to the Layout: Untitled window, right-click, and select **Paste**.
4. With the Arrow tool, choose **Edit ► Select All**. An entirely new drop-down menu, **Layout**, appears. Choosing **Layout ► Ungroup** (see Figure 7.38) allows you to keep, delete, or move the elements of these two separate analyses as you see fit.

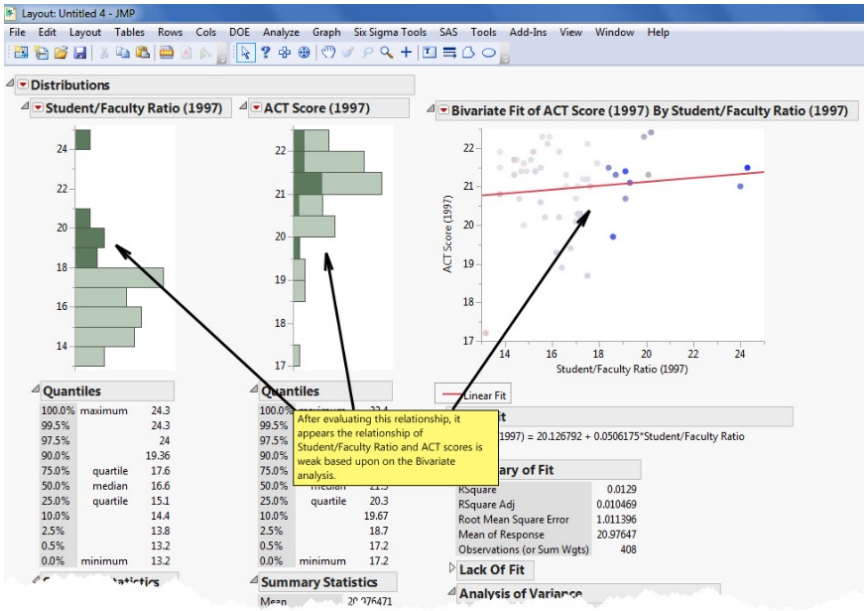
Figure 7.38 The Layout Menu



a. Use the Selection tool to change or delete individual elements.
b. Use the Arrow tool to move things around.

c. Use Lines, Annotate, and other tools as you would in other platforms (see Figure 7.39).

Figure 7.39 Layout with Multiple Platforms and Annotations



Note: With the Layout menu, you may also Align the contents into an efficient layout or Group (or regroup) separate elements together.

7.9 Using Scripts to Save or Share Work

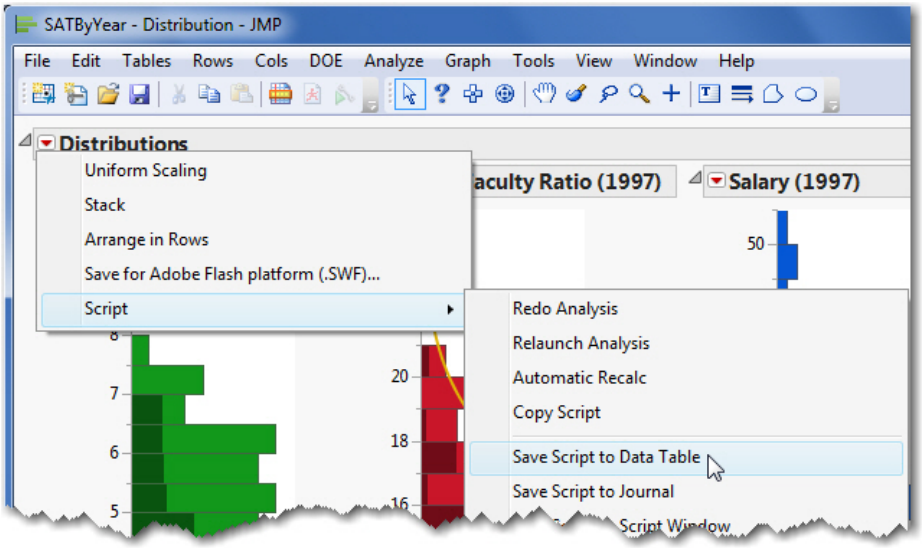
In the previous section, we learned that Layout helps us capture static elements of our results, which is useful when you are pasting them into another document. However, you'll often want to save or share your results in a manner that allows you to generate the same results again. JMP Scripts capture steps, procedures, or actions which allow you to re-run your past work, perhaps on new data. For example, you may want to go back to an analysis you've done and possibly add additional rows to your data table and re-run the analysis or even share your work with another JMP user who can pick up where you left off.

One of the things that even experienced JMP users might not realize is that JMP has a built-in scripting language that is always available in the background, which can capture the work you've done so that you can easily reproduce it. It is available by simply clicking on the menu.

From the red triangle of any platform or graph is a **Script** option. When selecting the **Script** option, you see a submenu.

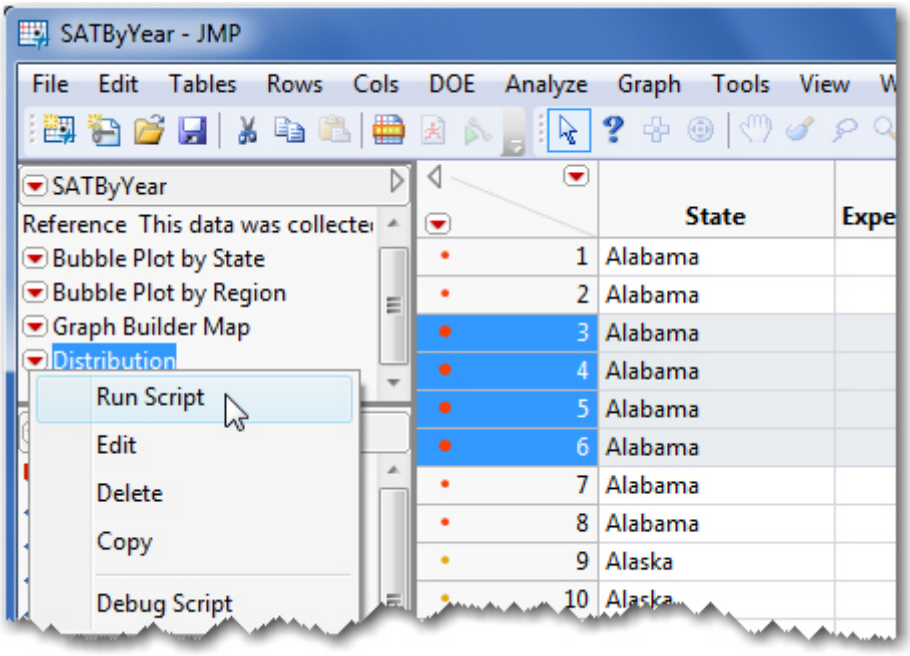
1. **Save Script to Data Table** translates this work into a script and places it in the Table panel of the Data Table (see Figure 7.40).

Figure 7.40 Saving a Script to a Data Table



2. Go to the Data Table. You now see an item with the title of the platform you were using in the Table panel with a red triangle (Distribution in this case). Select the red triangle next to Distribution and **Run Script** (see Figure 7.41). You should now see exactly what you developed before. You can save and share these scripts (within the Data Table, in this case) or rename them to something more descriptive.

Figure 7.41 Running the Script



7.10 Summary

In this chapter, we learned about communicating with graphs and about sharing JMP graphs and files and moving them into other applications. We only scratched the surface in this chapter, but we tried to address the most common and basic functionality. As mentioned, there are excellent reference materials available, both within JMP and from books dedicated to these topics. For more information, see the Bibliography section in this book.

Chapter 8 Getting Help

8.1 The Help Tool

8.2 The Help Menu

8.3 The JMP Starter

8.4 External Resources

We designed this book to be a quick overview of the most commonly used features of JMP. As you begin to use JMP more frequently or dig deeper into its analytics, you will inevitably have questions or find places where you'll want to learn more. This chapter is designed to help you navigate the extensive resources available to help you answer those questions. We begin with JMP's wide range of built-in resources.

Within JMP lives a wide variety of documentation, tutorials, and other support materials. It is highly recommended that you learn about these, because they were designed to assist you while you are using the software. These tools offer the most immediate answers.

Should you need additional assistance, other resources include outstanding technical support, teaching resources, live and on-demand Webcasts, user groups, blogs, and even social networking initiatives. JMP hosts dozens of events each year, including its annual Discovery Conference. JMP training courses are also available through SAS Education; many books, ranging from elementary to advanced, are referenced in the bibliography at the end of this book.

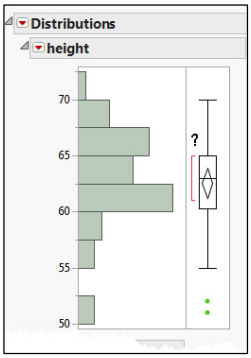
8.1 The Help Tool

When you have any question about what you're looking at in a JMP window, what a graphic or result is, or how it was generated, the Help tool (also known as the question mark tool or the **?** icon) in the Toolbar or the **Tools** menu is a great place to start.

The Help tool provides a direct link to the documentation concerning the item in question.

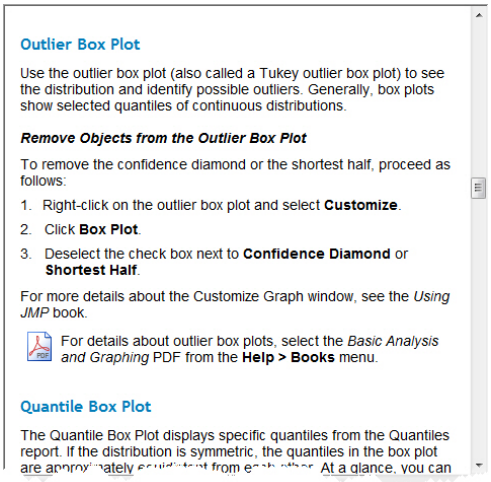
1. Select the question mark from the toolbar and move the transformed "**?**" cursor to the item in question (see Figure 8.1).

Figure 8.1 The JMP Help Tool



2. Click on the item in question, and JMP reveals the specific place within the documentation that addresses the concepts or output at hand (see Figure 8.2).

Figure 8.2



Outlier Box Plot

Use the outlier box plot (also called a Tukey outlier box plot) to see the distribution and identify possible outliers. Generally, box plots show selected quantiles of continuous distributions.

Remove Objects from the Outlier Box Plot

To remove the confidence diamond or the shortest half, proceed as follows:

1. Right-click on the outlier box plot and select **Customize**.
2. Click **Box Plot**.
3. Deselect the check box next to **Confidence Diamond** or **Shortest Half**.

For more details about the Customize Graph window, see the *Using JMP* book.

For details about outlier box plots, select the *Basic Analysis and Graphing* PDF from the **Help > Books** menu.

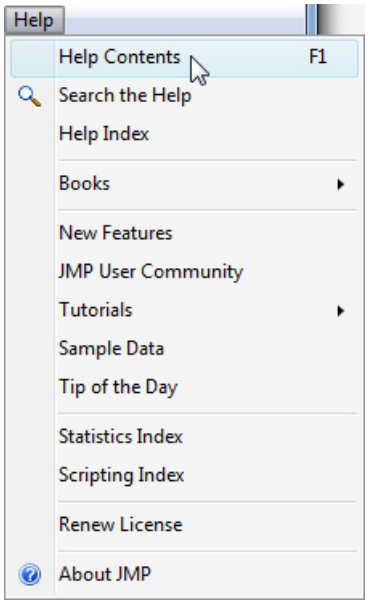
Quantile Box Plot

The Quantile Box Plot displays specific quantiles from the Quantiles report. If the distribution is symmetric, the quantiles in the box plot are approximately equidistant from each other. At a glance, you can

8.2 The Help Menu

From the **Help** menu (see Figure 8.3), you can access specific help topics along with all of JMP's documentation, tutorials, sample data, and shortcuts to every statistical routine in JMP. Let's define each of these items in the menu.

Figure 8.3 The JMP Help Menu



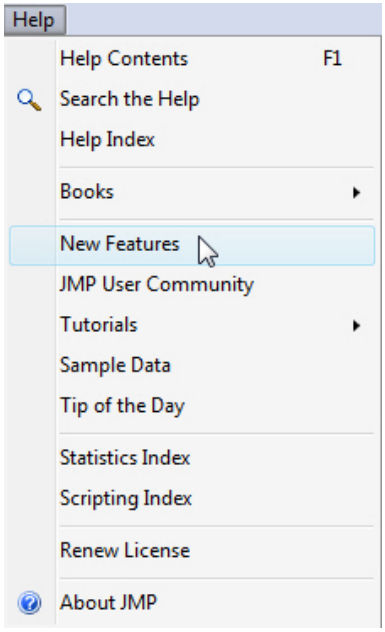
Contents, Search, Index

Each of these first three items is designed for navigating the user documentation:

- **Help Contents** provides a Table of Contents view.
- **Search the Help** provides a hypertext search that allows you to search for all the content related to a topic or phrase.
- **Help Index** lists the key terms and allows you to find a topic in the user documentation.

Note: If you are an existing user who has updated your version, there is a handy summary of new features to the version you are using. Select **Help ▶ New Features** (see Figure 8.4).

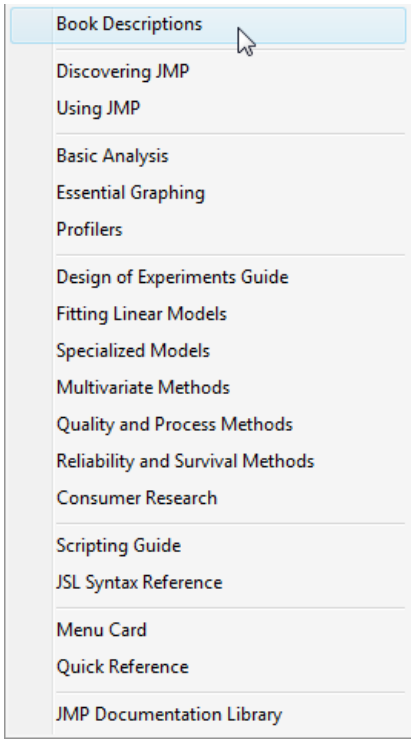
Figure 8.4 New Features in JMP



Books

This menu links to PDF versions of each of JMP's user manuals (see Figure 8.5). More than 4,300 pages of JMP reference material are right at your fingertips. You can search and print any of these manuals. Handy quick-reference cards that you can print are also included. The *Discovering JMP* book is a logical next step for the reader of this book, as it covers the basics in slightly greater depth.

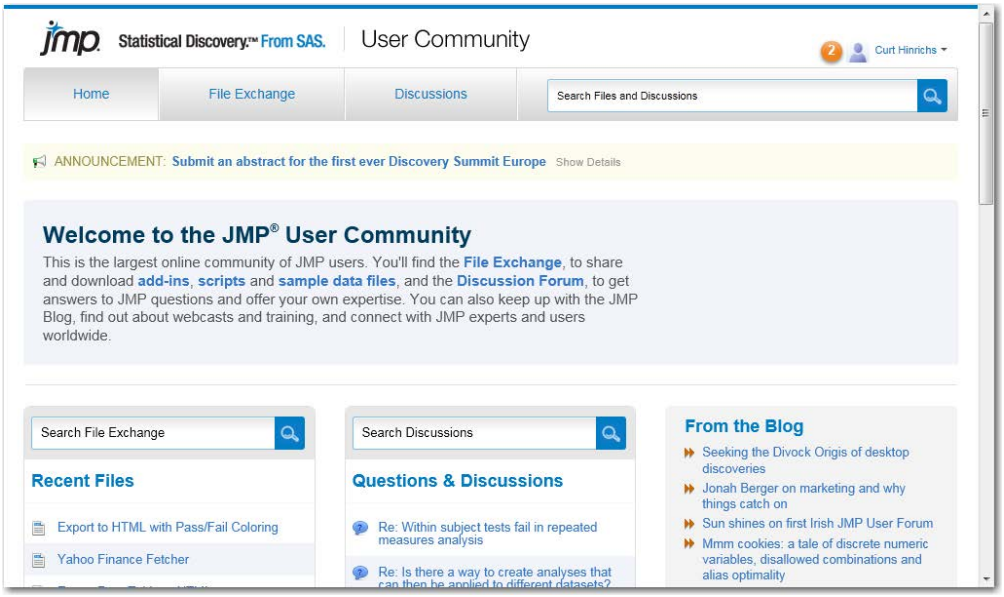
Figure 8.5 Documentation Included with JMP



JMP User Community

This menu item links to the JMP User Community (Figure 8.6), which is a portal where you can access a wide array of resources such as the File Exchange, JMP Discussion Forum, the JMP Blog, Learning Library, Technical Support, and the Knowledge Base. This is also where you can connect with other users through user groups or learn about upcoming events.

Figure 8.6 The JMP User Community

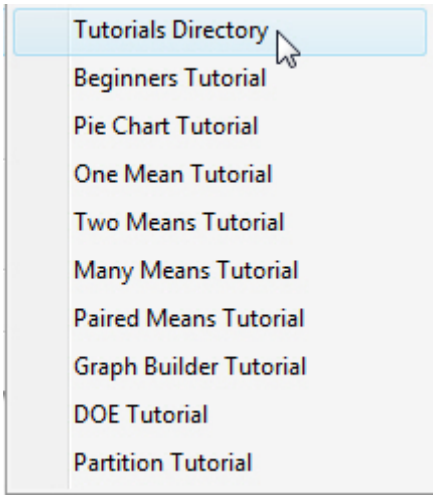


- The File Exchange is a database of user-developed scripts, helpful tools, and custom add-ins.
- The JMP Discussion Forum is where you can ask the experts about perplexing performance.
- Web Resources includes the Learning Library where you can find brief guides for common statistical methods.
- The Knowledge Base where common issues and their solutions are posted is located in the Support section.

Tutorials

The tutorials cover the more commonly used statistical tools in JMP (see Figure 8.7). Each of these can be completed quickly to help you not only master the JMP steps, but also review the concepts involved. Each tutorial walks through the concept at hand, illustrating how to perform the analysis with JMP while explaining what each step does.

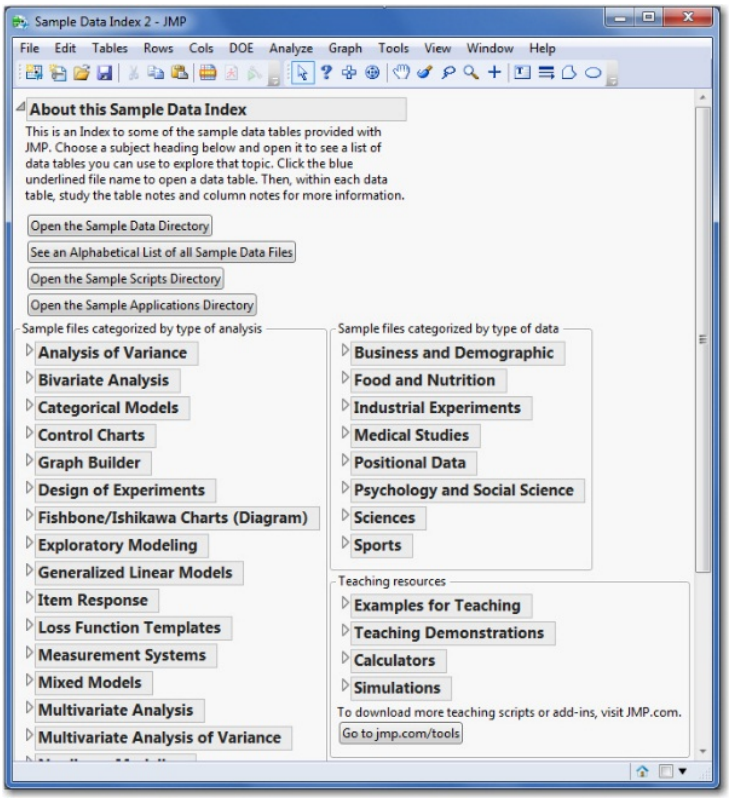
Figure 8.7 JMP Tutorials



Sample Data Index

The Sample Data Index contains all of the data used in this book as well as in the JMP documentation (see Figure 8.8). It provides a convenient place to access data sets that you might want to employ to see how JMP performs a specific method or type of problem, or to use for teaching purposes. It also contains other examples that might be of interest and provides a guide for how your data should be structured for a particular analysis.

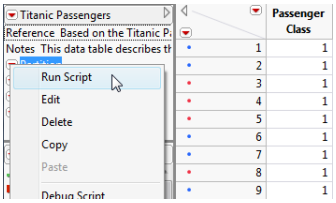
Figure 8.8 Sample Data Index



The sample data contains over 400 data sets. Many of these contain scripts that will generate an appropriate analysis, including those illustrated in the JMP documentation. Click on a gray triangle to disclose the data tables under a category of interest. To generate an analysis using a script, open a data table and locate a script in the Data Table's top left panel.

- Click on the red triangle next to a script and select **Run Script** (see Figure 8.9).

Figure 8.9 Running a Script from a Data Table

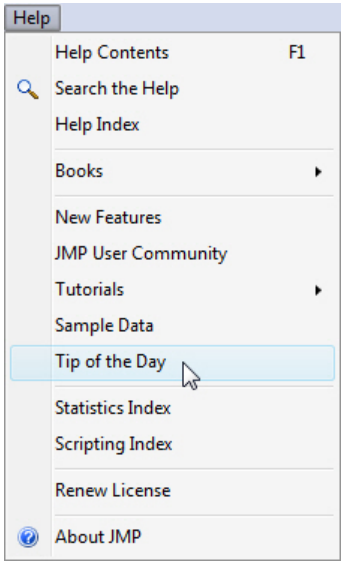


Note: The **Open the Sample Data Directory** and **See an Alphabetical List of all Sample Data Files** buttons in this window allow you to quickly locate and access a data table, including those used in this book.

Tip of the Day

JMP has summarized the most common and important FAQs into a collection of 52 *Tip of the Day* boxes (see Figure 8.10). Often these are somewhat hidden features, but worthy of special mention in this form.

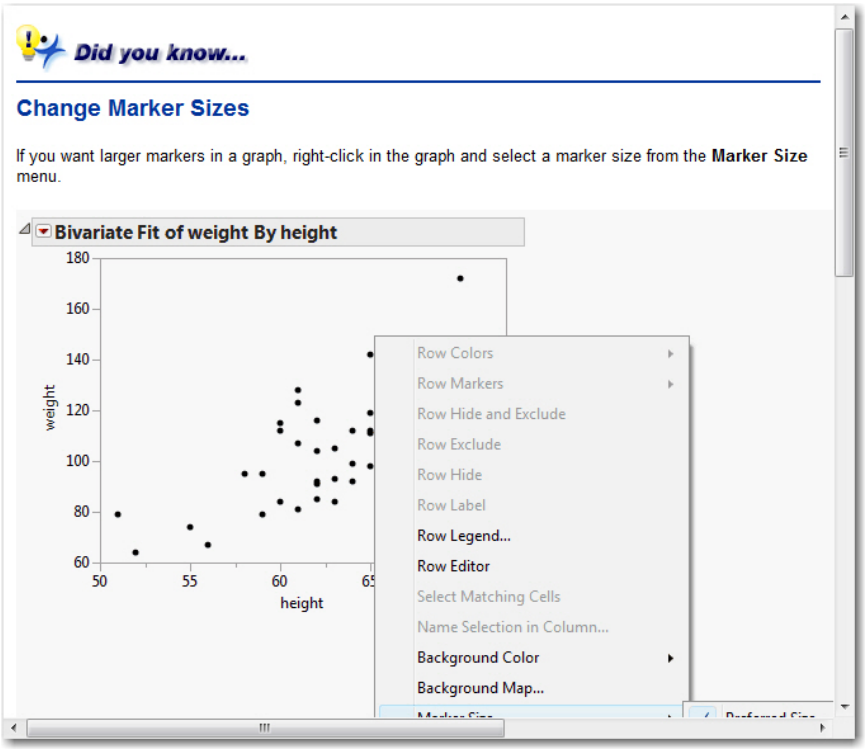
Figure 8.10 Tip of the Day



After JMP is installed, Tip of the Day boxes will appear each time you launch JMP (see Figure 8.11). We encourage you to review these at least once, because many of them will save you valuable time and frustration in the long run. If you do not want these to appear at launch, you can remove

this default by un-checking the box at the lower left corner of the window or by selecting **File ▶ Preferences ▶ General** and unchecking the **Show the Tip of the Day at Startup**.

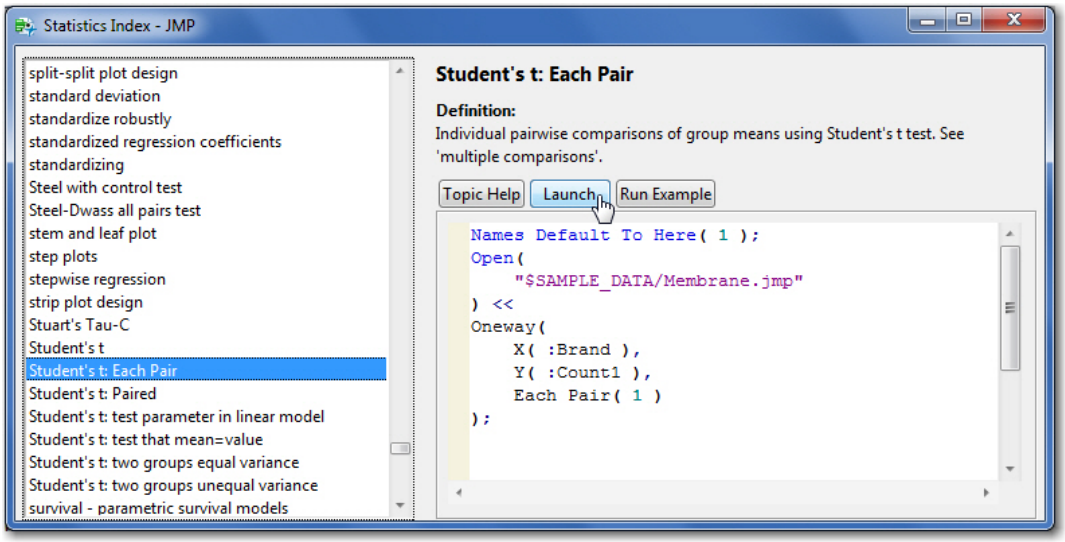
Figure 8.11 Tip of the Day with JMP Launch



Statistics Index

The Statistics Index is the next item on the Help menu. Select **Statistics Index** when you know which statistical method you need but not where to find it in JMP (see Figure 8.12).

Figure 8.12 Statistics Index

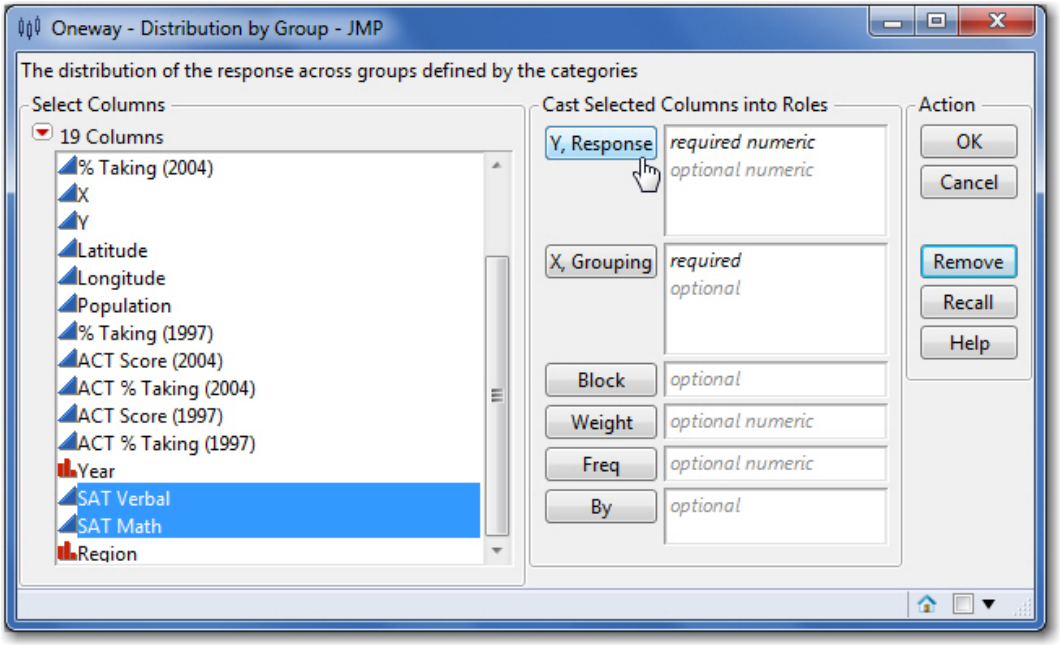


This feature provides the following information:

- A comprehensive and alphabetical index of statistical tools included in JMP.
- Brief definitions of statistical terms with links to more comprehensive information through the **Topic Help** button. In some cases, an **Example** button is available to illustrate the technique.
- A direct shortcut to the JMP window or platform which generates a desired statistic. Select **Help** ▶ **Statistics Index** ▶ *[choose statistic]* > **Launch** (see Figure 8.13).

Note: When using the indexes as a shortcut, you need to first have your data open or follow the prompts to open it.

Figure 8.13 Launching a Platform from the Statistics Index



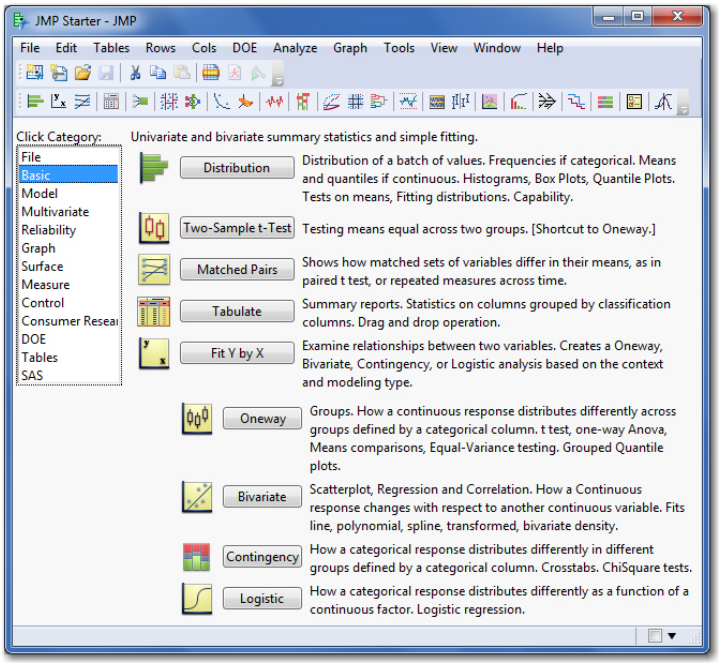
A Scripting Index follows the Statistics Index which provides a comprehensive library of JMP scripting functions with links to examples and help. The Scripting Guide (**Help ▶ Books ▶ Scripting Guide**) is an excellent resource for learning how to create custom applications using JMP Scripting Language (JSL).

8.3 The JMP Starter

The JMP Starter window provides an alternate means to navigate JMP along with a useful way to learn about the software. As discussed in Chapter 1, this book focuses on the native menus which appear at the top of the JMP window. For some users who may be familiar with traditional statistical software, the JMP Starter may be a helpful and natural interface to JMP. For these users, we cover the JMP Starter essentials in this section. The JMP Starter consists of 13 categories with corresponding topics within each category.

- Select **View ▶ JMP Starter** (see Figure 8.14). Click through the items in the **Click Category** box on the left and see the corresponding topics within each category to the right. Notice that it organizes topics slightly differently and provides a description of the graphs and techniques within each option.

Figure 8.14 The JMP Starter



- Click on one of the topic buttons and a corresponding window launches to generate that output (similar to the Statistics Index in the Help menu).

There is extensive overlap with the native menus we've been discussing to this point, but the JMP Starter provides a little more description of and within each topic.

8.4 External Resources

When you can't find answers within JMP's built-in resources, there are many external resources available. JMP has a very useful website, provides exceptional technical support, and offers many events (including the annual Discovery Conference, regional user groups, and free webcasts). Comprehensive JMP training is available through SAS Education at SAS Regional Training Centers, at company sites, and through the web. In this section, we introduce you to some of these resources.

The JMP Web Site

Think of the JMP website as the source for the latest information and a portal to the resources we discuss in this section. The website is a gateway to the **JMP User Community** which was described earlier, white papers, blogs, webcasts, and to downloads, such as maintenance upgrades. To learn about the latest JMP events and information, go to: http://www.jmp.com

JMP Webcasts

JMP webcasts are offered each week for new users of the software at no cost. JMP also provides less frequent webcasts on more advanced features of JMP. These are live events hosted by JMP systems engineers and developers. Recorded or "On-demand" versions are also available for your convenience. Go to: http://www.jmp.com/about/events/ondemand/

JMP Technical Support

JMP Technical Support provides support to registered users from e-mail and phone. JMP Technical Support is staffed by experts who not only know the software, but who are also well-versed in the methodology itself.

Tech Support by e-mail: **support@jmp.com**
Tech Support by phone: 1-919-677-8008

JMP Japan Tel: +81 3 6434 3782
JMP China Tel: +86 21 61633088
JMP Germany Tel: +49 6221 / 415 – 200
JMP France Tel: +33 (0) 1 60 62 12 12
JMP Italy Tel: +39 02 8313 4460
JMP UK Tel: +44 (0) 845 402 9907

You can also find many Frequently Asked Questions (FAQs) and known issues at the Knowledge Base: http://www.jmp.com/forms/knowledge_base.shtml

SAS Training for JMP

SAS Education offers several courses on JMP ranging from basic to advanced. These courses are offered at SAS Training Facilities throughout the world, at customer sites, and through the web. The courses offered at the time of publication include:

- Applied Statistics for Engineers
- ANOVA and Regression
- Categorical Data Analysis
- Analyzing Multidimensional Data
- Data Exploration
- Applied Statistics for Scientists
- Introduction to Categorical Data Analysis
- Analyzing Marketing Data: Going Beyond Tabs and Bar Charts
- Predictive Modeling Using JMP Pro
- Lean Six Sigma Green Belt Project Leader
- Quality by Design (QbD)
- Statistical Process Control
- Measurement System Analysis
- Modern Screening Designs
- Design and Analysis of Mixture Experiments
- Classic Design of Experiments
- Custom Design of Experiments
- Custom Designs for Industrial Experiments
- Product Reliability
- Analysis of Dose-Response Curves
- Stability Analysis
- Introduction to the JMP Scripting Language
- Application Development Using the JMP Scripting Language

For more information about these courses or to register, go to: http://support.sas.com/training/.

User Groups

There are several regional user groups that have regular meetings. User groups are a convenient way to connect with other JMP users and to learn about some of the software's latest developments. For more information, go to http://www.jmp.com/about/events/usergroups/

Appendix A • Integrating with SAS

A.1 Working with SAS Data

A.2 Working with SAS Programs

When you use JMP, you are using one in a family of SAS products. JMP is designed as a nimble and fast desktop data discovery tool for use by a broad audience. At the beginning, SAS software was created to provide analytics on mainframe computers when few other options were available. Over the years, as problem sizes grew, so did SAS software. 91 of the top 100 companies on the 2013 FORTUNE Global 500® list use SAS software on every size and type of problem.

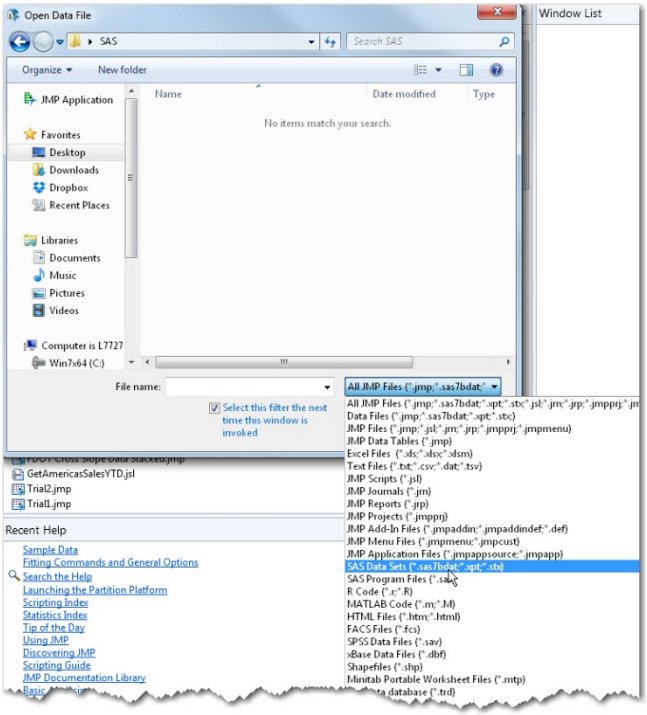
For many years, JMP has been an integrated client to SAS, not just a stand-alone product as described in this book. This means that as your problem size grows or extends beyond that which is suitable for JMP alone, you can migrate smoothly from desktop JMP to other tools from SAS. When analyzing SAS data with JMP, you first need to import that data into JMP, which we will describe in the next section.

A.1 Working with SAS Data

SAS data files can be stored either on your local machine or on a network drive. You open them using the **File** menu.

Select **File ▶ Open**. Select **All JMP Files** to disclose the file types**.** You see that JMP can read a variety of SAS data and file types (see Figure A.1). Locate the SAS file and select **Open**.

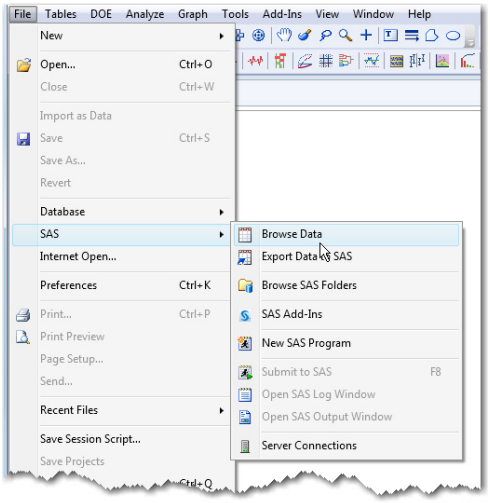
Figure A.1 Select SAS Data Sets



Many organizations connect different data sources using SAS to automate and streamline data integration. If you have an individual copy of SAS or if your organization has a SAS Metadata Server, then JMP can connect to data in these SAS frameworks.

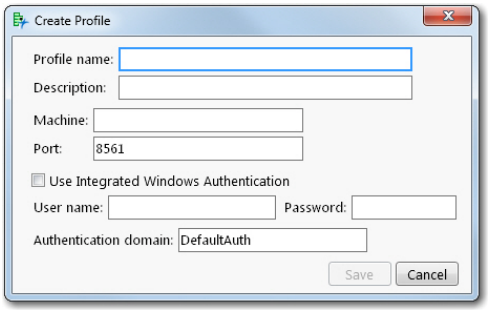
If you have a metadata sever or you have SAS installed locally, you can select **File ▶ SAS ▶ Browse Data** (see Figure A.2).

Figure A.2 Browse SAS Data



You are prompted to log into the SAS Metadata server (see Figure A.3). This is a powerful feature of SAS that provides a permission and security framework to keep data safe.

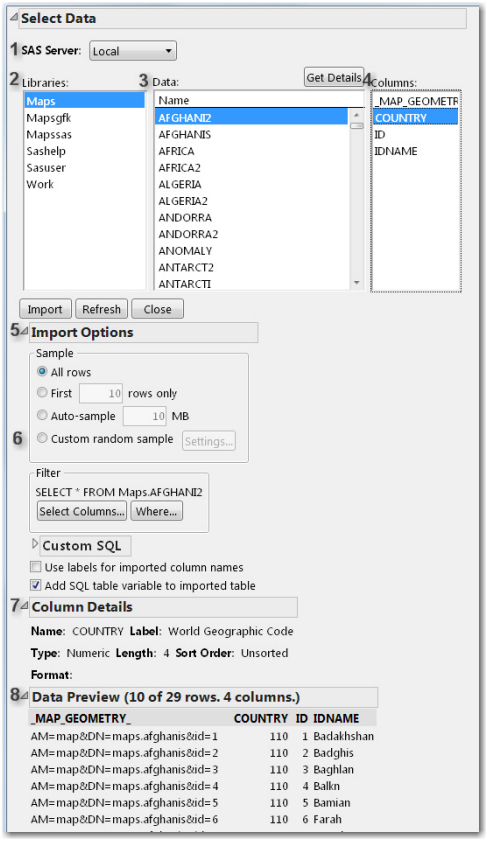
Figure A.3 Create SAS Profile



Type your user name and password and click **OK**.

There is a convenient data import window that has many useful features (see Figure A.4).

Figure A.4 Data Import Window



1. Local and network SAS **Servers** are shown in the SAS Server column.
2. SAS **Libraries** that you have permission to access are shown.
3. **Data** Tables within those libraries that you have permission to access are shown.
4. **Columns** within those data tables that you have permission to access are shown.
5. An interactive **Import Options** panel allows writing of SQL queries that can be run on the data. This is supported under the **Custom SQL** report tab.
6. A **Custom Random Sample** panel allows random sampling against the data. This is a critical feature when you are importing and managing large data sizes in JMP.
7. The **Column Details** panel shows details about the column(s).
8. A **Data Preview** panel details the file size in rows and columns before it is imported to JMP. This is also valuable when accessing especially large data sources that might be too large to fit on your desktop computer.

A SAS Metadata Server also provides tools to prepare and report data using server resources that JMP can leverage. We describe a few highlights in the next section.

A.2 Working with SAS Programs

SAS supports its own programming language that provides a rich set of tools, scaling to large and complex problems and bigger computer hardware.

When JMP has access to SAS, JMP can leverage these tools to extend its capabilities in several ways.

Opening a SAS Program

You may open an existing SAS program just as you would open any data file. Use **File ▶ Open** and then specify SAS Program in the file of type submenu (Figure A.5). Locate the program and then select **Open**. This will open the program within a SAS Program Editor window where you may further edit the program or simply submit the program to SAS (Figure A.6).

Figure A.5 Open a SAS Program

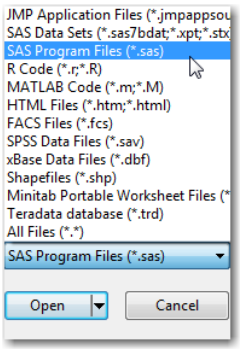
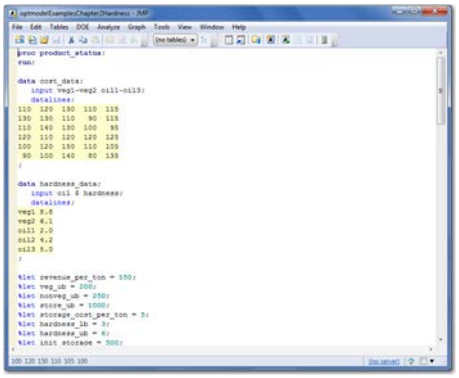


Figure A.6 SAS Program Editor



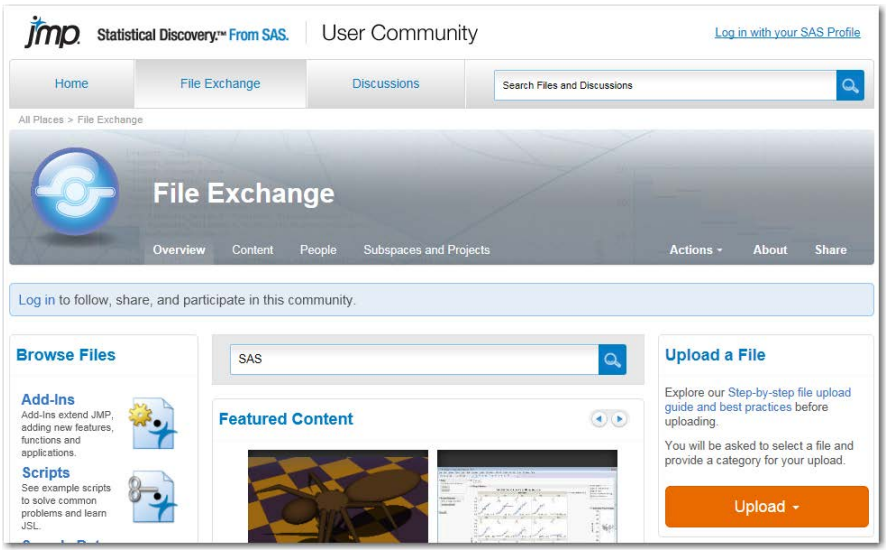
To submit to SAS, right-click within the window and select **Submit to SAS**.

SAS Add-ins

SAS Add-ins are SAS Programs that are packaged in a user-friendly JMP dialog window. SAS Add-ins allow you to take advantage of SAS' advanced analytics or custom applications conveniently and accessibly without needing to program in SAS. In this way, JMP can work as an easy-to-use client to SAS, assuming SAS is installed and connected.

A variety of SAS Add-ins are available for download from the File Exchange located at the JMP User Community (Figure A.7). Go to **Help ▶ JMP User Community** and click the **File Exchange Tab**.

Figure A.7 JMP File Exchange

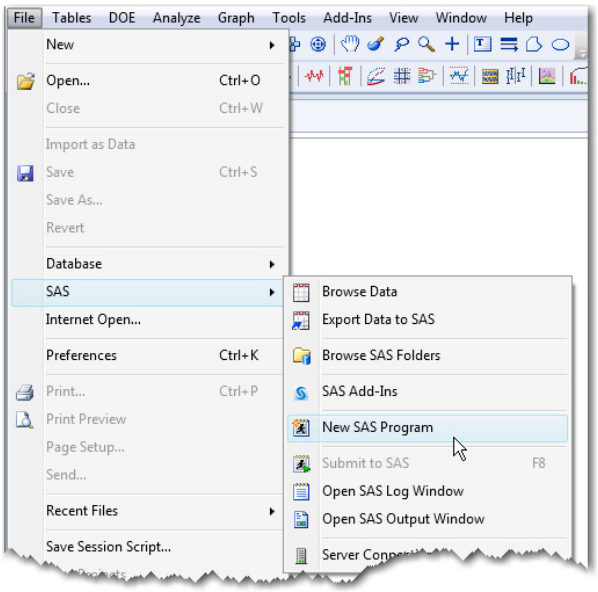


Note: You can create your own SAS Add-ins, which can be real time savers, but we recommend that you consult with a SAS expert.

Writing a SAS Program

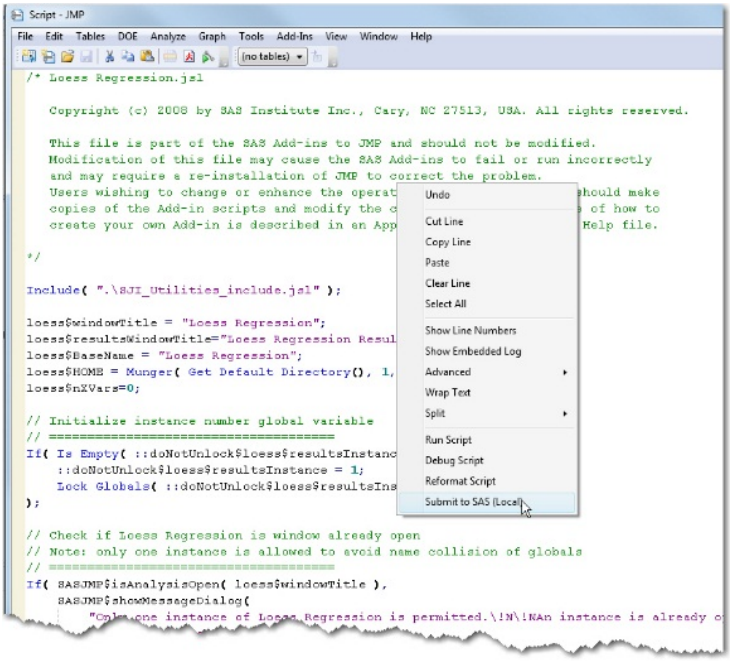
SAS programs can be written in JMP and submitted to SAS. In JMP, select **File ▶ SAS ▶ New SAS Program** (see Figure A.8).

Figure A.8 Write a SAS Program



A blank coding window appears. Type your program in the window. To submit the program to SAS, right-click in the coding window and select **Submit to SAS**. See Figure A.9, which contains a sample SAS program for illustration.

Figure A.9 Sample SAS Program



If you want to begin learning about SAS programming, we recommend *The Little SAS Book: A Primer* by Lora Delwiche and Susan Slaughter or *SAS for Dummies* by Stephen McDaniel and Chris Hemedinger.

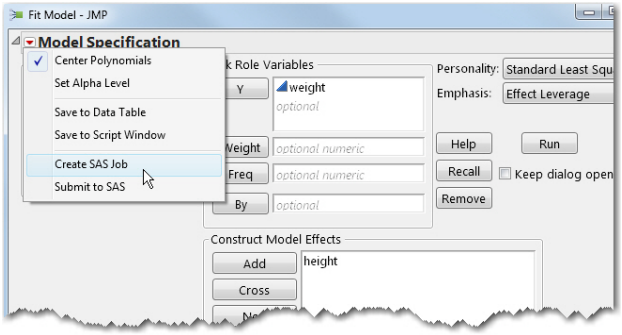
Automatically Generating a SAS Program

Some limited support for automatically generating a SAS program within JMP is available. Within the Fit Model platform (and ARIMA in Time series) in JMP, SAS program code can be automatically generated and then submitted by JMP. Let's try it out:

Select **Help ▶ Sample Data**, then select **Examples for Teaching ▶ Big Class**.

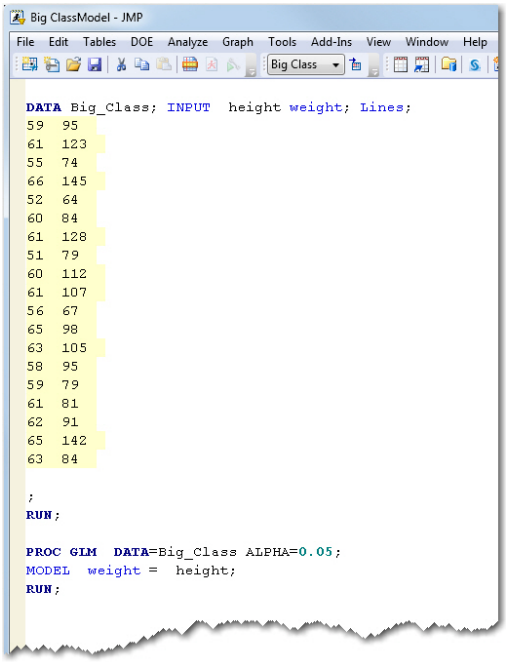
Within JMP, select **Analyze ▶ Fit Model**. Select **weight** for the Y role, select **height,** and click **Add**. Then, select the red triangle/hot spot in the upper left of the launch window and select **Create SAS Job** (see Figure A.10).

Figure A.10 Create a SAS Job from Fit Model



This creates the SAS program in a separate window for the model you specified (see Figure A.11).

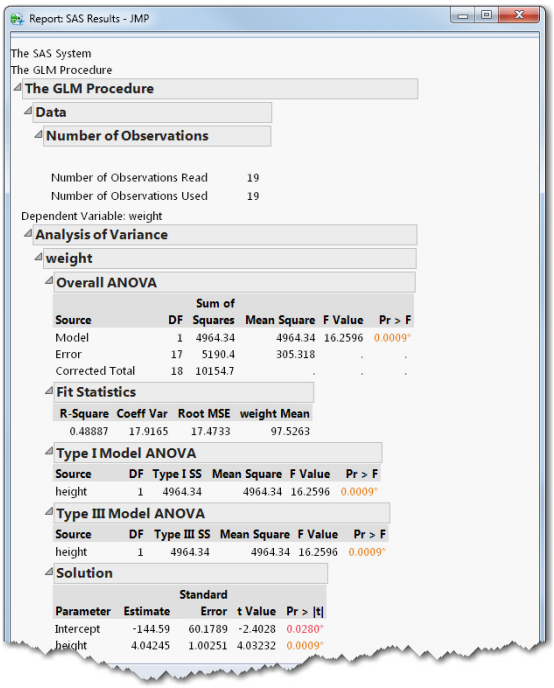
Figure A.11 SAS Code



This SAS program can now be submitted to SAS by right-clicking in the window and selecting **Submit to SAS**.

The program runs in SAS and returns the results to JMP (see Figure A.12).

Figure A.12 SAS Report in JMP Report Window



Of course, in this simple model, JMP could easily generate the results, but when you want to run a very large or complex model, SAS can provide more options and will provide more detailed results.

Note: JMP and JMP Pro also provide a technique using Partition to generate SAS scoring code that can be utilized by SAS predictive modeling products. The idea is that a candidate predictive model can be developed with JMP, then scaled up and enhanced for production purposes using enterprise SAS products.

Using the **Big Class** dataset above, select **Analyze ▶ Modeling ▶ Partition**. Select **weight** for the Y role and select **age, sex and height** for the **X, Factor** role and click **OK**. Then select **Split** (or **Go** if using validation) until the desired model is found. Then select the red triangle for Partition of weight and choose **Save Columns ▶ Make SAS DATA Step**.

Why SAS? Why JMP?

SAS is a highly scalable analytics and data management environment that can handle very small problems up to massive ones. JMP is designed to be easy to use and fast, but your data tables need to fit within the memory (or RAM) available on the desktop when you are using it. As noted in Chapter 2, JMP 11 can handle data table sizes up to half the size of your available RAM. When problem sizes exceed these limits, SAS provides a convenient alternative and hosts a robust set of capabilities to handle truly big data. SAS can process any data size into the terabytes and beyond.

SAS is also a very rich and robust environment for analytics and data management. When larger-sized problems appear, or if you are thinking about production-type data management, security, and reporting, SAS provides the tools and means to handle these needs seamlessly.

Appendix B • Understanding Results

In the process of analyzing data, you will encounter terms in the results that reference vital information. Throughout Chapters 5 and 6, we discussed many of the relevant terms and results in the context of the application and analysis we were performing. This appendix is designed to provide a basic description of the common terms used in each of those platforms to help you understand your own results. We also point out where you can find complete descriptions in JMP's documentation. In most cases, you can simply use the "?" help tool and select the term in question to obtain more information. In other cases, by holding your mouse cursor over statistics, you can see Hover Help, which provides context-specific assistance in interpreting a statistic.

While a graph can be worth a thousand words and is more easily interpreted, the statistical results generated from the Analyze menu contain numerical results and terms that might be less clear. It is important that you understand the basic ideas behind the results. This section offers a basic and general reference rather than a comprehensive one. To increase your confidence in interpreting statistical results, we encourage you to seek advice from an experienced data analyst or reference the books that ship with JMP (see the Help menu) or those in the bibliography.

This appendix covers key terms and related concepts in the book, in alphabetical order. For the purpose of this appendix, the terms *column* and *variable* are interchangeable in these definitions.

Bivariate Plot: Also known as a *scatter diagram or scatterplot* where each point in the plot expresses both an *X* and *Y* value. These two-dimensional plots are used when comparing one continuous column with another continuous column. See **Help ▶ Books ▶ Basic Analysis ▶ Chapter 5: Bivariate Analysis.**

Box and Whisker Plot: Also called a *box plot* or an *outlier box plot*. A graphical presentation of the important characteristics of a continuous variable. Box plots display the interquartile range of the data (the "box"), the spread (the whiskers), and potential outliers (disconnected points). Box plots are useful for describing variables that have a skewed distribution and for comparing two or more distributions.

Confidence Interval: An interval within which we expect a value to fall with a given certainty. For a 95% confidence interval, we are 95% certain, or confident, that a new value will fall within this interval.

Contingency Table: A table showing the observed frequencies of two nominal variables, with the rows indicating one variable and the columns indicating another.

The table reports a chi-square statistic. See **Help ▶ Books ▶ JMP Stat and Graph Guide ▶ Contingency Table Analysis**.

Count: The total number of members in a group.

Correlation: A measure of relationship between two or more variables. It is a relationship where changes in the value of one variable are accompanied by changes in another variable or variables. For example, a correlation of 1 indicates that as one variable increases, the other variable increases the same amount. A correlation of -1 would indicate that as the value of one variable increases, the other variable will decrease by the same amount. See **Help ▶ Books ▶ Basic Analysis ▶ Chapter 5: Bivariate Analysis.**

Degrees of Freedom: Also abbreviated DF, degrees of freedom are associated with many statistical estimates. Intuitively, degrees of freedom are the number of freely varying observations in a data-based calculation. The larger the sample size, the larger the degrees of freedom and the stronger the inferences we can draw about population parameters. See **Help ▶ Books ▶ Basic Analysis ▶ Oneway Analysis**.

Distribution: The values of a single column or variable in terms of frequency of occurrence, spread, and shape. A distribution can be observed (based on data) or theoretical. Some examples of theoretical distributions are normal (bell- shaped), binomial, and Poisson. Distribution is also JMP's univariate platform.

***F* Ratio:** In an ANOVA), the ratio of between-group variability to the within-group variability(See One Way Analysis of Variance). The *F* ratio is used, in conjunction with Prob > *F*, to test the null hypothesis that the group means are equal (that there is no real difference between them). In general, larger *F* ratios indicate significant differences between at least two means. See **Help ▶ Books ▶ Fitting Linear Models ▶ Stand Least Squares Report and Options.**

Frequency: The number of times a categorical value is observed, a type of event occurs, or the number of elements of a sample that belong to a specified group. It is also called *count*.

Interquartile Range: A measure of variability or dispersion for a continuous column calculated as the difference or distance between the 25th and 75th percentiles (the first and third quartiles, respectively). Fifty percent of values fall within the interquartile range and is expressed by the box in a box and whisker plot.

Logistic Regression: A type of regression technique where the *Y*, dependent variable is nominal or ordinal and at least one *X*, independent variable is continuous. Logistic models (sometimes called *logit models*) are used to predict the probability of occurrence of an event based on the values of one or more continuous variables. See **Help ▶ Books ▶ Fitting Linear Models ▶ Logistic Regression with Nominal or Ordinal Responses.**

Maximum: The largest value observed in the sample.

Mean: A measure of location or central tendency of a column of continuous data. It is the arithmetic average computed by summing all the values in a column and dividing by the number of non-missing rows.

Median: A measure of location or central tendency of a continuous column of data. It is the middle value in an ordered column, which divides a distribution exactly in half. Fifty percent of the values are higher than the median and 50% are lower.

Minimum: The smallest value observed in the sample.

Multiple Regression: An analysis involving two or more independent *X* variables as predictors to estimate the value of a single dependent variable. The dependent *Y* variable is usually continuous, but the independent *X* variables can be continuous or categorical (provided that at least one of them is continuous). The model is usually estimated by the method of standard least squares. See **Help ▶ Books ▶ Fitting Linear Models ▶ Stepwise Regression Models.**

One-Way Analysis of Variance (or One-Way ANOVA): A procedure involving a categorical *X*, independent variable and a continuous *Y*, dependent variable. One-way ANOVA is used to test for differences in means among two or more independent groups (though it is typically used to test for differences among at least three groups because the two- group case can be analyzed with a *t*-test). When there are only two means to compare, the *t*-test and the *F*-test (see *F*-test) are equivalent. See **Help ▶ Books ▶ Fitting Linear Models ▶ Examples of Model Specifications and Their Model Fits.**

Outlier: An observation that is so extreme that it stands apart from the rest of the observations; that is, it differs so greatly from the rest of the data that it gives rise to the question of whether it is from the same population or involves measurement error. One common rule of thumb for an outlier is any value that is 1.5 times the interquartile range if the distribution is approximately normally distributed.

Partition: The Partition platform iteratively separates data according to a predictive relationship between a *Y* and multiple *X* values, from strongest to weakest, forming a tree structure. Partition searches through the data table to find values within *X* columns that best predict the outcome of *Y*, your column of interest. Partition is a data mining or predictive modeling technique. See **Help ▶ Books ▶ Specialized Models ▶ Partition Models.**

Prob > *F:* In ANOVA, the probability of obtaining (by chance alone) an *F*-value greater than the one calculated if, in reality, there is no difference in the population group means. Prob (or "p" values) of 0.05 or less are often considered evidence that a model fits the data. See **Help ▶ Books ▶ Basic Analysis ▶ Bivariate Analysis ▶ Fit Line and Fit Polynomial ▶ Linear Fit and Polynomial Fit Reports ▶ Analysis of Variance Report.**

Prob > *t:* A *p*-value or measure of significance for a *t*-test. For a one-sample test or for a test of differences between two means, it is the probability of obtaining a value more extreme than the hypothesized value, if the null hypothesis were true. Prob > *t* values of 0.05 or less are usually

considered significant. See **Help ▶ Books ▶ Basic Analysis ▶ Bivariate Analysis ▶ Fit Line and Fit Polynomial ▶ Linear Fit and Polynomial Fit Reports ▶ Analysis of Variance Report.**

Quantiles: Values that divide an ordered set of continuous data (from smallest to largest) into equal proportions. Related terms are *deciles* (dividing data into 10 parts) and *quartiles* (dividing data into four parts, or quarters). Values in the 97th percentile, or quantile, are equal to or larger than 97% of all values in the distribution.

Quartiles: Values in a continuous column of data that are first ordered (from smallest to largest) and then divided into four quarters, each of which contains 25% of the observed values. The 25th, 50th, and 75th percentiles are the same as the first, second, and third quartiles, respectively. See also **Quantiles**.

Regression: A statistical procedure that shows how two or more variables are related, which is represented in the simple case by a fitted line in a bivariate scatterplot. The fitted line, along with its regression equation, allows one to predict values of *Y* based on observations of *X*. The simple form of this equation is expressed as $y=mx+b$, which determines the extent to which one variable changes with another.

RSquare: A measure of the adequacy of a model defined as a proportion of variability that is accounted for by the statistical model. RSquare provides a measure of how well future outcomes are likely to be predicted by the model. See **Help ▶ Books ▶ Basic Analysis ▶ Bivariate Analysis ▶ Fit Line and Fit Polynomial ▶ Linear Fit and Polynomial Fit Reports ▶ Summary of Fit Report.**

Standard Deviation: A measure of variability or dispersion of a data set calculated by taking the square root of the variance. It can be interpreted as the average distance of the individual observations from the mean. The standard deviation is expressed in the same units as the measurement in question. It is usually employed in conjunction with the mean to summarize a continuous column.

Standard Least Squares: A method of fitting a line to data in a bivariate plot or multiple regression model. Least squares is used where there is one continuous Y and at least one continuous X column in your model. Least squares fits a model that minimizes the total sum of squares in the data, hence the name "least squares." The least squares method is used extensively for prediction and for calculating the relationship between two or more variables. See **Help ▶ Books ▶ Fitting Linear Models ▶ Standard Least Squares Report and Options**.

Sum of Squares: A measure of variation of the model to the observed data. This is calculated by squaring the errors (the vertical distance between each data point and the model fit) and summing those values. We square the values to obtain a positive sum of the errors regardless of whether the observed values are above or below the fit. The best model fit is one where the total sum of squares is minimized.

Appendix C • JMP Shortcuts

This appendix provides the essential menu steps necessary to generate a desired result with JMP. JMP launch windows require that you specify the right data and modeling types in order to generate a desired result. Should you have questions about modeling types or the options displayed in the windows, refer to Chapter 2.

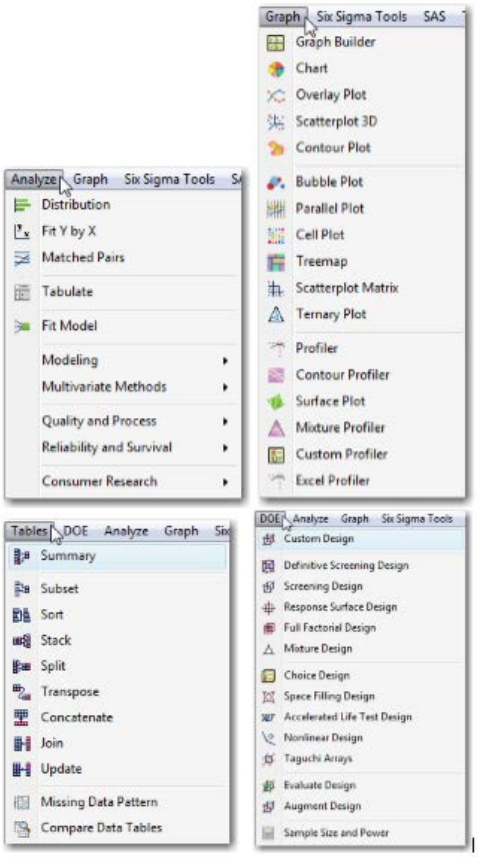
Some launch windows contain options (for example, **Charts, Fit Model**) that should be selected before you execute your results. These steps contain a comma (,) followed by a black inverted triangle (▼) to indicate a drop-down menu within the window for required settings.

A semicolon (;) indicates that there is a secondary step in the process that occurs only after you have executed some base results. The ▼ > symbols indicate a selection is required from a hot spot (or red triangle) drop-down menu that appears in the base results.

There is often more than one way to complete a task (e.g. Chart and Graph Builder); we have utilized the most direct means to achieve theme here.

Some of the tasks included in this section are not covered in the book, but you can reference these tools in JMP's built-in documentation at **Help ▶ Search**.

Figure C.1 JMP Menus



Task	Menu Selection
Adding Labels	*Click on column heading;* **Cols > Label/ Unlabel**
ANOVA -One Way -Two or More Factors	 **Analyze > Fit Y by X; ▼ > Means/ Anova** **Analyze > Fit Model**
Bar Chart	**Graph > Chart**
Basic Charts	**Graph > Chart**
Bivariate	**Analyze > Fit Y by X**
Box Plots -One Level -Two or More Levels	 **Analyze > Distribution; ▼> Outlier Box Plot** **Analyze > Fit Y by X; ▼ > Display** **Options > Box Plot**
Bubble Plot	**Graph > Bubble Plot**
C-Chart	**Analyze > Quality and Process > Control Charts > C**
Chart	**Graph > Chart,▼ *type***
Chi-Square	**Analyze > Fit Y by X**
Color or Mark by Column	**Rows > Color or Mark by Column**
Column Info Dialog	*Click on column heading;* **Cols > Column Info**
Concatenate	**Tables > Concatenate**
Contingency Platform	**Analyze > Fit Y by X**
Control Charts	**Analyze > Quality and Process > Control Chart >** *Select* ***type***
Correlation	**Analyze > Multivariate Methods > Multivariate**
Covariance	**Analyze > Multivariate Methods > Multivariate; ▼ > Covariance Matrix**
CUSUM	**Analyze > Quality and Process > Control Chart > CUSUM**
Data Filter	**Rows > Data Filter**
Data Mining	*see* Partition
Density Ellipses -Bivariate -Scatterplot Matrix	 **Analyze > Fit Y by X; ▼Density Ellipses** **Graph > Scatterplot Matrix; ▼ > Density Ellipses**

Task	Menu Selection
Descriptive Statistics	**Analyze > Distribution; ▼Display** **Options > More Moments**
Design of Experiments	**DOE**
Distribution / Univariate	**Analyze ▶ Distribution**
Distribution Fitting	**Analyze > Distribution; ▼> Continuous Fit (***select distribution* **or** "All"**) or Discrete Fit (***select distribution***)**
Excel Files, to Open	**File > Open >** *Specify Excel file format*
Experimental Design	*see* Design of Experiments
Exponential Smoothing	**Analyze > Modeling > Time** **Series; ▼ > Smoothing Model**
Fit Line	**Analyze > Fit Y by X; ▼ > Fit Line**
Fit Polynomial	**Analyze > Fit Y by X; ▼ > Fit** **Polynomial (***specify degree***)**
Fit Y by X Platform	**Analyze > Fit Y by X**
Forecasting/Time Series	**Analyze > Modeling > Time Series**
Formula Editor	***Click on column heading;*** **Cols > Formula** (or to create new column) **Cols > New; ▼ Column Properties > Formula**
Frequency Distribution	**Analyze > Distribution**
Full Factorial Design	**DOE > Full Factorial Design**
Gauge Chart	**Analyze > Quality and Process > Variability/Attribute Gauge Chart**
Goodness-of-Fit (Test for Normality)	**Analyze > Distribution; ▼Continuous Fit > Normal; ▼(***under* **Fitted Normal) > Goodness of Fit**

Task	Menu Selection
Graphs -Chart -Graph Builder -Overlay Plot -Control Chart -Bubble Plot -Scatterplot -Scatterplot 3D	 **Graph > Chart** **Graph > Graph Builder** **Graph > Overlay** **Analyze > Quality and Process > Control Chart > *type*** **Graph > Bubble Plot** **Analyze > Fit Y by X** **Graph > Scatterplot 3D**
Graph Builder Platform	**Graph> Graph Builder**
Histograms	**Analyze > Distribution**
Histogram Color	*right-click on histogram* **>** *select* **"Histogram color"**
Holt-Winters	**Analyze > Modeling > Time Series; ▼ > Smoothing Model > Winters Method**
Horizontal Bar Chart	**Graph > Chart, ▼ Horizontal**
IR Chart	**Analyze > Quality and Process > Control Chart > IR Chart**
Joining	**Tables > Join**
Kruskal-Wallis Test	**Analyze > Fit Y by X; ▼ > Nonparametric > Wilcoxon Test**
Least Squares Regression	*see* Regression, Simple or Multiple
Line Chart	**Graph > Chart, ▼ Line Chart**
Logistic Regression	*see* Regression, Logistic
Mapping data	**Graph > Graph Builder, Drag Shape column into Shape Map box**
Mosaic Plot	**Analyze > Distribution; ▼ > Mosaic Plot**
Mosaic Plot with Two Columns	**Analyze > Fit Y by X**
Moving Averages	**Analyze > Modeling > Time Series; ▼ > Smoothing Model > Simple Moving Average**
Moving Range Chart	**Analyze > Quality and Process > Control Chart > IR,** *check* **Moving Range (Average)** *box*

Task	Menu Selection
Multiple Comparisons	**Analyze > Fit Y by X; ▼> Compare Means (***specify comparison***)**
Multiple Regression	*see* Regression, Multiple
Multivariate Platform	**Analyze > Multivariate Methods**
Oneway ANOVA	**Analyze** > **Fit Y by X; ▼> Means/Anova/Pooled t**
Outlier Box Plot	*see* Box Plot
Overlay Plot	**Graph** > **Overlay**
p Chart	**Analyze** > **Quality and Process** > **Control Charts** > **P**
Parallel Plot	**Graph** > **Parallel Plot**
Pareto Diagram	**Graph** > **Pareto Plot**
Partition	**Analyze** > **Modeling** > **Partition**
Phase Chart	**Graph** > **Control Chart** > *select type, specify column in* **Phase** *role*
Pie Chart	**Graph** > **Chart, ▼ Pie Chart**
Pivot-Table	*see* Tabulate
Point Chart	**Graph** > **Chart, ▼ Point Chart**
Power Calculations	**DOE** > **Sample Size and Power**
Prediction Profiler	**Analyze** > **Fit Model; ▼** > **Factor Profiling** > **Profiler**
Predictive Modeling	*see* Partition, Profiler, Fit Model
Profiler	*see* Prediction Profiler
Process Control	*see* Control Charts
R-Chart	**Analyze > Quality and Process > Control Chart > R**
Recursive Partitioning	*see* Partition
Regression, Simple -with Line Fit -with Polynomial Fit -with Spline Fit	 **Analyze > Fit Y by X; ▼ > Fit Line** **Analyze > Fit Y by X; ▼ > Fit Polynomial** **Analyze > Fit Y by X; ▼ > Fit Spline**
Regression, Logistic	*see* Regression, Simple or Multiple
Regression, Multiple	**Analyze > Fit Model**
Regression Trees	*see* Partition
Residual Analysis	**Analyze > Fit Model; ▼ > Row Diagnostics > *select type***
Response Surface	**DOE > Response Surface Design**
Run Charts	**Analyze > Quality and Process > ControlChart > Run Chart**
Sample Size Calculations	**DOE > Sample Size and Power**

Task	Menu Selection
S-Chart	**Analyze > Quality and Process > Control Charts > x-bar, check** "S" **box**
Sample Data Directory	**Help > Sample Data**
Scatterplot -3D Scatterplot -Matrix -with Line -with Polynomial Fit -with Spline Fit	**Analyze > Fit Y by X** **Graph > Scatterplot 3D** **Graph > Scatterplot Matrix** **Analyze > Fit Y by X; ▼ > Fit Line** **Analyze > Fit Y by X; ▼ > Fit Polynomial** **Analyze > Fit Y by X; ▼ > Fit Spline**
Screening Design	**DOE > Screening Design**
Simple Regression	*see* Regression, Simple
Sort	**Tables > Sort**
Spearman's Rho	**Analyze > Multivariate Methods > Multivariate; ▼ > Nonparametric Correlations > Spearman's *p***
Stem-and-Leaf	**Analyze > Distribution; ▼ > Stem and Leaf**
Stepwise Regression	**Analyze > Fit Model, ▼Stepwise**
Subset	**Tables > Subset**
Summary Summary Statistics Analyze > Distribution	**Tables** > **Summary** **Analyze** > **Distribution**
t- or z-Test -One Sample -Two Sample -Paired t-Test	 **Analyze** > **Distribution; ▼** > **Test Mean** **Analyze** > **Fit Y by X; ▼** t Test **Analyze** > **Matched Pairs**
Tables -Sort -Subset -Join -Concatenate -Summary	 **Tables > Sort** **Tables > Subset** **Tables > Join** **Tables > Concatenate** **Tables > Summary**
Tabulate Platform	**Analyze > Tabulate**

Task	Menu Selection
Test for Equal/Unequal Variances	**Analyze > Fit Y by X; ▼ > Unequal Variances**
Test for Proportions -One Proportion -Two Proportions	 **Analyze > Distribution; ▼ > Test Probabilities** **Analyze > Fit Y by X**
Test for Normality (Goodness-of-Fit)	**Analyze > Distribution; ▼ > Continuous Fit > Normal; ▼(*under* Fitted Normal) > Goodness of Fit**
Time Series Plot	**Analyze > Modeling > Time Series**
Tree Map	**Graph > Tree Map**
Two or More Factor ANOVA	**Analyze > Fit Model**
Tukey Box Plot	*see* Box Plot
U-Chart	**Analyze > Quality and Process > Control Chart > U**
Univariate/Distribution Platform	**Analyze > Distribution**
Variability Chart	**Analyze > Quality and Process > Variability/Gauge Chart**
Wilcoxon Rank Sum Test	**Analyze > Fit Y by X; ▼ > Nonpara- metric > Wilcoxon Test**
Wilcoxon Signed Rank Test	**Analyze > Distribution; ▼ > Test Mean,** *check* **Wilcoxon Signed Rank** *box*
XBAR Chart	**Graph > Control Chart > XBAR**

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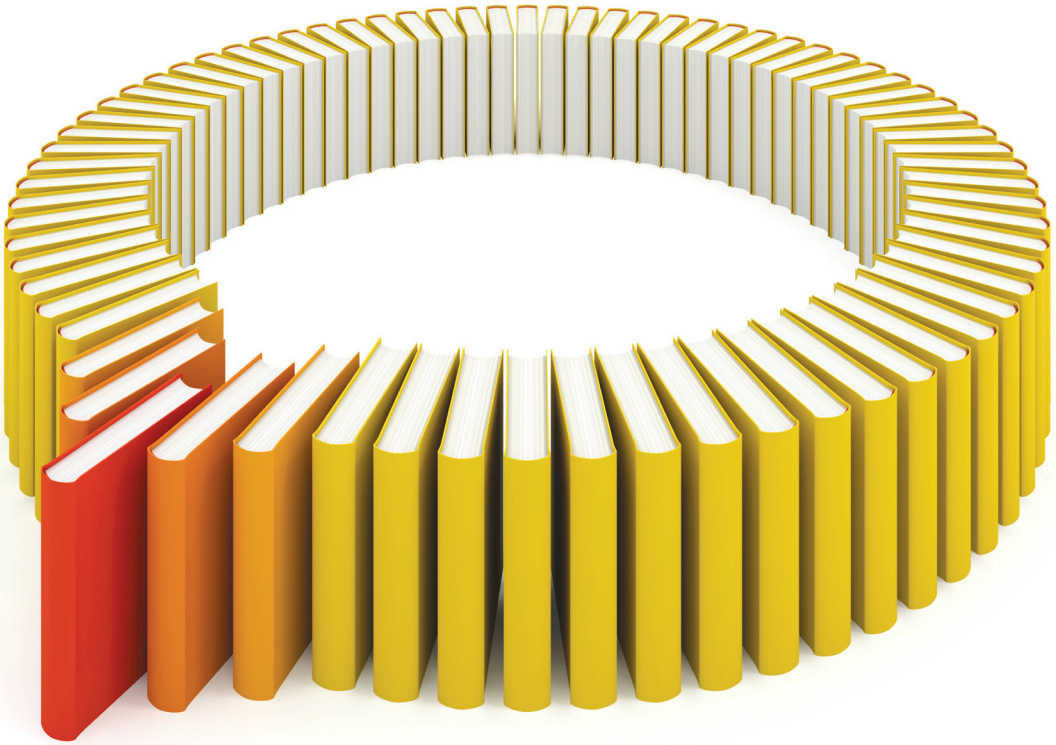

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